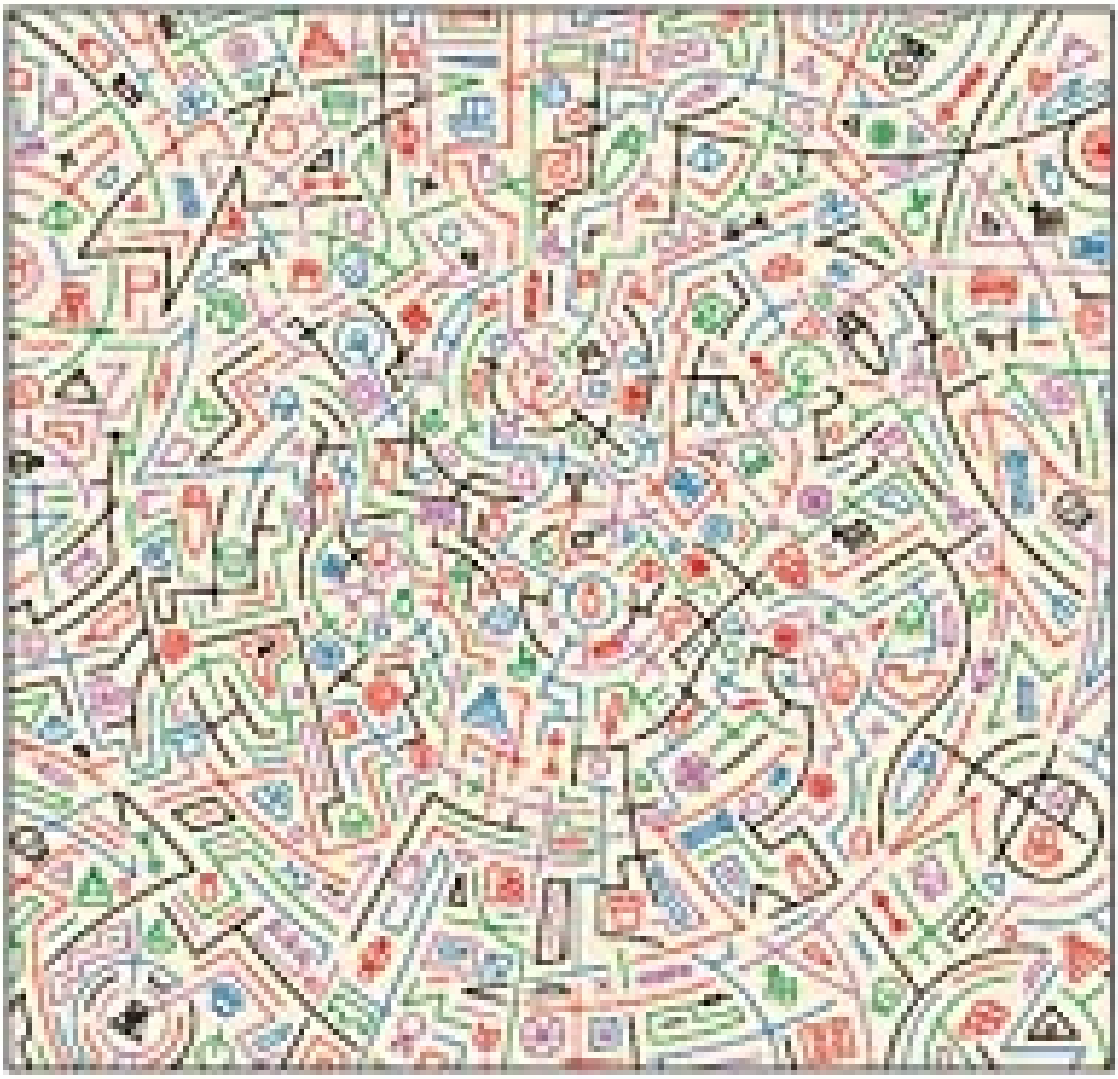
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Network Science in Archaeology

Tom Brughmans and Matthew A. Peeples



Network Science in Archaeology

Network Science in Archaeology provides the first comprehensive guide to a field of research that has firmly established itself within archaeological practice in recent years. Network science methods are commonly used to explore big archaeological datasets and are essential for the formal study of past relational phenomena: social networks, transport systems, communication, and exchange. The volume offers a step-by-step description of network science methods and explores its theoretical foundations and applications in archaeological research, which are elaborately illustrated with archaeological examples. It also covers a vast range of network science techniques that can enhance archaeological research, including network data collection and management, exploratory network analysis, sampling issues and sensitivity analysis, spatial networks, and network visualization. An essential reference handbook for both beginning and experienced archaeological network researchers, the volume includes boxes with definitions, boxed examples, exercises, and online supplementary learning and teaching materials.

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CONTENTS

List of Tables *page* viii
List of Boxes ix
Acknowledgments x
Introduction to the Online Resources Associated with This Book xii

1 Introducing Network Science for Archaeology 1
1.1 What Are Networks and What Is Network Science? 1
1.2 Where Does Network Science Fit in Archaeology? 11
1.3 Trends in Archaeological Network Research 16
1.4 How to Read This Book 20
1.5 Summary 23
Further Reading 24

2 Putting Network Science to Work in Archaeological Research 26
2.1 Introduction 26
2.2 Approaches to Networks in Archaeology 27
2.3 Material Culture Networks 28
2.4 Movement Networks 36
2.5 Spatial Proximity Networks 42
2.6 Visibility Networks 44
2.7 Applications Many and Varied 47
2.8 Datasets and Exercises 52
2.9 Summary 62
Further Reading 62

3 Network Data 64
3.1 What Are Network Data? 64
3.2 Network Data Formats 67
3.3 Types of Networks 72
3.4 Longitudinal Network Data 92
3.5 Best Practice Guidelines 97
3.6 Summary 98
Further Reading 99
Exercises 99

4 **Exploratory Network Analysis** 103
4.1 What Is Exploratory Network Analysis? 103
4.2 Which Analytical Method Should I Use? 104
4.3 How Do I Interpret My Analytical Results? 105
4.4 Exploratory Network Analysis Methods 107
4.5 Don't Stop Here! 143
4.6 Summary 145
Further Reading 146
Exercises 146

5 **Quantifying Uncertainty in Archaeological Networks** 149
5.1 Error and Uncertainty in Networks and Beyond 149
5.2 Missing or Poor Quality Information in Archaeological Networks 155
5.3 A General Approach to Uncertainty for Archaeological Network Data 162
5.4 Coping with Uncertainty 186
5.5 Summary 188
Further Reading 189
Exercises 190

6 **Network Visualization** 193
6.1 A Picture Is Worth a Thousand Words 193
6.2 Node Placement and Graph Layout 200
6.3 Visualizing Node and Edge Properties and Attributes 209
6.4 Visualizing Communities and Groups 220
6.5 Visualizing Networks through Time 225
6.6 Interactive Visualizations 230
6.7 Case Study: Visualizing Networks in the US Southwest 231
6.8 Summary 234
Further Reading 235
Exercises 236

7 **Spatial Networks and Networks in Space** 237
7.1 What Are Spatial Networks? 237
7.2 Archaeological Spatial Networks 239
7.3 Planar Networks 241
7.4 Exploratory Network Methods Designed for Spatial Networks 244
7.5 Spatial Network Models 246
7.6 Case Studies 254
7.7 Summary 259
Further Reading 260
Exercises 261

8 **Uniting Theory and Method for Archaeological Network Research** 263
8.1 The Potential of Relational Thinking in Archaeology 264
8.2 Developing a Trajectory of Archaeological Network Research 266

8.3 Relational Theories Here, There, and Everywhere 271
8.4 Onward and Upward 280

Appendix A Answers for Exercises 281
Appendix B Software 291
Glossary and Graph Theoretic Notation 294
References 311
Index 344

TABLES

2.1 An overview of the use of network data in archaeology *page* 30

4.1 The top three scoring settlements according to degree, closeness, betweenness and eigenvector centrality (a) for the basic Roman road network (Fig. 4.13a) and the inclusion of isolates in this network, (b) within a 50-kilometer buffer, and (c) by connecting them to the nearest neighbor in the basic network 142

BOXES

2.1 Ceramic similarity networks for design and technological style in the southern Appalachian region *page* 34
2.2 Space syntax approaches to a city neighborhood in Roman Ostia 40
2.3 Architectural inscriptions and stone monuments in the Maya region 48
4.1 Exponential random graph models (ERGMs) 117
5.1 Explanation of Spearman's ρ 161
5.2 Boxplots 169
6.1 Be kind to the color blind! 210
7.1 Space matters 239
7.2 A model of settlement hierarchy in Bronze Age Crete 253

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INTRODUCTION TO THE ONLINE RESOURCES ASSOCIATED WITH THIS BOOK

This book provides a detailed introduction to the concept of network research in archaeology as well as a guide to many specific network data management and analytical methods and models. The text of the book itself has been designed to stand alone and can be read and used without reference to any other external resources. We realize that many readers may want more guidance to help them get started doing actual analyses. To meet this need, we have created an elaborate Online Companion to the book (Peeples and Brughmans, 2022), which provides downloadable archaeological network datasets and an extensive Markdown document for the R programming language (R Core Team, 2021). This R Markdown document goes through examples of network data management and basic analysis techniques for all of the common methods and models covered in the book and also provides unique custom tools designed for more complex analyses such as assessments of uncertainty. Along with this, the Online Companion also contains all of the code and descriptions necessary to replicate the analyses in this book in R (or in a few cases, other software) as well as code needed to recreate data-based figures. The HTML version of the Online Companion is available at https://archnetworks.net and the raw R Markdown document and associated data, images, and code are also available on GitHub: https://github.com/mpeeples2008/ArchNetSci. These resources will be periodically updated by the authors and also include a public feedback function to allow users to ask questions, contribute data or methods, and develop resources for classroom teaching or self-teaching. We hope to build a community of active archaeological network researchers around these resources.

1

Introducing Network Science for Archaeology

1.1 WHAT ARE NETWORKS AND WHAT IS NETWORK SCIENCE?

Networks are nothing more than a set of entities and the pairwise connections among them. This simple definition encompasses a tremendous amount of variation from communication systems like the internet to power grids to neurons in the brain to road systems and flights between airports to our own social networks defined through familial ties, acquaintance, or any manner of interaction one could imagine. Over the last 20 years or so, academic interest in networks and the complex properties of network systems has grown by leaps and bounds. This has been mirrored by a growing excitement by the public in general (see best-selling works including Barabási and Frangos 2014 and Watts 2004). It is not uncommon these days to see networks and network visuals used as explanatory tools in news stories or popular articles shared across social media (another kind of network) exploring the complicated connections among characters in television shows, books, or people and organizations involved in news stories. Everyone, it seems, is excited about networks and networks are everywhere.

So, what is the big deal? Why have networks captured so much academic attention if the basic concept of a network is seemingly so simple? The real power of networks for researchers lies in their explanatory and predictive power across a wide variety of social and natural phenomena. There is a long tradition of social network analysis in the social sciences, and in particular sociology, going all the way back to the 1930s (see Freeman 2004 for a historical account). This work has shown that formally defining and measuring the properties and structure of social relationships often reveals features of social systems that are otherwise hidden if we only consider the attitudes and attributes of the people or other units involved. Since the late 1990s, a different set of network concepts have also taken hold among

researchers interested in complex systems in physics, biology, and related fields (see Newman 2010). The excitement in this realm is largely focused on the availability of massive datasets and the emerging realization that networks comprising phenomena as diverse as the internet, the human brain, and even human and animal social networks are apparently governed by some common organizing principles and sometimes exhibit similar dynamics. Network science is a rapidly growing interdisciplinary field sitting at the intersection of these traditions of research, which promises to provide new insights along the edges of traditional disciplinary inquiry.

By way of example, showing the potential power of networks and what we can learn from them, we can turn to a classic study focused on one famous political dynasty, the House of Medici in the early Renaissance. Over the course of the first few decades of the 15th century, the House of Medici in the Republic of Florence rose from one of a number of wealthy families vying for power to a dynasty wielding unprecedented political, economic, and religious authority for centuries. The Medicis' influence eventually extended well beyond Florence, producing three Popes and numerous other high-ranking officials across the Italian peninsula and Europe. So, what explains the meteoric rise of the Medici dynasty? Was it purely their wealth? While the Medici family was among the wealthy families in Florence, there were many other rivals who equaled or surpassed them. Were the Medicis simply master strategists? To the contrary, historic accounts from the period describe Cosimo de' Medici in particular as enigmatic, reactive, and passive in dealings both public and private, with no apparent specific overarching goals (Padgett and Ansell 1993:1262–1263). Why, then, did the Medici dynasty rise so dramatically when so many others fell?

In the early 1990s John Padgett and Christopher Ansell set out to answer this question in an innovative and influential historical study focused on understanding the potential role of networks in the rise of the Medicis' political power and social influence (Padgett and Ansell 1993). Relying on the detailed work of historians outlining the business and personal dealings of the Florentine elite, Padgett and Ansell were able to reconstruct networks of marriage, economic relationships/business co-ventures, and patronage among the prominent 15th-century Florentine families (Fig. 1.1). This research revealed something surprising. While most of the prominent families were mutually connected in a single dense set of complex and overlapping relations, the Medici family consistently fell in a more intermediate position for different kinds of relations. Indeed, the Medicis had both more diverse connections (they married and created business ventures with many different families) and they tended to interact with families that were not otherwise interacting. Padgett and Ansell argue that this allowed the Medici family to develop

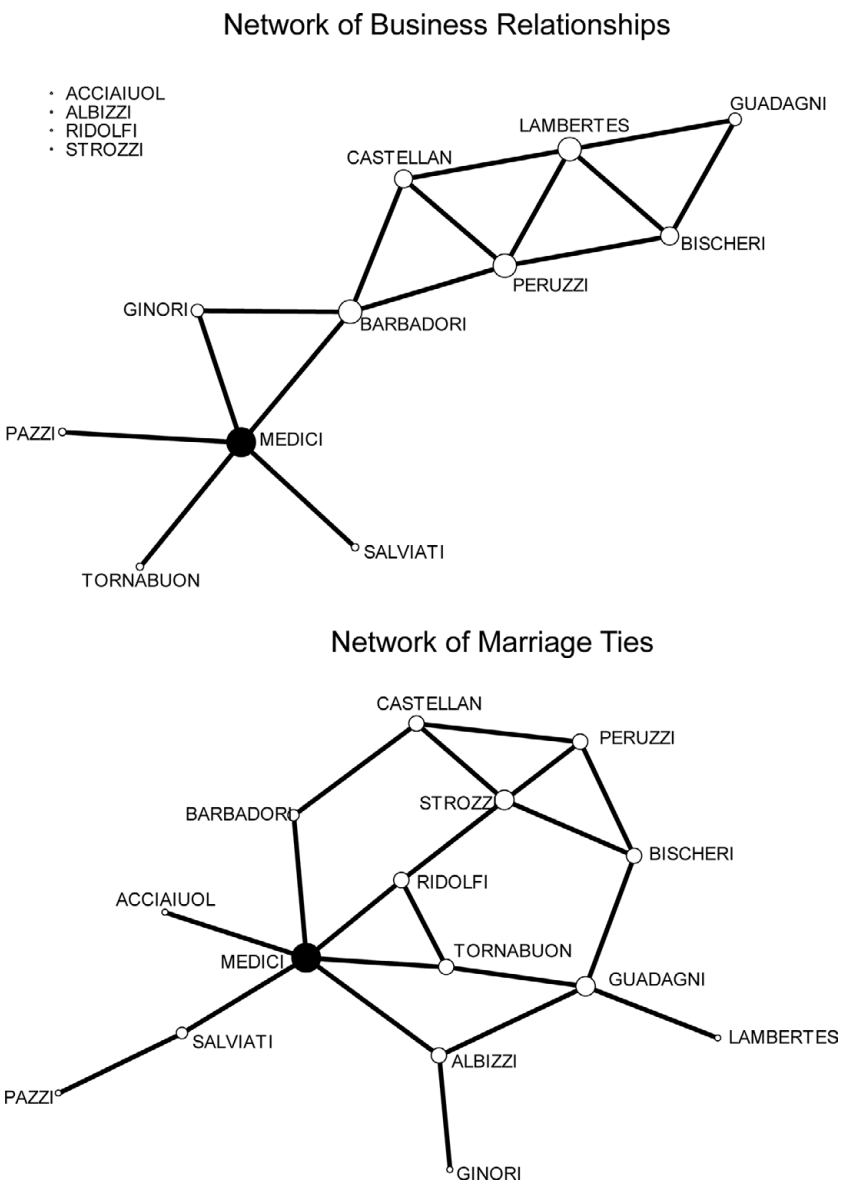


FIG. 1.1 Network diagrams showing business relationships and marriage ties among prominent families in 15th-century Florence based on data published by Padgett and Ansell (1993). Note that the Medici family in both networks has both more connections than other families and connects many families that are otherwise not connected.

a coalition where everyone involved was connected *through* them, and thus they became the center of their socioeconomic sphere. Padgett and Ansell use historical accounts to convincingly argue that the individual decisions in this process leading toward the dominance of Medici authority were largely unintentional and were the product of dynamics common to a wide variety of network systems. Only after they were well established did the Medici family learn the true potential of what they had built.

The rise of the Medici family underscores a few important features about the nature of networks in general. First, networks and positions within them matter in a real, material sense. Beyond this, thinking about and formally tracking relations can often reveal surprising patterns that would otherwise be difficult or impossible to recognize through analyses focused on the attributes of the people or entities involved alone. The formal study of networks can tell us much about the relative importance and influence of the actors within that network as well as the processes behind significant changes in interaction over time. For archaeologists, the use of network concepts and network methods pushes us to think about the *relationships* driving social change in addition to exogenous processes that have often been given priority. The application of network science approaches to archaeological data has a great deal of potential to develop new insights into old questions as well as a whole body of well-developed and interesting research questions that are new to archaeology.

In this book, our goal is to both introduce network science and a wide variety of network methods for an archaeological audience, and also to make an argument for the importance of *relations* and *relational data* for understanding many natural and social phenomena that are of interest to archaeologists. In the remainder of this chapter, we set the stage by providing some basic definitions and concepts as well as a brief overview of the history of networks in archaeology, the place of network science in archaeological research, and the organization of this book.

1.1.1 *Basic Concepts*

For the purposes of this book, we define a **network** as a formal system of interdependent pairwise relationships among a set of entities (or actors). Networks are often represented and visualized as *graphs* with the actors in question depicted as a set of *nodes* or *vertices*, and the relationships among them drawn as lines, typically referred to as *edges* or *ties* (see Fig. 1.2). In this book we will use the term **node** to

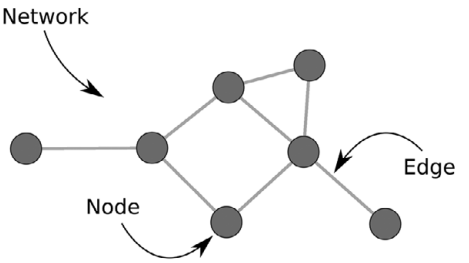


FIG. 1.2 Example network showing nodes and edges.

refer to the entities in a network and the term **edge** to refer to relationships between node pairs. In practice, many network researchers use the term *network* to refer to various kinds of network representations (like graphs and other mathematical notations), and for the sake of simplicity, we follow that general rule here. We do wish to note, however, the subtle but extremely important distinction between a network as a system of relations and a network representation as a formal abstraction of that system (see Chapter 3 for an in-depth discussion of the connection between networks, network data, and network representations).

A *network* is a formal system of interdependent pairwise relationships among a set of entities.

Networks involve the formal definition of *nodes* as the entities in question and *edges* as the relationships among them.

A *network representation* is a formal abstraction of a network created for the purposes of visualization or analysis. In this book, network representations are simply referred to as networks.

In this book we consistently use the terms *network*, *node*, and *edge*. However, alternative terms are common in other disciplines:

- A *network* is often called a *graph* in mathematics and computer science.
- A *node* is often called a *vertex* in physics, mathematics, and computer science, and an *actor* in sociology.
- An *edge* is often called a *link* in computer science, a *bond* in physics, and a *tie* or a *relationship* in sociology.

The nodes in a network can represent almost any kind of entity, from individuals or larger collectives (lineages, villages, corporations, nations, etc.) to objects,

geographic locations, or even events. Likewise, the kinds of relationships that can be used to define edges are virtually endless, ranging from all manner of direct social interactions to biological relationships; flows of information, influence, or goods; shared participation in organizations or trends; and geographic proximity. Although the nature of nodes and edges in any given network representation can be quite variable, the important point is that they are defined using consistent criteria across all actors and relations within a given context.

Formal network research typically involves both the visualization of networks as well as the calculation of a broad range of statistics designed to quantify various aspects of network structure and node or edge position. **Network structure** is a general concept referring to the form and properties of a network. Are all actors connected or are there many isolated nodes? Is there a tendency for clustering or subgroup formation, or are all nodes connected in one dense set of relations? Are there intermediate nodes between otherwise separate clusters or are clusters wholly isolated from one another? Are some nodes more central to certain kinds of flows than others? There are a variety of network analytical tools designed both to represent such features of network structure (and many, many others we will discuss in the coming chapters) and to explain variation in such features at both the local (node/edge) and global (network-level) scales.

Network structure refers to the general properties of a network including the overall patterns of relations, the presence/absence and nature of subgroups, the variation in the positions of actors within that network, and a broad range of other potentially salient features of organization.

1.1.2 *The Relational Perspective*

So, what makes networks special? Creating a network representation of some system of pairwise relationships can often be quite informative in and of itself. Network visuals are striking and can reveal important organizational principles of a system that are not otherwise apparent (see Chapter 6). We argue, however, that the true utility of network approaches lies in the **relational perspective** fundamental to the study of networks, that is, the underlying assumption that the nature and structure of *relationships* among actors are as important (or in some cases, more important) for understanding and predicting the behavior of actors in a network than the attributes of those actors themselves (see Chapter 8 for an extended argument). For example,

the relational perspective suggests that in order to understand the role and importance in the Roman Empire of the province of Baetica in southern Spain, we cannot simply refer to the mineral wealth or agricultural potential of the region. Rather, we must consider how the region's position in the Roman transport system and the social relationships of its inhabitants that tied it to the capital of Rome were likely crucial in making it one of the most important and affluent regions in the Roman empire that eventually provided the first non-Italian emperors. Network methods and models help us formally describe and analyze such relational patterns.

Over the last two decades many archaeologists and material cultural specialists working in a variety of contexts have begun to shape what they see as a new direction in investigations of objects and identities sometimes glossed as "the relational turn" across many areas in the social sciences and humanities (see Harrison-Buck and Hendon 2018; Selg and Ventsel 2020; Van Oyen 2016). In our view, this relational turn does not represent a single paradigm but generally groups works that focus on the primacy of relations not just as drivers of social change but as constitutive components of persons and objects themselves. Within such perspectives entities and relations cannot be wholly separated and the agency of nonhuman entities is explicitly considered. Such relational perspectives draw on diverse theories and concepts including interpretive models like Actor-Network Theory (Latour 2005), entanglement theory (Hodder 2012; Hodder and Mol 2016), relational notions of personhood (Strathern 1988), and assemblage theory (DeLanda 2016), among many approaches. Some researchers have begun to consider the potential connections between such theoretical models and formal network methods and data (Knappett 2011, 2018, 2020; Knutson 2021; Van Oyen 2015, 2016) though empirical evaluations of such perspectives have been rare as of yet. In this book, we use the concept of the "relational perspective" in a somewhat narrower sense, focusing explicitly on the material role that relations in networks play in generating outcomes for actors within those networks. There is clearly overlap between formal network methods and broader notions of relationality that will likely continue to be explored (e.g., chapters in Donnellan 2020).

The *relational perspective* at the core of formal network approaches is the notion that the structural properties of networks and variation in the positions of nodes and edges in a network are just as important for explaining or predicting the behavior of the actors of that network as the attributes of the social actors themselves.

1.1.3 *Network Science and Network Theory*

One criticism often directed toward network research is that it is simply a loosely defined set of methods and mathematical tools and does not constitute an explicit approach with its own unique theoretical underpinnings, research agendas, and insights (see discussion in Borgatti and Halgin 2011b; Borgatti et al. 2009). Are networks simply tools to get a job done or do networks also offer a fundamentally new theoretical perspective? In this book, we argue that we can have it both ways to a certain extent (see also Peeples 2019). Network approaches can be profitably used as analytical tools to address a number of traditional archaeological research concerns (a hammer and some nails), but network approaches also offer exciting novel research agendas beyond the realm of traditional archaeological questions (the plans to build a fancy new gazebo). In this book, we attempt to walk the fine line between these two perspectives, exploring both the practical methodological aspects of the network approach as well as what we see as the deeper theoretical insights the approach has to offer. We suggest that network perspectives and network methods have the potential to open up archaeological investigations to a broad array of important topics that have, as of yet, seen considerably less attention than they deserve (see also discussions in Brughmans 2013b; Mills 2017; Peeples 2019).

So, what then is the "network science" where this book gets its title? Here we borrow a useful definition from the inaugural issue of the journal *Network Science*. Brandes et al. (2013:2) define **network science** as "the study of the collection, management, analysis, interpretation, and presentation of relational data." That seems simple enough: network science offers specific methods and tools to deal with relational data consisting of entities and relations, and relations are important for understanding a broad range of phenomena. Network science provides tools to collect the data necessary to create formal network representations and explore and interpret network structures. For example, in one recent study, Golitko and Feinman (2015) used network science methods and visualization tools to explore the procurement and distribution of pre-Hispanic Mesoamerican obsidian (see also Golitko et al. 2012). This involved collecting data on the frequency of occurrence of objects made from different obsidian sources at a number of important sites and creating a network representation based on the shared frequencies of objects from those sources. In this network representation, sites were defined as nodes and strong similarities in site assemblages based on obsidian sources were represented as edges. They subsequently used this network to explore the relative centrality (importance) of specific sites and areas for directing flows of obsidian across the region, producing results that led to new archaeological insights.

As this brief example illustrates, network science *methods* are certainly useful, but we argue that where the rubber really meets the road when it comes to network

science is when we use these methods to explicitly explore network theories. **Network theories** are formalizable and testable expressions of dependencies (or contingencies) among nodes, edges, attributes, outcomes, or global network structures, or any combination thereof (see Chapter 8). In other words, network theories are formal statements about how one part of a network system or one kind of relationship in a network can influence the development, spread, or decline of some other salient feature of that system or the actors within it (see Chapter 3). As with any theoretical concept in archaeology, network theories can come from traditional archaeological concerns or may be derived from expectations based on the properties of networks themselves. In either case, the application of network theories in archaeology (or any field for that matter) typically involves two basic components: (1) the development of a model for abstracting network concepts and techniques to study a real-world phenomenon as network data and (2) a formal evaluation of the network dependencies, contingencies, and/or relational processes described by a network theory using network data.

This process of abstraction is, of course, not unique to network research. As archaeologists we study diverse phenomena involving past human behavior, but we typically cannot study these phenomena directly. Instead, we must always abstract the phenomenon we are interested in exploring using archaeological concepts and develop tools for representing such concepts using archaeological data. Network approaches to archaeology are no different. For example, let's say we are interested in exploring how the position of a settlement within a regional transportation system influenced the growth of that settlement. In this case, the general archaeological concept we are interested in exploring is the movement of people and resources across a region using transportation corridors, and the implications of this movement for settlement growth. The notion of the "position" of a settlement in relation to such flows can be abstracted using the network concept of "centrality," which refers to a broad set of approaches used to describe the relative importance of nodes for directing or receiving flows across a network (see Chapter 4 for further discussion). In order to represent this concept using archaeological data we could then define a simple point-to-point (settlement-to-settlement) network using sites as nodes and roads connecting them as edges, perhaps with some additional considerations of the length or formality of road segments. From here we have got our network data (derived from archaeological data) and the path ahead is relatively straightforward. We can calculate and evaluate relative differences in network centrality and compare these to attributes of settlements including their size or rates of growth to evaluate our relational theories.

As the discussion above suggests, modeling and abstracting archaeological data into network data is therefore a fundamentally archaeological thing to do. It involves a constant dialogue among archaeological data, disciplinary knowledge, archaeological theory, and network concepts (see also Section 1.2). In archaeological applications of

network science, network theories can include theories about how a network structure evolves, how processes and flows take place in relation to the network structure, how all aspects of the network affect the behavior and opportunities of the nodes and edges, or many more relational questions. Network theories can describe how relational aspects of the past phenomenon of interest functioned, or they can be theoretical arguments about why it is appropriate to use network concepts and data to abstract and represent a given phenomenon. Both of these are network theories. For example, in Golitko and Feinman's (2015) study described above, the authors theorize that top-down control of obsidian production and distribution by major settlements resulted in the important positions of these settlements in the obsidian distribution network: this is an archaeological theory about relational aspects of a past phenomenon. They also suggest that network centrality measures are appropriate representations of these relative importance positions of major centers: this is a theoretical argument about the appropriateness of using particular network methods and representations to address the question at hand in a given context.

To put it simply, **network science in archaeology** is the study of network models and network theories developed for an archaeological research context, and formally expressed and tested using network methods. Although network science techniques without explicit network theories may sometimes offer useful analytical explorations, the ability of such methods to lead to new insights into past human behavior is significantly enhanced *when theory and method are combined*. We cannot emphasize enough that network science can only make unique contributions to our understanding of past human behavior when archaeologists let their use of network science be guided by the specific nature of archaeological research contexts, critical evaluations of archaeological data, and careful considerations of relational theories (see Chapter 8).

Network science is the study of the collection, management, analysis, interpretation, and presentation of relational data.

Network theories are formal and testable expressions of dependencies among nodes, edges, attributes, outcomes, or global network structures or any combination thereof. They express why and how relationships matter in a certain research context.

Network science in archaeology is the study of network models and network theories developed for an archaeological research context, and formally expressed and tested using network science methods.

1.2 WHERE DOES NETWORK SCIENCE FIT IN ARCHAEOLOGY?

So far we have covered what network science is and how it differs from other approaches. But how does it fit with what we do as archaeologists? Where should we position network methods and models in our research process and thinking? In this section we will provide an abstract overview of how network science might be incorporated in a generalized archaeological research process. We intentionally do not go into many specifics here. In Chapter 2 we provide a wide range of archaeological examples of the use of network science for studying past relational phenomena to give you a sense of the current landscape of archaeological network research.

Figure 1.3 offers a graphical representation of our argument regarding the place of network science in the archaeological research processes. The first thing to notice is that *doing network science in archaeology* is necessarily *doing archaeology*. Network science applied to archaeological research is a subset of archaeological research: it does not happen in isolation, it is not immune to the limitations of archaeological data nor does it replace archaeological theory. Just like any other formal or informal approach applied to our discipline, network science cannot be considered a black box positioned outside our discipline – a black box into which

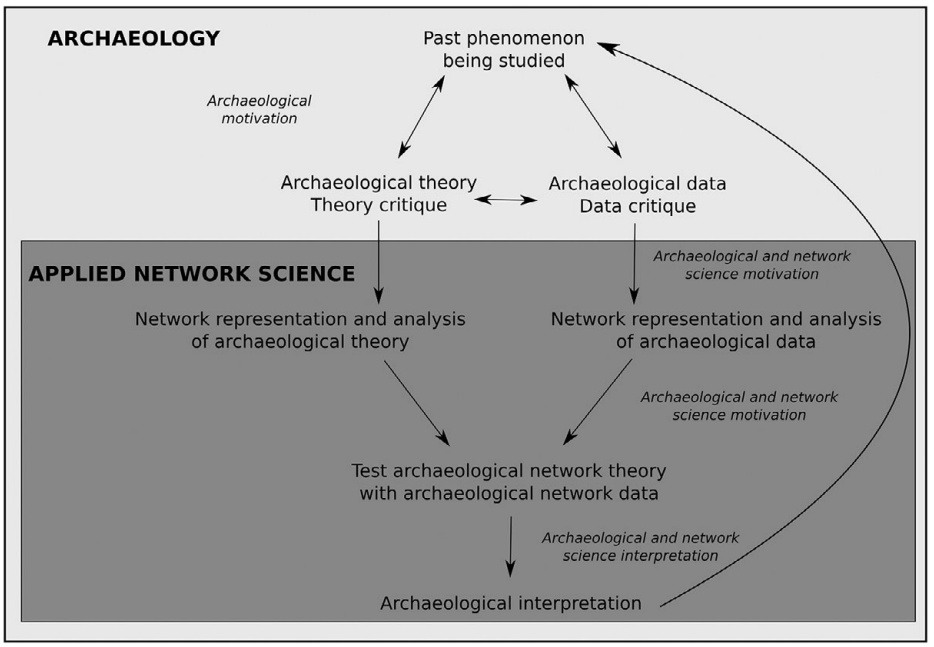


FIG. 1.3 Generalizing abstraction of a typical archaeological network research process.

archaeological theories and data are inserted, and out of which a ready-made archaeological interpretation comes. Every step of the process of network science demands archaeological argumentation and a keen awareness of the specific properties and limitations of archaeological data. Why is network science necessary in this archaeological research context? What limitations do archaeological data impose on my use of network science techniques or models? Can my archaeological theories be appropriately represented using network science concepts and tools? Failing to address these questions is bad network science *and* bad archaeology. Doing critical network science research inevitably requires us to do critical archaeological research. In this book we will guide you through the process of doing **critical archaeological network research**, by continually emphasizing the archaeological questions you should be asking yourself as part of critical network science.

All archaeological research aims to help us better understand a particular past phenomenon related to human behavior. This is why in the abstract research process shown in Figure 1.3 we have placed the past phenomenon under study at the top. Archaeologists collect data to explore this past phenomenon and we formulate theories to describe that past phenomenon, with data and theory engaging in a constant dialogue. For example, if we wish to study the movement of individuals through ancient Pompeii (Fig. 1.4), we must first collect and critically assess information about past excavations of the town. Perhaps this data collection will suggest areas where we might be missing parts of the town plan or other data, triggering fresh excavations or other additional research. In our evaluation of these data, we will identify structures that will help us reconstruct the town plan. Critical assessments of this archaeological information will allow us to attach varying degrees of uncertainty to each element of this reconstruction. From here, we can develop expectations about how Romans entering the town might have moved around within it: for example, based on our knowledge of Roman towns in general, we might suggest that the forum had a gravitational pull or funneling effect on all movement through the town.

So far we have merely described archaeological research: there is nothing particularly "networky" about the previous paragraph. We could start introducing network science into this research process at this point in two ways: *modeling archaeological data* and *modeling archaeological theories*. To *model our data* of Pompeii's town plan with a network representation, we might consider representing each road junction as a node and each road as an edge (Fig. 1.4) to create a spatial road network (see Chapters 3 and 7). The structural properties of the network as a whole and of each node's position can subsequently be studied using exploratory network analytical techniques and metrics (see Chapter 4). We might similarly

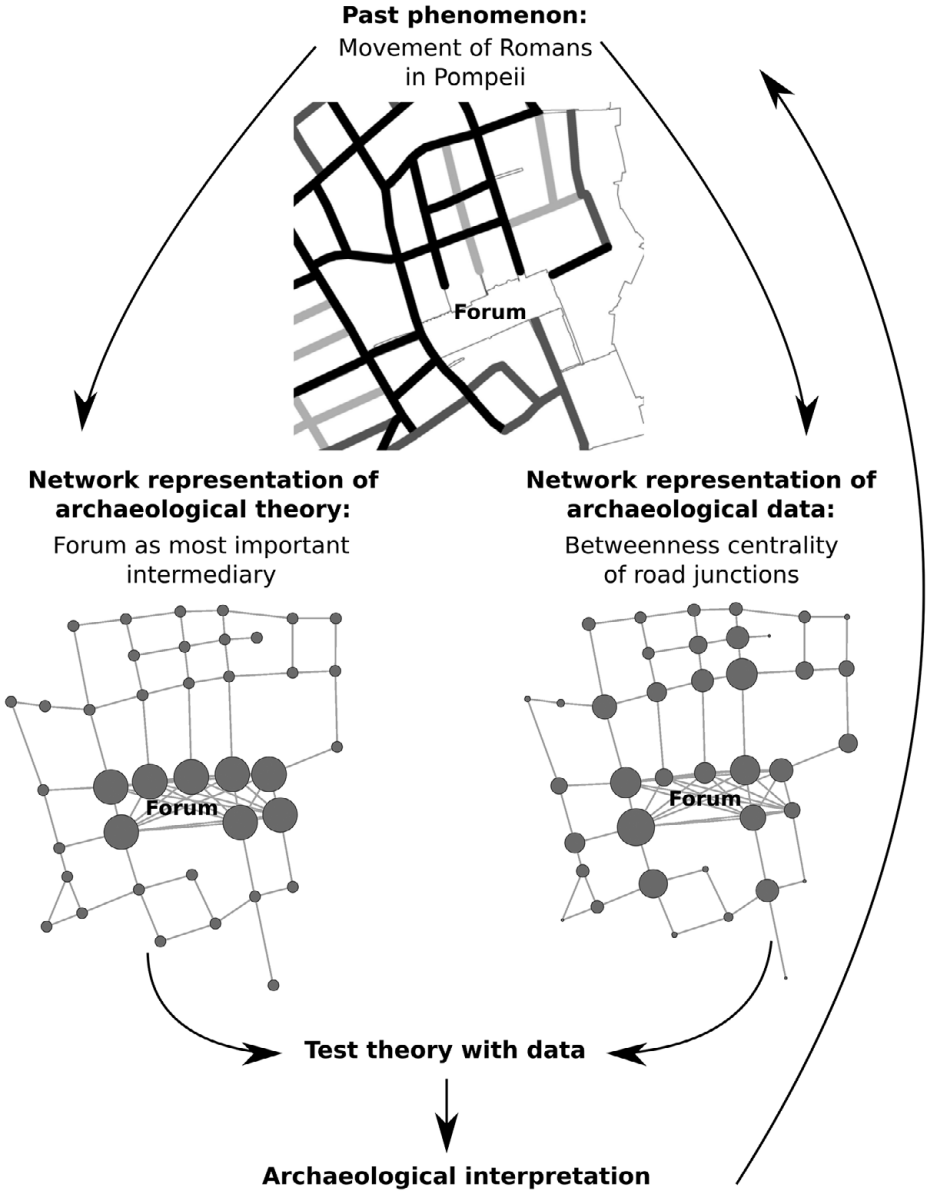


FIG. 1.4 Abstract example of an applied archaeological network research process using a subset of the Pompeiian town plan (excerpt from Poehler 2017: Fig. 6.2). The past phenomenon under study is the movement of people in Pompeii. Our archaeological data can be analyzed as network data by representing road junctions as nodes and road segments as edges. Network data can also be used to represent the archaeological theory of the importance of the forum junctions for mediating the movement of people by representing hypothesized importance of junctions as node attribute values (here represented by node size). The properties of the dataset can be analyzed using an exploratory network analysis technique that is an appropriate representation of the theorized process: betweenness centrality (a measure of the importance of a node sitting on intermediate paths across the network [see Chapter 4], here represented by node size). Comparing theorized node attribute values with betweenness centrality scores allows us to test the theory and interpret the results to gain insights into the past phenomenon.

model our theory of movement through Pompeii by attaching an attribute representing the theorized importance of each junction to all nodes in this network and assigning higher importance values to junctions along the forum as compared to other junctions in the town plan (Fig. 1.4). We can then again use network metrics to study this formal network representation of our theory: What are the relationships between our attributes of interest (e.g., distance from the forum) and the structural positions of nodes in the network? Network science allows us to formally express such archaeological network theories, and it allows us to explore what the implications are of our theory for its constituent parts: the roads, junctions, and the Romans that moved over them.

From here, network science tools and techniques can be used to evaluate archaeological theories with archaeological data. A crucial prerequisite for this step is critical evaluation of both archaeological theory and archaeological data: you need to make a convincing argument that a particular archaeological theory is testable with a particular archaeological dataset; you need to explicitly design an analysis that enables this test to be performed; and you need to evaluate the reliability of the archaeological evidence for addressing this theory. Network science offers formal tools to implement such analyses. To facilitate this process, consider asking yourself the following questions:

- What past processes are of interest and how can they be represented using my archaeological dataset?
- What archaeological or relational theory might explain the patterns in my archaeological dataset?
- How could the theory of interest be represented and evaluated using archaeological evidence?
- What are the limits of the data and/or what further archaeological data would I need to collect in order to fully evaluate this theory or represent the network process of interest?

Once we are ready to formally test network theories using archaeological data, there are several ways we might proceed. In many contexts, we can evaluate a network theory directly by representing our archaeological dataset as a network. This approach could be used to test our theory about the importance of the forum and the two main roads in Pompeii. We could use betweenness centrality (see Chapter 4) to calculate in our network representation of archaeological data how important every junction and road was as an intermediary in the movement of people (Fig. 1.4). We could subsequently compare these centrality values from our analysis with those of our formal representation of our archaeological theory to

determine whether the forum does indeed have a higher centrality as predicted by our theory. Formal network theories can also be tested without explicitly creating a network representation from archaeological data. For example, Bentley and Shennan (2003) evaluated network theories focused on cultural transmission using a non-network representation of data patterns in pottery designs from Neolithic Europe. Specifically, they developed expectations for the emergence and popularity of material cultural styles over the course of cultural evolution under different kinds of transmission based on a general stochastic network model and then tested this model using assessments of the frequencies of pottery motifs through time. As this suggests, one can evaluate network models without conducting "network analyses" using archaeological data in the most conventional sense.

Even though formally testing theories in the way we outline here is a constructive way to improve our understanding of past human behavior, do not despair if this is not immediately possible in your research context. Indeed, network science approaches have much to offer archaeological research even if we are not yet prepared or able to directly test formal theories with data. For example, we can use network science concepts and models to formally represent or explore the implications of our archaeological theories in abstract, or we can create exploratory network representations of our archaeological data to get a general sense of structural patterning in our archaeological data before defining our theories or questions. In this way, network science approaches to archaeology can help us explicitly think through and develop our theories, help us better understand and perhaps refine or reformulate our questions, and help us develop predictions of the archaeological patterns we would expect to see as the outcome of our theories in case one day the data necessary to test these theories are available. Network science tools similarly help us explore our archaeological data, understand underlying patterns, and perhaps guide data correction or new data collection.

With the abstract research process described here we aim to illustrate where network science fits in archaeological research on the most general level. Such a network research process in practice should be part of a multi-method approach to understanding past human behavior: there is no reason why every aspect of archaeological network research should be dominated by exclusively network data and tools. Network science offers tools that are sometimes appropriate because they allow the archaeologist to do something they want to do or could not do any other way. But if non-network statistical or spatial analysis techniques are more appropriate for representing an archaeological data pattern, then these should be used alongside network methods.

Network science can only make constructive contributions to archaeological research when it is inserted into the archaeological research process in appropriate places. What counts as appropriate is determined by the theories being evaluated, data critique and research context, and a critical understanding of the network science concepts and tools themselves.

> Doing *critical archaeological network research* requires archaeological argumentation at every step of the network science application process. Why is network science necessary in this archaeological research context? What limitations does my archaeological data critique impose on my use of network science? Can my archaeological theories be represented using network science concepts and tools?

1.3 TRENDS IN ARCHAEOLOGICAL NETWORK RESEARCH

Network research has a long history in the physical, behavioral, and social sciences, but it is only recently that network approaches have gained a major foothold in archaeology. The number of published works focused on the formal analysis of networks in archaeology has increased nearly threefold over the last 10 years (Fig. 1.5) and we see indications that archaeological network research is still on the rise. In this section, we briefly outline the history of network research in archaeology, including what we see as some of the most important recent trends. This brief overview only scratches the surface, however, and we refer readers interested in the history of networks in archaeology to several recently published overviews (Brughmans 2013b; Brughmans and Peeples 2017; Mills 2017; Peeples 2019) that cover various topics in greater detail.

Network approaches as they exist today owe their origins to three major traditions of research: (1) graph theory, (2) social network analysis, and (3) complexity science. Although there is considerable overlap among these areas of research, each also has its own unique research agendas, methods, and tools. Importantly, network methods and models have found their way into archaeology several times somewhat independently over the last 50-plus years or so, inspired by each of these three major traditions.

Graph theory is the mathematical field focused on studying the formal structure of pairwise relationships among entities, often in the form of algebraic matrices. The beginnings of graph theory go back all the way to 1736 when the mathematician Leonhard Euler wrote the first formal theorem (see Biggs et al. 1976). Graph

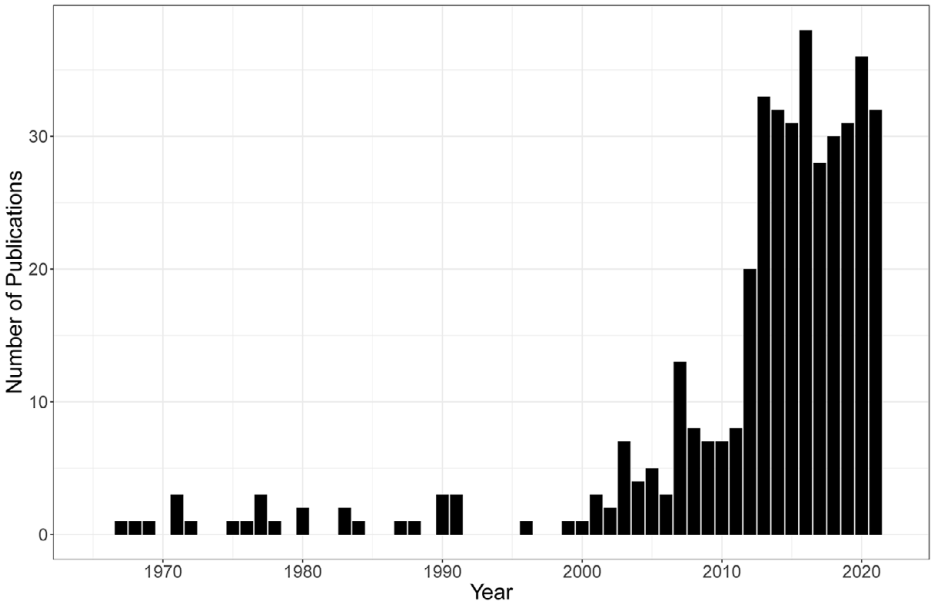


FIG. 1.5 The number of published formal archaeological network studies per year between 1965 and 2021 (using data updated from Brughmans and Peeples 2017).

theory is, in many ways, the mathematical foundation of all other network approaches and has widespread applications in the physical sciences generally for the quantification and explorations of connected systems. In the 1960s and 1970s, a number of geographers began to pick up on graph theoretic concepts and methods to describe and analyze patterns of human land use and settlement (e.g., Chorley and Haggett 1967; Pitts 1965, 1978). By the late 1970s, largely inspired by these geographic studies, a few archaeologists also began to attempt to apply graph-based models to explore archaeological data. The earliest formal graph theoretic studies in archaeology were conducted by John Terrell (1976, 1977) and Geoffrey Irwin (1978), both working in Oceania. These studies relied on the creation of simple graphs based on the geographic proximity of settlements to serve as useful sources of hypotheses about potential interaction among island communities. Indeed, in these early studies, graphs were not seen as "real" networks representing a past reality in any sense but instead as null models that could be used to evaluate archaeological data. Although graph theory remained popular in the archaeology of the Pacific (Hage 1977; Hage and Harary 1983, 1991; Hunt 1988) for a number of years, graph-based methods never took off in archaeology as they did in other closely related fields. This is perhaps, in part, because graph-based models for network data are largely focused on documenting network structures but offer little

guidance in terms of testable models and expectations for the outcomes of such network structures. Between the 1980s and the early 2000s, a small number of graph theoretic studies were published in archaeology focused on many different parts of the world, but these amounted to far fewer than one publication per year (e.g., Broodbank 2000; Jenkins 2001; Peregrine 1991; Rothman 1987; Santley 1991).

Social Network Analysis (SNA) refers to a long tradition of exploring formally defined social relations that has its origins in a number of developments in the early and mid-20th century, including the study of kinship and social structure (sociometry; Moreno 1934; see also Freeman 2004) and the Manchester School of British social anthropology emerging in the 1950s (e.g., Barnes 1954; Bott 1955, 1957; Mitchel 1969; Nadel 1957). Researchers working in this realm were interested in exploring the complex relationships between social structure and social differentiation, and began to develop a number of formal tools for describing variation in social structure toward that end. Many of these early methods shared much in common with contemporary approaches to graph theory, and indeed, early social network scholars began to collaborate with mathematicians to develop these ideas further (e.g., Barnes and Harary 1983; Harary et al. 1965). By the 1970s, SNA had emerged as a distinct paradigm focused on developing both a general theory of structural relations as well as increasingly diverse quantitative tools designed to analyze social structures using network data. SNA today is a vibrant field with numerous dedicated journals and conferences and a dizzying array of methods and applications. Research under the heading of SNA is most often focused on exploring the importance of structural relations among individuals (though not entirely so) and in particular the social processes driving structural tendencies in networks.

Interestingly, SNA was suggested as a potentially useful approach for exploring patterns of interaction among archaeological settlements by Cynthia Irwin-Williams (1977) almost as early as the very first applications of graph-based methods described above. Although SNA methods rapidly gained popularity in the social and behavioral sciences broadly in the 1970s and 1980s, it is not until recently that we see any substantial impact in archaeological research. Beginning only a little over 10 years ago, we have begun to see the proliferation of a number of archaeological studies explicitly inspired by SNA models and methods coming largely out of sociology (e.g., Bernardini 2007; Birch and Hart 2018; Golitko et al. 2012; Hart 2012; Hart and Engelbrecht 2012; Mills et al. 2013a, 2015; Peeples and Haas 2013). Several recent projects represent collaborations among archaeologists and sociologists (such as the Southwest Social Networks Project; Mills et al. 2013a, 2018; see Section 2.8.2). Archaeologists have used tools for quantification and visualization developed in SNA to address archaeological questions at a variety of

scales. For example, Mills and colleagues (2013a, 2013b, 2015) in a series of studies have used SNA methods and theories to explore the impact of a major period of migration on the nature of regional social relations across a large portion of the US Southwest. Although many archaeological forays into SNA largely represent the use of these methods to address traditional archaeological concerns, there have also been several recent studies that explicitly test SNA theories in an attempt to engage in debates beyond the typical borders of archaeology (e.g., Birch and Hart 2018; Borck et al. 2015; Lulewicz 2019; Peeples and Haas 2013), a trend we hope will continue.

Complex network approaches (or complexity science approaches to networks) are the most recent major tradition in network research to develop. This approach largely developed out of work in physics and computer science, going all the way back to the 1960s, but began to emerge as a distinct research tradition in the early 2000s (Newman 2010, 2011). Complex network approaches are focused on exploring the emergence of nontrivial properties of networked systems that are not a property of the individual nodes or relations in that network. Many of the most influential findings in this body of research are focused on the generalizability of networks. In other words, complex network approaches are often focused on the features that are common across all manner of networks. For example, a great deal of work has focused on identifying and explaining the emergence of small-world structure in networks. Small-world networks are networked systems where most nodes are not connected to each other, but almost every node is reachable from almost every other node across only a few steps (Watts and Strogatz 1998; see Chapter 4). Small-world structures have been argued to emerge in a wide variety of real-world networks, suggesting that the principles driving their emergence may be a function of certain kinds of networked relationships generally rather than any specific context or the attributes of nodes or edges in those networks. Complex network approaches are often focused on explaining "global" or graph-scale network properties such as these rather than individual positions, which are more often the focus of SNA studies.

Interest in complex network approaches in archaeology rose in the early 2000s along with a general interest in complex systems. Much of this interest spurred from an influential volume called *Complex Systems and Archaeology* published in 2003 (Bentley and Maschner 2003), which included a number of case studies that involved testing complex network models using archaeological data. In recent years, a number of archaeologists have begun to collaborate with complexity scientists from other fields. For example, the archaeologist Carl Knappett has collaborated with physicists Ray Rivers and Tim Evans on a series of network models focused on

the Bronze Age in the Aegean. This work involved the creation of a series of complex network models to assess the potential patterns of interaction and the robustness of maritime networks to a variety of disturbances (Knappett et al. 2011; Rivers 2016; Rivers and Evans 2013; Rivers et al. 2013).

Over the last decade or so it has been possible to speak of a distinct and growing body of **archaeological network research** drawing on each of the major traditions briefly described above. This trend is due in part to the increasing availability of software tools for analyzing and visualizing network data (e.g., UCInet, Pajek, Gephi, and many packages for the R and Python platforms; see discussion of software in Appendix B) but is also partly a response to the increasing popularity of network models in the broader social and natural sciences. Archaeologists have been quite cosmopolitan in applications of network science, drawing on the massive body of interdisciplinary network research to develop useful new methods for addressing old archaeological questions as well as a whole suite of new questions we were not asking even a few years ago; many of these will be elaborately discussed in the next chapter.

As we write today there are numerous sessions and papers on archaeological networks at just about every major national and international conference as well as one annual community-led conference (The Connected Past: http://connectedpast.net) explicitly dedicated to the explorations of networks in archaeology and history. Every year there are numerous dissertations and theses applying network science to archaeological data, as well as book-length treatments of network research and journal special issues (e.g., Brughmans et al. 2016; Collar et al. 2015; Knappett 2011, 2013, 2014; Lozano et al. 2017). Recent work includes numerous exciting applications of network scientific tools to archaeological data and a burgeoning literature focused on the unique challenges of applying network methods and models to archaeological data (e.g., Brughmans et al. 2016). Importantly, archaeological network practitioners have already begun to breach the boundaries of archaeology and regularly present at conferences like the International Network for Social Network Analysis (INSNA) Sunbelt conference focused on the study of social networks generally. We see these all as positive developments and suggest that the future bodes well for network science in archaeology (see also Peeples 2019).

1.4 HOW TO READ THIS BOOK

We decided to write this book because both of us have frequently been approached by archaeologists with a general interest in network techniques or a sense that

relational data may help them address interesting questions, but with no notion of where to begin. There are numerous recently published historical overviews, perspective pieces, and calls for the increased importance of networks in archaeology (e.g., Brughmans 2010, 2013b; Collar et al. 2015; Mills 2017; Peeples 2019), but to date, there is no single manual or guide that helps you along the path from archaeological question to archaeological data to network data to network analysis to useful results. Our goal with this book is to provide an overview of the state of the art in archaeological network studies including examples and detailed discussions of the kinds of questions that can be addressed as well as the unique challenges of deriving and analyzing network representations from archaeological data to help readers make these connections. We have attempted to keep this book practical in that we address specific problems and provide solutions for issues that archaeological network analysts are likely to face. At the same time, we have devoted substantial attention to describing what particular methods do (though we have largely relegated mathematical formulas to the glossary) and, perhaps most importantly, when it would or would not be appropriate to apply a given method to a given problem. Throughout this book we present new key concepts and terms in **bold** text where they first appear and provide definitions in boxes as well as in the glossary at the end of the book.

Even in a field as young as archaeological network studies, we cannot hope to be comprehensive. Instead, we have chosen a broad range of topics selected to highlight the diversity of approaches already in the literature and what we see as many of the most important potential future directions. We want to give our readers all the basic tools necessary to collect, manage, and analyze archaeological network data and address substantive questions on their own. We illustrate these approaches throughout with archaeological case studies and shorter examples explored in boxed vignettes. Beyond this, we want readers to come away with a general understanding of network methods and models that will allow them to confidently venture out into the broader network science literature beyond the borders of archaeology to find new and exciting theories, methods, and research questions. We have also created a detailed online companion to this book that provides datasets, additional examples, and the information necessary to replicate the analyses in this book.

We have written this book to be both a useful general introduction to the world of archaeological networks as well as a handbook for readers who have more experience with these approaches. *If you only have one hour* to find out what archaeological network research is, then read this chapter (congratulations, you've made it this far) and Chapter 2 along with the summaries of all other chapters.

Chapter 2 provides an extended discussion of the kinds of evidence that archaeologists have used to create network representations and several examples of what has been done with such networks. *If you have a little more time* and want to get an idea of how to apply network science to a range of archaeological cases, we suggest you continue by reading Chapter 3, which focuses on both the process through which archaeological data are collected and abstracted to network data as well as the general features of networks as analytical objects and network data. The first three chapters of the book also cover a number of important topics on the appropriate selection of methods and models for a range of common research problems in archaeology. *If you are already familiar with network science and archaeological applications* and want help identifying and applying specific methods in your own archaeological research, then read the middle chunk of the book: Chapters 4–7. These chapters focus on a series of analytical topics: exploratory network analysis, sampling and uncertainty assessment, network visualization, and spatial networks. Some of the methods presented are already commonly applied to archaeological network research, whereas others represent novel directions for future studies (in particular, Chapter 5). We do not expect that all readers will work through these middle chapters line by line, but instead, this portion of the book can be considered more of a reference and guide to good practice for a range of common and not so common techniques. For each of these middle chapters we have provided brief case studies illustrating the major concepts and approaches as well as a series of exercises designed to let you evaluate your own comprehension (with answers and worked examples provided at the end of the book as well as in the online companion). *If you want to engage with the mathematical implementations* of these techniques, in addition to reading Chapters 4–7, follow up on the references in the further reading lists and see the equations in the glossary. Finally, Chapter 8 provides our own perspective on the potential *future(s) for network science in archaeology* including where we think the field can/should go both methodologically and theoretically to overcome a wide range of challenges. This chapter also returns to the point made here in the introductory chapter about the importance of combining network methods and explicit network theories, but we discuss this issue in more practical terms building on the concepts introduced in the rest of the book. We hope the last chapter serves as both a jumping-off point and an inspiration for readers to expand on what they have learned here; archaeological network science is a young field and there is a need for new creative approaches to explore and expand its limits.

Because software for managing and analyzing network data tends to change quite rapidly, we have decided not to tie the main text of this book to any particular software package (we provide brief descriptions of some current common software

packages in Appendix B). Instead, we present the basic theoretical justifications for and "nuts and bolts" descriptions of the approaches covered here and refer readers to the *associated online companion* for the specific analytical details. This online companion is quite elaborate and provides tutorials and examples of the analyses discussed in this book primarily using the R programming language (R Core Team 2021). These examples will be periodically updated by the authors and, importantly, also include a public commenting feature so that those who use this book can contribute, ask questions, and even develop related exercises designed for courses or for self-teaching. We hope the online appendix and the associated open source data, code, and tutorials will develop into a thriving ecosystem of archaeological network practitioners. In order to help that happen, we have also designed the online companion to be a clearinghouse for open archaeological network datasets. We have populated this resource with many of our own published datasets, including those used in this book, and we hope others will do the same.

A number of datasets and case studies are used repeatedly throughout the book to illustrate network concepts and methods, and Section 2.8 introduces the data used. These have been selected to offer geographically and temporally varied examples, as well as to demonstrate how network methods might be used in very different research contexts or to study diverse phenomena. They include Roman roads, ceramic design and technology from the US Southwest, Medieval sites in the Himalayas, archaeological publications, and Iron Age and Roman sites in southern Spain.

1.5 SUMMARY

- Our goal in this book is to both introduce network science and a wide variety of network methods for an archaeological audience and also to make an argument for the importance of relations and relational data for understanding many natural and social phenomena that are of interest to archaeologists.
- A network is a formal system of interdependent relationships among a set of entities.
- A network representation is a formal abstraction of a network created for the purposes of visualization or analysis. In this book, network representations are simply referred to as networks.
- Network representations involve the formal definition of nodes as the entities in question and edges as the relationships among them.
- Network structure refers to the general properties of a network including the overall patterns of relations, the presence/absence and nature of subgroups, the

variation in the positions of actors within that network, and a broad range of other potentially salient features of organization.

- The relational perspective at the core of formal network approaches is the notion that the structural properties of networks and variation in the positions of nodes and edges in a network are important for explaining or predicting the behavior of the actors of that network in addition to the attributes of the actors themselves.
- Network science is the study of the collection, management, analysis, interpretation, and presentation of relational data.
- Network theories are formal and testable expressions of dependencies among nodes, edges, attributes, outcomes, or global network structures or any combination thereof. They express why and how relationships matter in a certain research context.
- Doing critical archaeological network research requires archaeological argumentation at every step of the network science application process. Why is network science necessary in this archaeological research context? What limitations does my archaeological data critique impose on my use of network science? Can my archaeological theories be represented using network science concepts and tools?
- Network science can only make constructive contributions to archaeological research when it is inserted into the archaeological research process in appropriate places. What defines those appropriate places is determined by the archaeologist's theories, data critique and research context, as well as by a critical understanding of the network science concepts and tools themselves.
- Network approaches as they exist today owe their origins to three major traditions of research: (1) graph theory, (2) social network analysis, and (3) complexity science. Each of these has influenced archaeological research independently several times over the last 50 years or so.

Further Reading

The following resources provide detailed accounts of the history and applications of archaeological network research using a broad array of examples.

Brughmans, Tom 2010 Connecting the Dots: Towards Archaeological Network Analysis. *Oxford Journal of Archaeology* 29(3):277–303.

2013 Networks of Networks: A Citation Network Analysis of the Adoption, Use and Adaptation of Formal Network Techniques in Archaeology. *Literary and Linguistic Computing, The Journal of Digital Scholarship in the Humanities* 28(4):538–562.

Brughmans, Tom, Anna Collar, and Fiona Coward 2016 *The Connected Past: Challenges to Network Studies in Archaeology and History*. Oxford University Press, Oxford.

Brughmans, Tom, Barbara J. Mills, Jessica L. Munson, and Matthew A. Peeples 2023 *The Oxford Handbook of Archaeological Network Research*. Oxford University Press, Oxford.

Brughmans, Tom, and Matthew A. Peeples 2017 Trends in Archaeological Network Research. *Journal of Historical Network Research* 1(1):1–24.

Collar, Anna, Fiona Coward, Tom Brughmans, and Barbara J. Mills 2015 Networks in Archaeology: Phenomena, Abstraction, Representation. *Journal of Archaeological Method and Theory* 22(1):1–32.

Knappett, Carl 2011 *An Archaeology of Interaction: Network Perspectives on Material Culture and Society*. Oxford University Press, Oxford.

2013 *Network Analysis in Archaeology: New Approaches to Regional Interaction*. Oxford University Press, Oxford.

2014 Avant-propos. Dossier: Analyse des réseaux sociaux en archéologie. *Nouvelles de l'archéologie* 135:5–8.

Mills, Barbara J. 2017 Social Network Analysis in Archaeology. *Annual Review of Anthropology* 46:379–97.

Peeples, Matthew A. 2019 Finding a Place for Networks in Archaeology. *Journal of Archaeological Research* 27:451–499.

In addition to the above archaeological resources, the following general network texts provide excellent introductions to the history of network research, network science, and some of the most common applications.

Barabási, Albert-László, and Jennifer Frangos 2014 *Linked: How Everything Is Connected to Everything Else and What It Means for Business, Science, and Everyday Life*. Basic Books, New York.

Borgatti, Stephen P., and Daniel S. Halgin 2011 On Network Theory. *Organization Science* 22(5):1168–1181.

Brandes, Ulrik, Garry Robins, A. N. N. McCrainie, and Stanley Wasserman 2013 What Is Network Science? *Network Science* 1(1):1–15.

Coscia, Michelle 2021 The Atlas for the Aspiring Network Scientist. www.networkatlas.eu.

Freeman, Linton C. 2004 *The Development of Social Network Analysis: A Study in the Sociology of Science*. Empirical Press, Vancouver.

Knoke, David H., and Song Yang 2008 *Social Network Analysis*. 2nd ed. SAGE, Los Angeles.

Scott, John, and Peter J. Carrington 2011 *The SAGE Handbook of Social Network Analysis*. SAGE, London.

2

Putting Network Science to Work in Archaeological Research

2.1 INTRODUCTION

The purpose of this chapter is to give you the basic lay of the land in the world of archaeological network research in order to provide context for the remainder of the book. As we saw in Chapter 1, although archaeologists have applied graph-theoretic and network analytic methods toward archaeological questions for more than 50 years, it is really only in the last 10 years or so that such approaches have become common. Archaeological network science is still quite a young subdiscipline and is constantly changing. There are likely to be some "growing pains" as we all figure out how to best adopt, adapt, and develop network methods appropriate for archaeological data and archaeological questions. This is perhaps not too different from where specializations like GIS were in archaeology 15–20 years ago (see Connolly and Lake 2006; Wheatley and Gillings 2002).

Archaeologists have necessarily been very creative when making use of network methods, and this is reflected in the great diversity of applications that is expanding year by year. Indeed, it is startling to see how much has changed between reviews of network research in archaeology published only a few short years apart (cf., Brughmans 2010, 2013b; Brughmans and Peeples 2017; Mills 2017; Peeples 2019). Despite all of the diversity and rapid growth, when we take stock of the ever-expanding list of publications and applications of archaeological network research, there are a few specific areas that have now seen considerable research and where network methods are starting to realize important contributions toward addressing archaeological questions and even broader questions in the social and behavioral sciences. We briefly introduce and discuss these in turn in the sections below. Given the exciting rate of growth and creativity in the field, we expect (and perhaps even hope) that this discussion will be somewhat outdated just a few years from now.

In this chapter, we first provide a basic discussion of the kinds of network data that archaeologists use (leaving the details on data formats and management for Chapter 3) and then describe some of the most prominent themes, theories, and topics in archaeological network research in greater detail, including material culture networks, movement and transportation networks, spatial proximity networks, visibility networks, as well as a selection of other less common applications, many and varied. This is not meant to be a comprehensive literature review nor is it meant to discuss the specific methods and models in great detail (that is what the rest of the book is for). Instead, what we hope to do here is to introduce the kinds of questions that archaeologists are asking and answering with network methods and models, and to give a couple of illustrative examples of different kinds of data and approaches to spark your imagination. Notably, although we divide this discussion largely along the lines of the kinds of data used to generate networks, there is considerable overlap among these approaches: comparisons among networks generated using different kinds of evidence is an important area of future research. We hope this brief overview will give you a better sense of the archaeological applications of networks to aid in your understanding of the sometimes very technical concepts and methods introduced in the following chapters.

We conclude this chapter with a brief introduction to the case studies and primary archaeological network datasets that we have used throughout the book as examples. These provide illustrations of many of the different kinds of networks highlighted in this chapter. They are used in different parts of the book to illustrate analytical approaches, to provide visualizations, and as practical exercises. These data are also available for you to download in the online companion to this book in a variety of formats. We encourage you to follow along or try your own analyses.

2.2 APPROACHES TO NETWORKS IN ARCHAEOLOGY

Archaeology is an incredibly diverse field, covering a wide range of phenomena and specialties: from studies of ancient bones and tools to reveal the origins of our species to the reconstruction of the maritime voyages that led to the occupation of the Pacific islands to the analysis of the acoustic properties of dining halls in Medieval European castles. Archaeological network research also reflects this diversity. We use network data to represent structures in archaeological data and relational theories and we use them to represent the abstractions we formulate about a wide range of past relational phenomena. As this suggests, there is no single "archaeological network data" nor is there one way of using network data to address

archaeological questions. In Chapter 3 we discuss the gathering, management, and presentation of network data in detail, but to provide context for this chapter, we first list a range of examples of the kinds of phenomena that archaeologists have used in network studies as well as the diverse ways nodes and edges have been defined in this work. Table 2.1 lists some of the most common applications of network data along with a reference to a published example (for other examples, see Collar et al. 2015, table 1). This is by no means a complete list but it provides some indication of the range of phenomena that have been explored and the kinds of evidence used to explore them. We have grouped these examples in Table 2.1 into the basic categories we use to organize the remainder of this chapter but also note that these approaches often intersect and blend together in practice.

2.3 MATERIAL CULTURE NETWORKS

By material culture networks here we simply mean formal network representations generated using any kind of material cultural data. This could be counts of objects or buildings by type, technological or aesthetic attributes of objects or spaces, geochemical or other archaeometric data, images of objects or features, or anything else that we as archaeologists can record or do with the physical manifestations of human behavior. As you might imagine, this is an extremely broad category that encompasses quite a few different approaches. There are, however, a few areas where such work has been concentrated in recent years: networks based on geochemical and related provenance data, networks based on material cultural frequency data, and networks based on shared technological styles or practices involved in the production or use of objects or features themselves. As we will illustrate in the discussion below, although the kinds of questions such networks have been used to address are quite variable, there is considerable overlap in the arguments and methods used to link material patterns to network data across different kinds of material evidence.

The fundamental premise underlying most recent (and not so recent; Irwin-Williams 1977) archaeological network studies relying on material cultural evidence is that similarities in the materials recovered at locations of interest provide *some kind* of evidence of connections among those places or objects. As we typically cannot directly observe interactions in archaeological contexts, we must use proxy evidence to define our networks, but exactly how material patterns equate to relational connections is an extremely important and complex question. Sindbæk (2013), for example, notes that even if we know the origin of an object and its ultimate

destination we typically lack information about the links along the way that connect A to B. Further, similarities in the materials made and/or used in different contexts could be generated through a variety of social processes from exchange to population movement to emulation or shared historical origins (Mills 2016; Peeples et al. 2016:63–67). Thus, although many archaeological network studies rely on traditional methods for the analysis of such material cultural networks, archaeologists also often conceptualize the nature of the network and the meaning of individual ties somewhat differently. Specifically, edges in material cultural networks in archaeology are often treated as statements of the *probability* that ties or relationships existed between nodes rather than binary social ties in the strict sense.

The most common approach to building material cultural networks in archaeology relies on the presence/absence or frequency of objects tabulated according to some classification system. This could be, for example, ceramic or lithic tool type or style counts, information on site feature types, information on the geochemically identified sources of objects, information on the attributes of specific objects, or all manner of other count or occurrence data that archaeologists record. This frequently involves creating ties among archaeological contexts that share some specific class of object, attribute, or source (Mizoguchi 2009, 2013; Sindbæk 2007a) and sometimes further weighting network edges based on the number of co-occurrences (e.g., Blake 2013, 2014; Coward 2010, 2013). The presence of the same class of object at a pair of sites is commonly referred to as a relationship of copresence. Where systematic frequencies of objects, attributes, or sources are available, this approach is often extended to what are called *general similarity networks* (Ösborn and Gerding 2014), which involve defining and/or weighting edges based on the proportions of object classes included in pairs of contexts (see discussion in Section 3.3.5). The argument typically used to justify material cultural similarity networks such as these is that the strength of similarity of objects or attributes recorded for two contexts provides an indication of the strength of shared affiliation between those contexts. Although shared affiliations may not always equate to direct interactions, it is argued that people in contexts with very similar materials were more likely to have interacted than people in contexts with very different materials (see Mills 2016). As we will see below, exactly how connections based on copresence or similarity are interpreted varies considerably based on the specific questions at hand and the nature of the underlying data.

Network representations generated using objects tied to specific origins or production sources using chemical compositional analysis or other lines of evidence (e.g., known production centers or factories in the case of historic archaeological research) provide an excellent context for applying a wide variety of network

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Table 2.1 *An overview of the use of network data in archaeology*

Category	Phenomenon	Nodes	Edges	Published example
Material Culture	Co-occurrence	Contexts and artifact categories	Occurrence of artifact category in context	Östborn and Gerding 2014
	Similarity	Contexts or sites	Similarity metric based on artifacts	Mills et al. 2013a
	Stylistic attributes	Artifact design attributes	Co-occurrence or similarity in stylistic features/attributes	Cochrane and Lipo 2010
	Geochemical	Sources or contexts	Presence or frequency of sources at sites	Bernardini 2007
Movement	Roads	Settlements or crossroads	Roads/paths	Isaksen 2008
	Hydrology	Confluences	Waterways (navigable)	Apolinaire and Bastourre 2016
	Least-cost path	Landscape locations	Pathways of traversal	Herzog 2013
	Accessibility	Discrete architectural spaces	Access between discrete spaces	Cutting 2003
	Axial map	A discrete axial line	Intersections among axial lines	Turner et al. 2005
Spatial Proximity	Relative neighborhoods	Locations in geometric space	Relative distances and configurations among locations	Jimenez-Badillo 2012
	Proximal point analysis	Locations in geometric space	K-closest neighbors	Terrell 1977
	Gravity models and related approaches	Locations in geometric space	Hypothesized interaction based on distance and attractors	Rivers et al. 2013

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Visibility	Line of sight	Locations of observations or observers	Line of sight	Brughmans and Brandes 2017
	Shared viewshed (second order visibility)	Locations of observations, not the observed locations	Shared viewsheds between viewpoints	Turner et al. 2001
Other Approaches	Food webs	Species	Trophic relationships (is eaten by)	Crabtree et al. 2021
	Textual networks	Entities mentioned in texts	Relationships mentioned in texts	Munson and Macri 2009
	Citation networks	Publications	Directed citations	Borner et al. 2004
	Coauthorship	Authors	Coauthorship	Brughmans and Peeples 2017
	Museology	Catalogue entries	Collection metadata connections	Griffin and Klimm 2023
	Illicit antiquities trade	Buyers and sellers	Transactions	Tsirogiannis and Tsirogiannis 2016
	Semantic networks, graph database	Subject and object	Predicates	Vitale and Simon 2023

methods and models. Network terminology and network visuals have long been used to represent flows of materials among sources and destinations even where formal network analytical tools have not been applied (e.g., Evans 1989; Pengilley et al. 2019; Webb 1974). When we can formally describe the volume and/or direction of flow of materials from place to place using a network, however, we can move beyond simple distributional models and assumptions to explore how the structural positions or properties of geographic locations are patterned in relation to one another or in relation to other outcomes of interest (e.g., does site size or longevity relate to network centrality?).

The questions archaeologists have focused on when exploring geochemical or source provenance networks have often revolved around exchange and the dynamics of economic interaction. For example, Golitko, Feinman, and others in a series of studies focused on obsidian exchange across the Maya world (Golitko and Feinman 2015; Golitko et al. 2012) used similarities in the frequencies of geochemically sourced obsidian objects at settlements as a proxy for the network flows of economic activity among Maya polities through time. They used these source similarity networks to argue, among other things, that the changing structure of economic interaction may have been a key factor in the decline of certain Maya centers and regions. Understanding the dynamics of economic systems in this way can further aid in explorations of the network drivers of other kinds of adaptive behaviors. For example, Gjesfjeld and Phillips (2013; Gjesfjeld 2015; Phillips 2011) explore the circulation of obsidian and ceramics among islands along the Kuril archipelago off the coast of Russia and Japan using network methods. They argue that changes in the topology of networks of exchange provide information on the underlying adaptive strategies that hunter-gatherer populations used in the region to cope with changing profiles of subsistence risk in this unpredictable environment (Gjesfjeld and Phillips 2013).

In addition to exploring economic activity, networks generated using geochemical and other source data have also been used to track the emergence and development of communities and other large social groups or identities at various scales. The underlying argument here is that groups of individuals or settlements that share common spheres of frequent interaction may have represented identifiable social entities of various sorts in the past. For example, Ladefoged and colleagues (2019) track the use of obsidian among a series of settlements built by Polynesian people arriving in Aotearoa (New Zealand) more than 700 years ago. They note changing patterns of obsidian use through time that do not map neatly onto simple geographic expectations of interaction as a function of distance. Instead, the authors document the emergence of network clusters or communities, many of which appear to closely correspond with contemporary Māori iwi (tribal)

territories. Similarly, Bernardini (2007) uses network methods to track the direction and volume of flow of ceramics among settlements along the Hopi mesas in northern Arizona through time. He uses centrality metrics to demonstrate that the people of each mesa engaged in different kinds and scales of ceramic production and circulation locally, which is in line with Hopi traditional histories of each mesa but stands in contrast to the typical treatment of the region by archaeologists as a relatively homogenous whole. In this way, network communities derived from sourced material data are used to posit geographic scales of frequent social interaction and, by extension, the likely scales of social groups in the past.

Material frequency networks have also often been used to explore the dynamics of social identities and population changes through time and across space. Blake (2013, 2014), for example, builds networks using a broad array of easily traceable *foreign* objects found at a series of Bronze Age sites in west-central Italy in order to explore the most common patterns of long-distance interaction. She notes that clear and cohesive network subgroups are identifiable in these networks and that these subgroups presage the location and extent of the Etruscan and Latin regions later in time. This, she argues, suggests that networks of long-distance interaction may have been instrumental in the development and crystallization of these later ethnic identities. In another series of studies Mills and colleagues (Mills et al. 2013a, 2013b, 2015, 2018) explore networks of ceramic similarity in the US Southwest in order to understand the changing structure of interaction and social identification across a series of well-documented long-distance migrations. In this work, the authors show that network relations often predict the direction and timing of later population movements and also track how migrations can result in dramatic transformations of regional social networks of interaction.

Material culture networks also have considerable potential for the study of cultural evolution and social transmission. Cochrane and Lipo (2010), for example, explored patterns of decoration on Lapita pottery from the South Pacific using similarity network methods based on shared design attributes in an attempt to evaluate various hierarchical and nonhierarchical models for the transmission of designs among Lapita communities through time and across space. Their research suggests that patterns of similarity in Lapita designs can best be explained by horizontal transmission rather than a cladistic model and that formal network methods provide a useful means for identifying and quantifying such patterns of transmission. Despite the applicability of such models and the success with similar network approaches in ethnographic contexts (Heinrich and Broesh 2011) and simulation studies (Kandler and Caccioli 2016), direct empirical archaeological studies of cultural transmission using network methods have been relatively rare (but see Romano et al. 2020 for a discussion of future prospects; see also Chapter 8).

Another area of material cultural network research that has seen considerable development in recent years focuses on the use of technological characterizations of objects and practices to analyze common production techniques as a proxy for social interaction, social proximity, and/or social learning. This work builds on a long and robust literature in archaeology focused on communities of practice, the chaîne opératoire approach, and related concepts in the anthropology of technology (e.g., Dietler and Herbich 1998; Dobres and Hoffman 1994; Gosselain 2000; Lemonnier 1992; Van Oyen 2016; Wenger 1998). Although this area of research is diverse, in general such approaches are similar in that they posit a connection between evidence for shared contexts of learning and evidence for interaction more broadly (Roux 2019, 2020; Watts 2013). In other words, groups of people that produce objects using similar techniques (especially when the techniques involved are idiosyncratic and difficult to learn from finished products) tend to be groups of people who frequently interact or share common historic origins.

The approach to building networks using data on similarities in techniques is similar to the approaches used for material culture more generally (similarity networks or co-occurrence networks), but the theoretical connection between interaction and shared technology usually facilitates a more direct argument about the relationship between material pattern and social proximity. For example, Watts (2013) conducted a detailed analysis of lithic reduction techniques used in a series of projectile points from the Tonto Basin in Arizona and identified what he argued was the work of individual knappers and then simulated networks of interaction among settlements based on where the work of different knappers ended up. Similarly, Peeples (2011, 2018; Peeples et al. 2021) created networks of similarity based on a detailed technological characterization of cooking pots from the western Pueblo region of the US Southwest and argued that similarity in technological attributes provides evidence for strong patterns of interaction and shared historic relationships among the communities where similar pottery was produced and used (see Box 2.1 for an additional example).

Box 2.1 CERAMIC SIMILARITY NETWORKS FOR DESIGN AND TECHNOLOGICAL STYLE IN THE SOUTHERN APPALACHIAN REGION

It is sometimes possible to create multiple distinct network representations indicative of different kinds of social processes and interactions using the same set of objects. For example, Lulewicz (2019) created similarity networks based on the temper and design styles of 9th- through 17th-century

ceramics from the southern Appalachian region in the United States to explore the relational processes involved in the emergence and development of the Etowah chiefdom. In this research he suggests that different attributes of the same objects, in this case ceramic temper type and ceramic decorative style, can be used to produce distinct networks with different relational interpretations. He argues that technological decisions involved in temper choice are low-visibility attributes that are unlikely to be shared beyond tight-knit groups of ceramic producers and, thus, that patterns of similarity in temper likely represent frequent and close interactions. He then argues that surface treatment and decorative style is a high-visibility attribute of ceramics that could more easily be observed and shared and more likely represents intentional social signaling. As Figure 2.1 illustrates, the networks representing frequent interactions (temper) are strongly geographically partitioned by region through time, whereas the networks representing social signaling (surface treatment) are generally more integrated across regions with fewer dense clusters. This suggests that these distinct forms and contexts for interaction generated different kinds of relational structures across the study area through time. This example reveals the interpretive power and utility for material network studies that track and compare multiple distinct relational processes (see Peeples 2018; Upton 2019 for further examples).

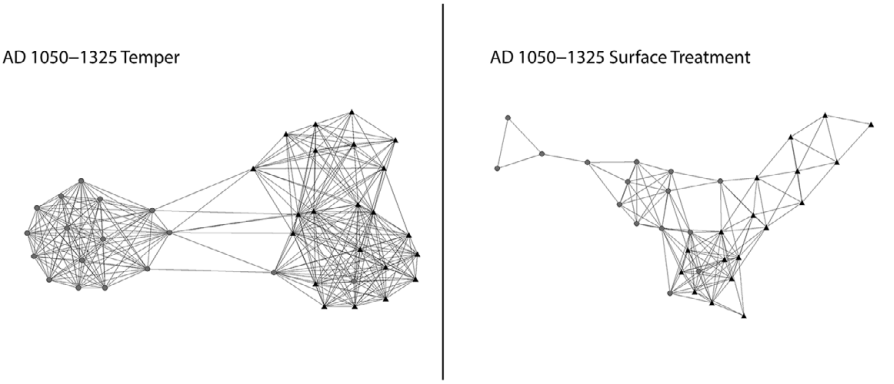


FIG. 2.1 Ceramic similarity networks for the southern Appalachian region ca. AD 1050–1325 based on shared temper type (left) and surface treatment (right). Sites in the southern portion of the study area are shown in black and sites in the northern portion are shown in gray. Figure courtesy of Jacob Holland-Lulewicz.

As the examples discussed above suggest, connections in material cultural networks are typically evaluated among entities of interest (sites, regions, other contexts) based on some quantified measure of overlap in the assemblages for those contexts. In such cases, the objects themselves are often not the analytical focus in their own right but simply the material proxy for defining and tracking connections among other social actors. In a few recent studies (e.g., Knappett 2011; Mol 2014) researchers have also shown that network methods and models provide an ideal context for applying object-centered analyses to explore how the social and the material intersect. For example, Mol (2014:121–124) discusses how systems of exchange for cultural valuables like the famous *kula* system from the Trobriand Islands (see Malinowski [1922] 2014; Weiner 1992) can be better understood if we simultaneously consider how the activities and properties of both people and objects co-produce systems of value. Methods and models that focus on objects as active participants in the emergence and dynamics of social networks are increasingly common in recent years and this is a trend that we expect will continue (see also Van Oyen 2016; chapters in Donnellan 2020)

2.4 MOVEMENT NETWORKS

Another particularly common topic in archaeological network research is the study of observed or theorized transport systems and movement-related phenomena. Such studies typically explore the movement opportunities between pairs of places, the overall structure of a transport system, the relative positions of towns/places, and the risks and rewards associated with these positions. Such methods allow us to explore the most frequently used paths, the strategic positions of places in landscapes, and the network distance (as opposed to geographic distance) among places. Movement networks also aim to assess the implications of the structure of a landscape or transport system for enabling or constraining the flow of goods, people, or information.

Road and trail systems lend themselves particularly well to formal representation as networks (e.g., Santley 1991), typically through the representation of roads as edges that connect at crossroads or settlements/structures. In some cases, large parts of past road systems are documented through written sources and archaeological excavations. This is the case for much of the

Roman road network, which has been extensively studied through the remains of stretches of roads across the former Roman Empire and finds of milestones, but also a range of textual sources including the Antonine Itineraries, the Vicarello goblets, and the Tabula Peutingeriana (e.g., Carreras and de Soto 2013; Isaksen 2007, 2008). Shawn Graham (2006) used the Antonine Itineraries to create a representation of the Roman road network as reflected by this particular historical source. He used this network representation to explore how fast information would spread to all cities in the road network and whether this speed differed between the Roman provinces. The ORBIS project (Scheidel 2014) similarly created a network model of the Roman world that included roads, navigable rivers, and simulated main maritime routes, as well as costs for traversing each route segment. This allows for exploring distances and financial costs of flows over this hypothetical representation of the entire Roman transport system, assuming different modes of transport (on foot, oxcart, horse), seasons of travel, or priorities (speed, money, distance).

In contexts where historic documentation of roads and trails is not as widely available, approaches like remote sensing have been used to great effect to identify and document paths and formal trails at broad landscape scales (Friedman et al. 2017; Ur 2003). For example, Menze and Ur (see Menze and Ur 2012) analyze network centrality among settlements in northern Mesopotamia connected by more than 6,000 kilometers of trails that were identified using remote sensing focused on recently declassified CORONA satellite imagery (see Ur 2003). They further use these networks to model potential patterns of interaction by distance along these trails.

Networks defined based on formal trails, paths, or roads need not be limited to regional scale studies, however. For example, Pailes (2014) explored the network structure of houses and walkways between them within the Classic Period Hohokam village of Cerro Prieto in southern Arizona to evaluate the relationship between network position and socioeconomic status. Such an approach can also be applied in well-defined portions of larger urban settings (e.g., Wernke 2012). Crawford (2019) recently explored pedestrian movement along roads in a thoroughly mapped portion of the city of Ostia using a toolset called Urban Network Analysis originally developed by urban planning researchers focused on movement and the provisioning of resources in contemporary cities. Her work demonstrates a means for modeling how pedestrian

movement in urban contexts can be constrained or facilitated by the nature and status of buildings along roads and paths.

Riverine transport systems (or hydrological networks) are often studied using similar network approaches, where river stretches tend to be represented as directed edges connecting confluences represented as nodes (see Connolly and Lake 2006). Many archaeological studies of past riverine systems were influenced by Pitts' (1965, 1979) pioneering studies of the Medieval river trade network in Russia. These include Peregrine's (1991) study of the central location of Cahokia on the Mississippi River drainage, Isaksen's (2007, 2008) study of riverine and road networks in Roman southern Spain, Apolinaire and Bastourres (2016) study of pre-Hispanic riverine settlement systems in Argentina, and Duffy's (2020) study of metal procurement and exchange along riverine paths in the Bronze Age of the Carpathian Basin.

As the examples above suggest, road and river networks are commonly used to study phenomena related to economic exchange. These transport networks can be considered the media that allow for goods, money, ideas, or people to be exchanged between different places. Processes of exchange can be formulated that take place on top of these transport network structures: reciprocity, redistribution, market exchange (Polanyi 1957; for a useful graphical overview, see Renfrew and Bahn 2008:361, 375). For example, a theorized down-the-line exchange process could be explored by studying the distribution of finds over a transport network: Do we observe a decreasing volume or diversity of a type of object with network distance away from its production place? Note we do not consider as-the-crow-flies distance but *distance over the network edges* (both approaches are of course complementary). One example of this is the distribution of Hellenistic and Roman ceramic tableware over a theorized terrestrial network structure in the eastern Mediterranean (Brughmans 2010). In this same research context even more complex exchange models acting on top of a network structure have been explored. For example, Brughmans and Poblome (2016a, 2016b) explored the extent to which ceramic tableware distributions could be explained by a market economy model (represented as a network model with variable integration), whereas Brughmans and Pecci (2020) explore processes of redistribution of Roman amphorae over the ORBIS model of the Roman transport system (Scheidel 2014). The increasing number of highly detailed studies of material provenance through isotopic data promises even

more opportunities for highly informed studies of material culture flows and exchange.

In the case of river and road networks both nodes and edges are spatially embedded and follow a specific course through physical space. However, in other cases the edges between pairs of locations are unknown or only hypothesized and have to be derived using models. This is the case for movement networks derived from least-cost path analyses, where particular paths between locations are simulated with reference to the effort for individuals to move over a physical landscape. Network representations of least-cost paths through a landscape can be studied using network methods (Hill et al. 2015; Verhagen et al. 2013; White and Barber 2012) and can also be compared with spatial network models representing alternative theories of movement (Bevan and Wilson 2013; Herzog 2013). Similar approaches can be used to study the traversal of seascapes (Leidwanger 2013; Slayton 2018; Warnking 2016) and can form the basis for network representations of sea travel (Arcenas 2015; Scheidel 2014).

Many of the approaches mentioned so far in this section on movement are most frequently applied at large landscape or regional scales. Space syntax is a useful approach for exploring movement-related phenomena on smaller spatial scales. Space syntax was initially developed in architectural and urban studies (Hillier and Hanson 1984) and later adopted and expanded in several directions by archaeologists interested in exploring and comparing the formal configurations of individual buildings or larger urban landscapes. In particular, archaeologists have frequently applied the formal access analysis approach that comes out of the architectural space syntax tradition. This approach is used to define and visualize formal patterns of access, focused on either physical movement or visibility within buildings or larger built environments, by carefully documenting the mutual "reachability" of discrete spaces within those environments.

In such access studies, a network is defined to evaluate connections from one discrete space to another through dendritic networks of traversable or visible connections (e.g., connections between rooms via a doorway or mutual visibility between discrete spaces). Conventions differ in different contexts for whether the discrete spaces or connections/intersections along paths between them are defined as nodes or edges, respectively. In either case, such access networks allow for the consideration and comparison of structural properties of spatial arrangements such

as the mutual reachability of individual spaces, the relative depth of spaces (number of edges that must be crossed to reach that space), or the relative importance of a given connection in structuring movement across the built environment. Such formal considerations of access and the cultural logics of space have been argued to provide insights into a broad range of topics including social stratification, social organizational or demographic change, the provisioning of urban services, public versus private spaces, the dynamic life histories of spaces and structures, and social identification (e.g., Branting 2007; Brusasco 2004; Clark 2001; Cutting 2003; Fairclough 1992; Ferguson 1996; Fladd 2017; Foster 1989; Grahame 2000; Wernke 2012).

Box 2.2 SPACE SYNTAX APPROACHES TO A CITY NEIGHBORHOOD IN ROMAN OSTIA

The ancient city of Ostia, located where the river Tiber flowed into the Mediterranean, was a key harbor provisioning Republican and Imperial Rome. Its urban layout is exceptionally well documented, allowing archaeologists to explore how people might have moved through and experienced this urban environment. Hanna Stöger (2014) applied a range of space syntax techniques for this purpose to one of the city street blocks: Insula IV ii. First, the insula was represented as a graph for access analysis, in which each space (such as rooms and corridors) was represented by a node, and passages from one space to another as edges (Fig. 2.2a). This allows, among other things, the identification of spaces that facilitate access to many other spaces (i.e., nodes with a high degree). Second, a so-called route matrix of axial lines was created, representing the lines in the space that allowed for the longest views (Fig. 2.2b). The result can be interpreted as reflecting particularly likely paths of movement. Third, a visibility graph analysis (VGA) was generated to explore how these longest visibility lines connect to each other and are integrated (Fig. 2.2c). Both axial lines and VGA reveal the southern courtyard as an area with a higher potential for social interaction. This example presents a three-way approach common in space syntax, where dedicated analytical techniques allow for the study of movement, organization, and visibility. In combination, this approach presents a rich picture of how architectural features may have structured human experiences in an ancient built environment.

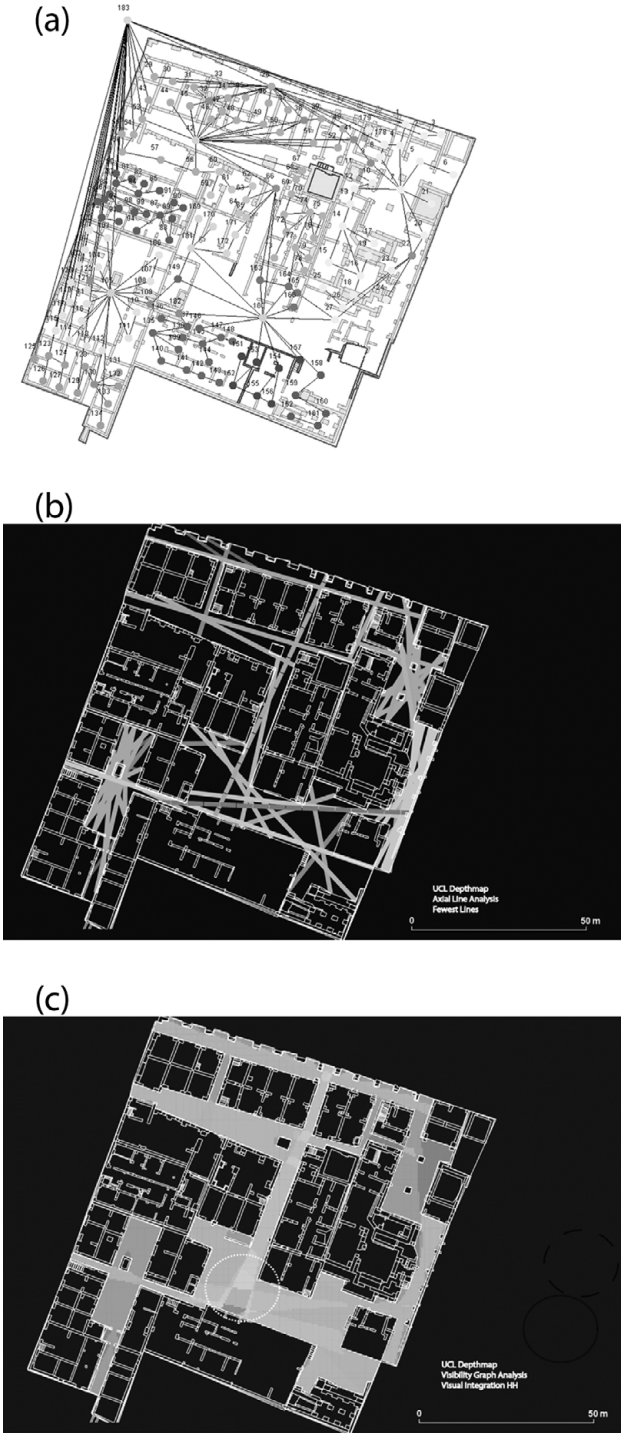


FIG. 2.2 Spatial relationships in Ostia's Insula IV ii represented as (a) an access analysis graph, (b) a route matrix based on axial lines, and (c) a visibility graph analysis (VGA) (reprinted with permission from Stöger 2014:figs. 3, 7, 8).

2.5 SPATIAL PROXIMITY NETWORKS

One of the first areas where network methods were applied in archaeology was the study of spatial proximity as a driver for interaction and social relations across networks. Analyses of spatial relationships and organization are, of course, some of the most common kinds of archaeological analyses at scales from the individual test pit all the way up to the region or beyond. Proximity network methods are well suited to facilitating the exploration not just of distributions and similar spatial patterns but also of the relational structures of connections among spaces and places and the role such topological patterns may have played in the emergence or development of interactions across space and in the identification of particularly important actors or paths for directing or receiving flows across those networks. Importantly, in their most basic form, proximity network methods only require that we have locations for the places, spaces, or objects that are to be the subject of network analyses, and the distances between them. Such methods can therefore be readily applied to a wide range of existing archaeological or geographic data. In general, proximity-based network models have been used either as rough proxies for potential interaction across a region of interest, as a simple basic hypothesis of interaction to be compared to other lines of material evidence, or as an approach toward modeling optimal network configurations for some particular kind of social or economic process of interest.

John Terrell's (1977) use of *K-nearest neighbor* networks for explorations of inter-island interactions in the Solomon Islands was an early influential application of proximity networks and graph-based methods that went beyond simple visualizations. K-nearest neighbors here simply refers to a network where discrete spaces are defined as nodes and connections are drawn to the *K* closest other nodes in that network (see Section 7.5.4). It is useful for exploring theories stating that transmission or interaction is most likely between nearby places and that the number of other places with which connections can be maintained is limited. Terrell called his approach *Proximal Point Analysis* (PPA), a name that stuck and is more commonly used among archaeologists to describe similar methods. Although this approach did not immediately take hold in archaeology broadly, PPA methods have been regularly applied in the study of island and maritime networks, particularly in Oceania (Hage and Harary 1991, 1996; Irwin 1978) and also more recently by Broodbank (2000) in his study of inter-island community interactions in the Early Bronze Age Cyclades (see also Mol 2014:42–47). Collar (2013) further applied this approach to the study of the diffusion of the cult of Jupiter Dolichenus throughout the Roman Empire using PPA as a simple model of

spatially constrained interaction that makes relatively few assumptions about the social processes driving interaction. A number of related spatial proximity models have been defined that are not focused exclusively on distance but on the configurations of nodes in relation to one another (see Section 7.5). This includes, for example, relative neighborhood graphs where nodes are connected if they are at least as close together as they are to any other node. Studies of transportation systems often rely on models like Delaunay triangulation, which connects nodes among spatially explicit territories when they share a side in a Voronoi or Thiessen polygon (e.g., Fulminante 2012, 2020).

A number of other popular proximity models rely not simply on which nodes are closest to each other or in a particular relative configuration but also on the actual distances among them. Such an approach could be used to model potential network processes based on the constraints of the movement technology underlying the network process under consideration. For example, if exploring maritime travel among islands there may be some distance threshold beyond which marine technology makes single trips unfeasible, so a network among islands could be created with binary ties defined between pairs of islands that are within that distance threshold from each other (see Rivers et al. 2013:133). Although there is some utility in defining ties as simply present or absent to allow for the identification of network topologies and node and edge structural positions, defining a cutoff in this way may limit certain kinds of investigations. In such cases, it may make sense to treat the distances (or inverse distances) among nodes as edge weights (see Irwin 1978). These distances need not be limited to simple Euclidean (as the crow flies) distances, either. For example, Hill and colleagues (2015; see also Mills et al. 2013a) defined connections among sites with overlapping distance buffers representing one day's travel on foot (out and back) for a series of contemporaneous settlements in the US Southwest and compared these networks to those defined based on ceramic similarity.

There is also a wide variety of more complex spatial proximity models that have been applied in archaeological contexts, many of which build on methods coming out of econometrics, geography, and spatial ecology. Gravity models, for example, are econometric models used to predict flows across a network of entities in relation to the relative distances among those entities and their attractive properties, which can include population size or other factors. This model has proven useful in predicting the volume and direction of flows across a range of spatialized economic processes (Anderson 2011) and has also seen regular use in archaeology. For example, Rihll and Wilson (1987, 1991) used gravity models to explore the emergence and development of Greek city-states. Knappett, Evans, Rivers and others (e.g., Evans and Rivers 2017;

Knappett et al. 2008; Rivers et al. 2013) have expanded on this work to use extended gravity models and a bespoke model called ARIADNE to explore potential maritime connectivity among Middle Bronze Age sites in the Aegean and to identify which particular network configurations would be more or less affected by specific kinds of disturbances. Other models include the exponentially truncated power functions that Menze and Ur (2012) use to model expected flows of materials among settlements in Northern Mesopotamia, and the entropy maximization function that is frequently applied in spatial ecology. The latter was used by Davies and colleagues (2014) to explore the geographic and political factors involved in changing settlement distributions across the Bronze and Iron Age in the Khabur Triangle region of Syria. There are many other potentially useful models already in the literature, and exploring those which have been useful in geography, spatial ecology, economics, and related fields would likely reap huge benefits for archaeologists as well.

As the examples discussed here suggest, spatial proximity networks are often used to evaluate the provisioning of resources across some context as a function of space. Importantly, spatial proximity models are also often used as a foil or null model for other kinds of network analyses. For example, Mills and colleagues (2013a) created a two-mode network (see Section 3.3.4) among sites in the US Southwest with edges evaluated between sites and the obsidian sources from which recovered materials were obtained. In this study Mills and colleagues (2013a) defined edges among sites and sources that were either under- or overrepresented from what we would expect based on a model of geography and topographic constraints alone. Specifically, they calculated the cost-equivalent distance from each site to each source and used a simple distance decay model to estimate what obsidian sources would be represented at a given site if proximity were the only consideration. Using this network they were able to demonstrate dramatic changes in the way obsidian was obtained through time with settlements largely conforming to geographic proximity expectations prior to the migration and with dramatic deviations from geographic proximity expectations after the migration. Comparisons of spatial expectations and material evidence for patterns of interaction like this are relevant for a broad range of traditional archaeological questions (see discussion in Chapter 8).

2.6 VISIBILITY NETWORKS

Network methods have frequently been used in archaeological research to represent visibility-related phenomena. These are phenomena where the ability for past

individuals to visually observe particular natural or cultural features is theorized to have influenced their behavior. Examples include past visual signaling networks, the ability to visually control approaching individuals, armies or crucial natural resources, changes in the observed proportion or nature of the surrounding landscape, and visual focal points or vistas in urban environments or even at regional scales.

Visibility networks can be generated in a number of ways, and by far the most common is the representation of lines of sight as network data (De Floriani et al. 1994). In line-of-sight networks the set of nodes represents the observation and observed locations and the edges represent lines of sight. A pair of nodes is connected by an edge if a line of sight starting at the eye level of an observer at one observation point can reach the observed point, that is, if the line of sight is not blocked by a natural or cultural feature. A pioneering archaeological application of line-of-sight networks is the work by David Fraser (1980, 1983:379–387) on chambered cairns in Orkney in the United Kingdom. Fraser argued that past power relationships of dominance and subservience might be expressed in the dominant and intervisible positions of chambered cairns, a theory he explored through a range of spatial network analytical measures.

The edges of line-of-sight networks are most commonly derived from binary lines of sight or binary viewsheds; binary since a feature is either visible or not. Additional information could be attached as edge attributes to further nuance the treatment of edges. First, we could argue that deriving line-of-sight networks from binary observations between observation locations represented by single points is highly sensitive to the selection of point locations and the resolution of the digital elevation model (DEM). When observation points represent past settlements or other large features, it might be more appropriate to sample point locations at each digital elevation model cell within the feature areas. This allows for calculating the number or proportion of points in one area that can observe a number or proportion of points in the other area, and vice versa. The results can be bundled as edge weights revealing how much of the observed area is visible in each direction. Second, the length of a line of sight could be included in the analysis, allowing for the edges to be explored as distance bands, to distinguish between different features becoming visible over different distances, similar to Higuchi viewsheds (Wheatley and Gillings 2000). Third, when line-of-sight networks are derived from probabilistic viewsheds (Fisher 1991) to account for the errors in the creation of the digital elevation model, the edges should not be binarized but rather carry their probability as an edge attribute to allow for interpreting the most and least probable network patterns (for an example, see Brughmans et al. 2015).

Line-of-sight methods are often used to study hypothesized past signaling networks, where information could be passed on through smoke, fire, or flag-based signaling (e.g., Earley-Spadoni 2015; Swanson 2003). Most such archaeological studies perform a visual evaluation of binary line-of-sight networks to evaluate whether a set of site locations could have functioned as a signaling network (e.g. Ruestes Bitrià 2008). Because a signaling network represents the ability to pass on information, path-based network measures such as the average shortest path length or betweenness centrality are appropriate in this research context (see Section 4.4.3). In their study of a hypothesized visual signaling network between monumental Bronze Age Nuraghi towers in Sardinia, De Montis and Caschili (2012) used betweenness centrality to identify important intermediaries for the flow of signaled information. Although such network and GIS methods allow us to represent what a theorized signaling network might have looked like and analyze how it might have functioned, it is extremely rare to obtain archaeological evidence that confirms that a set of sites functioned as a signaling network. However, new spatial network methods have been developed to evaluate the probability that a theorized signaling network could have functioned effectively and could have been established purposefully, based on configurations and exponential random graph models (Brughmans and Brandes 2017; Brughmans et al. 2014, 2015; for an example, see Section 4.4.3).

A few alternative network approaches for studying visibility-related phenomena beyond line-of-sight networks also have potential. These include the network data representation of total viewsheds (first proposed for natural landscapes by O'Sullivan and Turner 2001), and the two-mode and one-mode network representations of cumulative viewsheds (Brughmans and Brandes 2017). These methods allow for exploring the structure of visibility patterns on a landscape-wide scale, for example, the identification of landscape areas that are highly visible from the same set of landscape areas (i.e., the identification of clusters or "communities" in landscape visibility networks that are strongly connected). Such approaches involving landscape-wide visibility networks derived from total and cumulative viewsheds are rare, however, because they remain extremely demanding in terms of computation time and output storage (but see Brughmans et al. 2017). Bernardini and Peeples (2015) presented a landscape-wide analysis combining both viewshed and line-of-sight networks to identify what they call "sight communities" by first characterizing prominent peaks within the viewshed of individual settlements and then defining connections among settlements that shared views of the same prominent peaks.

A further approach to visibility networks has been developed in architecture and landscape planning and is known as an axial map. It is a prominent tool in space

syntax and is commonly used for the study of pedestrian movement patterns in architectural space (Hillier and Hanson 1984; Turner et al. 2005). An axial map is created by drawing a set of axial lines through space on a two-dimensional plan of a built environment. An axial line is any line between intervisible nodes of the polygons representing architectural features (see Turner et al. 2005 for a more elaborate definition). A network is subsequently constructed from this by representing the intervisibility lines as nodes and the intersections of lines as edges, and can be studied using network science measures. Axial maps can be extended by weighting the edge by the angle at which intervisibility lines cross, a so-called angular analysis (Turner 2001).

2.7 APPLICATIONS MANY AND VARIED

The major sections above barely scratch the surface of the extremely diverse field of archaeological network research. In this section we discuss a few of the less common (but no less useful) approaches to applying network methods and models to archaeological data and questions. Although the sheer diversity of approaches is impressive, this coverage is still by no means exhaustive. We hope this brief discussion will inspire you not only to apply some of the approaches to building and analyzing archaeological networks described here but also to develop your own custom approaches to problems and questions in your own area of research. We see archaeological network research as a rapidly maturing field that has the potential to make unique and valuable contributions to both archaeology and network research at large. The sky is the limit.

Biological networks: A large number of biological and biochemical phenomena are commonly studied using network methods including protein-protein interactions, connections between neurons in the brain, and genetic networks. A type of biological network with particular potential for the study of past communities is a food web. These are directed networks where nodes represent species and directed edges represent the trophic flow of energy from prey to predator. Food webs represent the predatory interactions and flows of energy in ecosystems. Archaeologists have pioneered the application of this concept to better understand the position and role of the human species in ecosystems through time (Crabtree and Dunne 2023; Crabtree et al. 2015, 2021). Beyond this, network methods are now also being applied to explore the structure of kinship using human skeletal materials in formal cemeteries (Johnson 2016, 2023). As patterns of phenotypic similarity are influenced by biological relatedness in complex ways, network

methods provide a useful formal tool for evaluating variation in kinship structure within and between mortuary contexts.

Textual sources: For historical periods, we have the opportunity to combine material and textual sources in our network studies. Such sources range from archives and letters for more recent periods to inscriptions, graffiti, stamps, signatures, and clay tablets. In his study of Roman brick production in the Tiber Valley, Shawn Graham (2006) combined textual evidence from brick stamps with a geochemical study of clay sources to produce a network of brick producers, brick-firing kiln owners, and lands on which kilns are located. The sociopolitical interactions of Classic Maya societies were studied using network methods, using the inscriptions on stone monuments detailing diplomacy, warfare, kinship, and other relationships (Munson and Macri 2009; Scholnick et al. 2013). Moreover, historical and archaeological network research share a lot of the same data-related challenges and research interests (see, e.g., Lemercier 2005, 2014, 2015), and many historical network studies can clearly inspire archaeological work.[1]

Box 2.3 ARCHITECTURAL INSCRIPTIONS AND STONE MONUMENTS IN THE MAYA REGION

Munson and Macri (2009) provide a fascinating example of the potential for textual (epigraphic) networks based on archaeological data. They compiled a large database of Maya place names and other inscriptions from stone monuments to create a network where pairs of sites are connected when they are mentioned in an inscription at another site. Since place-name glyphs often occur as part of longer texts, it is also possible to code the nature of the phrase where the place was invoked to indicate what type of interaction it may entail. Does the phrase provide evidence of an antagonistic relationship or a diplomatic one? Is a connection one of dynastic lineage or between rulers and subordinates? With this kind of information it is possible to explore similarities and differences in the structural properties of networks based on different kinds of relations among settlements across the region. Figure 2.3 shows the full Classic Maya epigraphic network with edge colors representing different kinds of relationships and with nodes color-coded by region. Loops indicate self-referential statements where the site name is found on an inscription at that site (Fig. 2.3).

[1] An elaborate list of resources can be found on the Historical Network Research website: http://historicalnetworkresearch.org (accessed 22 December 2019).

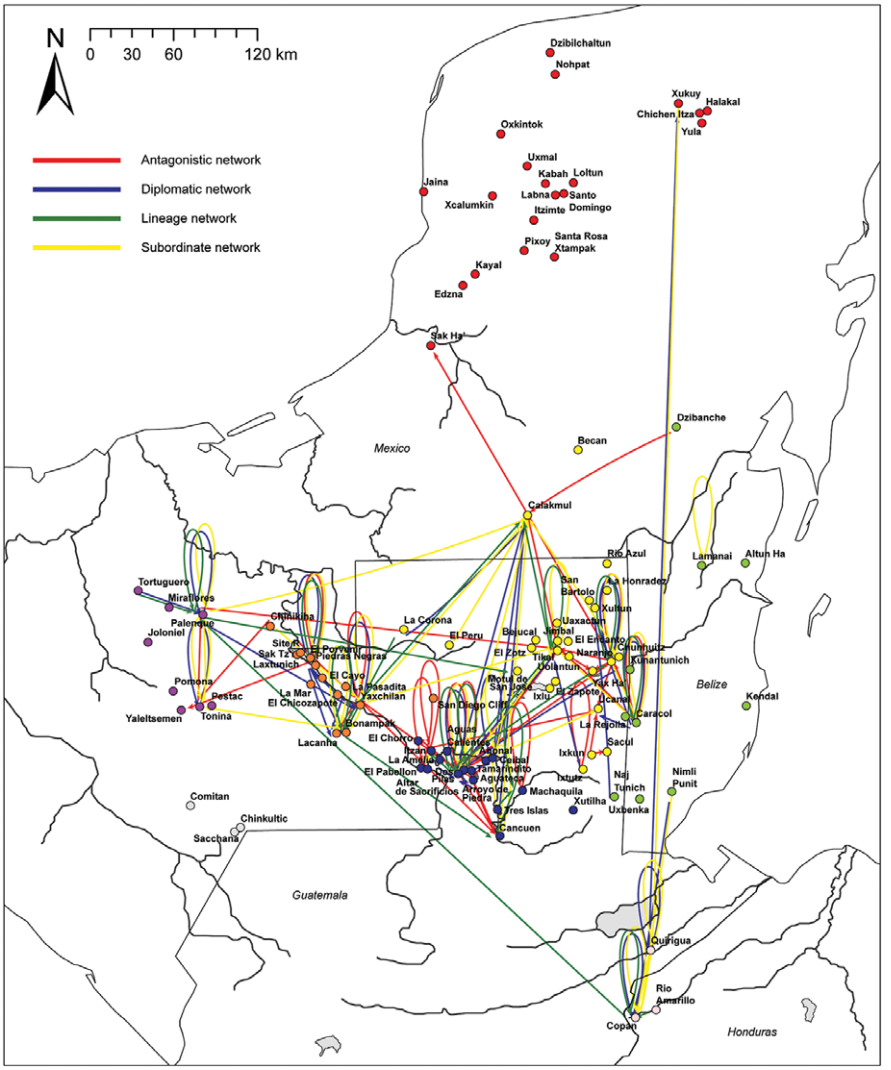


FIG. 2.3 Epigraphic network for Classic Maya sites with edges color-coded by the nature of each mention, reproduced with permission from Munson and Macri 2009 Fig. 2.

Knowledge networks: Network methods are often used to study citation patterns in scientific literature, collaboration between academics, and the creation and spread of new knowledge (White 2011). Citation and coauthorship network studies have been performed to explore the adoption of formal network methods in archaeology (Brughmans 2013a) and the community of archaeological network researchers (Brughmans and Peeples 2017). A larger-scale approach was taken by

Sinclair (2016), who explored the discipline of archaeology as a whole and its many sub-disciplines. Mickel and Meeks (2015) focused on a much smaller group of scholars in their network and topic modeling study of how knowledge is shared between team members at the Çatalhöyük excavations. Mickel, Sinclair, and Brughmans (2023) provide an overview of the study of knowledge networks in archaeology.

Dark networks: Some relational phenomena are hard to trace due to their illicit nature, such as terrorist networks or arms trade transactions. Sometimes these so-called dark networks concern past phenomena or archaeological resources. For example, the illicit trade of antiquities between individuals and museums can be traced and algorithms can be developed to reconstruct unknown traders and transactions (Tsirogiannis and Tsirogiannis 2016). In a similar vein, Graham and Huffer (2023) use network analytic approaches to explore the factors that facilitate the trade of antiquities online including the role of recommendation platforms and social media in promoting the sale of legal and illegal archaeological materials and human remains.

Diagrams and ontological networks: Network approaches are great tools for communicating relational theories or archaeologically documented relationships. For example, a network visualization was used to communicate a new theory of connectivity between Late Pre-Colonial island communities on East Guadeloupe (French West Indies) as reflected through material data types and visibility (Brughmans et al. 2017). Hodder and Mol (2016) use network visualizations and layout algorithms to revisualize and reinterpret so-called tanglegrams representing theorized relationships between material data categories and past practices. Ontologies used to map archaeological linked open data tend to be represented as network diagrams, allowing users to explore how concepts are related according to a particular ontological code (Vitale and Simon 2023).

Simulations: Simulation can be a powerful tool for helping us to evaluate, explain, or compare network dynamics or the potential generative processes underlying network formation in a range of contexts. For example, Kandler and Caccioli (2016) develop a simulation framework for exploring the factors that influence the adoption and spread of innovations among individuals within a networked system and how different kinds of network topologies promote or hinder that spread. Agent-based simulations have also been used to help researchers better understand the potential underlying relational processes that generate aggregate network patterns (for a comprehensive introduction including a chapter on networks, see Romanowska et al. 2021). For example, Watts and Ossa (2016) create an agent-based simulation of exchange, modeled after the Hohokam region

of southern Arizona, which allows them to explore different kinds of market and nonmarket exchange systems. They then compare the results of this model under different assumptions to an empirical network of material similarity to evaluate which sort of exchange system provides the best fit with their observations. Beyond this, agent-based models can even be used to evaluate the assumptions of archaeological material cultural networks generally. For example, Gravel-Miguel and Coward (2023) present an agent-based model that simulates artifact production, exchange, and discard, and they then analyze the resulting assemblages using methods commonly applied to real archaeological assemblages. This procedure allows them to recognize which kinds of robust patterns in simulated behavior are identifiable in the resulting simulated archaeological sample and which are not. This provides an extremely important check on assumptions built into our methods.

Museology: A museum's central resource to enable it to share knowledge and ideas is the material culture held in its collection. The objects on display in any museum are commonly vastly outnumbered by those held in its collection, not accessible to the public. Network methods can offer interesting ways of unlocking entire digitized collections to curators and the public alike, enabling their exploration from a relational perspective by focusing on how the many objects are related to each other. For example, Griffin and Klimm (2023) drew on the nearly half a million digitized objects in New York's Metropolitan Museum of Art collection, and explored it from a relational perspective. They created a two-mode network in which objects are connected to a medium (a material or a technique) to trace how the use of materials and techniques vary across time and space. In another example, Larson, Petch, and Zeitlyn (2007) focus on the people behind the collection of Oxford's Pitt Rivers Museum. They trace the connections between curators, donors, and objects, exploring, for example, networks of museum donors who were also fellows of the Royal Geographical Society or who held a diploma in anthropology from the University of Oxford.

Networks as tools: Network and graph theoretic methods have also long been used as tools to aid in other kinds of more traditional archaeological analysis. Indeed, the earliest applications of network visualizations and analysis of which we are aware were focused on seriation (Jelinek 1967; Kendall 1969, 1971a, 1971b) and the exploration of spatial organization (Clarke 1968; Doran and Hodson 1975). The more recent surge in interest in archaeological applications of network methods has also seen developments along these lines. For example, Munson (2015) used network methods and correspondence analysis to explore the complex microscale connections among contexts and stratigraphic sequences in a temple in

the Maya lowlands to identify synchronous construction episodes over a 1,600-year-long period of occupation (see Concas et al. 2019 for other approaches to seriation using similar methods). In another study, Merrill and Read (2010) used graph-theoretic methods and matrix algebra to define overlapping sets of objects that showed consistent spatial co-associations within a complicated open Mousterian site in the Levant. Their method allowed them to identify intrasite activity areas even where different activities overlap. In another publication Plutniak (2021) used graph-theoretic concepts and topological properties of networks defined based on refits of fragmented artifacts in stratigraphic context to evaluate and compare post-depositional processes. By considering the relationships among refit artifacts between layers, he was able to define a means for systematically assessing the likely degree of mixing for excavated sites. We expect that we will continue to see the use of network and graph-theoretic methods as tools for addressing other kinds of more traditional archaeological questions.

2.8 DATASETS AND EXERCISES

Throughout the remainder of this book we provide several brief case studies exploring the methods and models discussed in detail as well as exercises exploring these same methods. In order to make these examples easy to follow from chapter to chapter, we will rely on a small number of real archaeological network datasets, drawing on our own published work and data generously provided by colleagues. We have (mostly) kept the example datasets reasonably small so that network graphs and statistics are easily interpretable in print, often by subsetting much larger network datasets. In all cases we have also provided the larger version of each dataset to facilitate experimentation for advanced users. These data are available along with extensive metadata documenting them in the online companion and are briefly described here.

2.8.1 *Roman Road Network*

The development of an elaborate road system is one of the most enduring legacies of the Roman Republic and Empire. Areas that came under Roman control were connected to Rome and important provincial centers through entirely new roads as well as redeveloped existing roads. From roughly the 2nd century AD onward this resulted in an integrated terrestrial transport network connecting North Africa, the

Middle East, and western and southern Europe. Much of the subsequent development of transport systems in these regions built on this Roman system. The core structure of the present-day road networks of the Old World still reflects this system developed by the Romans.

Network science offers the ideal tools to represent and explore the Roman road system. Do all roads lead to Rome? Which roads lead you there faster and what were the important transport hubs? How do the structures of different provinces' road networks differ?

Throughout this book we will use the Roman road network to illustrate a range of network methods (Fig. 2.4). Our primary source for roads of the entire Roman world is the Barrington Atlas of the Greek and Roman World (Talbert 2000) and their digitization by the Ancient World Mapping Centre (2012). In many of our examples we will focus in particular on the roads of the Iberian Peninsula, which have been digitized in great detail by Pau de Soto (de Soto and Carreras 2021). In our analyses of the Roman road network ancient settlements are represented as nodes and the existence of a road between two settlements is represented by an edge. We also include the length of a road as an edge attribute (additional information about the edge recorded along with the network data).

2.8.2 *Southwest Social Networks/cyberSW Project Ceramic Similarity Networks*

The Southwest Social Networks (SWSN) Project (and subsequent cyberSW Project) is a large collaborative effort focused on exploring methods and models for network analysis of archaeological data to better understand patterns of interaction, population movement, and demographic change across the US Southwest and Mexican Northwest through time (ca. AD 800–1800; Borck et al. 2015; Giomi et al. 2021; Mills et al. 2013a, 2013b, 2015, 2018; Peeples and Haas 2013; Peeples and Roberts 2013; Peeples et al. 2016). During the interval considered by this project the region was inhabited largely by sedentary agricultural populations (though more mobile populations were also present throughout this period) with communities as large as several thousand people at the peak. The region is blessed with excellent archaeological preservation, a fine-grained chronology anchored by dendrochronological dates, and nearly 150 years of focused archaeological research.

These data have been used to address a number of relational questions revolving around population dynamics, social change, and the network drivers of settlement success or failure. What is the relationship between migration and transformations in social networks? What are the risks and rewards of different kinds of network

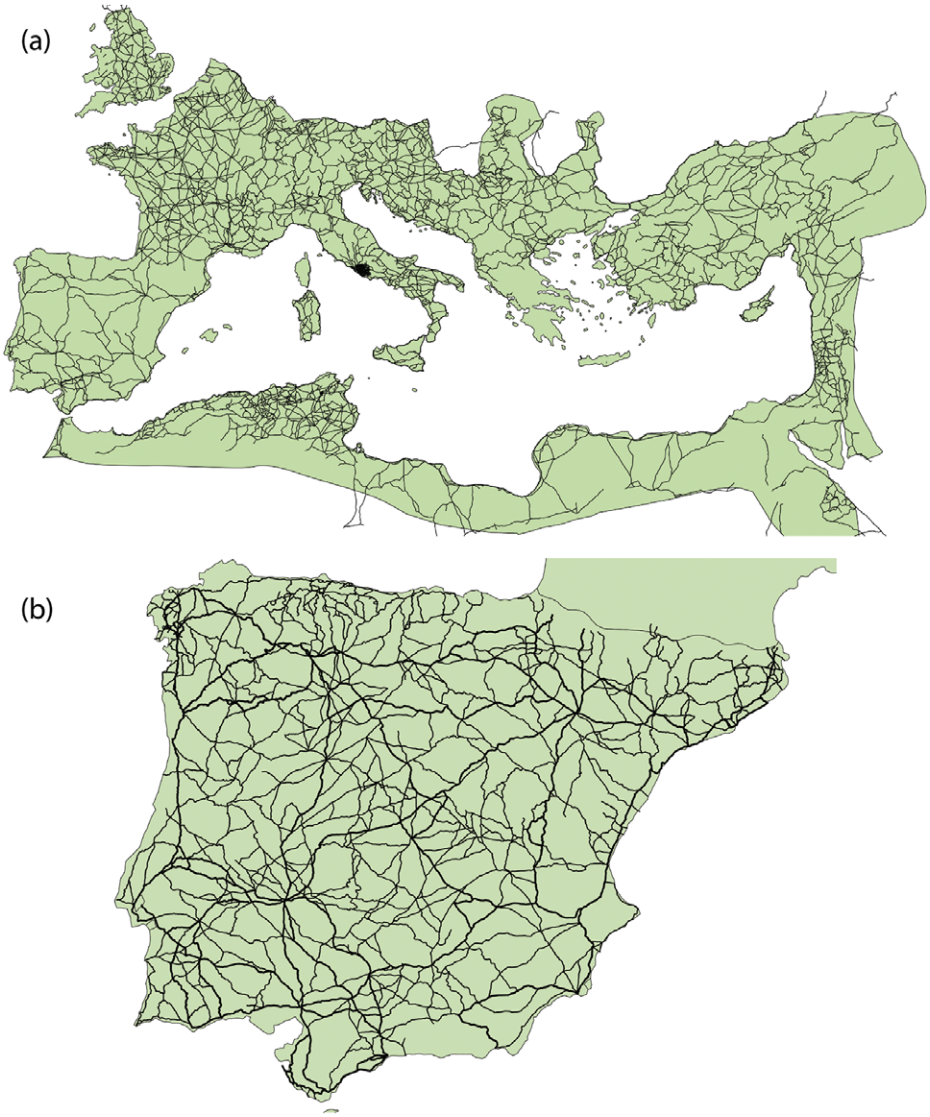


FIG. 2.4 Datasets used for our case studies on the road networks of (a) the Roman Empire as a whole (source: Ancient World Mapping Centre 2012) and (b) a highly detailed representation of the Roman road network on the Iberian Peninsula (source: de Soto and Carreras 2021).

positions? How does material similarity relate to social similarity? How does network position influence the ability of a site or region to weather severe climate challenges?

To address questions such as these and many others, the project team has gathered a massive database with information on the location and size of tens of

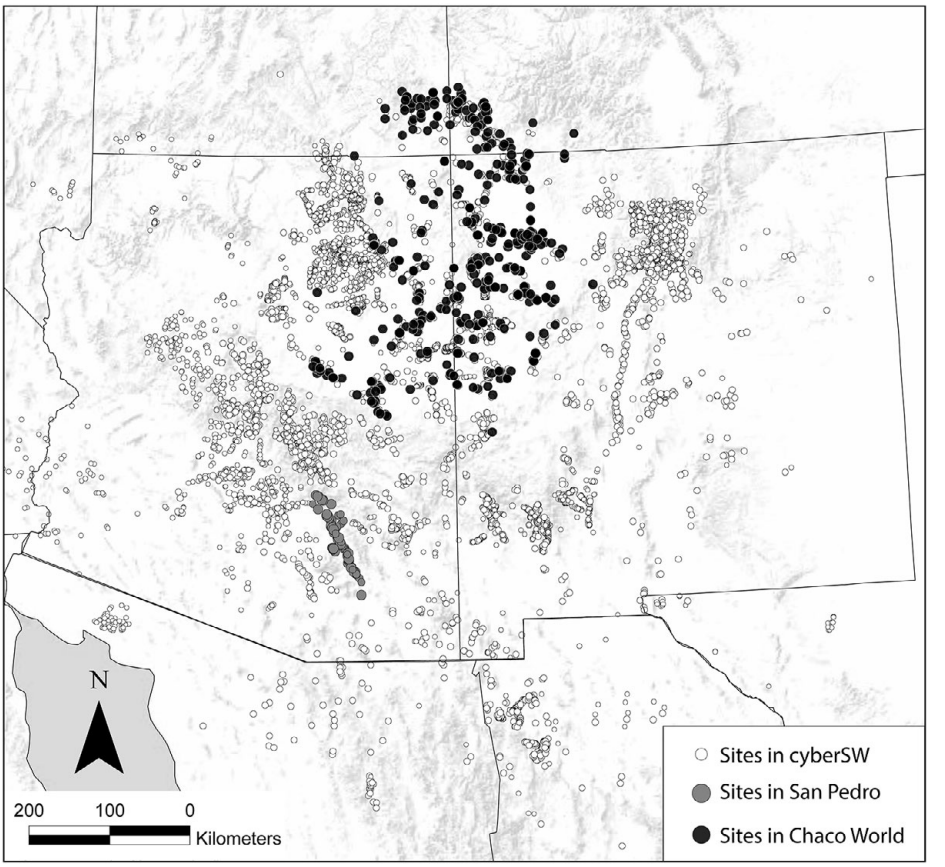


FIG. 2.5 Map of the cyberSW project study area showing all sites in the database with the San Pedro Valley and Chaco World subsets of the database shaded.

thousands of archaeological sites and ceramic and other material cultural typological frequency data consisting of millions of objects associated with those sites to explore how patterns of material similarity, exchange, and technology change across time and space in the study area (Fig. 2.5). These data as well as tools needed to analyze them are available in an online platform called cyberSW (cyberSW.org). This online platform even allows you to explore these data directly in your internet browser. The size and complexity of the SWSN/cyberSW data make it a particularly good example to use for discussing the decision processes involved in visualizing and analyzing large networks.

In several sections of this book we also use two subsets of this larger dataset: the San Pedro Valley and the Chaco World. The San Pedro Valley in southern Arizona is a well-studied portion of the SWSN study area (see Clark and Lyons 2012; Gerald

2019) that was an early focus of network methodological exploration by the team (Mills et al. 2013b). This data subset includes detailed ceramic typological frequency for all known major settlements across this region during the late pre-Hispanic period (ca. AD 1200–1450). The Chaco World is a large-scale social and political system that spanned much of the Colorado Plateau ca. AD 800–1150. This settlement system was marked by the construction of massive public architectural features known as great houses and great kivas. This subset of the database includes information on architecture and ceramic typological data for a large portion of the known Chacoan architectural complexes throughout the US Southwest. The Chaco World has been a major focus of the SWSN/cyberSW project (Giomi and Peeples 2019; Giomi et al. 2021; Mills et al. 2018).

In these networks as we use them in this book, individual settlements are treated as nodes and edges are defined and weighted based on similarities in the ceramic wares recovered at those settlements. Ceramic data used to generate networks are apportioned into a sequence of 50-year chronological intervals using methods described in detail by Roberts and colleagues (2012) so that we are able to explore change through time. Site locations and other site attribute data are also considered in some examples in the remainder of the book.

2.8.3 *Cibola Region Ceramic Technological Similarity Network*

The Cibola region along the Arizona and New Mexico border in the US Southwest is a large and diverse physiographic region spanning the southern edge of the Colorado Plateau and the ancestral homeland of the contemporary Zuni (*A:shiwi*) people. Peeples and colleagues (Peeples 2011, 2018; Peeples et al. 2021) have explored patterns of technological similarity and communities of practice in this region at a series of sites dating ca. AD 1100–1350 through explorations of corrugated ceramic cooking pots. Corrugated pots, which are produced across much of the US Southwest from at least the 9th through the 14th century, are coiled ceramic vessels where the coils used to make the vessel are never fully smoothed. Thus, these ceramics retain substantial amounts of evidence of the specific techniques used to produce them.

These ceramic technological data have been used to address many questions revolving around the relationship between technological patterns and social processes. How do patterns of technological similarity relate to patterns of interaction? What is the relationship between spatial and social distance? What are the similarities and differences in patterns of interaction suggested by comparisons of different kinds of material culture?

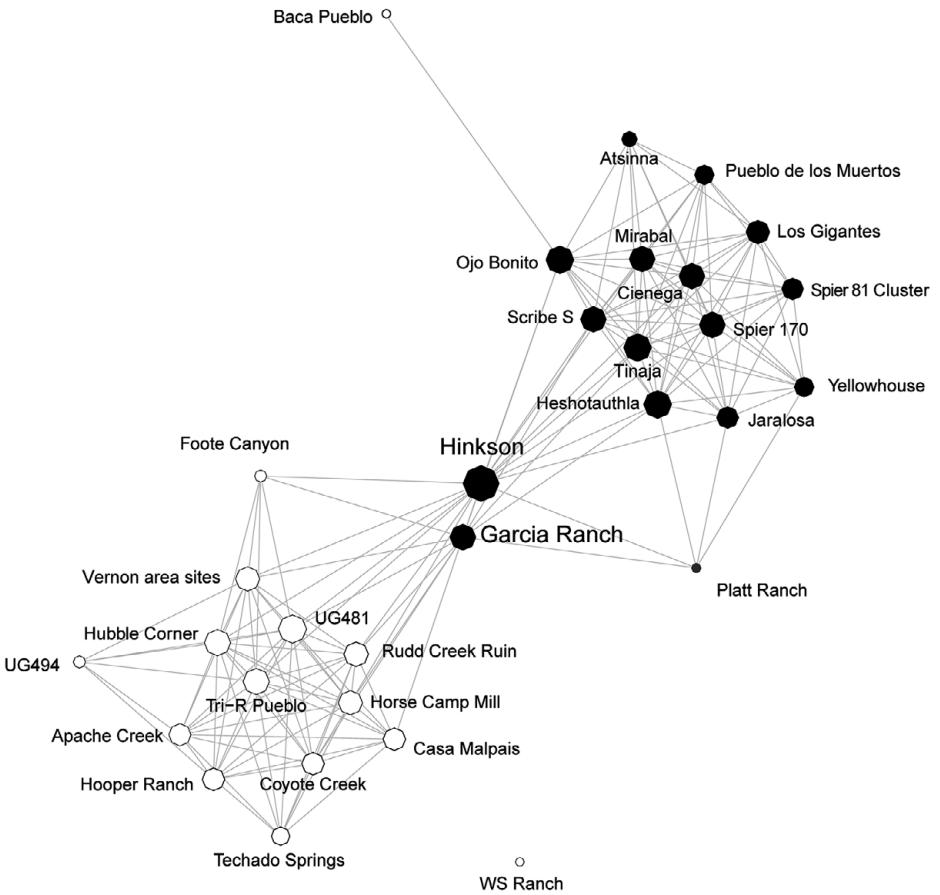


FIG. 2.6 Network graph showing connections among Cibola region settlements based on strong similarities in the technological attributes of corrugated cooking pots recovered at each site. Sites are color-coded by region where sites in the northern half of the study area are shown in black and sites in the southern half are shown in white.

In this book we use data on ceramic technological production techniques to generate similarity networks originally published by Peeples (2011, 2018). In these networks each settlement is treated as a node with similarity metrics (see Chapter 3) defining the weights of edges between pairs of sites based on an analysis of a number of metric and coded attributes of individual ceramic vessels (Fig. 2.6). Sites are temporally divided into two periods and will also be used to explore changes in network topology through time. In addition to these material cultural data, we also have additional site attributes such as location and the types and frequency of public architectural features.

2.8.4 *Himalayan Visibility Network*

Hundreds of forts and small fortified structures are located on mountaintops and ridges in the central Himalayan region of Garhwal in Uttarakhand, India. Despite being such a prominent feature of the history of the region that is interwoven with local folklore (Garhwal means "land of forts"), this fortification phenomenon has received very little research attention. It might have had its origins during the downfall of the Katyuri dynasty in the 11th century and continued up to the 15th century when the region was consolidated by the Parmar dynasty and possibly even later as attested by Mughal, Tibetan, and British aggressions.

Why were so many fortified structures located in such exceptional locations in this mountainous region? One key theory is that they enabled visual control over the surrounding landscape and could jointly have functioned as a communication system through visual signals such as smoke, fire, flags, and light. The combined use of GIS-based visibility analysis and network science methods is ideal for exploring this theory (for an example, see Rawat et al. 2020).

In this book we will use this research context as an example of spatial networks and, more specifically, visibility networks (Fig. 2.7).This is made possible thanks to the survey of forts in the region performed in the context of the PhD project by Nagendra Singh Rawat (2017). We use a catalogue of 193 sites (Rawat et al. 2020, Appendix S1), and use the case of Chaundkot Fort and its surroundings as a particular case study. Chaundkot Fort is theorized to have been one of the key strongholds in the region and is also the only one to have been partly excavated (Rawat and Nautiyal 2020). In these case studies we represent strongholds as nodes, and a probable line of sight between observers located at a pair of strongholds is represented by a directed edge. The length of each line of sight is represented by an edge attribute.

2.8.5 *Archaeological Publication Networks*

Our knowledge and stories of past human behavior are as much shaped by the material remains we excavate as they are by the actions and interactions of the archaeologists who study them. Aspects of these actions and interactions are formally represented in publications. Such papers can be coauthored, reflecting scientific collaboration networks and communities of practice. Authors cite other authors' works to indicate explicitly that they were influenced by it or that it is related to the paper's subject matter.

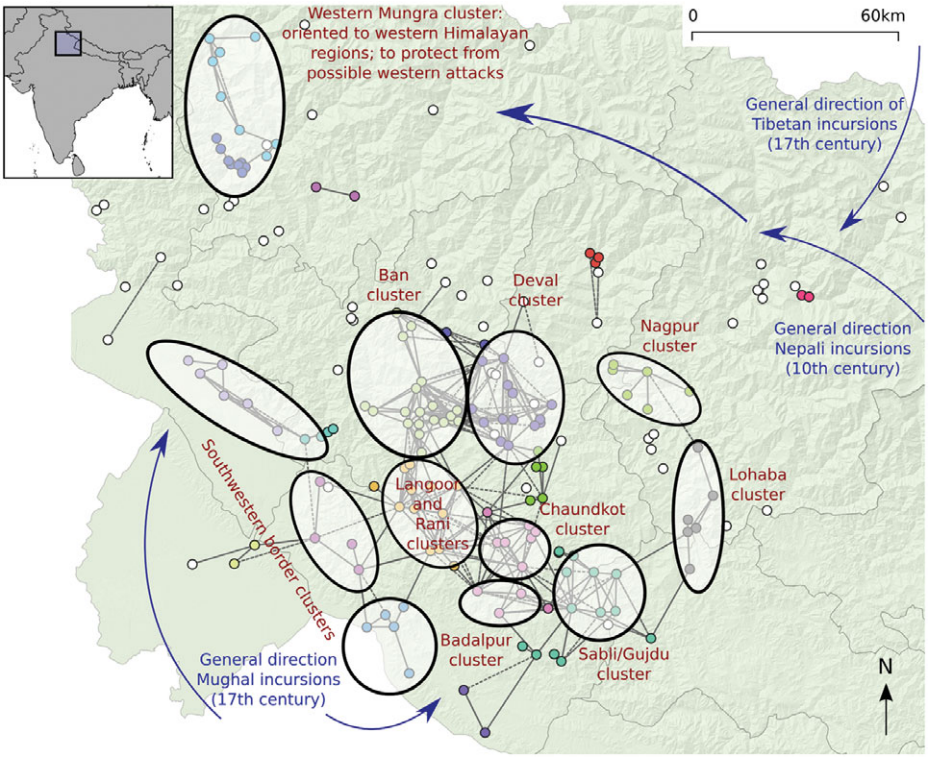


FIG. 2.7 The 193 strongholds (nodes) connected by lines-of-sight up to 25 kilometers in length (at which distance large fire and smoke signals would have been visible). Node colors represent communities of nodes identified through the Louvain modularity method (see Section 4.4.6) only for lines of sight up to 15 kilometers (from Rawat et al. 2021).

In previous work, we have turned the tools of archaeological network science on archaeological network researchers themselves (Brughmans 2013a; Brughmans and Peeples 2017). We studied the coauthorship and citation practices of the more than 250 publications that have applied formal network methods to archaeological research topics from 1967 to the present. From a list of publications, we can generate an undirected coauthorship network by representing individual authors as nodes, and connecting a pair of authors with an edge if they have been coauthors on one or more papers. These edges are further valued based on the number of papers that a pair of authors coauthored (Fig. 2.8). Moreover, a directed citation network can be made from the bibliographies of this list of publications. In a citation network, each node represents an individual publication that is connected to all other publications in its bibliography with a directed edge. The edge goes from the citing publication to the cited publication, so it represents the source and

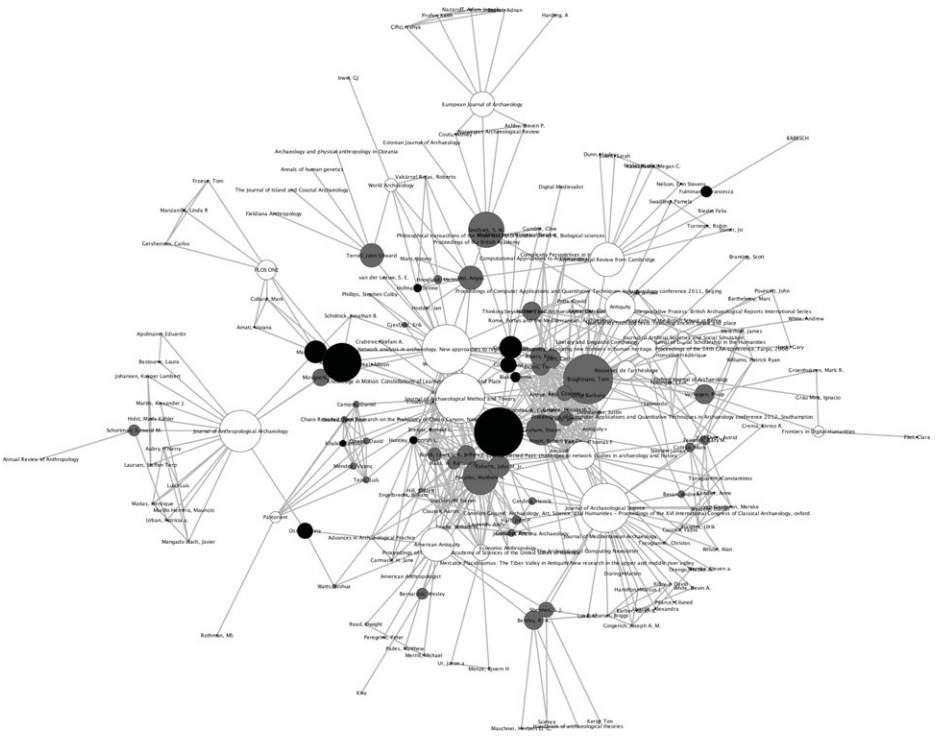


FIG. 2.8 Two-mode archaeological publication network, representing a set of individual authors as nodes that are connected to nodes in a set of publication venues (journals, books, proceedings) in which they have published (from Brughmans and Peeples 2017: fig. 10).

direction of academic influence as explicitly expressed in publication. We use networks of archaeological network research publications throughout this volume to illustrate concepts such as the acyclic structure of citation networks.

2.8.6 *Iron Age Sites in Southern Spain*

The Guadalquivir river valley in the south of Spain between present-day Seville and Córdoba was densely urbanized in the late Iron Age (early 5th century BC to late 3rd century BC). Many settlements were dotted along the rivers and the southern part of the valley (Fig. 2.6), and this settlement pattern was focused on nuclear settlements sometimes referred to as *oppida*. Some of these reveal defensive architecture and many are located on elevations. Previous studies of Iron Age settlements in the region have explored possible explanations for their locations (Brughmans et al. 2014, 2015; Keay and Earl 2011). Given their elevated locations, one theory that has received

considerable attention was intervisibility. Could small settlements surrounding oppida be seen from them, and could oppida be located partly to allow for visual control over surrounding settlements? Did groups of Iron Age settlements tend to be intervisible, forming communities that were visible on a daily basis? Were there chains of intervisibility that allowed for passing on information from one site to another via visual smoke or fire signals, and did these chains follow the other key communication medium in the area, the navigable rivers?

These questions have been explored in previous research using GIS and network methods, using a dataset of 86 sites (more about this dataset and research topic: Brughmans et al. 2014, 2015; Keay and Earl 2011). Figure 2.9 illustrates some of the results of this study, showing lines of sight connecting pairs of Iron Age settlements at distances up to 20 kilometers at which large fire and smoke signals would be visible. To account for errors in the Digital Elevation Model (DEM), a probabilistic line-of-sight analysis was performed that introduces random errors into the DEM that can have a blocking or enhancing effect on the lines of sight. Figure 2.9 shows intervisibility connections that were present more than 50 percent of the time in the probabilistic analysis, that is, the lines of sight that are most probable (see Brughmans et al. 2014 and 2015 for more details). The locations of these 86 sites and the network displayed in Figure 2.9 are also available as Appendix A in

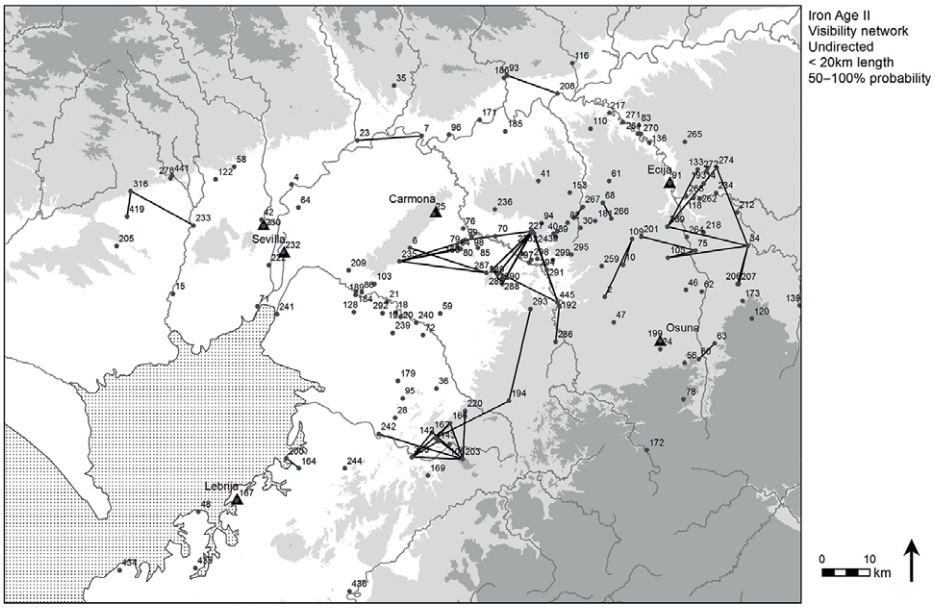


FIG. 2.9 The lower Guadalquivir river valley with the 86 Iberian (Iron Age II) sites used in the case study. Note the clustering of sites around the rivers. Lines of sight with greater than 50 percent probability shown. Source: Brughmans et al. 2014:fig. 6b.

Brughmans et al. (2014). These locations are used in Chapter 7 of this book to illustrate spatial network models that explore different geographical structures that might underlie the settlement pattern.

2.9 SUMMARY

- Network methods and models have been applied to a broad range of archaeological questions and datasets.
- Material cultural networks are typically defined based on the copresence or similarities in objects or features.
- Material cultural networks are often defined based on geochemically sourced objects, typological occurrence or frequency data, and technological characterizations of objects.
- Movement and transportation networks include networks generated based on roads, trails, rivers, and other presumed paths of travel.
- Movement networks are often used to explore questions about the transportation and provisioning of people and resources across a landscape or access among spaces at various scales.
- Spatial proximity networks refer to networks generated based on the relative location or configurations of nodes in relation to one another.
- Spatial proximity models have been used to address archaeological questions regarding likely patterns of interaction and also often serve as null models for other kinds of social network analyses.
- Visibility networks are typically defined based on line-of-sight relationships among nodes of interest.
- Visibility networks have been used to explore intervisibility and communication networks, monitoring, and shared cultural connections to visible landmarks.
- There are many more questions and datasets that have been used to generate networks in archaeology.
- Datasets used in this book are described in detail.

Further Reading

Introductions to all of the types of archaeological networks featured above and many more can be found in the following handbook.

Brughmans, Tom, Barbara J. Mills, Jessica L. Munson, and Matthew A. Peeples (editors) 2023 *The Oxford Handbook of Archaeological Network Research*. Oxford University Press, Oxford.

The following edited volumes and special issues provide many applied examples of archaeological network research in a diverse range of research contexts.

Brughmans, Tom, Anna Collar, and Fiona Coward 2016 *The Connected Past: Challenges to Network Studies in Archaeology and History*. Oxford University Press, Oxford.

Collar, Anna, Tom Brughmans, and Barbara J. Mills (editors) 2015 The Connected Past: Critical and Innovative Approaches to Networks in Archaeology. *Journal of Archaeological Method and Theory* 22(1).

Knappett, Carl 2013 *Network Analysis in Archaeology: New Approaches to Regional Interaction*. Oxford University Press, Oxford.

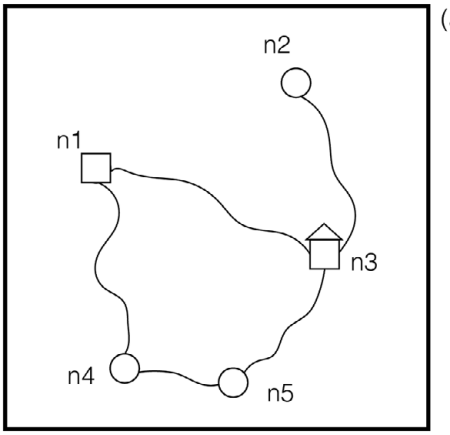
3

Network Data

In Chapter 1, we defined a *network representation* as a formal abstraction created for the purposes of visualization or analysis. Such network representations are created using *network data*. In this chapter, we define network data, its diverse types and data formats, and we give a wide range of examples of how it can be used to represent abstractions of archaeological data and relational theories. We conclude this chapter with best practice guidelines for how to go about collecting, documenting, storing, and sharing your network data.

3.1 WHAT ARE NETWORK DATA?

Network data consist of a set of nodes and at least one structural variable (represented by a set of edges) connecting pairs of those nodes. A variable is a term used to describe, for analytical purposes, any feature or element of data that can change or vary. For example, if we wish to explore the types of sites in a set of Roman towns, then their designation as colony, municipium, or town could be represented in a variable "site type," since it varies between sites (Fig. 3.1a). In this case, "site type" is a site-specific variable because it describes a feature of individual sites, and in network research it could be considered a **node variable** (sometimes referred to in the social networks literature as a composition variable or node attribute) (Fig. 3.1b). All other archaeological data we have recorded for these sites, such as the material remains excavated at them, could equally be represented as node variables. Network data additionally include **structural variables**: descriptions of features or elements of pairs of nodes, where each node can be part of multiple node pairs. For example, if we know that some pairs of sites were connected by



(a)

(b)

Node variable

Node	Site type
n1	municipium
n2	town
n3	colony
n4	town
n5	town

(c)

Structural variable

Dyad	Road
n1,n2	0
n1,n3	1
n1,n4	1
n1,n5	0
n2,n3	1
n2,n4	0
n2,n5	0
n3,n4	0
n3,n5	1
n4,n5	1

FIG. 3.1 Network data representation of sites represented as nodes, with a certain site type represented as node variables, and connected by roads represented as structural variables.

ancient Roman roads, then the presence or absence of roads between all pairs of sites (called dyads in network terminology; see Section 4.4.2) could be represented in the structural variable called "roads" (Fig. 3.1c). Structural variables can also be defined based on comparisons of node variables. For example, edges can be defined between pairs of sites that share some node variable or attribute.

The formal representation of structural variables is what defines network data. It enables the study of a wide range of dependencies within and between these variables that allow us to formally express and explore relational theories:

- **Dependencies between structural variables:** Is the presence/absence of some roads dependent on the presence/absence of others? For example, in Figure 3.1, the road from site n4 to site n5 could have been created to enable more efficient movement between these sites: in this relational statement, the existence of a road is proposed to be dependent on the previous structure of all other roads because the previous configuration did *not* facilitate efficient movement between n4 and n5.
- **Dependencies between structural and node variables:** Are pairs of sites connected by a road because they are of the same site type or because they have a certain site type? For example in Figure 3.1, is the colony site n3 connected by the highest number of roads *because* it is a colony? Alternatively, are the towns n4 and n5 connected to each other *because* they are towns (i.e., because they have the same value of their node variable)? In the first relational question, the

existence of roads (a structural variable) is proposed to be dependent on the site type of one site (a node variable), and in the second example the existence of a road (a structural variable) is proposed to be dependent on the similarity of a feature of a pair of sites (a node variable).

- **Dependencies between node variables:** What associations exist between the site type of different sites? What associations exist between the site type and the other information we have available about these sites, such as the artifacts excavated at them? Network data shares this type of dependency with other data types, and exploring correlations and associations between nonstructural variables is the bread and butter of much data analysis.

Network data consist of a set of nodes and at least one structural variable (represented by a set of edges) connecting pairs within this set of nodes.

Node variables or node attributes are data of any form (categorical, presence/absence of a particular feature, measurements, etc.) pertaining to individual nodes within a network.

Structural variables are descriptions of features or elements of pairs of nodes, where each node can be part of multiple node pairs.

A crucial requirement implied by this definition of network data is that the *boundaries of nodes, edges, and variables need to be defined* for analytical purposes. For example, one has to be able to define the limits of a site, a path, and a site type. All archaeologists know this is potentially problematic, but that does not stop any of us from constantly drawing boundaries in all our quantitative and qualitative work. We talk about sites and pottery types not necessarily because we believe they were meaningful categories in the past, but because they are useful analytical tools to allow us to better study the past and perform comparisons with the work of others. Moreover, the requirement to specify boundaries is one that network data representation shares with all other types of formal data representation. All techniques developed for other data types to add uncertainty or fuzziness to uncertain boundary identifications can equally be used with network data (see Peeples et al. 2016:67–70; Chapter 5). One can also approach this uncertainty by trying out different ways to define boundaries and exploring the impact on the network created. For example, if the key results of your network analysis change between two plausible definitions of your node boundaries, then we know this definition is critical to your study (see Chapter 5 for a further discussion of this issue).

3.2 NETWORK DATA FORMATS

Although we commonly view a network as a set of points connected by lines, there are in fact a variety of ways in which network data can be recorded and presented. Some of these are frequently used mathematical notations of network data; others are more often used simply to import network data into network science software. In this section we will introduce some of the most important data formats. They will be illustrated through a subset of the Himalayan Medieval strongholds network (see Section 2.8.4), focused on the large fort of Chaundkot (triangle in Fig. 3.2a) and the smaller strongholds in its vicinity, many of which can be seen from Chaundkot at distances shorter than 15 kilometers as represented by the edges (Fig. 3.2a).

Although we have largely restricted mathematical notation and algorithms to the glossary, there is some basic notation that we will have to use throughout the book.

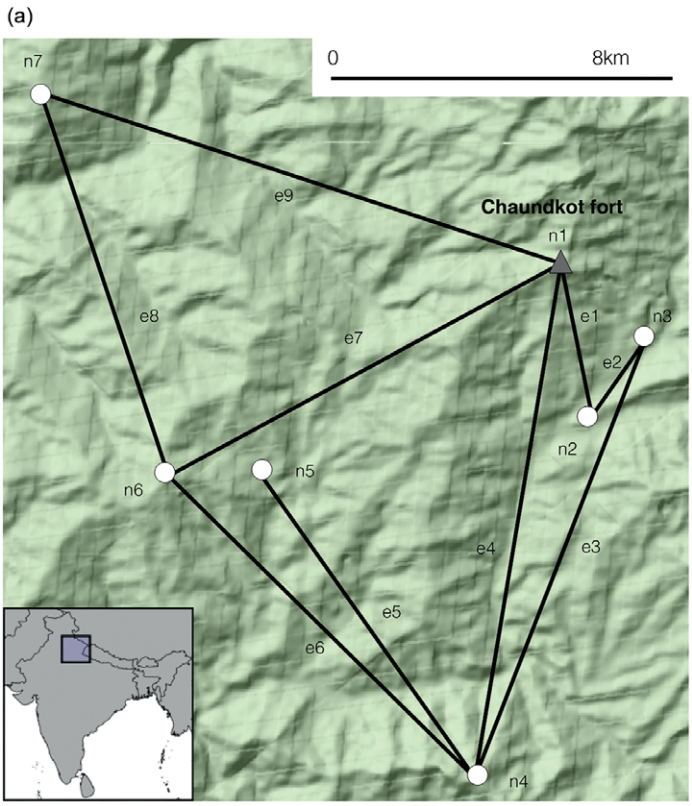
- N represents a set of nodes and g indicates the number of nodes. This set of nodes can thus be written as $N = (n_1, n_2, n_3, \ldots n_g)$. The lowercase notation n_1 is used to refer to individual nodes.
- E represents a set of m edges, and this set can be written as $E = (e_1, e_2, e_3, \ldots, e_m)$. Again, the lowercase notation e_1 is used to refer to individual edges.
- Since each edge connects a pair of nodes, an edge can also be written as (n_1, n_2) and the edge list can be written as $E = ((n_1, n_2), (n_1, n_3), (n_1, n_4), \ldots, (n_i, n_j))$.
- A network is represented by G (which relates to the *graph* in terminology often used in mathematics) and contains a set of nodes and edges, thus, $G = (N, E)$.

3.2.1 *Edge List*

The edge list is a very quick and easy way to capture network data. It simply lists the edges in the network one by one by node ID: $E = ((n_1, n_2), (n_1, n_3), (n_1, n_4), \ldots, (n_i, n_j))$. The edge list of the example network in Figure 3.2 can be written as:

$$E = ((n_1, n_2), (n_1, n_4), (n_1, n_6), (n_1, n_7), (n_2, n_3), (n_3, n_4), (n_4, n_5), (n_4, n_6), (n_6, n_7)).$$

An alternative way of writing the edge list is to treat it as a long data table with one edge per row as in Figure 3.2b. The edge list in its vertical format is a common input format for network science software. Indeed, the edge list is a great format for the initial collection of network data, in particular when this is done manually, as only the edges that are present need to be recorded. As many real-world networks are



(d) Adjacency matrix

	n1	n2	n3	n4	n5	n6	n7
n1		1	0	1	0	1	1
n2	1		1	0	0	0	0
n3	0	1		1	0	0	0
n4	1	0	1		1	1	0
n5	0	0	0	1		0	0
n6	1	0	0	1	0		1
n7	1	0	0	0	0	1	

(e) Incidence matrix

	e1	e2	e3	e4	e5	e6	e7	e8	e9
n1	1	0	0	1	0	0	1	0	1
n2	1	1	0	0	0	0	0	0	0
n3	0	1	1	0	0	0	0	0	0
n4	0	0	1	1	1	1	0	0	0
n5	0	0	0	0	1	0	0	0	0
n6	0	0	0	0	0	1	1	1	0
n7	0	0	0	0	0	0	0	1	1

FIG. 3.2 A subset of the Himalayan Medieval strongholds network (see Section 2.8.4).

sparse, meaning that there are a large number of absent edges, this can often greatly reduce the workload. As we discuss further below, where edges have weights, direction, or other attributes associated with them, these can be recorded as additional columns in the edge list.

3.2.2 *Adjacency List*

The adjacency list consists of a set of ordered lists, where the first node in each list is connected to all subsequent nodes in that same list. It is therefore more concise than the edge list (in which each relationship has its own row in a tabular format), but unlike the edge list it does not result in lists of equal length (each row in an edge list typically has two values, representing the pair of nodes). For example, the first list entry in the adjacency list in Figure 3.2c is n_1, n_2, n_4, n_6, n_7; representing Chaundkot fort (n_1) and the four other strongholds it is connected to via lines of sight. This network format can help to speed up data collection but is often quite awkward to use for further analyses so is typically converted to another format. Its conciseness is useful for saving space and reducing file sizes.

3.2.3 *Adjacency Matrix*

The adjacency matrix is one of the most common and versatile network data formats for data analysis in network science (in sociology it is sometimes referred to as the sociomatrix). It is a matrix of size $n \times n$, with a set of rows equal to the number of nodes, and a set of columns equal to the number of nodes. The node names or identifiers are typically used to label the columns and rows. When a pair of nodes is connected by an edge (i.e., when they are *adjacent*), then the corresponding cell will have an entry. For example, from fort Chaundkot the stronghold n_2 can be seen, hence the two cells in the adjacency matrix corresponding to (n_1, n_2) and (n_2, n_1) have a value of 1 (Fig. 3.2d). As discussed below, where there are weights or values associated with specific edges, these can be indicated in the relevant cell instead. Notice that the adjacency matrix is symmetric because it is in this example not treated as a directed network (see also Section 3.3.2). This means that (n_1, n_2) and (n_2, n_1) are considered the same, and as a result the top and bottom halves of the adjacency matrix will be the same. These two halves are separated by the diagonal, representing edges connecting each node with itself. In networks where such self-loops are not allowed or are not meaningful (as in the case of the line-of-sight network), this diagonal will be empty (they are shaded in Fig. 3.2d). In some cases for undirected networks, only the upper or lower triangle of the matrix will be presented. For small networks or subsets of networks exploring the adjacency matrix directly like this can sometimes be informative. In larger networks this becomes intractable and it is often difficult to even fit a matrix on a single page or screen.

The order of the rows and columns is arbitrary, but often row and column ordering (often called *permutation*) can be used for data management purposes (e.g., to order alphabetically on node label), analysis, or to visually represent information. For example, the rows and columns can be ordered according to the number of connections each node has or to highlight particularly dense clusters in the network. In fact, matrices are excellent network visualization formats, especially for dense networks, since they allow for highlighting the absence of edges, whereas most other representation formats tend to highlight the presence of edges (see also Chapter 6). When manually creating a network dataset, the adjacency matrix will probably not be your format of choice given it requires all absent edges to be written as well and is generally unwieldy to work with manually for all but the smallest networks. However, data can be exported in matrix format from your database management system through queries, and most network science software packages allow for creating and exporting adjacency matrices from imported edge lists or other formats. Adjacency matrices are memory intensive for your computer, though, and can be inefficient for the study of very large networks.

3.2.4 *Incidence Matrix*

A second common matrix format for the representation of network data is the incidence matrix. Whereas the adjacency matrix is a symmetric square matrix because it has an equal number of rows and columns, the incidence matrix will typically be rectangular because it is used to represent sets of entities or events in the rows and columns (although it can be symmetric if the number of entities and events happen to be the same). The incidence matrix can be used to record the *incidence* of nodes with edges (Wasserman and Faust 1994:152). In such cases, the nodes are listed in the rows and the edges in the columns, where an entry is made in a cell if the corresponding node and edge are connected. This means each column in the incidence matrix has two entries. For example, Chaundkot fort has a line of sight e_1 emanating from it, hence the cells corresponding to (n_1, e_1) and (n_2, e_1) have entries (Fig. 3.2e).

Incidence matrices are more frequently used for representing so-called *two-mode* or *bipartite* networks, in which edges between two distinct sets of nodes are recorded. In this case the number of entries per column does not have to equal 2 given that the columns do not represent edges but instead some other feature or attribute or event that defines a second set of nodes. Since the number of nodes in both sets are not necessarily equal, the incidence matrix offers an appropriate

representation of this type of network data (two-mode networks will be elaborately discussed in Section 3.3.4).

3.2.5 *Node and Edge Information*

The four network data formats described above allow us to capture structural variables: the nodes and edges. However, archaeological network research is an applied discipline in which these nodes and edges are used to represent particular things, for example, sites, artifact types, proximity, or lines of sight. We typically have some additional information about the nodes and edges, and, depending on our research aims, it could be useful to attach this information to the nodes and edges.

A common format for doing so is to create a node or edge table, in which each row represents a node or edge and every column holds one type of information about the node or edge. Figure 3.3 shows the node and edge tables of the subnetwork of Himalayan strongholds. In this example, the node table (Fig. 3.3a) includes the latitude and longitude coordinates of each fort, its name, the type of fort and landscape location, its elevation, and the rock formation it is located in. The edge table (Fig. 3.3b) includes the distance of each line of sight in meters. As noted above, edge-specific information can be directly appended to the edge list format (Fig. 3.3c). By reading these tables into our network analysis software and attaching the data to structural variables, we can use this additional information for network visualization (e.g., represent all major forts as gray triangles, Fig. 3.3d) or to

(a) Node table

Node	Latitude	Longitude	Name	Type	Landscape	Elevation (m)	Rock Formation
n1	78,9010	29,9781	Chaundkot	Fort	Ridge	2040	Nagthat
n2	78,9066	29,9500	Govin	Fortalice	Saddle	1296	Nagthat
n3	78,9184	29,9645	Kim	Fortalice	Isolated hill	1437	Nagthat
n4	78,8835	29,8848	Chhetra	Fortalice	Ridge	1979	Chandpur
n5	78,8383	29,9403	Guraad	Fortalice	Saddle	1423	Chandpur
n6	78,8180	29,9398	Garhi_4	Fortalice	Ridge	1786	Chandpur
n7	78,7920	30,0085	Gahad_4	Fortalice	Ridge	1558	Chandpur

(b) Edge table

Edge	Distance (m)
e1	3131,3
e2	1866,28
e3	9465,01
e4	10504,52
e5	7577,5
e6	8684,77
e7	9013,67
e8	8128,5
e9	11059,9

(c) Edge list plus table

Edge	Nodepair	Distance (m)
e1	n1,n2	3131,3
e2	n2,n3	1866,28
e3	n3,n4	9465,01
e4	n1,n4	10504,52
e5	n4,n5	7577,5
e6	n4,n6	8684,77
e7	n1,n6	9013,67
e8	n6,n7	8128,5
e9	n1,n7	11059,9

(d) Network



FIG. 3.3 (a) Node table and (b) edge table of (c) the network shown in Figure 3.2.

explore dependencies between the structural variables and the node and edge information (see Section 3.1).

3.3 TYPES OF NETWORKS

Although all network data consists of structural variables between nodes, sometimes it is useful to represent additional information about these variables, such as their strength or directionality. Other times it might be appropriate to use types of structural variables that have to follow specific rules, such as only allowing for edges to cross from one set of nodes to another set of nodes. In this section we give an overview of the most commonly used network data type, focusing in particular on those network types that have been used widely in archaeological network research.

3.3.1 *Simple Networks*

We call network data a **simple network** (or simple graph) if we only have a set of nodes and a set of edges connecting them, with no additional information about the edges or specific rules they need to follow. Simple networks are unweighted and undirected one-mode networks. A network is considered **connected** if all node pairs can be connected by a path, which means that no nodes are isolated (unconnected) and there are no separate components (for paths and components, see Sections 4.4.5 and 4.4.6). A network is considered **empty** if it contains no edges. At the other extreme, if all node pairs are connected, then it is considered **complete** (i.e., all edges that theoretically could exist in the network are present). Although empty and complete networks are useful tools for analytical comparison, almost all archaeological applications of network science work with empirical or theoretical networks that lie somewhere on the spectrum between these two extremes. Network researchers sometimes describe networks as falling along a spectrum between **dense networks** where nearly every possible edge is present and **sparse networks** where very few of the possible edges are present. These terms are, of course, relative and do not refer to any particular cutoff but these concepts are often tied to assessments of network properties like density (discussed in Section 3.3.1). Archaeological co-occurrence or similarity networks (see Section 3.3.5) tend to be dense networks where most of the possible relations are active or have at least some non-zero weight whereas archaeological road networks tend to be relatively sparse where most nodes are not directly connected (Fig. 3.4).

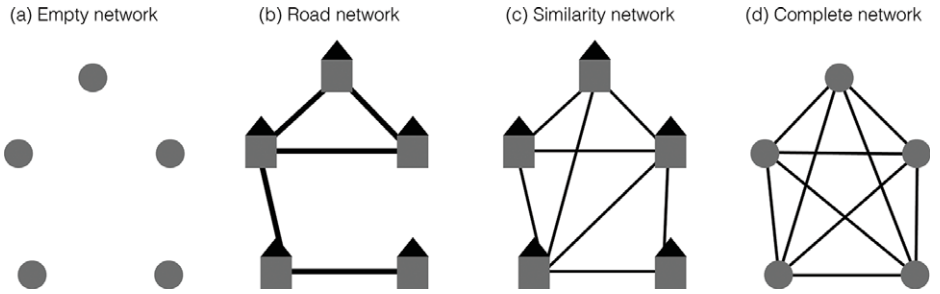


FIG. 3.4 The spectrum between empty and complete networks. Archaeological road networks tend to be closer to empty networks than archaeological similarity networks.

A *simple network* consists of a set of nodes and a set of edges with no additional information about them.

An *empty network* contains no edges.

A *complete network* contains all possible edges.

A *dense network* contains a high proportion of the possible edges for a set of nodes.

A *sparse network* contains a small proportion of the possible edges for a set of nodes.

3.3.2 *Directed Networks*

Sometimes relationships are directional, meaning they have an orientation. For example, the flow of a river is directed downstream. In such cases we can incorporate this information in our network data by distinguishing between the source and the target of an edge. If a section of a river flows from confluence A to confluence B, then A can be represented as the source node and B as the target node. In fact, many networks used in archaeological network research can be considered directional: roads (Fig. 3.5a), rivers (Apolinaire and Bastourre 2016; Peregrine 1991), lines of sight (from the observer to the observed; Brughmans et al. 2015; Earley-Spadoni 2015), citation (from the newer publication to the older publication; Brughmans and Peeples 2017), least-cost paths (uphill or downhill; Herzog 2013), semantic networks (subject-predicate-object; Vitale and Simon 2023), food webs (X is eaten by Y; Crabtree 2015; Crabtree et al. 2021), or even material culture networks representing flows of material from source to destination (Bernardini 2007).

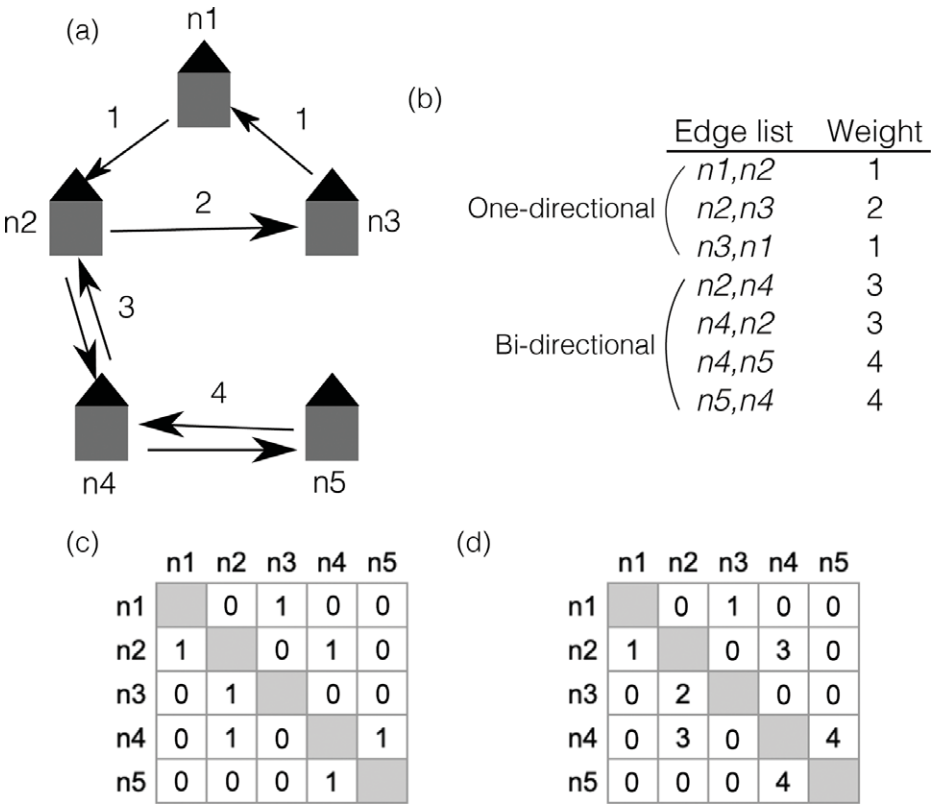


FIG. 3.5 (a) A weighted directed road network including three one-way streets, with (b) its edge list with values, (c) its asymmetric binarized adjacency matrix, and (d) its asymmetric weighted adjacency matrix.

A **directed network** (sometimes called a *digraph*) consists of a set of nodes and edges connecting them for which the orientation or direction is specified (directed edges are sometimes called arcs). It follows that in directed networks (n_1, n_2) does not necessarily equal (n_2, n_1). An edge list in directed networks can therefore have more rows than in an undirected network with the same number of nodes (Fig. 3.5b). The adjacency matrix of a directed network also differs from its undirected version: it tends to be asymmetric, meaning the top triangle above the diagonal will differ from the bottom triangle because each captures a different orientation. Edges are typically recorded from column to row in the adjacency matrix of a directed network (Fig. 3.5c).

Some directed networks tend to be **unidirectional**, like the direction of a river stream, in which the river flows in only one direction. Hence, the edges in the opposite direction cannot exist and the number of edges between each pair of nodes

can be no more than one. Citation networks are another example of networks that are typically unidirectional: publication A can cite an older publication B, but the older publication B cannot in turn cite the younger publication A. Other directed networks can be considered **bidirectional** if movement in both directions is a possibility. For example, a two-lane road can allow for traffic from junction A to junction B and vice versa. Crucially, bidirectional networks do not require all node pairs to be connected by two edges (one for each direction); otherwise, we might as well represent them as simple networks. For example, a road network can include one-way streets as well as two-way streets. Acyclic networks are a specific type of directed network used for citations, genealogies, and hydrology, discussed in detail in Section 4.4.4.

A *directed network* consists of a set of nodes and edges connecting them for which the orientation or direction is specified.

3.3.3 *Signed, Categorized, and Weighted Networks*

Not all relationships are the same: some are strong and others are weak, some are frequently used and others not, some are highly probable and others not, some are long and others short, some are positive and some are negative. Relationships can carry a value, and network science offers a range of approaches to incorporate this information into our network data. In this section we briefly outline various common approaches to incorporating signs, categorical classifications, and weights or values into relational data and network representations.

A **signed network** is a network where the edges carry additional information in the form of a positive or negative sign (sometimes referred to as the edge's *valence*). A positive sign is typically represented as "+" and a negative sign as "−". Edges in signed networks can either be present and positive, present and negative, or absent (Fig. 3.6). It is therefore crucial not to confuse the binary nature of signs with the

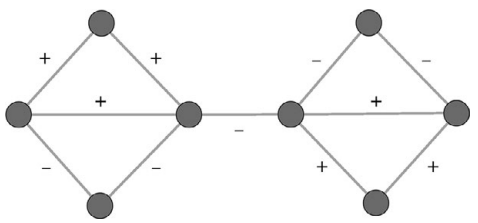


FIG. 3.6 An example of a signed network.

presence or absence of the edges themselves; rather, signs are values of present edges. Signs can be used to represent binary and opposed values of edges such as "positive" and "negative," "friend" and "enemy," or "steals from" and "gives to." Empirical work focused on signed networks has shown that the structural features of relations characterized by negative and positive ties often differ within the same network. For example, Szell and Thurner (2010) observed interactions among players in a massive multiplayer online computer game and coded interactions among friends (positive edges) and enemies (negative edges). In their study they found that positive relations were more likely than expected by chance to entail tight-knit groups of nodes characterized by reciprocal interaction (or closed triads; see Section 4.4.2), whereas negative relations were typically represented by sparse connections. Thus, adding signs to edges provides new avenues for exploration and new relational theories and structural relationships to evaluate (such as structural balance; see Facchetti et al. 2011). Importantly, although the specifics are beyond the scope of this book, signed networks also sometimes require somewhat different analytical assumptions and methods, so they should be applied with care (Chiang et al. 2014; Fenoaltea and Meng 2021).

We are unaware of any examples of archaeological networks that explicitly use signed networks and signed network methods. There are, however, examples of archaeological networks with ties coded by content where such methods might be particularly relevant. For example, Munson and Macri (2009; see Box 2.3) create epigraphic networks among Classic Maya settlements with edges defined based on mentions of settlements in monumental inscriptions (i.e., if settlement A is mentioned in an inscription at settlement B, there would be a connection from B to A). The context of the text of each inscription is further used to code each relation as antagonistic, diplomatic, subordinate, or based on lineages. Although Munson and Macri (2009) do not explicitly use signed network methods, they do find different properties and correlations among networks defined based on these different kinds of relations. This example would provide an excellent context for exploring signed network approaches to compare, for example, antagonistic and diplomatic ties in terms of structural balance. This could be used to test relational theories about the expected topological properties of networks based on positive or negative edges.

As this discussion of signed networks suggests, it can often be useful to code relations in a network using additional information about the nature or content of the interaction. In some cases, however, it may not make sense to think of relations as either positive or negative (opposites) but rather to use some other simple nominal categorization (like the Munson and Macri 2009 example discussed above). Although such an approach has some features in common with signed

networks, some of the special features, methodological tools, and theories designed for dealing with signed networks (which often assume edges with oppositional properties and evaluate balance among signed relations) are not relevant. Although there is not a commonly used name for networks with such coded relations, we call them **categorized networks** here. Categorized networks are networks where pairwise relations are coded according to some nominal categorical scheme that does not represent an opposition. One classic example of such a coding scheme comes from sociologist Mark Granovetter's "strength of weak ties" argument (Granovetter 1973). In this work, he argues that small, well-defined groups tend to be connected by strong ties, whereas relations between groups tend to be weak but exercise significant cross-group power. He defines ties as either "strong" or "weak" not based on any network properties but based on an external assessment of the nature, content, and time involved in each relation. This perspective offers a concise relational theory that can be represented as a categorized network provided that we can provide a satisfactory means for defining and recording "strong" and "weak" ties. This theory further predicts specific differences we would expect for the topological properties of strong and weak ties, respectively.

Such an approach to categorizing network relations would certainly be relevant in many archaeological contexts, although applications thus far are rare. The Munson and Macri (2009) example described above provides a good example of an application where categorizing relations in such a way resulted in new relational insights. For example, they demonstrate that different kinds of relations operate at different modal distances and also have different topological properties. For example, lineage relations showed a higher degree of centralization (concentration of ties in a small number of nodes; see Section 4.4.7), suggesting that the dominant patterns of familial relations among Maya rulers may have been limited to a smaller number of settlements than other kinds of relations. Where relations can be similarly categorized, we argue there is considerable potential for insights such as this that would be missed were edges not coded.

In many cases, we may also have some means for defining the strength or value of edges in a network based not on a simple binary or categorical classification but on ordinal or continuous data. In such cases we can use **weighted networks** (sometimes called *valued networks*) in which each edge carries a value. The edge value can be recorded in an edge table. For example, the edge table in Figure 3.3b records the length in meters of each line of sight between pairs of strongholds, which can be used as an edge value. The adjacency matrix of a weighted network can include edge values by replacing the binary 0 and 1 values representing the presence or absence of edges with the edge values (Fig. 3.5d). Weighted networks

can be used to represent the strength, frequency, or intensity of a relationship typically by using numerical continuous or ordinal values. Examples of weighted edges in archaeological network research abound. They include the frequency of contact between island communities (Amati et al. 2018; Evans et al. 2012), the physical distance between sites (Leidwanger 2013), the length of a line of sight (Brughmans and Brandes 2017), and the artifact similarity value of a pair of site assemblages (Mills et al. 2013a; Peeples and Haas 2013).

Note how these valued networks allow us to incorporate a wide range of types of information from the qualitative "weak" and "strong" values of categorized networks to the use of ordinal and continuous values. Such edge values can be taken into account when analyzing the structure of the network, allowing us to ask questions such as "What sets of sites form clusters with highly similar artifact assemblages?" or "What cities are easily reachable from all other cities in light of the length of all roads connecting them?" Moreover, if edge values are key to the dependencies studied, then a valued network should not be binarized (i.e., the values of the edges should not be reduced to a mere presence or absence of links) given that the unweighted and weighted versions of network analysis techniques will often yield different results (see Chapters 4 and 5; Peeples and Roberts 2013).

Weighted networks are relatively common in archaeological network research, and weights have been defined using a range of different data types and assumptions (see discussion in Prignano et al. 2017). For example, Coward (2013; see also Sindbaek 2007a, 2007b, 2013 for a similar example) creates a network based on co-occurrence in material cultural categories and weights such ties based on the number of categories that co-occur. This model for weighting edges assumes that contexts with more kinds of items in common can be conceptualized as having stronger relations than those with fewer or no co-occurrences. Weighted networks are also often used to encode information about spatial proximity and travel time. For example, the ARIADNE model is an archaeological network model of movement among Bronze Age sites in the Aegean Sea, and edges over land and water among settlements are weighted using distance, travel costs, and modeled estimates of probable resource flow based on settlement size (Evans et al. 2012; Knappett et al. 2008; see also Amati et al. 2018). In this model, edge weight is modeled to represent the likely frequency and volume of interaction among settlements given the properties of those settlements and their relative locations. As these two brief examples highlight, weighted networks require researchers to develop not only a means for representing variability in edges through network data, but also a clear model for interpretation of the meaning of variability in weight.

A *signed network* is a network where the edges carry a positive or negative sign indicating some opposed property of relations in the network.

A *categorized network* is a network where edges are classified according to some nominal category that does not necessarily represent an opposition (differentiating it from a signed network).

A *weighted network* is a network in which the edges carry a nonbinary value that indicates the strength of a given relationship.

3.3.4 *Two-Mode Networks and Affiliation Networks*

In the network types introduced so far, we consider one set of nodes and a single structural variable related to those nodes. However, in some research contexts we study distinct sets of entities such as artifact types and sites, individuals and their region of origin, or authors and the publications they coauthor. In these cases we can also consider structural variables measured between the two different sets of entities such as the *presence of* artifact types at sites, individuals *originating from* a particular region, or authors *writing* a particular publication. These are all examples of relationships crossing from one type of entity to another.

Such networks containing two sets of nodes and a structural variable *between* them (represented as edges) are called **two-mode networks** (in the sociology literature, and bipartite networks/graphs in physics and mathematics). Each distinct set of nodes is considered a so-called **mode** (multimodal networks with three or more classes of nodes are also possible but outside the scope of this book; see Melamed et al. 2013). In sociology, two-mode networks are often used for studying the **affiliation** of individuals with organizations, such as the presence of professionals on the boards of companies or the attendance of scholars at conferences (referred to as **affiliation networks**). Two-mode networks offer an excellent way to represent and explore the consequences of common membership in groups or copresence in places. Many researchers have highlighted the duality of actors and events in affiliation networks (Breiger 1974; Field et al. 2006; Simmel 1955), meaning that events mediate connections among actors and actors mediate connections among events (i.e., when one actor is involved in two events, those events are also connected and vice versa). In archaeological network research, two-mode networks are commonly used to represent artifact assemblages using co-occurrence or similarity to define edges.

A *mode* is a distinct set of nodes.

Two-mode networks are networks where two separate categories of nodes are defined with a structural variable (edges) only between these categories.

Affiliation in social network analysis refers to data relating to the presence/absence or weight of relationships between two sets of nodes in a network.

Affiliation networks are networks where nodes and edges are defined based on affiliation data.

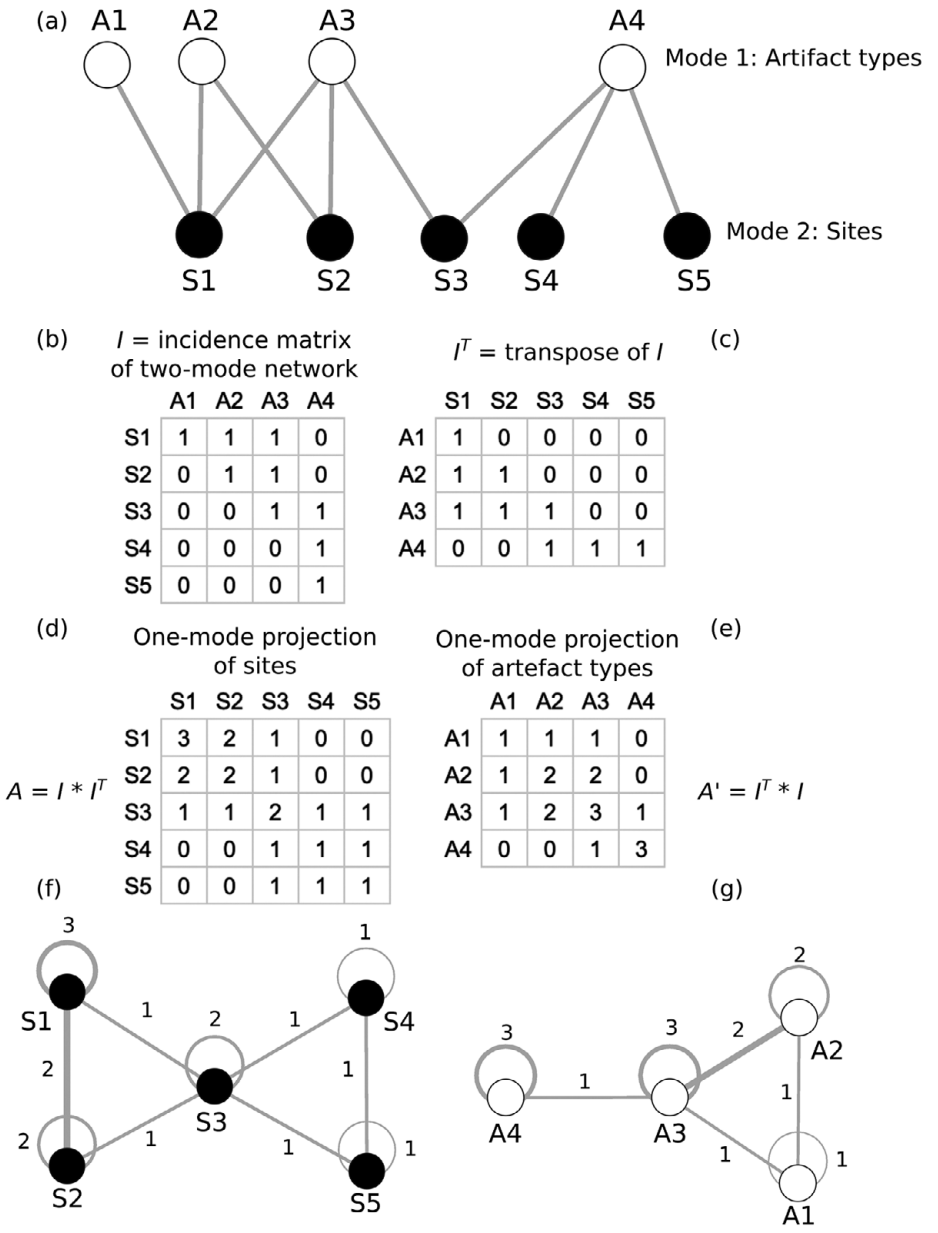
Researchers have used various rationales for relying on affiliation data to study social relations, and the specific approaches differ based on how types of nodes themselves are conceptualized (Wasserman and Faust 1994:296–298). In many cases, it is simply easier or more convenient to directly observe shared affiliations or group membership rather than complex and ever-changing networks among actors. For example, studies of so-called dark networks that are focused on the covert activities of terrorist groups often rely on affiliation data as a proxy for underlying networks of interactions among actors as direct data on connections are not available (e.g., Roberts and Everton 2011). In many ways, this picture mirrors the case we are faced with as archaeologists as we often must use material culture as a proxy for the social processes and connections we typically cannot directly observe (see Peeples et al. 2016).

Beyond the requirements of data collection, there are many substantive reasons why we would be interested in affiliations. Many studies treat shared affiliations among actors (or conversely shared actors among events) simply as direct evidence of a social relationship worth characterizing. In other words, the set of shared affiliations is seen as important in their own right rather than as a means for identifying other underlying social structures among actors or events. In other cases, shared affiliations are seen instead as the preconditions necessary for developing social ties among actors that are the actual phenomenon of interest. For example, people who attend more of the same events have greater *opportunity* to develop other important social relationships that may be of interest (see Feld 1981). In the other direction, shared affiliations can also be seen as *indicators* of other social relations that exist outside those affiliations. Under this model, similarities in the affiliations of two actors (or events involving similar sets of actors) are seen as reflections of the underlying social ties that are often *difficult to observe directly* (even by the actors themselves). As Borgatti and Halgin (2011a) note, these different perspectives on the relationships among social ties, social similarities, and

affiliations necessitate somewhat different approaches to network projection and analysis (see discussion below). Perhaps most importantly, affiliation networks allow us to directly consider connections among actors and events and the interplay between different kinds of nodes in ways not possible using one-mode methods alone.

A crucial feature of a two-mode network is the existence of a structural variable *between* the members of the two modes and the absence of structural variables *within* each mode. This means that in the case of a co-occurrence network we can create an edge between an artifact type and a site if that artifact type was excavated at that site, but we cannot create an edge between an artifact type and another artifact type. At first sight, this might seem like a limitation of this network type: Surely there are interesting relationships to be studied *within* this set of artefact types? In fact, it is not a limitation at all when we consider a two-mode network that represents the relationships within both modes as being mediated through the other mode: artifact types are related to each other precisely because they were excavated at the same sites; in turn, sites are related to each other precisely because the same artifact types were excavated at them. The reason why the rule that no edges can be created within one mode needs to be strictly enforced when creating two-mode networks is because this will allow for revealing such relationships *within* modes. This is achieved through projection and matrix algebra.

A two-mode network is often represented using an incidence matrix (see Section 3.3.4), in which one mode is listed in the rows and another mode is listed in the columns. The incidence matrix therefore shows the structural variable *between* the two modes. When we are interested in deriving the structural variables *within* each mode, then we can use the incidence matrix to create a one-mode **projection**. By multiplying the incidence matrix with its transpose (switching rows with columns) we obtain a weighted adjacency matrix with the same number of rows and columns representing the one-mode projection of the nodes in the *rows* of the incidence matrix: $A = I * I^T$. If we instead multiply the transpose with the incidence matrix, then we obtain the one-mode projection of the nodes in the *columns* of the incidence matrix: $A' = I^T * I$. Each cell A'_{ij} in the one-mode projection has a numerical value representing the number of nodes of the different mode that are connected to both i and j. For example, in Figure 3.7a sites S1 and S2 both have two artifact types in common (A2 and A3), so in the one-mode network of sites (Fig. 3.7f) the value of the edge between S1 and S2 is 2. The diagonal of this new symmetric adjacency matrix will further represent the total weight of all connections a given node has (or weighted degree centrality; see Section 4.4.1). This type of projection by matrix multiplication is typically referred to as simple weighting.



	A1	A2	A3	A4
S1	1	1	1	0
S2	0	1	1	0
S3	0	0	1	1
S4	0	0	0	1
S5	0	0	0	1

	S1	S2	S3	S4	S5
A1	1	0	0	0	0
A2	1	1	0	0	0
A3	1	1	1	0	0
A4	0	0	1	1	1

	S1	S2	S3	S4	S5
S1	3	2	1	0	0
S2	2	2	1	0	0
S3	1	1	2	1	1
S4	0	0	1	1	1
S5	0	0	1	1	1

	A1	A2	A3	A4
A1	1	1	1	0
A2	1	2	2	0
A3	1	2	3	1
A4	0	0	1	3

FIG. 3.7 Two-mode networks explained.

Although this is the most common procedure, it is far from the only option. For example, projections from two-mode scientific collaboration networks often use an alternative scheme where connections among coauthors are weighted based on the number of collaborators to adjust for the impact of papers with long author lists

(where collaborators are presumably less closely connected than in one-on-one collaborations; see Newman 2001). To our knowledge such alternative projection models have not been explored in depth in archaeological network research. This is perhaps a topic worth investigating in the future.

Network *projection* refers to the procedures used to convert a two-mode network into two one-mode networks, each focused on a distinct category of nodes.

How do we read such one-mode projections and how should we go about interpreting them in an archaeological research context? We will use the example of artifact co-occurrence on sites to further illustrate the one-mode projection procedure (Fig. 3.7a). An incidence matrix can be used to represent the presence or absence of artifact types (mode 1 in the columns) on sites (mode 2 in the rows) (Fig. 3.7b). If an artifact type has been excavated at a site, then the corresponding cell will have a value of 1. We can transpose this incidence matrix by switching the rows and columns (Fig. 3.7c). When the incidence matrix is multiplied by its transpose, then we have created a one-mode projection of sites (Fig. 3.7d, f). This one-mode network connects a pair of sites with an edge if one or more of the same artifact types has been excavated at them. We can therefore interpret patterns in this network as revealing the *similarities of sites' artifact assemblages*. Can we identify clusters of sites that have particularly similar or different assemblages? What does this mean in light of our data critique or theoretical framework? Similarly, we can multiply the transpose of the incidence matrix with the incidence matrix to create a one-mode projection of artifact types (Fig. 3.7e, g). This one-mode network connects a pair of artifact types with an edge if they were both recovered at the same site one or more times in the sample. We can interpret this network as revealing the *distributions of artifact types*. Do some groups of artifact types show very similar or different distribution patterns? Again, how should we critically interpret these patterns in light of data and theory critique?

In many ways two-mode networks and affiliation data are a good fit for many kinds of common archaeological data. For example, Ladefoged et al. (2019) created a two mode network using sourced obsidian for a series of sites in Aotearoa (New Zealand) with edges drawn between sites and the specific obsidian sources represented in the assemblages of those sites. Since both sites and obsidian sources can be geographically situated, this provided a convenient means for assessing obsidian procurement patterns and territories across space as well as a means for assessing

similarities among both sites and obsidian sources independently by means of the one-mode projections of those data. In other studies Elliot Blair (2015, 2023) examined both two-mode and one-mode networks defined based on mortuary features at the 17th-century Mission Santa Catalina de Guale (on St. Catherines Island off the coast of the US state of Georgia) and the beads recovered in those individual mortuary contexts. Blair created network visualizations in both one and two modes and found that juxtaposing both kinds of networks aided in the identification of network structure and communities of consumption across the mission cemetery. Although these studies illustrate the advantages of two-mode networks as well as the one-mode projections they can be used to generate, such approaches have been relatively rare in archaeology so far. We hope that such methods increase in prominence in the coming years.

3.3.5 *Similarity Networks*

Similarity networks simply refer to one-mode networks where nodes are defined as entities of interest with edges defined and/or weighted based on some metric of similarity (or distance) defined based on the features, attributes, or assemblage associated with that node (see also Östborn and Gerding 2014). Such an approach is frequently used in archaeology to explore material cultural networks where nodes are contexts of interests (e.g., sites, excavation units, houses) and edges are defined or weighted based on similarities in the relative frequencies of artifacts or particular classes of artifacts recovered in those contexts. This approach to constructing material cultural networks was first proposed in archaeology in the very early forays into network approaches by Cynthia Irwin-Williams (1977), but it was not until recently that similarity networks really took off (see Blair 2015; Borck et al. 2015; Freund and Batist 2014; Golitko and Feinman 2015; Golitko et al. 2012; Gravel-Miguel 2016; Hart 2012; Hart and Englebrecht 2012; Hart et al. 2019; Jennings 2016; Lulewicz 2019; Mills et al. 2013a; Peeples 2018; Peeples and Haas 2013; Peeples and Roberts 2013; Terrell 2010; Weidele et al. 2016). These approaches have been less common in other network fields (where direct connections are more often directly observable) but have recently been applied in biological and healthcare-related networks to identify groups of patients with similar conditions or attributes (Barkhordari and Niamanesh 2015; Brown et al. 2016; Pai and Bader 2018; Pai et al. 2019) or to identify similarities in genes sequences or proteins to infer function (Cho et al. 2016; Franzén et al. 2016; Valavanis et al. 2010).

Similarity networks are networks where edges are defined or weighted based on a quantitative metric of similarity or distance based on node attributes or artifact assemblages.

In some respects we could think of a similarity network as a special case of an affiliation network as they typically begin with the same starting point, a two-way table with nodes as rows and the assemblage of objects that will be used to define similarity as columns. For example, Mills and colleagues (2013a; see also Section 2.8.2) created ceramic similarity networks for a series of sites in the US Southwest by defining individual sites as nodes and assessing similarities among the ceramic assemblages present at each. They began with a two-way table (incidence matrix) with sites as rows and with ceramic wares as columns and counts of ware by site in each cell. They then used a modified version of the Brainerd-Robinson metric of similarity, which provides a measure of total similarity based on the proportional representation of ceramic ware categories found at each site, to create a symmetric adjacency matrix where every row and column represents an individual site and each cell represents the similarity value between a particular pair of sites (scaled between 0, or no similarity, to 1, or perfect similarity, in this case). The resulting adjacency matrix was then treated as a weighted network where the similarity scores defined the weight of the edges. In this way we could think of the application of a similarity metric on an incidence matrix as another means of one-mode projection (see Fig. 3.8).

The vast majority of recent archaeological similarity networks use the Brainerd-Robinson similarity metric. This is probably because it was already a common tool in archaeology for comparing nominal data and archaeological assemblages prior to the application of network methods and because most of the early archaeological similarity networks used this method. This is certainly not the only approach, however, and different similarity metrics may perform better or worse for different kinds of data or questions. The Brainerd-Robinson metric, as we call it in archaeology, has actually been independently invented several times in many different fields (versions of this similarity metric or related distance metrics have been called city-block distance, taxi cab distance, Manhattan distance, the sociological dissimilarity index, and Minkowski distance $p = 1$). This metric is well suited to situations where counts of objects by nominal category by context are systematic and can be meaningfully considered as percentage data.

Despite the near total concentration on Brainerd-Robinson in archaeological similarity networks, there are many other possible similarity and distance metrics

(a) Artifact type frequency data

	Type 1	Type 2	Type 3	Type 4
S1	14	2	0	3
S2	22	10	3	0
S3	8	4	15	3
S4	16	6	3	1
S5	25	12	15	1

(b) Similarity metric calculation (Brainerd-Robinson)

$$S_{ab} = \frac{200 - \sum_k | P_{ak} - P_{bk} |}{200}$$

k = all ceramic wares
P_{ak} = Percentage of ware k at site a
P_{bk} = Percentage of ware k at site b

(c) Similarity matrix

	S1	S2	S3	S4	S5
S1		0.73	0.47	0.76	0.60
S2	0.73		0.49	0.93	0.78
S3	0.47	0.49		0.55	0.70
S4	0.76	0.93	0.55		0.83
S5	0.60	0.78	0.70	0.83	

Ties > 0.70 defined as present

(d) Weighted network graph

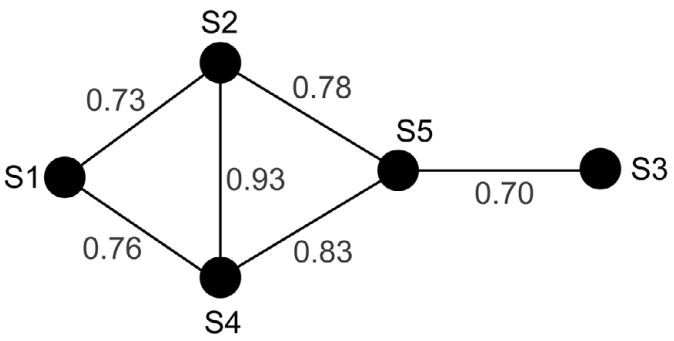


FIG. 3.8 Similarity networks explained.

that could be used instead (see Habiba et al. 2018 for a discussion of many potential options). These include the Morisita Index, which also defines similarity in terms of relative representation of categories but is based on raw counts rather than proportions and can sometimes perform better when there is considerable variability in sample size between nodes. Watts and Ossa (2016) use this measure in their similarity networks focused on Hohokam settlements to model economic interaction strategies. Cochrane and Lipo (2010) use Hamming distances to define distances among binary (presence/absence) style data of characters by taxa in their study of Lapita pottery design. Johnson (2016, 2023) used Mahalanobis dissimilarities among individuals to create networks based on craniometric similarities for a series of Tiwanaku-affiliated sites to explore potential patterns of kinship within cemeteries.

Although we are unaware of any archaeological examples as of yet (other than basic experimentation; see Peeples 2017), it would also be possible to use measures like χ^2 distance to define and weight edges among nodes. This metric is the basis of correspondence analysis, a method frequently used in archaeological seriation, and defines distance weighting the contributions of individual categories inversely to their frequency (see Peeples and Schachner 2012). In other words, rare categories have a bigger impact on the resulting distance metric than common categories. In some cases this may be desirable. If the data column data used to generate similarities include mixed measurement levels (e.g., presence/absence, ordinal, interval, nominal, ratio), metrics specifically designed to deal with this like the Gower similarity index may be appropriate (see Peeples 2018). If data are only presence/absence, then measures like the Jaccard coefficient or the simple matching coefficient could also be used. Beyond this, it would also be possible to use coordinates provided by other ordination methods like principle components analysis or correspondence analysis as a measure of the relative similarity of nodes (see Munson 2015 for a related argument). This approach could be particularly useful when you are interested in defining similarities in terms of only the strongest dimensions of variation in an incidence matrix or axes from an ordination that have a particular behavioral interpretation. Habiba and colleagues (2018) present an experimental analysis using a broad range of common and uncommon similarity metrics on the Southwest Social Networks Project dataset (see Section 2.8.2) to illustrate how the choice of metric can influence results and substantive interpretations. This is all to say, then, that the range of possible approaches to defining similarity networks using archaeological data is much broader than the current literature suggests. We argue that this is an issue that deserves further exploration in the future. As different similarity and distance metrics have different properties,

choosing one approach over another should be a decision informed by the nature of the data at hand and the specific questions to be asked of that data rather than simply relying on existing approaches.

So far we have described adjacency matrices produced by applying similarity or distance metrics to an incidence matrix as weighted networks where the similarity value between each pair of nodes is directly treated as the weight of the corresponding edge. In many cases this is how such similarity networks are analyzed (e.g., Mills et al. 2013a; Peeples and Haas 2013), but in other studies, researchers want to define some threshold or quantile that defines ties as present and similarity values below that threshold as absent (e.g., Golitko et al. 2012; Hill et al. 2015). In some cases, although networks are *analyzed* based on the full weighted network, binary edges are defined for the purposes of visualization (Mills et al. 2013a, 2018; see Chapter 6). Most modern network analysis software these days allows for the direct analysis of weighted networks in this way, but not every network metric has a weighted version. Thus, the decision on whether and how to binarize continuous similarity data is an important one and should be made with regard to the particular question and network relational theories at hand. Peeples and Roberts (2013) demonstrate that networks created based on the same similarity data with different thresholds for defining ties can often have dramatically different node-, edge-, and graph-level properties in addition to dramatically different network visuals (see Fig. 6.16; see also Weidele et al. 2016 for further discussion of related approaches to deal with this issue). Peeples and Roberts (2013) suggest that you should not discard the information provided by fully weighted edges unless you have a theoretical justification for doing so. For example, if you are interested in evaluating only the strongest similarities represented in a network (or conversely only the weakest), then binarizing continuous similarity metrics may make sense. Hill and colleagues (2015) created binary ties in their ceramic similarity networks in the US Southwest to explore how the strongest patterns of ceramic similarity relate to their model of spatial proximity. In this study, the strongest ties as they defined them demonstrated temporal dynamics that were obscured when only considering the full weighted network, and, thus, binarizing ties for this analysis paid dividends (Hill et al. 2015; see Golitko and Feinman 2015 for another study using both weighted and binary ties for different purposes).

As noted briefly above, even when analyses are based on the full weighted similarity network, it is often useful to create binary ties for the purpose of visualization (see also Section 6.3.5). Thus, one of the most frequent questions we get from researchers and students first working with such similarity network data is: How, exactly, should we define a threshold for the presence or absence of an edge

between a pair of nodes? Although there is no one-size-fits-all answer to this question, we provide some useful heuristics here. One useful place to start is with simple exploratory data analysis. Produce a histogram of all of the values in your similarity matrix (be sure to exclude the diagonal). Are there any obvious breaks in the distribution suggesting strong or weak ties? If so, that may be an appropriate threshold to try. If your purpose is only visualization, it may be useful to simply try a range of values to see what works for producing a visual that helps to inform other analyses. If you get a tangled hairball of a network (see Chapter 6) at one threshold, try another higher threshold or another kind of visualization. In some studies (e.g., Mills et al. 2018) ties were defined based not on an absolute value but a quantile from the distribution of values (such as the top 15 percent of similarity scores are defined as ties). It is important to remember in this case that such approaches place an artificial bound on measures like network density (see Section 4.4.1) and this will need to be taken into account in any subsequent analyses. In some cases it may be possible to define a useful threshold using other statistical and simulation-based approaches. For example, Peeples (2011, 2018; Peeples et al. 2021) in his networks of technological similarity among settlements in the Cibola region of the US Southwest based on cooking pots simulated 10,000 random versions of the underlying ceramic data and calculated similarity matrices for each of those random runs. He then defined ties as present when they were at least one standard deviation above the median *expected* value across all of the random runs (see Chapter 5 for additional information on such resampling procedures). Thus, ties in this case are defined as relations that have significantly higher similarities than would be expected by chance. In your own work you may need to try many methods, but your decision should be informed by theory and a careful consideration of your data.

3.3.6 *Ego Networks*

Sometimes we focus our research efforts on how a particular entity (node), such as a specific archaeological site, stood in relation to other entities. What can we say about the communities of the different occupational phases of this site, and how was their behavior affected by interactions with neighboring communities? What is the relational environment within which this particular site was embedded? In such cases it is not crucial to draw on evidence of all known sites in the area and time period, given that our research questions refer exclusively to a particular site's connections. But in other cases, archaeologists are forced into formulating such site-specific research questions because the archaeological record is patchy, and we

can formulate certain research questions only about certain sites. Archaeologists often find themselves in this situation, and not just for the study of sites but any entity relevant for studying past human behavior.

When we aim to understand the relational environment within which an entity is embedded, because it is relevant for our research questions or because data collection challenges dictate this focus, then archaeological network research can make use of so-called **ego networks**: a type of network that includes a focal node (the so-called *ego*), the set of nodes the ego is connected to by an edge (the so-called *alters*), and the edges between this set of nodes. The ego network is a subnetwork of larger networks including all entities in the population, and the data collection requirements for creating ego networks are therefore much more limited: we only need to obtain information about a handful of nodes in the population and the edges between them. Such a data collection strategy can be dictated by the preservation of the archaeological record (i.e., only for the ego and its alters is sufficient archaeological data available to enable network research). But ego network construction could equally guide archaeological data collection strategies themselves.

Ego-centered archaeological network research focuses on the local opportunities awarded to a particular entity thanks to its local structural position. How do the local relationships within which the ego is embedded allow for mediating information between its direct neighbors? What role does the ego play in its immediate environment? In cases where the ego networks of multiple nodes in an archaeological network can be created, we can explore what nodes fulfill the same structural roles or are similarly embedded locally. We can subsequently refer to additional archaeological information and our theoretical framework to explore what these roles might mean, or what other commonalities nodes of the same local role we are informed about. Take the example of the ancient road system in Figure 3.9. The ego networks of towns n1 and n3 are identical and reveal to us that they each form an equal part of a triangle: the road connections these two towns are locally embedded in teach us that they both have equal opportunities for mediating local traffic along these roads. However, the ego network of town n2 looks very different. It is also an equal part of the triangle of roads that connects n1 and n3, but in addition it is also the town that provides access to roads leading to town n4. The local structural role played by town n2 is different: it has opportunities towns n1 and n3 do not have because it can mediate traffic to and from n4.

An *ego network* includes a focal node, the set of nodes the ego is connected to by an edge and the edges between nodes in this set.

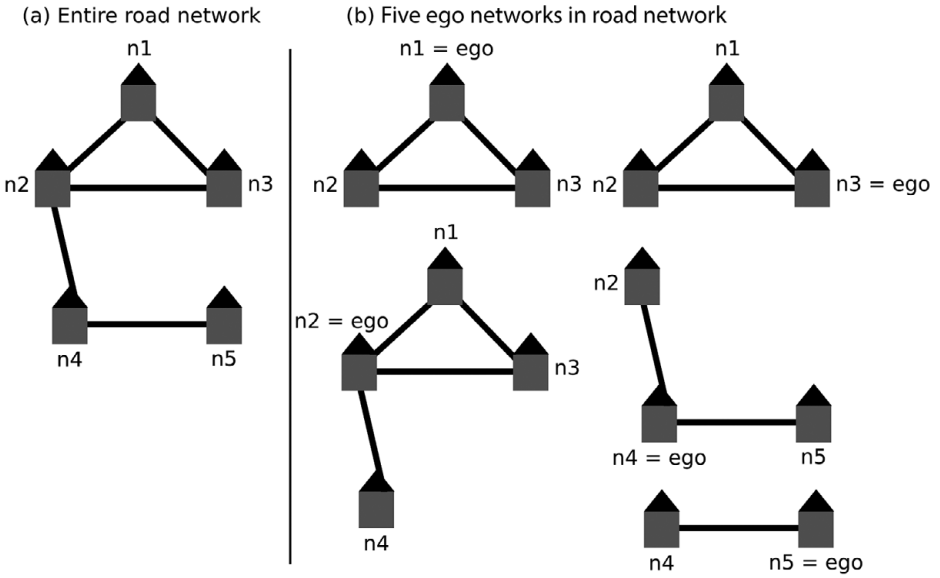


FIG. 3.9 (a) Road network with its (b) five ego networks. Note n1 and n3 play the same structural role in their respective ego networks.

3.3.7 *Multilayer Networks*

In the simplest terms, **multilayer networks** are networks where a single set of nodes is connected by two or more sets of edges that each represent a different kind of relationship among the nodes (see Kivelä et al. 2014 for a discussion of many different uses and many different types of multilayer networks). In a social network this could be, for example, a set of individuals with relations based on email contact as one layer and based on familial relationships on another layer. The basic idea behind multilayer networks is that the addition of multiple layers provides a richer description of overall patterns of interaction and network topology than a monoplex (single-layer) network would. This is a relatively new area of network science in general but we see potential here. Although the details are beyond the scope of this manual, there are more and more new methods available every year for calculating network graph-, node-, and edge-level statistics (see Chapter 4) considering multilayer networks as well as new interlayer metrics and methods that are exclusive to multilayer networks. Social relations do not occur in a vacuum and it is natural to consider how one kind of interaction or relational proxy might be influenced by another (see Szell and Thurner 2010). As archaeologists frequently deal with multiple kinds and categories of material culture and want to capture similarities and differences among them (e.g., Lulewicz 2019; Peeples 2018),

multilayer networks seem to be a natural fit for many archaeological questions. To our knowledge only one archaeological study so far has made explicit use of multilayer network methods. Upton's (2019) dissertation explores the impact of migration on networks of interaction and social signaling in the late prehistoric period in the central Illinois River valley using multilayer networks to document how different kinds of networks of interaction and influence co-construct one another. This study demonstrates the potential of such approaches for archaeological research in other contexts, and we hope others will continue to develop these methods further.

A *multilayer network* is a network where a single set of nodes is connected by two or more sets of edges that each represent a different kind of relationship among the nodes.

3.3.8 *Combining Types of Networks*

In this section so far we have focused on particular types of network, one at a time. In practice, however, it is quite common for these different forms of networks to be combined. For example, we could create a one-mode directed weighted network where edges have both value and direction and all nodes are of the same mode. Further, we could create a two-mode undirected weighted network where nodes are classified into two distinct modes and ties have values but not direction. We could further define a directed signed network where edges are classified as positive and negative and relations flow in only one direction. It is possible to create essentially any combination of the types and features described in this section to create all sorts of networks that provide different kinds of information about different kinds of relational processes. Importantly, picking the appropriate type of network to build is not arbitrary. Selecting the appropriate model must be done with consideration of both the relational theories and questions you hope to evaluate and the availability and quality of data available to you (see further discussion in Chapters 2 and 8).

3.4 LONGITUDINAL NETWORK DATA

So far we have introduced network data as static snapshots of relationships at some unspecified point in time. However, much of network science (especially in the last

few decades) is concerned with the study of how network structures change through time, what processes drive this change, how the processes themselves evolve, and how a study of past network structure and the history of structural change can be used to forecast future structures or transitions. Such topics can be addressed by creating longitudinal (or temporal) network data. The typical approach to collecting longitudinal network data is to perform observations at time intervals and record relationships at each observation moment. This results in an ordered set of network snapshots documenting network change through time. Taking a longitudinal approach allows for exploring how stable the network structure is through time, and to differentiate between parts of the network that are subject to high or low temporal variation. These approaches are key to archaeological network research. Many archaeological network studies focused on change through time have tended to take this "film strip" approach of *lining up multiple network snapshots* to evaluate network dynamics and temporal dependencies.

To enable a longitudinal perspective in archaeological network research, we first need to assign at least a relative chronological sequence of network states (i.e., "A is older than B"). The estimated or observed dating can be derived from either the nodes, the edges, or both. For example, in a visual signaling network where sites are connected by lines of sight, the chronological sequence would likely be derived from evidence that sets of sites were occupied at the same time. The lines of sight themselves cannot be dated as such, because they are merely the result of human observations between pairs of locations that have to have been contemporary. In such cases where the nodes provide chronological information but edges do not, we expect each network snapshot to only include those nodes that are contemporary, and subsequent snapshots can consist of different sets of nodes (i.e., some nodes can be added, removed, or all could be replaced entirely); the presence of edges in each snapshot is then determined by the presence or absence of the nodes to which they are related (Fig. 3.10a). In a line-of-sight network, for example, no edge will be created for an observed line of sight between site A and site B if at the period of time in question site A was not occupied. In a sense, the dating of the edges in this example is derived from the dating of the nodes. Major ancient roads, on the other hand, could serve as examples where dating is derived from relationships. The construction period of a major new road between a pair of preexisting towns offers evidence of the creation of a new edge between these towns. In such cases where only the edges provide chronological information, we have to be able to state that all nodes were contemporary such that the set of nodes will not change between the snapshots, but the edges will (Fig. 3.10b). In rare cases where both nodes and edges can be

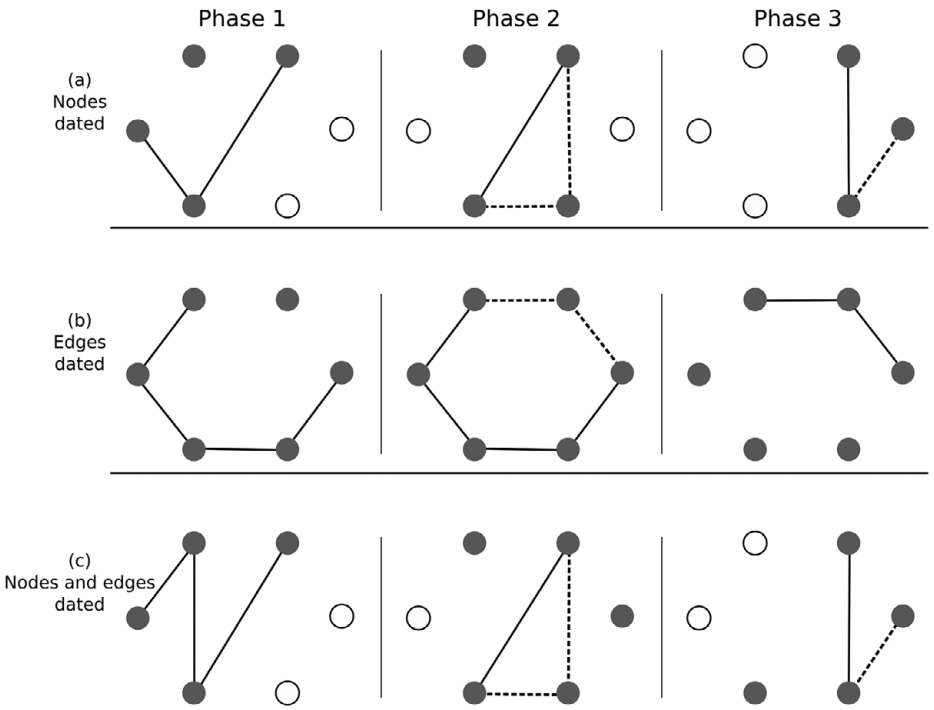


FIG. 3.10 Three examples of different dating sources. Gray nodes are contemporary; white nodes are not contemporary; and newly created edges are represented as dashed lines. (a) When only nodes are dated, incident edges of noncontemporary nodes will be removed with the removal of nodes. (b) When only edges are dated, the set of nodes tends to remain the same throughout the phases.

independently dated, the resulting sequence of snapshots can change in both the number of nodes and the number of edges (Fig. 3.10c).

Chronological dating of archaeological materials will always be difficult, but the issues surrounding chronologies are well known in archaeology. As such, creating longitudinal network data in archaeological network research should rely first and foremost on archaeological data critique and theory. It is not a network science problem but an archaeological one, and archaeologists have developed a range of approaches to identify contemporary archaeological materials and assemblages. The most common approach by far is to assign dated entities to predefined chronological categories or phases based on some external information. This often consists of creating a sequence of network snapshots defined using archaeological phase classifications or other dates. Phases either could be temporal "bins" (e.g., 500–400 BC or AD 1200–1450) or could be commonly used cultural phases (e.g., Roman republican period, Roman imperial period, Pueblo III, Pueblo IV).

For example, Golitko and colleagues (2012) created a series of networks to explore the exchange of obsidian in the Maya region, dividing the assemblages of individual sites by phase using temporal assignments for particular assemblages from those sites made by excavators (Classic Period, Terminal Classic, Early Postclassic, and Late Postclassic; see similar approaches in Gjesfjeld 2015; Hart and Engelbrecht 2012; Lulewicz 2019). Such a temporal sequence can also be generated using absolute dates where available. For example, Coward (2010) created a series of networks representing 1,000-year-long intervals for a set of Epipaleolithic to Neolithic sites in the Near East. In this study, nodes were defined as individual radiocarbon-dated contexts within sites (rather than sites as a whole) and were assigned to the intervals that overlapped with the dates for that particular context. As is the case with any phase-based archaeological analysis, the temporal resolution of the networks is defined by the resolution of the dates available and the chronological schema used.

Beyond these phase-based approaches, there have also been a number of methods developed specifically to apportion (subdivide) the assemblages of artifacts or features at long-lived sites into shorter intervals that represent discrete portions of an occupation. These approaches are not, of course, limited to network studies but have proved useful in many recent archaeological network studies. For example, Roberts and colleagues (2012) defined a procedure for chronologically apportioning ceramic assemblages into smaller temporal intervals using dates assigned to individual ceramic types and some distributional assumptions about the popularity of those types through time. Specifically, this approach combines the date range for the site and the date ranges assigned to ceramic types through cross-dating, and assumes a normal popularity curve to estimate the probability that a given ceramic object was deposited in a given interval (see also Fentress and Perkins 1988 for a related approach). Such methods not only allow researchers to potentially consider finer chronological resolution but also allow for a more systematic comparison among sites with partially overlapping occupations for the interval where they do overlap. Other similar approaches toward developing regional chronologies and apportioning assemblages such as empirical Bayesian estimation (e.g., de Pablo and Barton 2015; Ortman 2016; Ortman et al. 2007) and aoristic approaches (e.g., Crema 2012) have proved useful in demographic and spatial analyses and show potential for addressing similar chronological resolution issues for archaeological network research (see Mills et al. 2018).

Beyond the methodological questions regarding longitudinal networks, there are a number of other important theoretical questions that archaeologists have yet to fully grapple with. In particular, how do models and methods designed to explore

social networks of interaction in contemporary settings relate to the kinds of data and the chronological resolution that we often have in the archaeological record? What does it mean to create a network that represents patterns of similarity or intervisibility or spatial proximity during an interval that is dozens or hundreds or even thousands of years long? Is it appropriate to use methods originally designed for tracking instantaneous interactions for periods of time that represent multiple generations? Are there adjustments that we need to make to methods or are there limitations we should place on interpretations? How can we systematically evaluate the impact of chronological resolution on our networks (Peeples et al. 2016)? Moreover, most of the empirical network studies of the use of longitudinal data outside archaeology tend to focus on change in networks through time on the order of days, weeks, months, or a few years at most (e.g., Ryan and D'Angelo 2018; Suitor et al. 1997; Wissink and Mazzucato 2018). Archaeology rarely offers such temporally detailed observations, instead typically considering long-term change over years, decades, centuries, or more. This is undoubtedly an area where archaeologists could make major contributions, as the discipline with access to observations of long-term changes in human behavior. What techniques can be used to explore long-term changes in relational phenomena? Do such phenomena reveal different patterning over long-term timescales as compared to short-term ones? We suggest that these questions deserve greater attention in archaeological network science in the future.

Finally, there are a number of statistical network modeling approaches that offer considerable potential for archaeological network research with a temporal dimension. Exponential random graph models (Lusher et al. 2013) and stochastic actor-oriented models (Snijders et al. 2010) explore whether a single observed network state or subsequent network states could have been the result of a tendency toward the creation of particular network patterns. There have also been methods developed to identify and quantify the degree of change in network states across longitudinal networks (Park and Sohn 2020) and methods for exploring changing structures in networks with changing composition (when nodes join or leave over time as is common in archaeological networks; see Huisman and Snijders 2003). We suggest that these should be areas of exploration in the coming years.

A *longitudinal network* is a network that includes information about the change of network structure through time.

3.5 BEST PRACTICE GUIDELINES

Creating your own network data from your archaeological data or theories requires some preparation and practice. Although much of this will be software-specific, the following are a few general tips and guidelines that you can keep in mind whenever you plan to work with network data.

- Create an experiment or analysis design and identify the archaeological data needed (ask yourself the questions listed in Section 1.2).
- Develop a data management or collection structure that (1) allows you to easily record data, (2) will enable the analyses envisaged, and (3) will allow for the data to be extracted (either directly or through queries) in a network data format (see Section 3.3). Graph databases might be useful for this, but any data management system will enable the creation of some network data formats as long as this was prepared for when deciding on the data collection/digitization strategy.
- Consistently use unique identifiers for nodes and edges. When adding additional information to nodes and edges, always collect this in node and edge tables that use these unique identifiers.
- Creating small network datasets manually is possible, but they always should be thoroughly reviewed to identify data entry errors.
- When creating network data manually, it will be easier to start by making an edge list or some other concise data format rather than a matrix (i.e., "start long and go wide"). An edge list only requires edges that are present to be written, whereas a matrix requires all cells to be written. Many network software packages allow for importing edge lists and most also enable exporting networks as matrices should this be needed.
- Always allow for replication of the network analysis results you publish by (1) publishing your network data along with your results, (2) documenting your network data with metadata, (3) publishing the archaeological data or theoretical work the network data are derived from, and (4) publishing the code or methodological pipeline that was followed to create the network data.
- Network metadata should include the following: creator, date created, reuse license, sources used for the creation of the dataset, acknowledgment of other works used, number of nodes, number of edges, files included in dataset, description of files, format of dataset, software used to create the dataset, and methodological pipeline followed to create the dataset.

3.6 SUMMARY

- Network data consist of a set of nodes and at least one structural variable (represented by a set of edges) connecting pairs within this set of nodes.
- Structural variables are descriptions of features or elements of pairs of nodes, where each node can be part of multiple node pairs.
- Network data allow for the formal representation and study of three types of dependencies: (1) dependencies within structural variables, (2) dependencies within and between structural and node variables, and (3) dependencies between and within node variables.
- The creation of network data requires that boundaries of nodes, edges, and variables need to be unambiguously defined for analytical purposes only.
- Network data can be used in archaeology to represent a wide range of entities of research interest as nodes (e.g., confluences, authors, sites) and the relations between pairs of those entities as edges (e.g., lines of sight, proximity, artifact co-occurrence).
- The most commonly used network data formats include the edge list, adjacency list, adjacency matrix, and incidence matrix.
- Additional information about nodes and edges can be stored in tables.
- A simple network consists of a set of nodes and a set of edges with no additional information about them.
- An empty network contains no edges.
- A completely connected network contains all possible edges.
- A directed network consists of a set of nodes and edges connecting them for which the orientation or direction is specified.
- When edges between a pair of nodes can only have a single orientation, then the network is unidirectional; if edges with both orientations are possible, it is bi-directional.
- A signed network is a network where the edges carry a positive or negative sign.
- A categorized network is similar to a signed network but edges can be assigned to more than two nominal categories and the categories need not be opposed.
- A weighted network is a network in which the edges carry a nonbinary value.
- Two-mode networks are networks containing two sets of nodes and a structural variable between them.
- A mode is a distinct set of nodes.
- Two distinct one-mode projections can be made from a two-mode network, by multiplying its incidence matrix with the transpose of its incidence matrix: a one-mode projection of the nodes in the rows and a one-mode projection of the nodes in columns.

- An ego network includes a focal node, the set of nodes the ego is connected to by an edge, and the edges between this set of nodes.
- Longitudinal network data or temporal network data can be created if chronological information is available about the nodes, edges, or both.
- Nodes (or edges) are commonly classified in cultural, typological, or chronological categories or bins, although more diverse ways of attaching chronological information to network data exist.
- Network data creation for archaeological network research requires practice and planning. Always keep the best practice guidelines in mind.

Further Reading

The following include introductions to and definitions of network data in their many forms.

Brandes, Ulrik, Garry Robins, A. N. N. McCrainie, and Stanley Wasserman 2013 What Is Network Science? *Network Science* 1(1):1–15.

Wasserman, Stanley, and Katherine Faust 1994 *Social Network Analysis: Methods and Applications.* Cambridge University Press, Cambridge.

More examples of the diverse archaeological uses of network data are in the following.

Brughmans, Tom, Barbara J. Mills, Jessica L. Munson, and Matthew A. Peeples 2023 *The Oxford Handbook of Archaeological Network Research.* Oxford University Press, Oxford.

Collar, Anna, Fiona Coward, Tom Brughmans, and Barbara J. Mills 2015 Networks in Archaeology: Phenomena, Abstraction, Representation. *Journal of Archaeological Method and Theory* 22(1):1–32.

Collar, Anna, Tom Brughmans, and Barbara J. Mills (editors) 2015 The Connected Past: Critical and Innovative Approaches to Networks in Archaeology. *Journal of Archaeological Method and Theory* 22(1).

Knappett, Carl 2013 *Network Analysis in Archaeology. New Approaches to Regional Interaction*. Oxford University Press, Oxford.

Leidwanger, Justin, and Carl Knappett (editors) 2018 *Maritime Networks in the Ancient Mediterranean World.* Cambridge University Press, Cambridge.

EXERCISES

Answers for these exercises can be found in Appendix A.

3.1) You are performing a study of a set of archaeological sites located near the confluences of a river system. You are interested in exploring how past

communities at these sites could have transported goods along the river or could have used the river as a medium to enable interaction between them. There are at least three different ways in which you could make a network data representation for this study:

a) Represent confluences as a set of nodes, the presence of a site near a confluence as a node variable, and the presence/absence of a navigable downstream river connection between each pair of confluences as a directed structural variable.
b) Represent sites as a set of nodes and the presence/absence of a navigable downstream river connection between each pair of sites as a directed structural variable.
c) Represent river segments between confluences as a set of nodes, the presence/absence of a confluence between each pair of river segments as a structural variable, and the presence of a site near a confluence as an edge variable.

Make a drawing using nodes and edges of the network data representation for the first of the three cases described above, assuming the river system has 10 river segments, 5 confluences, and 3 sites. Note that there is not one correct answer: any drawing with the right number of river segments, confluences, and sites will be correct.

Using your first drawing, now make a second network representation of this drawing following the second case described above (i.e., sites are nodes). Note that this second drawing will likely have a different number of nodes and edges.

Using your first drawing, now make a third network representation of this drawing following the third case described above (i.e., river segments are nodes). Note that this third drawing will likely have a different number of nodes and edges.

Questions

How do you interpret the differences between these three network representations? What different types of relationships of the same river system do they highlight? What kinds of archaeological research questions would you ask of each of these types of representations? If you were to carry out this study, which of the three network data representations would you prefer to work with and why?

3.2) Consider the citation network in Figure 3.11. It is a directed network in which the set of nodes represents publications and the directed edges represent

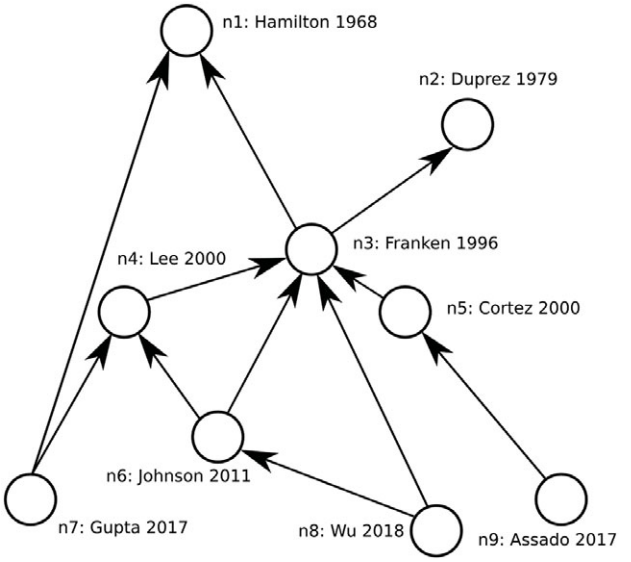


FIG. 3.11 Citation network used in the exercises.

citations from one publication to another. Represent this network in the following alternative network data formats:

a) A vertical edge list, including year of citation as an edge attribute
b) An adjacency list
c) An adjacency matrix
d) A node table including the publication author as a node attribute.

3.3) Consider the network in Figure 3.12. This is a two-mode network representing Roman governors and the provinces they governed between AD 79 and 118.

a) List the nodes that belong to mode 1.
b) List the nodes that belong to mode 2.

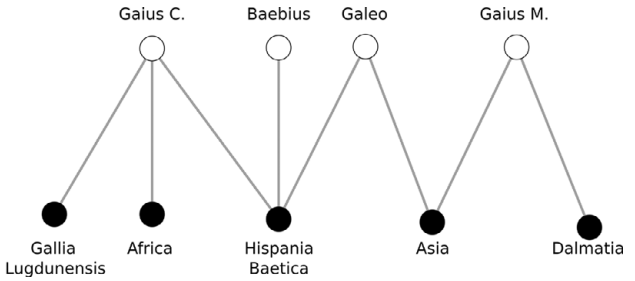


FIG. 3.12 Two-mode network used in the exercises.

c) Represent the incidence matrix of this two-mode network.
d) Draw the one-mode projection of mode 1.
e) Draw the one-mode projection of mode 2.
f) How would you interpret these one-mode projections? What kinds of archaeological research questions do they answer?

4

Exploratory Network Analysis

4.1 WHAT IS EXPLORATORY NETWORK ANALYSIS?

How dense is the network? What are the most centrally positioned nodes in the network? How long on average is a path through the network? Are there any cliques or communities in the network and which nodes are included? How does my network compare to other similarly defined networks in terms of local or global properties? These are the kinds of questions exploratory network methods can be used to answer.

Exploratory network analysis is simply exploratory data analysis (Tukey 1977) applied to network data. This covers a range of statistical and visual techniques designed to explore the structure of networks as well as the relative positions of nodes and edges. These methods can be used to look for particular structures or patterning of interest, such as the most central nodes, or to summarize and describe the structure of the network to paint a general picture of it before further analysis. Most archaeological network research involves an initial phase of exploratory analysis before any other work is done. Summarizing the structure of a network helps to determine where to invest research effort by identifying expected or surprising patterns. Fundamentally, exploratory analysis allows one to determine whether or not the research aim or question you have in mind can be addressed using a particular network dataset at all. Summarizing a network's structure can also help suggest what patterns might be interesting to study or what archaeological hypotheses could be tested with a dataset. Deriving summary descriptions further allows for comparing a studied network with other networks: Is this network particularly dense as compared to other similar networks?

> *Exploratory network analysis* is the use of statistical or visual analytical tools to explore the structure of networks by identifying and summarizing network patterning.

4.2 WHICH ANALYTICAL METHOD SHOULD I USE?

This chapter provides an overview of the archaeological use of exploratory network analytical techniques. Some techniques can be grouped depending on the analytical scale at which (or the type of network data to which) they are applied. Node-based measures describe structural properties of nodes, such as the number of edges a node has. Network-based measures describe structural properties of the entire network, such as a network's density. Dyad-based measures describe properties of node pairs. Weighted network measures are applied to weighted networks, and directed measures to directed networks. Other measures can be grouped depending on their ability to study phenomena commonly explored using network techniques: How central are nodes, and what are the communities in this network? In this chapter we will introduce some of the most popular exploratory network analysis techniques roughly grouped by method and phenomenon, and for each we introduce (where relevant) node-based, network-based, weighted, and directed versions of methods. The techniques we cover are widely applied in archaeological research but are by no means an exhaustive list. More techniques can be found in network science handbooks and reviews (in particular, Brandes and Erlebach 2005; Coscia 2021; Estrada 2012; Newman 2010; Wasserman and Faust 1994).

A huge number of exploratory network analytical techniques exist, which raises some questions. Which technique(s) should I use? Which techniques are most appropriate in light of my research aims? In short: Where do we start?

The best place to start is always with your archaeological research context. Some network techniques might be more useful than others in your research context because they allow you to answer your research questions or because they address phenomena that are key to your archaeological theories. Ask yourself the following questions:

- Am I trying to understand the position of individual nodes relative to other nodes in the network? If so, then node-based methods will likely be appropriate.
- Am I trying to understand how central or important particular nodes are? If so, then centrality measures will be of interest.
- Am I trying to understand the patterning of the entire network and its ability to perform or enable a certain function or process? If so, then graph-level methods and metrics will be of interest to you.
- Do I theorize people, goods, or ideas can flow from any node to any other node along paths through the network? If so, then dyad- and path-based methods will be of interest.

- Am I trying to identify sets of nodes in the network that have particularly strong relations with each other or that form otherwise defined groups? If so, then cohesive subgroups or community detection methods will be of interest to you.

In the context of archaeological research, you will probably answer "yes" to more than one of the above questions. It will often be useful to employ a diverse range of analytical techniques to understand patterning in archaeological data and inform your research decisions. Best practice in the archaeological use of quantitative methods dictates that methods are selected in light of their ability to answer your archaeological research questions. However, an important feature of exploratory network analysis is precisely to freely look for structure and to let your research be guided to further analyses by what you find. A preliminary exploration of a network dataset could reveal general patterns you already knew were there, but equally could reveal patterning that surprises you and can lead you to change your research questions or even gather new data. Exploratory network analysis can also help identify inaccuracies in your dataset that need correcting before you can proceed. It can identify gaps in the data, helping to guide further data collection before a quantitative network analysis is performed to answer your archaeological research questions.

Method selection should happen in light of research aims, questions, and archaeological theories. Ask yourself: Am I studying properties of the network, individual nodes, paths, and/or communities?

In addition, always perform an initial exploration of a network dataset using a range of different exploratory techniques.

4.3 HOW DO I INTERPRET MY ANALYTICAL RESULTS?

Just like analytical method selection, the interpretation of results in archaeological network research will be entirely dependent on the archaeological research context and theoretical framework. There is no uniform way to interpret a method's results when applied to different types of networks. A technical interpretation or description can be provided for every network method, but this alone will not lead to interesting insights into past human behavior. In social network analysis the types of networks studied tend to be standardized and well defined: nodes are usually social entities such as individual humans, communities, or corporations, and edges

are typically social relationships between them or opportunities for social interaction. In archaeological network research the types of networks studied are vastly more diverse and less standardized, ranging from social networks and visibility networks to material culture similarity networks and geochemical networks to physical transport networks and geographical proximity networks. This is a fair representation of the diversity of archaeological research itself, the range of specialisms it includes, and the diversity of past phenomena it studies. Clearly the interpretation of network analytical results will vary with the types of archaeological networks. But how precisely do we go about interpreting the number of edges connected to a node in a visibility network, for example, and does it differ from the interpretation of this measure in a material culture similarity network?

Through the case studies and archaeological examples in this book we will provide some inspiration for how to interpret a range of network analytical results for the main types of archaeological networks. Further inspiration can be provided from literature in other disciplines dealing with the same network types. But because new types of archaeological networks continue to be created, the most crucial advice for how to interpret network analytical results in a particular research context is to consider it *part of critical archaeological research* (see Chapter 1). Interpretations cannot be adopted directly from network science generally as such models only offer a way of representing and analyzing data and theories. Any critical archaeological research involves a constant back and forth between data critique, theoretical framework, and quantitative or qualitative methods. The interpretative process is an integral part of critical archaeological research when applying any quantitative or qualitative method, and network science is no exception to this.

To aid the critical interpretation of network analytical results in light of your archaeological research framework, we recommend you ask yourself the following questions:

- What type of network am I working with? How are nodes defined? How are edges defined? → Draw inspiration on how to interpret network analytical results from other archaeological and nonarchaeological studies using this same network type (see Chapter 2 for examples).
- What archaeological data (if any) were involved in the creation of this network? What was my data collection strategy? How complete is my dataset and what parts are missing? How have the data been interpreted by other archaeologists and what is my critique of such interpretations? How does the representation of this dataset as nodes and edges allow unique new insights into the studied past

phenomenon? → Let archaeological data critiques of this dataset inform and ground interpretation of analytical results.

- What is my theoretical framework? Do I theorize edges are a medium for the flow of resources or perhaps information? How does my theory specify the process of network change over time? How does my theory specify that my archaeological dataset is proxy evidence for specific past phenomena? → Let a well-specified theoretical framework inspire interpretation of analytical results.

4.4 EXPLORATORY NETWORK ANALYSIS METHODS

In this section, we outline some of the most common statistical measures, metrics, and techniques used to characterize local (referring to particular nodes or edges) and global (referring to the network as a whole) network positions, properties, communities, and topologies. We have organized this section to move from basic to more complex exploratory methods as the methods and concepts involved build on one another. This section also includes two case studies, the first on visibility networks in Cranborne Chase in southern England and the second focused on the Roman road network in the Iberian peninsula in order to provide a couple of concrete examples of how some of the exploratory measures we highlight here can be used.

4.4.1 *Degree and Density*

Arguably one of the simplest but also most useful methods for evaluating node positions and roles is to count how many edges each node has: this is known as the **degree** of the node. In an undirected unweighted network a node's degree is simply the sum of all edges incident to it (Fig. 4.1a). In a directed network we can distinguish between the number of incoming edges and the number of outgoing edges, known, respectively, as **indegree** and **outdegree** (Fig. 4.1b). In a directed network it is always interesting to explore both indegree and outdegree to identify the extent to which relationships are reciprocated; for example, some nodes might be hubs when considering incoming edges but if few of these edges are reciprocated they will not be hubs when considering outgoing edges. The incoming and outgoing edges will be interpreted differently depending on your archaeological theory and what the network represents, for example, the ability for a past individual to gather information and to share information. Strong differences in indegree and outdegree

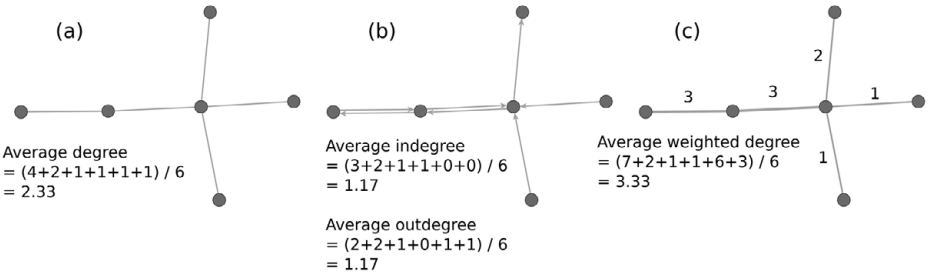


FIG. 4.1 Examples of calculating a network's average degree (a), average indegree and average outdegree (b), and average weighted degree scores (c).

might therefore be important to identify because they could indicate different roles in the network, such as an individual who is able to collect information from many other individuals but is not able to share information with many other individuals. In a weighted network the **weighted degree** score could take the weight of edges into account, by summing the weights of all edges incident to a node (Fig. 4.1c).

The *degree* of a node is the number of edges incident to this node.

The *indegree* of a node in a directed network is the number of incoming incident edges.

The *outdegree* of a node in a directed network is the number of outgoing incident edges.

The *weighted degree* of a node in a weighted network is the sum of weights of incident edges.

We can also calculate how many edges each node has on average, the so-called **average degree** of a network (Fig. 4.1a): the mean degree score of all nodes in the network. This network-based (global scale) measure is often used to compare different networks with each other. In a weighted network the average weighted degree is the mean of weighted degree scores in the network (Fig. 4.1c). In a directed network the average indegree is the mean of indegree scores and the average outdegree is the mean of outdegree scores (Fig. 4.1b). In directed networks the average indegree and average outdegree will always be the same, because the summed number of incoming edges always equals the summed number of outgoing edges: every outgoing edge must arrive somewhere. The average degree in a directed network further differs by a factor of two from the average indegree or outdegree in an undirected network (e.g., compare Fig. 4.1a with Fig. 4.1b). This is because in an

undirected network each edge will contribute to the degree score of *both nodes* incident to it, whereas in a directed network each edge will contribute to the indegree score of *only one node* incident to it.

> The *average degree* of a network is the sum of the degrees of all nodes in this network divided by the number of nodes.

When the degree scores of all nodes have been calculated we can also explore the **degree distribution**: the probability distribution of all node degree values (Fig. 4.2a). Examinations of this distribution allow for considerations of variation across nodes and also for the identification of global properties of the network itself. For example, if the degree distribution is positively skewed, most nodes have a low degree while a smaller number of nodes have considerably higher degree scores, which can result in a very high average degree (a so-called fat-tailed distribution). A power law degree distribution is an extreme version of this, where the degree distribution is such that the fraction of nodes with a certain degree varies as a power of the degree (Fig. 4.2b; Brinkmeier and Schank 2005:294; Newman 2010:247–249). When plotting such a power law degree distribution on double logarithmic axes it will show as a straight line (Fig. 4.2c; though arguing that a given distribution represents a power law requires further statistical assessment beyond simple visuals; see Clauset et al. 2009). The pattern of power law degree distributions (or approximations thereof) has been argued to be common in large real-world networks, such as the World Wide Web or the US power grid. Networks with power law degree distributions are known as scale-free networks, a class of networks frequently studied by physicists in particular (Barabási and Albert 1999) and which has also received a lot of attention by archaeologists (Bentley and Maschner 2003; Graham 2006b; Sindbæk 2007a, 2007b).

> The *degree distribution* is the frequency distribution of all nodes' degrees in the network.
>
> A *fat-tailed degree distribution* is a degree distribution with a long tail skewed toward a high degree.
>
> A *power law degree distribution* is one example of a fat-tailed distribution where the degree distribution follows a power law where the probability of a node with a certain degree is proportional to a power of that degree.
>
> A *scale-free network* is a network with a power law degree distribution.

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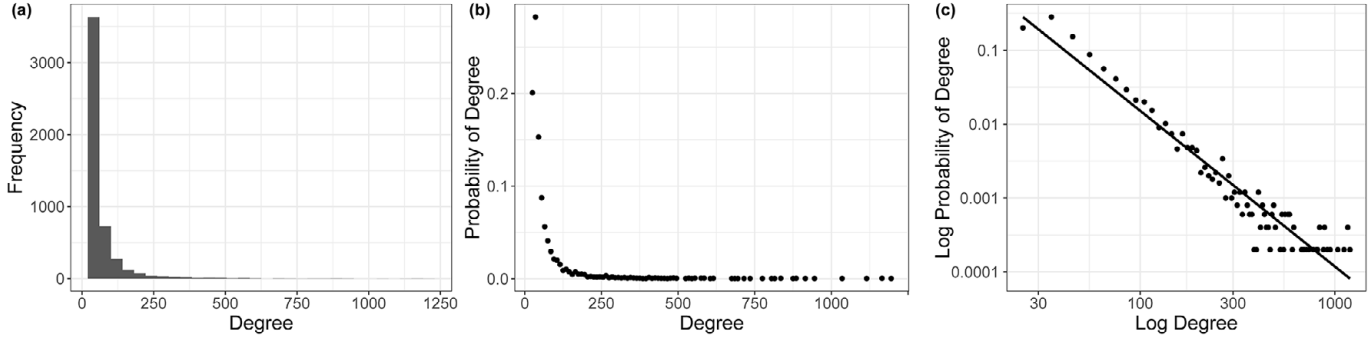


FIG. 4.2 Example of a degree distribution approximating a power law distribution based on simulated data. (a) Histogram of degree distribution, (b) plot showing the degree against the probability of degree, (c) plot showing the log degree against the log probability of degree with a line showing expectations for a power law distribution.

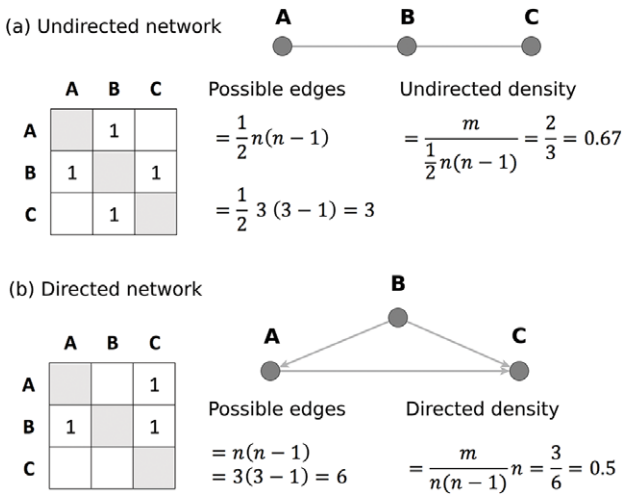


FIG. 4.3 Matrix and network representations of an undirected (a) and directed network (b), including the respective equations for calculating the number of possible edges and density in both cases.

The concept of **density** is closely related to degree. The density of a network is the fraction of edges that are present over the theoretical maximum number of edges that could be present in that network. This fraction gives a value between 0 and 1, where networks with a low score are considered sparse and networks with a high value are considered dense (see Section 3.3.1). In a simple undirected network where no self-loops are possible, the maximum number of edges can be calculated as follows (Fig. 4.3a): $1/2n(n-1)$, which means, half the number of nodes times the number of nodes minus one. This equation is most easily understood when looking at an adjacency matrix (Fig. 4.3a). The $1/2n$ refers to the fact that in an undirected network only one half of the cells in the adjacency matrix can be filled because it is symmetric, whereas the $(n-1)$ refers to the fact that the diagonal of the adjacency matrix has to be excluded because self-loops are impossible (or in other words, there are $(n-1)$ potential partners for each node to connect to). The equation for a directed network without self-loops is slightly different because the adjacency matrix is asymmetric and twice as many edges are possible (Fig. 4.3b): $n(n-1)$.

The *density* of a network is the fraction of the number of edges that are present to the maximum possible number of edges in the network.

4.4.2 *Subnetworks, Triads, and Clustering*

In the previous section we introduced some basic methods to describe the properties of individual nodes and summarize them on the level of the whole network. We will now switch our focus by introducing a set of methods that describe groups of nodes. These are so-called **subnetworks** (usually subgraphs in network science literature): a subset of the entire network, consisting of a set of nodes and a set of edges that connect these nodes. In addition to providing useful ways of describing the overall structure of the network, these methods allow for the identification of groups of nodes that share common features or that can affect each other in a particular way due to the configuration of relationships between them. All methods based on subgroups have a *defining characteristic*, such as the need for nodes in the subnetwork to be connected in components (see below). A subnetwork to which no node can be added without altering its defining characteristic is called a **maximal subnetwork**. Some of the methods introduced in the remainder of this chapter require subnetworks to be maximal, and we will explain this term in more detail below using the example of the *n*-clique method (see Section 4.4.6).

> A *subnetwork* (or subgraph) is a subset of the entire network, consisting of a set of nodes and a set of edges that connect these nodes.
>
> A *maximal subnetwork* is a subnetwork to which no node can be added without altering its defining characteristic.

The smallest group of multiple nodes is a set of two nodes, commonly referred to as a **dyad**. In a simple undirected network there can be only two states of a dyad: the pair of nodes either is connected by an edge or is not. In a directed network where no self-loops are allowed there are four possible dyad states (Fig. 4.4a). One step up from a dyad we have a **triad**: a set of three nodes and the edges between them. There are four possible states in a triad in undirected networks (Fig. 4.4b). Two of these are crucial for the concept of **transitivity** used below to measure the network's clustering. Let us consider the role of triads using an example of a social network where nodes represent individuals and the edges are acquaintance relationships. When a triad is **open** then there are two individuals who do not know each other but have a mutual acquaintance (A knows B and B knows C but A does not know C). When a triad is **closed** all three individuals are acquaintances (A, B, and C all know each other). Transitivity is a concept in network science to study the tendency of triads to be closed, or in social network terms the tendency for a friend of a friend to

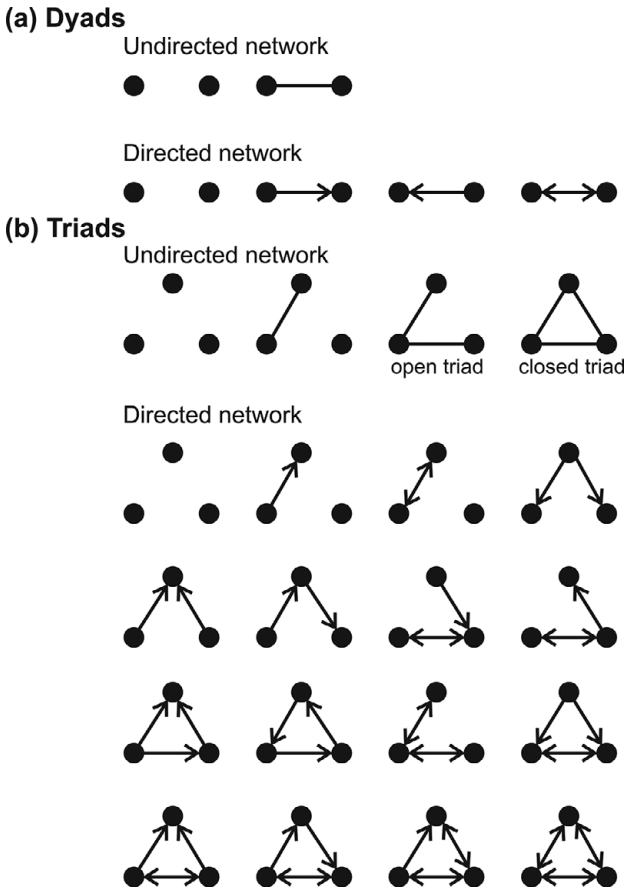


FIG. 4.4 Dyads and triads. (a) The number of states in undirected and directed dyads. (b) The number of states in undirected and directed triads.

be a friend. In a directed network there are many different ways in which transitivity can be expressed, and the total number of possible triad states in a simple directed network is 16 (Fig. 4.4b).

Triads and other types of subnetworks in network science literature (Estrada 2012:73–107; Lusher et al. 2013; Milo et al. 2002) are quite powerful network science tools because the overall structure of a network can be well described by identifying the types of subnetworks that appear and their frequencies. Such subnetworks are key ingredients in formal models representing theories of the key structuring elements of observed networks. For example, in the visibility network case study presented below in Section 4.4.3 the closed triad state is theorized by archaeologists to be an important structural feature in a network: it has been theorized that people in the past tended to locate new long barrows such

that closed triad intervisibility patterns were created. If this was the case, then we would expect to see many more of these closed triad states in the observed network than we would expect by chance (assessed using randomly generated networks with the same number of nodes and edges). An important first part of this kind of analysis is usually performing a **triad census**: counting how many of each triad state the network includes. A triad census can help to inform theories by revealing the relative frequencies of different triad states or whether certain states that would be expected are missing or overrepresented. In recent years, archaeologists have begun to apply more complex models to assess distribution of triads and other subnetwork configurations by comparing observed frequencies of such features to their counts in randomly generated networks. In order to create plausible random networks as a point of comparison, it is important that these random networks retain key properties of the observed network. This is a difficult analytical task, but methods known as exponential random graph models (ERGMs) are designed for just this purpose. A detailed discussion of ERGM models is outside the scope of this book but we expect that such approaches will be increasingly common in archaeological network research in the coming years (but see the introduction to ERGMs by Lusher et al. 2013 and their archaeological application: Brughmans and Brandes 2017; Brughmans et al. 2014; Cegielski 2020; Wang and Marwick 2021; see also Section 4.4.3).

A *dyad* is a set of two nodes and the edges between them.
A *triad* is a set of three nodes and any edges between them.
An *open triad* is a set of three nodes with two undirected edges between them.
A *closed triad* is a set of three nodes with three undirected edges between them.
A *triad census* is a set of counts of each triad state included in a network.

Measures designed to explore node and network clustering at the global and local levels have been developed that have subnetworks and particularly triads as their building blocks: the clustering coefficient is defined using subnetworks including each node's direct neighbors, and network transitivity is based on triads.

The **clustering coefficient** of a node takes the fraction of all the edges between the nodes it is directly connected to over the maximum number of edges that can exist between those neighbors (as defined by Watts and Strogatz 1998). This measure is sometimes called the local clustering coefficient, representing an average probability that a pair of neighbors of a node are themselves connected. One way of subsequently deriving the clustering coefficient of the network as a whole is to take

the mean of the clustering coefficient of all nodes. However, an alternative measure for network clustering is **network transitivity**, defined as three times the number of closed triads over the total number of triads in a network (Coscia 2021:136; Holland and Leinhardt 1971; Newman 2010:199–200). Although both network-based measures (network clustering coefficient and network transitivity) offer an indication of the level of local clustering in a network and of the probability that triads will be closed, they do not always provide similar results: a highly clustered network according to the clustering coefficient might not be so according to the transitivity measure (see Estrada 2012:82–83 for a detailed description of this problem). When applying these measures it is therefore advised to apply both (or provide theoretical arguments why one is more appropriate than the other in your research context) and compare the resulting scores with those obtained from random graphs with the same number of nodes and density.

A *node's clustering coefficient* is the fraction of all the edges between the nodes it is directly connected to over the maximum number of edges that can exist between these neighbors.

A *network's clustering coefficient* is the mean of all nodes' clustering coefficient scores in a network.

A *network's transitivity* is three times the number of closed triads over the total number of triads in a network.

4.4.3 *Case Study: Explaining Degree and Subnetworks in a Visibility Network*

In his study of the intervisibility network of long barrows in Cranborne Chase, Chris Tilley (1994:157–166) was particularly interested in the degree and clustering of long barrows (for more information on the data used in this case study, see Chapter 2). He argued that "it is of interest to note that those barrows with the highest degree of intervisibility with others form members of pairs of larger barrow groups" (Tilley 1994:158–159). Using degree- and subnetwork-based exploratory network analysis measures we can offer quantitative descriptions of this observation, which can be used in quantitative hypothesis testing and will allow quantitative comparison with other networks.

We see that the network consists of six components, a count that includes three isolated nodes (Fig. 4.5). The network density is 0.1 and its clustering coefficient is 0.43. These measures tell us the network is rather sparse and fragmented, with some

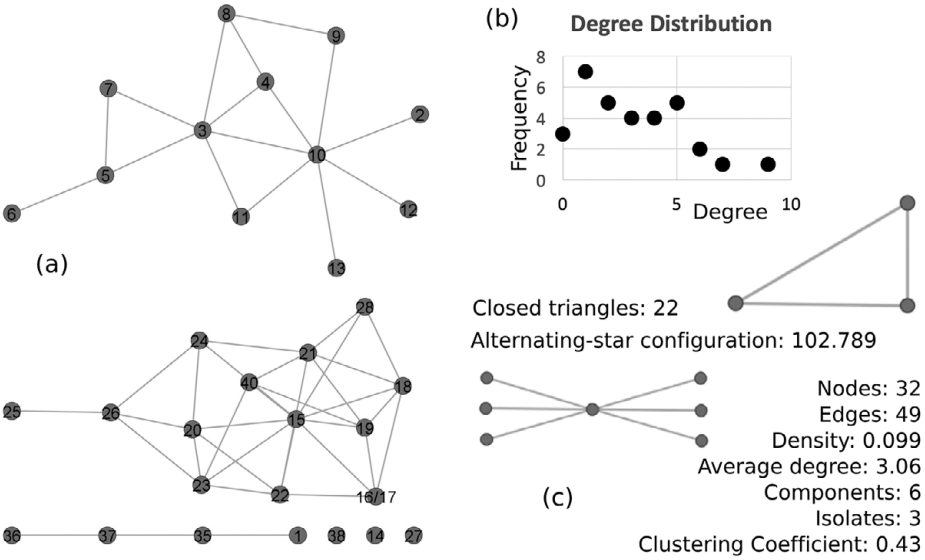


FIG. 4.5 (a) Cranborne Chase visibility network: nodes represent long barrows, edges represent intervisibility. (b) Degree distribution of visibility network. (c) Exploratory network statistics of visibility network.

denser clusters within the components. Such a structure is typical of visibility networks in landscape archaeology, in which long distances and changes in topography strongly limit the pairs of sites that can be intervisible.

On average each long barrow is intervisible with about three other long barrows. But when we plot this as a degree distribution, we notice a strong difference between the vast majority of long barrows that have a low degree (most frequently 1), and two long barrows that are intervisible with seven and nine other long barrows, respectively (Fig. 4.5b). Could there be a cultural explanation for this clustering and difference in node degree? Or could this pattern have emerged by chance?

A cultural explanation was hypothesized by Tilley (1994:159): "sites that were particularly important in the prehistoric landscape and highly visible 'attracted' other barrows through time, and sites built later elsewhere were deliberately sited so as to be intervisible with one or more other barrows." Such a theory can be formally represented and tested with network science, using the method of exponential random graph models (ERGMs) to assess how over- or underrepresented the proposed important configurations are (see Box 4.1; Brughmans and Brandes 2017; Brughmans et al. 2014; Lusher et al. 2013). Brughmans and Brandes (2017) created a model where the visually attracting nodes were represented by a

Box 4.1 EXPONENTIAL RANDOM GRAPH MODELS (ERGMS)

Exponential random graph models (ERGMs) are a class of statistical models that are designed to help represent, evaluate, and simulate ideas about network-generating processes and structural properties (for good introductions to the method and archaeological cases, see Amati et al. 2020; Brughmans et al. 2014; Lusher et al. 2013; Wang and Marwick 2021). These models allow us to formally represent our theories about how particular patterns of relationships (such as triangles or stars) or associations (such as mutuality or homophily by attribute) might emerge and persist in our networks and further to evaluate how well such theories account for our observed network data. Specifically, an ERGM can be used to generate large numbers of networks in a random process biased toward particular configurations and associations that represent our theories of interest. For example, if we are interested in evaluating a theory regarding preferential attachment, as was the case in the Cranborne Chase example above, we can design our ERGM to simulate random networks with a tendency toward star configurations (a configuration associated with preferential attachment). We can then compare the global properties of the random networks built with this generative process (preference for stars) to the global properties of our empirical network to evaluate how well that theory of tie formation explains our observed sample. Importantly, ERGMs are based on representations of local patterns that influence tie prevalence (e.g., a tendency toward triangle or star configurations or mutual connections) but these local patterns are evaluated in relation to the resulting global network properties that were not explicitly included in the model.

How then can we use ERGMs to evaluate whether our theory is a good theory for explaining the properties of a given network? If your ERGM is able to reproduce structural features of the observed network reasonably well, then it might have some explanatory value. But never forget this only means the theory is plausible, not that it is correct. Importantly, ERGMs also allow you to identify particular theories for network properties or edge prevalence that are unlikely to have generated the observed network. For example, if you build an ERGM that generates networks with a tendency toward mutual ties and that does not generate networks that mirror the properties of your observed sample, you might consider "mutuality" an unlikely generative process for creating that network. Thus, ERGM can push you to explore alternative theories for network generation, or it can force you to further specify your theory. Both should be welcomed. By identifying the range of plausible and unlikely network generative processes, ERGMs can help you develop a more robust explanatory framework for your observed network.

star-shaped configuration and where the tendency for the network to be clustered in groups was represented by a closed triangle configuration (Fig. 4.5). Counts of these configurations in the observed network were compared with those in simulated networks with a built-in statistical tendency toward establishing these configurations (i.e., a simulation model that formally represents Tilley's cultural explanation). Comparing these configuration counts showed the model could accurately reproduce the structure of the observed network, and therefore suggests Tilley's theory is an appropriate description of the observed network data (for more detail on this ERGM experiment, see Box 4.1 and Brughmans and Brandes 2017).

4.4.4 *Walks, Paths, and Distance*

It is common in archaeological network research to theorize or assume that the edges of an archaeological network can act as the medium for the flow of people, goods, or ideas. In network science such flows can be explored with path-based measures that represent how some resource can move from one node to another node in the network. Such assumptions of flows over the network are most appropriate for past social networks in which nodes represent individuals/groups that are able to share information if they are connected in the network, or for past transport systems where goods can be moved from one town to another via the roads and rivers connecting them. Assuming flows in many other archaeological networks, however, is not as straightforward. For example, what does a path through a network of archaeological assemblage similarity mean? In such cases path-based measures can still be used to explore purely structural features of the network and to compare these with other networks. To be able to appropriately interpret path-based results, however, the archaeologist will have to make explicit how they theorize edges in the network allow for flows of some sort. For example, one could theorize (as we commonly do) that artifact assemblage similarity of a pair of sites is interpreted as proxy evidence of more frequent interaction or exchange between past communities living at those sites. If this theory is stated explicitly and well supported, then path-based measures can be used and interpreted archaeologically (e.g., Mills et al. 2015). In such cases an edge in an archaeological similarity network is a representation of the possibility of interaction and exchange, and that cultural ideas of some sort can flow through the network over the edges.

There are many different ways in which flows or routes through a network can be represented, and here we will first introduce the most important types of representation that are used as the building blocks for the more complex measures

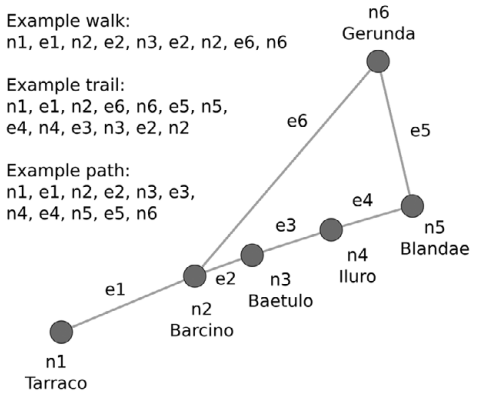


FIG. 4.6 Examples of paths, walks, and trails using a subset of the Roman roads network on the Iberian peninsula. Nodes represent Roman settlements, edges represent roads.

discussed further on in this chapter. A **walk** is the most general and is defined as any sequence of nodes connected through edges. In a walk the same edges and nodes can be repeated. A **trail** between a pair of nodes is more restrictive and does not allow for the repetition of edges. However, a single node can appear more than once in a trail. A **path** is the most restrictive type of route: it is a sequence of connected nodes and edges between a pair of nodes where no nodes or edges can be repeated.

Figure 4.6 illustrates the difference between these three types of flows using a subnetwork from the Roman road network on the Iberian peninsula. If Romans living in Tarraco (present-day Tarragona) wished to go to Gerunda (present-day Girona) over Roman roads, then they could take the following *walk* through the road network if they like sea views and had a lot of time to spare: n1, e1, n2, e2, n3, e2, n2, e6, n6. Note that our Romans passed twice through Barcino (present-day Barcelona) and the road between Barcino and Baetulo (present-day Badalona), and therefore this is considered a walk in network science. Our Romans now want to go from Tarraco to Gerunda and then continue their trip to Barcino where they will spend the night with relatives. But our Romans like to see some of the surrounding countryside and therefore do not want to use any single road twice. They could take the following *trail*: n1, e1, n2, e6, n6, e5, n5, e4, n4, e3, n3, e2, n2. Note that our Romans passed through Barcino twice but did not use any road twice, and therefore this is considered a trail in network science. If they wish to go from Tarraco to Girona and not visit any town or road twice, they could follow this *path* in which no nodes and edges are repeated: n1, e1, n2, e2, n3, e3, n4, e4, n5, e5, n6.

A *walk* is any sequence of nodes connected through edges that has that pair of nodes as end points. Nodes and edges can be repeated in a walk.

A *trail* is a walk between a pair of nodes in which no edges are repeated but nodes can be repeated.

A *path* is a walk between a pair of nodes in which no nodes and edges are repeated.

Walks in directed networks have to follow the directionality of the edges. For example, in Figure 4.7 a path from node 1 to node 4 has to cross three edges (n1, e1, n2, e2, n3, e3, n4), whereas the return journey from node 4 to node 1 crosses only two edges (n4, e4, n2, e5, n1). A cycle is a further type of walk that is particularly important in the case of directed networks. A **cycle** is a walk of at least three nodes that begins and ends in the same node, but in which no edges are repeated and no nodes are repeated except for the starting/ending node. Cycles are a common feature of transport networks such as the Roman road network (see, e.g., in Fig. 4.6 the cycle n2, e2, n3, e3, n4, e4, n5, e5, n6, e6, n2), but some types of directed networks such as genealogies, citation networks, and hydrological networks typically include no cycles at all: these networks are called **acyclic networks**. Such networks are typically directed and are often referred to in network science literature as *directed acyclic graphs* (DAGs), for example, a citation expresses the influence of one paper on another or a river flows from one junction to another. Such networks are also often **hierarchical networks** in that there is typically a means for nodes to be "classified through a relation of precedence" (Gemici and Vashevko 2018:1) such that edges flow from higher-level nodes to lower-level nodes

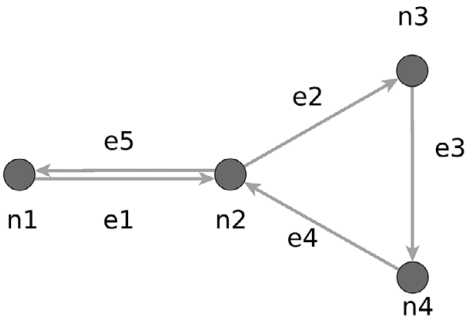


FIG. 4.7 An abstract directed network illustrating that the directionality of edges affects the length of paths. For example, the path from node 1 to node 4 crosses three edges, whereas the return path from node 4 to node 1 only crosses two edges.

and never the reverse. Hierarchical networks are sometimes also referred to as *trees* (see Section 7.3). For example, under typical circumstances a citation network will represent a hierarchical network in that new papers can only cite older works. Hierarchical networks are common in a number of settings including networks based on river or directed road junctures, food webs, or even certain social networks (see Clauset et al. 2008).

A *cycle* is a walk of at least three nodes that begins and ends in the same node, in which no edges are repeated and no nodes are repeated except for the starting/ending node.

An *acyclic network* is a network that includes no cycles.

A *hierarchical network* is a special class of network where nodes can be placed in an ordered set of levels where edges flow only from higher levels to lower levels with no cycles. Such networks are sometimes referred to as trees due to their stem and branch-like structure.

These basic types of walks through a network are used as the building blocks for more complex path-based measures. The most common of these is the **shortest path** between a pair of nodes (sometimes referred to in network science literature as the geodesic). The **length of the shortest path** is known as the topological distance between a pair of nodes, or simply the distance between nodes. This distance is expressed in terms of the number of edges crossed by the path. Returning to our example of the Roman road network (Fig. 4.6), we can say that our Romans' trip from Tarraco to Gerunda has a distance of two, because the shortest path n1, e1, n2, e6, n6 crosses two edges (e1 and e6).

A *shortest path* of a pair of nodes is the path between this pair of nodes with the shortest path length.

The *shortest path length* of a pair of nodes is the number of edges in the shortest path between the pair of nodes.

When we calculate the shortest paths between all pairs of nodes in a network we can obtain two exploratory network analysis measures that are commonly used to compare different networks: the network diameter and average shortest path length. The **diameter** of a network is the largest distance between any pair of nodes in the network. It quantifies how far apart the most separated two nodes in the network

are. In our example of the Roman road network (Fig. 4.6) the diameter is three, because the longest shortest path between any pair of Roman towns is three road segments (e.g., from Tarraco to Blandae). The **average shortest path length** is the sum of the shortest paths of all node pairs divided by the number of node pairs (i.e., the mean of shortest paths in the network). It quantifies how distant on average all nodes are to each other and is commonly used to compare different networks. The average shortest path length of our excerpt from the Roman road network (Fig. 4.6) is 1.47, which means that our Romans can reach any Roman town in this example road subnetwork from any other Roman town by crossing on average 1.47 roads.

The average shortest path length is one of a few commonly used methods to compare the structure of different networks. It is key to the definition of a **small-world network** structure, which captures the notion that in many real-world networks highly clustered communities exist that are connected to each other through relatively few links that significantly reduce the network's average shortest path length. This feature has been identified in many real-world networks (Watts 2004; Watts and Strogatz 1998) and has also influenced a lot of archaeological network research (Bentley and Maschner 2003; Graham 2006b; Sindbæk 2007a, 2007b).

The *diameter* of a network is the largest shortest path between any pair of nodes in the network.

The *average shortest path length* is the average of all shortest paths in a network.

A *small-world network* is a network in which the average shortest path length is almost as small as that of a uniformly random network with the same number of nodes and density, whereas the clustering coefficient is much higher than in a uniformly random network (a *uniformly random network* is defined as a network in which each edge exists with a fixed probability p).

What happens to paths when a network is disconnected? When some nodes are isolated and the network consists of multiple components (see next section), then not all nodes are reachable from all other nodes. In such cases not all pairs of nodes in a network can be connected by a path. Nodes that can be connected by a path are said to be **reachable**, and nodes that cannot be connected by a path are **unreachable**. Having unreachable node pairs in your archaeological network has implications for the more complex path-based measures. The distance between a pair of unreachable nodes is infinite or undefined, and so is the diameter of a network that

includes unreachable node pairs. For the average shortest path length of disconnected networks there are two straightforward outcomes: the sum of all shortest paths is divided either by the total number of node pairs or by the total number of node pairs minus those for which the shortest path is undefined (but often it is convenient to assign such distances to be infinite). An alternative pragmatic approach to path-based measures for disconnected networks is to calculate instead the diameter or average shortest path length of the largest connected component (in some software this is a default setting when applying path-based measures to disconnected networks). However, this does not reflect a structural feature of the network as a whole and can be extremely misleading when comparing these results with those of other networks. Archaeologists therefore need to be careful when applying such path-based measures to disconnected networks. An "infinite" or "undefined" result is not informative when comparing different networks; it does not reveal any path-based structural features of the network but merely points out its disconnected nature.

The following section will provide some ways forward by offering approaches that describe structural features of disconnected networks.

4.4.5 *Components and Bridges*

So far we have been mainly concerned with cases of **connected networks**: networks where all node pairs can be connected by a path, that is, networks where all node pairs are reachable. Some types of archaeological networks are typically connected networks. For example, in the Roman road network example we can at least in theory assume that all places were reachable from all other places through some sort of road, however minor. How else would all roads lead to Rome? A further example is the use of similarity measures like the Brainerd-Robinson coefficient for creating material culture similarity networks where some measure of similarity is represented between each pair of site assemblages, often due to the presence of a few ubiquitous artifact categories and the exclusion of sites for which no artifact assemblages are available. But other types of archaeological networks are often disconnected. A **disconnected network** is a network where not all node pairs are connected by a path, because of the existence of isolated nodes or multiple subnetworks that are not connected to each other. In visibility networks, for example, some places are often completely visually isolated and cannot be connected by an intervisibility edge to any other places. Similarly, in coauthorship networks of archaeological literature some groups of authors might be disconnected

from others in the network because they only ever coauthored with each other. The term **component** is used to refer to the subnetworks of the network that are not connected to each other: a subset of a network in which any pair of nodes can be connected to each other via at least one path, and where there can be no paths to any nodes outside this subset. More precisely, these are sets of nodes that share edges with each other but not with other sets of nodes in the same network. An alternative definition of a disconnected network is therefore a network that exists of multiple components.

A node pair is *reachable* if it can be connected by a path.

A *connected network* is a network where all node pairs are reachable.

A *disconnected network* is a network where not all node pairs are reachable.

A *component* is a subset of a network in which any pair of nodes can be connected to each other via at least one path, and where there can be no paths to any nodes outside this subset.

There are particular implications of disconnected networks for the structural positions of individual nodes and the flow of resources through the network as a whole. Consider a past visual signaling network used for military purposes, where information could be shared between intervisible places through fire, smoke, or flag signaling. If this visual signaling network is connected, then information can be shared equally among all places included in the signaling network. In case of an enemy attack at one side of this spatial network, for example, information about the attack could be signaled to all other places in the network. However, if multiple components exist, then the information in a component will remain in that component and cannot be shared with places in the other components. What would happen in the case of a targeted enemy attack on a crucial signaling location?

This example reveals why the identification of so-called cutpoints and bridges are powerful analytical tools in network science: both define nodes and lines critical to keeping a network connected (Fig. 4.8). A **cutpoint** is a node of which the removal results in a network with a higher number of components. In our visual signaling example this is equivalent to a targeted attack on a crucial signaling location, the neutralization of which would result in the signaling network fragmenting. A **bridge** is a similar concept applied to edges: it is any edge of which the removal results in a network with a higher number of components. This would be the equivalent in our visual signaling example of a physical obstruction being created between two signaling places such that one part of the communication network is cut off from the other part.

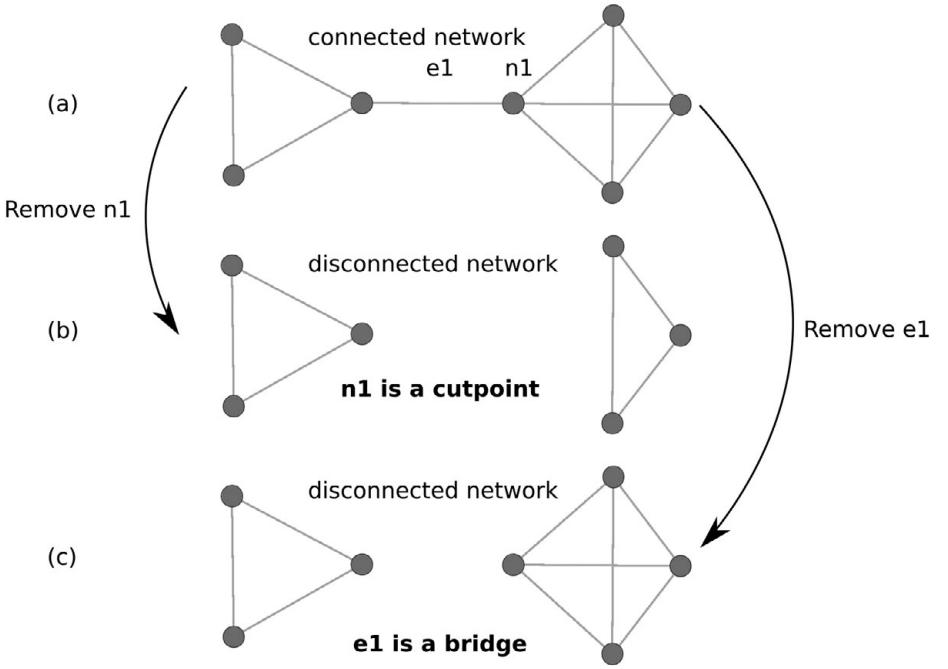


FIG. 4.8 The connected network in (a) can become disconnected into two components as in (b) by removing node 1: node 1 is a cutpoint. The same network (a) could become disconnected into two components in (c) by removing edge 1: edge 1 is a bridge.

Cutpoints and bridges can be used in archaeological network research to study network resilience or robustness, to explore the impact of node or edge removal as in our signaling network example, or to merely identify such key nodes and edges for controlling the flow of resources between different parts of a network. In a past transport network, for example, cutpoints and bridging edges had the opportunity to effectively control the movement of people and/or goods between different parts of the transport system, because they are necessary intermediaries or bottlenecks: all transported people and/or goods between different parts of the transport system will have had to go through this cutpoint and over this bridge. Cutpoints and bridges equally provide a useful measure of the connectivity of a connected network, if we count the number of nodes or edges that need to be removed to make a connected network disconnected.

A *cutpoint* is a node of which the removal results in a network with a higher number of components.

A *bridge* is an edge of which the removal results in a network with a higher number of components.

4.4.6 *Cohesive Subgroups, Cliques, and Community Detection*

In our discussion of subnetworks above we have focused on those with two or three nodes (dyads and triads). Subnetworks with three or more nodes are often used to identify **cohesive subgroups** in networks. These are sets of nodes among whom there are relatively strong or direct relationships, as compared to their relationships with nodes outside the set, for example, a set of past communities that have particularly high similarity of their artifact assemblages in a similarity network, or a set of settlements in a past road network that are directly connected to each other with good roads. In social networks they could represent tight-knit groups of friends, sometimes called cliques.

A **clique** as a network science concept is arguably the strictest method of defining a cohesive subgroup. It is any set of three or more nodes in which each node is directly connected to all other nodes. It can be alternatively defined as a completely connected subnetwork, or a subnetwork with maximum density. Revisiting our example of the Roman road network, we see that this network contains no interesting cliques of three or more nodes, because there are no sets of three or more nodes that are completely connected (Fig. 4.9a). However, if a direct road were created between Tarraco and Gerunda, this would create the following clique: Tarraco, Gerunda, Barcino (n1, n2, n6 in Fig. 4.9b).

Although a clique is a conceptually simple method, its applicability and usefulness can be restricted due to a few issues. A clique is a very strict definition of a cohesive subgroup in the sense that the absence of a single edge in a subnetwork would prevent the subnetwork from being a clique. The fragmentary nature of archaeological data used to create archaeological networks makes unknown or missing edges a common problem and can affect the usefulness of clique identification (see discussion in Chapter 5). Moreover, we can expect strong differences when identifying cliques in different types of networks. A typically dense network such as a material culture similarity network will have a high number of cliques but few of them will be considered meaningful due to the ubiquity of a few common artifact types, whereas a typically sparse road network will have far fewer cliques. A range of alternative methods for the identification of cohesive subgroups has been proposed based on reachability and subnetwork cohesion. We will introduce a few of them in this section (for exhaustive overviews, see Fortunato 2010; Wasserman and Faust 1994:249–290).

> A *clique* is a set of at least three nodes in which each node is directly connected to all other nodes in the set.

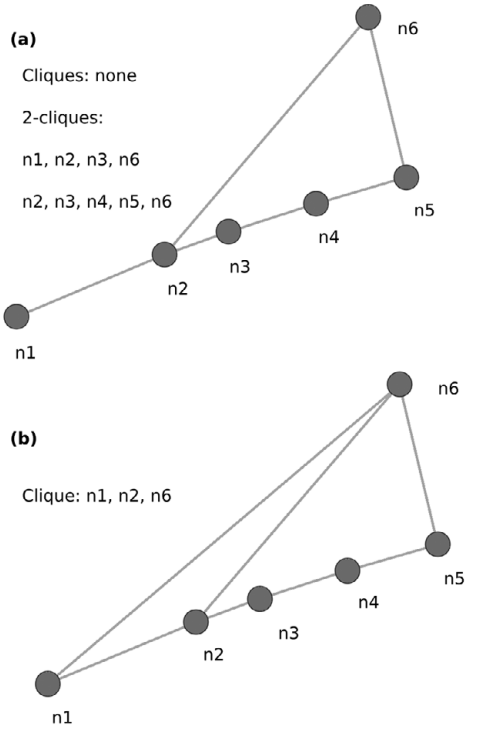


FIG. 4.9 (a) The identification of cliques and *n*-cliques in the Roman road subnetwork shown in Figure 4.6. There are no cliques in this network, but when a new road is created between Tarraco and Gerunda (n1 and n6) as in network (b), then the network contains one clique: n1, n2, n6.

Cohesive subgroup measures based on reachability are most appropriate in archaeological research contexts where edges are theorized to be the medium for the flow of resources, such as for past visual signaling networks or past transport networks. In these methods we require all nodes in the cohesive subgroup not to be connected to each other but rather to be close to each other in the network.

The network science concept of ***n*-clique** puts this into practice. An *n*-clique is a maximal subnetwork such that the distance of each pair of nodes in the subnetwork is not larger than *n*. When $n = 1$ in a so-called 1-clique all node pairs will be connected to each other directly, and 1-cliques are therefore normal cliques. However, as we increase the value of *n* we relax the strict requirements of a clique and allow for the identification of node sets that are part of the same maximal subnetwork because all members are reachable through a certain maximum number of edges. This is a maximal subnetwork because if any additional nodes were added the set would no longer have the characteristic of being separated by *n* or fewer steps. We can again look back at our Roman road network, where we can

identify the following 2-cliques: n1, n2, n3, n6 and n2, n3, n4, n5, n6 (Fig. 4.9a). From all nodes in these two sets all other nodes in the set can be reached in a maximum of two steps, but when we add another node to the set then this is no longer the case: the distance between some node pairs will increase to 3. What does this mean for ancient Romans? As an example we can interpret this in light of how information about the death of an emperor would be shared through this network: a courier from Rome will be able to share this information with the two Roman settlements by traveling from one settlement to another through a maximum of one other settlement (for the issues surrounding *n*-cliques and alternative reachability measures, see Fortunato 2010:84–85; Wasserman and Faust 1994:257–263).

> An *n-clique* is a maximal subnetwork such that the distance of each pair of its nodes is not larger than *n*.

Another set of methods of defining cohesive subgroups is based on the degree of nodes. These reflect the notion that members of a cohesive subgroup need to have a certain number of relationships in order to be considered cohesive. According to this logic, subgroups with many relationships are considered more cohesive than subgroups with few relationships.

The concept of ***k*-cores** is a particularly intuitive implementation of this notion. A *k-core* is a maximal subnetwork in which each vertex has at least degree *k* as defined within the subnetwork (i.e., not the degree of the node in the full network). Remember that by "maximal" we mean that no other node can be added to the subnetwork if it would alter the main characteristic of *k*-cores: having a minimum of degree *k*. We can apply this method to our example of the Roman road network to identify two cores (Fig. 4.10): a 1-core and a 2-core. Roman settlements part of the 2-core have at least two roads connecting them to other Roman settlements in the 2-core: a Roman messenger can travel to at least two other Roman settlements from each of these settlements in the 2-core. Notice also how the cores are *nested*: the 2-core is nested in the 1-core.

> A *k-core* is a maximal subnetwork in which each vertex has at least degree *k* within the subnetwork.

However, the interpretation of the Roman road network example reveals some limitations when applying *n*-cliques and *k*-cores to many types of archaeological

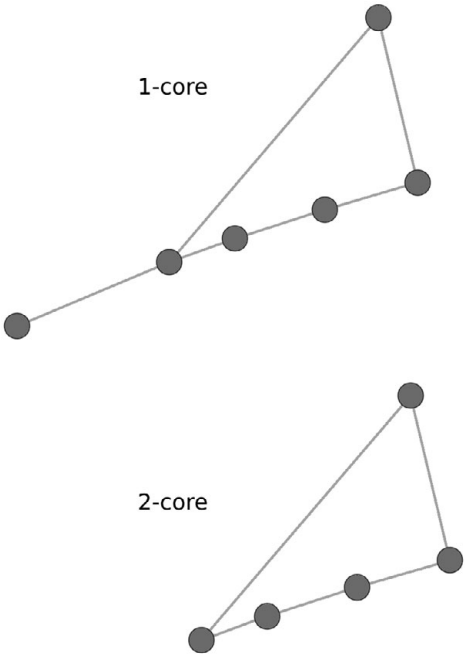


FIG. 4.10 The Roman road subnetwork of Figure 4.6 includes a 2-core of nodes with at least degree 2, which is nested inside (is part of) the 1-core of nodes with at least degree 1.

networks but particularly with spatially embedded networks: What about the distance between settlements, the length of roads? In spatial networks, a cohesive subgroup might be better defined as places connected in the network that are also physically close to each other. More generally, we could define cohesive subgroups with reference not merely to the density and number of relationships but also to some attribute value of the edges.

The concept of *m*-slices implements this notion and is similar to *k*-cores in that it identifies nested subgroups, but with reference to edge weight in weighted networks rather than node degree. An ***m*-slice** is a maximal subnetwork that only contains lines with at least *m* weight and their associated nodes.

We can apply this method to study spatial connections between our Roman road network example. So far we have been using an abstract network representation of this network, where pairs of Roman settlements are connected in the network if they have a direct road between them (see case study introduction in Chapter 2). In reality, however, the roads are far more winding and have different lengths (Fig. 4.11a). We will use *m*-slices to identify subgroups of settlements that are close to each other in physical space. In order to use *m*-slices in this way we will need to change our road segment edge weights from *distance* in kilometers to an equivalent

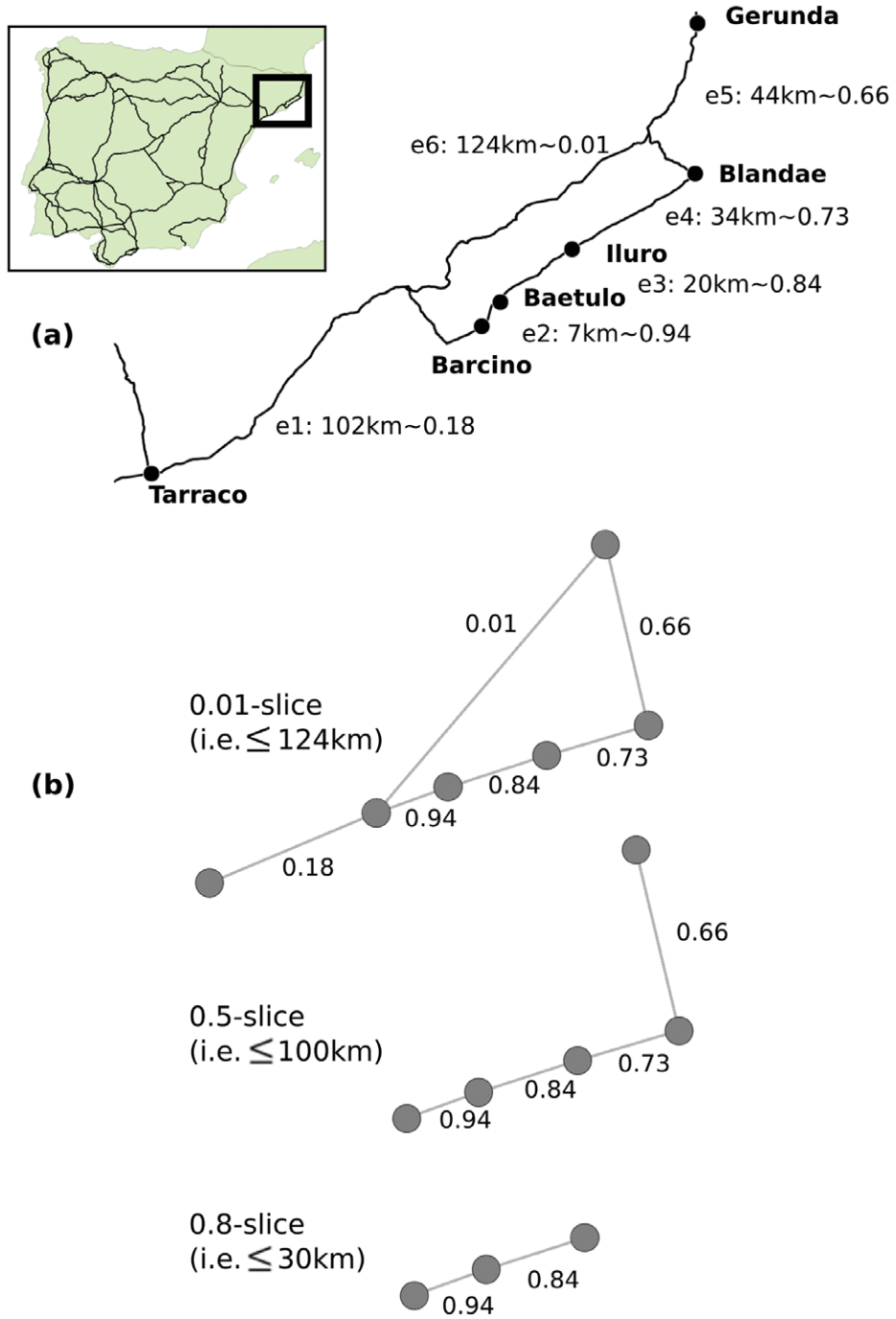


FIG. 4.11 (a) Geographical representation of the Roman road subnetwork used in Figure 4.6 with associated edge distances in kilometers. In order to apply *m*-slices, these distances are converted into a measure of closeness. (b) Three nested *m*-slices are shown including roads shorter than 124 kilometers, 100 kilometers, and 30 kilometers.

closeness measure. We do this by scaling the lengths of individual road segments to values between 0 and 1, inverting them (i.e., to a scale between −1 and 0), and adding 1 to obtain a measure of closeness rather than distance between each pair of Roman settlements in the network (a simpler alternative to this operation would be to change the definition of *m*-slices to identify edge weights equal to or *lower* than a certain distance). By changing the value of *m* we identify different nested slices: the coastal towns Barcino, Baetulo, and Iluro are physically closest to each other and connected in the network (Fig. 4.11b). Ancient Romans might have traveled more frequently between these towns in the 0.8-slice than between all towns in the complete network, some of which are far away from each other.

Thus, *k*-cores and *m*-slices are particularly powerful methods for exploring dense and valued networks, such as archaeological artifact similarity networks. They allow for identifying the nodes and subgroups with the highest degrees or edge values and gradually add lower-degree nodes and lower-value edges in an interactive data exploration process (for an archaeological example applied to a Roman ceramic tableware similarity network, see Brughmans 2010).

> In a weighted network an *m-slice* is a maximal subnetwork containing the lines with a weight equal to or greater than *m* and the nodes incident with these lines.

So far we have defined cohesive subgroups only with reference to local node-based or subgroup properties. But a large number of other commonly used network science measures exist that rather identify cohesive subgroups as having strong and dense relationships between nodes within the subgroup but few relationships between subgroups. A detailed description of these complex methods lies outside the scope of this textbook, and we limit ourselves here to listing the most popular methods (but see Fortunato 2010; Newman 2010:371–391; Radicchi et al. 2004; Wasserman and Faust 1994:267–270).

The oldest such measure is known as a **strong community LS-set**, which is a subnetwork such that the degree of all nodes within the subnetwork is higher than their degree outside the subnetwork (Lucio and Sami 1969). A weaker version of this method is the so-called **weak community LS-set**, where only the internal degree of the subgroup as a whole (rather than every individual node in it) needs to be greater than its external degree (Radicchi et al. 2004). Another method called the **Lambda set** implements the notion that a cohesive subgroup should be hard to disconnect by the removal of lines from the subnetwork (Borgatti et al. 1990). A further method introduced by Girvan and Newman (2002) builds on the idea of

identifying those edges between subgroups that are crucial for facilitating communication between them. It calculates the betweenness centrality (see Section 4.4.7) of each edge, removes the edge with the highest betweenness centrality value, and iteratively repeats this process. This method allows for the identification of nested communities, ranging from the top level where the entire network is connected to the bottom level where each node makes up its own community. Another popular community detection approach is the **walktrap algorithm**. This approach is designed to work for either binary or weighted networks and defines communities by generating a large number of short random walks and determining which sets of nodes consistently fall along the same short random walks. This approach was recently used by Mills and colleagues (2018) and proves to work particularly well in similarity networks.

The *most* commonly used methods for such community detection, however, are based on the concept of **modularity**: the observed density of links in a community should be significantly higher than the density we would expect in case the network was formed by a random process (Newman and Girvan 2004). A number of methods exist that aim to maximize modularity scores for communities in the studied network (for an overview and the problem of the inability of modularity measures to identify small communities, see Estrada 2012:203–206; Fortunato 2010:100–114; Newman 2010:373–382). A popular modularity-based community detection algorithm was developed by Blondel and colleagues (2008) and is known as **Louvain modularity**, after the affiliation of its authors. The method can be extended into a weighted version by considering the strength of edge weights in communities as well as their density.

Although they have rarely been applied in archaeology, this final group of community detection methods has great potential for identifying natural grouping in archaeological data networks or communities in representations of past relational phenomena. An example is Radivojević and Grujić's (2017) use of the Louvain modularity measure for identifying communities in copper supply networks in the prehistoric Balkans.

4.4.7 *Centrality*

Much of archaeological network research is concerned with identifying the most and least important nodes and edges in the network for different sorts of relational processes. But what does it mean to be important in a network? Network science offers numerous ways of conceptualizing "importance," including the following:

- Being connected to a high number of edges
- Being close to all other edges or nodes
- Being a crucial go-between for the flow of resources
- Being connected to many other important nodes.

The family of measures known as **centrality** methods implements many of these concepts of importance (and the related graph-level summary metric **centralization**). They offer a ranking of nodes or edges based on a measure of how well they are suited to play a certain role in the network. Different measures are better or worse for characterizing different kinds of network processes (see also Borgatti 2005).

But which centrality measure is the most appropriate one to use in your archaeological research context? It is crucial to realize that terms like *importance*, *centrality*, and *prominence* are loaded and ambiguous. Despite being intuitively understandable, they can be interpreted in many ways in network research, as the list above illustrates. Formal methods such as betweenness centrality and closeness centrality represent very different phenomena: the importance as an intermediary and the importance through proximity, respectively. Moreover, the formal implementation of these methods may include assumptions that are only relevant in selected archaeological theoretical frameworks. For example, betweenness and closeness are path-based measures and therefore only appropriate if you assume edges enable the flow of resources between three or more consecutive nodes.

It is therefore unacceptable to merely apply one or more centrality measures to an archaeological network without archaeological theoretical motivation and to interpret the results at face value as representations of node or edge importance. Most centrality measures will give highly different results and rankings of nodes, as they should, because they represent different meanings of the concept of "centrality" (see Fig. 4.12; see also Marten Düring's [2016] comparison of applying most centrality measures introduced in this section to fragmentary historical sources). When using centrality measures in archaeological network research, it is particularly crucial that you define explicitly and unambiguously what importance, prominence, or centrality means in your particular research context, and to select the most appropriate formal method in light of that definition:

- If you theorize important nodes are connected to many other nodes → Degree centrality, eigenvector centrality, and PageRank are appropriate.
- If you theorize important nodes are connected to many other important nodes → Eigenvector centrality, PageRank, hubs, and authorities are appropriate.

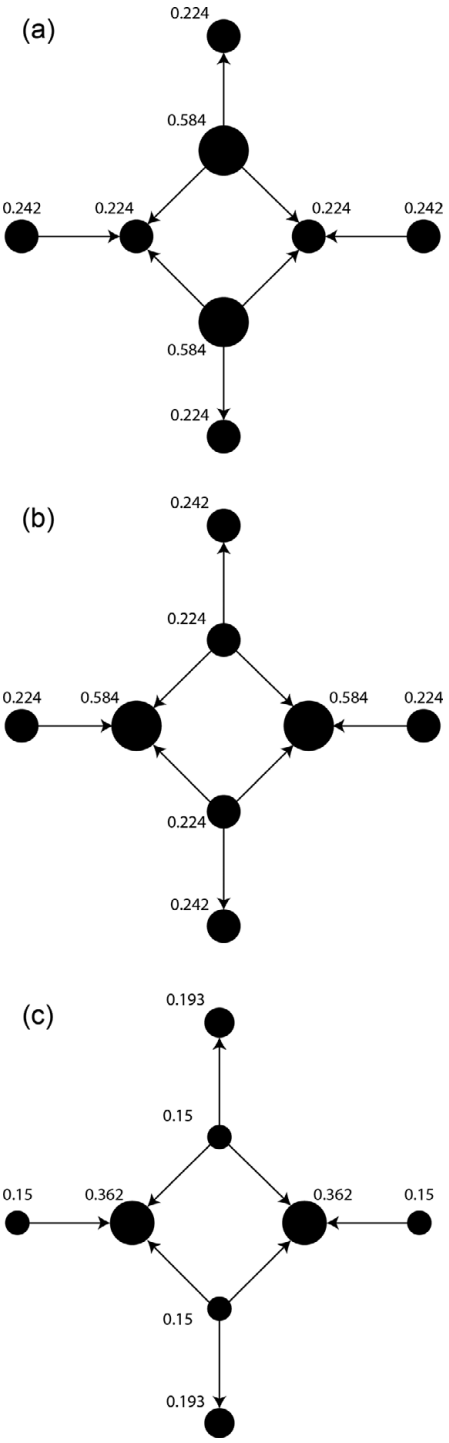


FIG. 4.12 Abstract network illustrating the differences between (a) nodes' hub scores, (b) authority scores, (c) and PageRank score. Note how different nodes are identified as important hubs and authorities, and how the authorities also score high according to PageRank.

- If you theorize that network edges are a medium for the flow of resources and that important nodes or edges are those that need to be passed frequently on all paths through the network → Betweenness centrality is appropriate.
- If you theorize that network edges are a medium for the flow of resources and that important nodes or edges are those that are close in network terms to all others → Closeness centrality is appropriate.

Centrality is a family of measures used to identify the most important or prominent nodes or edges in the network, depending on the different definitions of importance or prominence implemented in the network measure used.

What centrality, importance, or prominence means has to be defined explicitly in light of your archaeological theoretical framework before selecting and applying a particular centrality measure.

Centralization is the graph-level summary of centrality defined as the ratio of the sum of the differences from the maximum observed centrality to all other node-level centrality scores to the theoretical maximum possible sum of differences. Centralization measures the degree to which the node is focused on one or a small number of highly central nodes.

The simplest definition of an important node is one that has a lot of edges: a node's **degree centrality**. Because the specific maximum and minimum numerical results of a node's degree depend on the size of the network, the measure is often standardized (or normalized) when used as a centrality method. One approach to achieve this normalization is to divide a node's degree by the number of nodes minus 1 (i.e., the theoretical maximum degree of this node in a simple undirected network without self-loops). This results in a value between 0 and 1, where a node with a high value is considered highly central according to this particular definition of centrality. As stated above, the archaeological research context should explicitly inform how to interpret such results. For example, in a past social network a person with many contacts is central because they can spread information directly to many people. A node in an archaeological similarity network represents a site that is central because it shares evidence of artifact types with many other sites. However, in directed networks we can distinguish between indegree and outdegree as versions of degree centrality (see Section 4.4.1). For example, in a visibility network a node with high indegree represents a site that is highly visible from other sites and can be considered central as a visually prominent site in this cultural landscape. On the other hand, a node that has a high outdegree in this same visibility network

represents a site that is central because it can visually control many surrounding sites.

In such directed networks a further interesting elaboration of degree centrality can be made. We could identify, on the one hand, those nodes that are central because they point not only toward many other nodes but mainly toward many nodes *with a high number of incoming edges*. Such nodes are called **hubs** and are characterized by a high indegree with links to many authorities. On the other hand, we could identify those nodes that are central because they receive relationships not only from many other nodes but mainly from nodes *with many outgoing edges*. Such nodes are called **authorities** and are characterized by a high outdegree with links from many hubs (Fig. 4.12).

These node types are named hubs and authorities because they are usually applied to explore key actors in the flow of influence and information, typically in information networks such as the World Wide Web or citation networks. In an archaeological citation network a typical hub is a publication that cites a lot of other archaeological publications, including many important foundational publications. Typical hubs in citation networks are review publications (such as Trigger's 1989 *A History of Archaeological Thought*) and textbooks: all books in the Cambridge Manuals in Archaeology series are hubs in an archaeological citation network. A typical authority in such a network would be a publication that is cited very often but especially by review papers and textbooks, such as publications credited for paradigm shifts and the first introduction of new methods, theories, or data. Examples of authorities from the archaeological citation network include Arnold and Libby's (1949) paper on radiocarbon dating; *New Perspectives in Archaeology*, edited by Sally and Lewis Binford (1968); and Ian Hodder's (1986) *Reading the Past*.

A node's *degree centrality* is the number of edges incident to this node (i.e., a node's degree).

A node's *standardized degree centrality* is the number of edges incident to this node divided by the number of nodes in the network minus 1.

In a directed network, *hubs* are nodes with a high outdegree that have outgoing edges to many authorities.

In a directed network, *authorities* are nodes with a high indegree that have incoming edges from many hubs.

There is a range of other more complex measures that define a node's importance relative to the importance of other nodes it is connected to, and here we will limit

ourselves to briefly introducing two of the most commonly used methods. The **eigenvector centrality** measure was first introduced by Bonacich (1972) and can be calculated in three ways (described in full technical detail in Koschützki et al. 2005:48–50). Each of the methods to calculate eigenvector centrality follows the concept that high-centrality-scoring neighbors contribute more to a node's eigenvector centrality than low-centrality-scoring neighbors. Note how this differs from the case of degree centrality described above, where each neighbor contributes an equal amount to a node's degree centrality score. This results in a node's eigenvector centrality being proportional to the sum of the eigenvector centralities of the nodes it is directly connected to. A node's eigenvector centrality score will be high if the node has many neighbors, if it has neighbors with high eigenvector centrality, or both. See the case study in Section 4.4.8 for an example application and interpretation of this method to an archaeological case.

A measure that implements the importance of a node's neighbors in a node-ranking method for *directed* networks is **PageRank**. It follows the concept that a node's centrality score is proportional to the sum of the node's direct neighbors divided by their outdegree (Fig. 4.12). The measure recognizes that the directionality and the number of outgoing edges are crucial ingredients in determining a node's centrality in particular contexts, such as ranking pages on the World Wide Web. Indeed, PageRank is the brainchild of Brin and Page (1998), who later founded Google and included PageRank as an important ingredient in the first Google search algorithm that ranks web pages based on a text search. According to PageRank, for any webpage to simply be linked to other important websites such as Google is not sufficient to call it an important website in its own right, since Google links to a huge number of websites. Rather, the influence of a particular page on PageRank score is divided by the total number of websites that page links to. According to this logic, a website that is linked to by many other websites that have relatively few links will get a higher PageRank centrality score than a website that is only linked to by websites that link to a large number. In archaeological network research, PageRank can be appropriately applied to other directed information networks such as visual signaling networks and citation networks, as well as to past social networks.

The final two centrality measures we will introduce are *path-based* methods, where a node or edge is considered central if it occupies a crucial position to control the flows of resources throughout the entire network: closeness and betweenness centrality. Although we here present the application of these centrality concepts to rank nodes, they can equally be applied to rank edges based on their importance in mediating resources. Remember that calculating edge betweenness centrality was a

crucial step in the Girvan and Newman (2002) community detection method (see Section 4.4.6).

The **closeness centrality** of a node is the mean distance from this node to all other nodes. By calculating this measure for all nodes in the network a ranking is obtained to determine which nodes are the closest in network terms to all other nodes. Being close to all other nodes has important implications for a node's ability to collect and spread resources. In a past social network, for example, a person with high closeness centrality will be able to spread information efficiently to a large number of others, and will equally be well positioned to obtain new information from all others. Alternatively, in a past road network a settlement with a low closeness centrality is far away from all other settlements in network terms, which means a large number of road sections in the road network will need to be traversed in order to reach this settlement.

> The *closeness centrality* of a node is the number of other nodes divided by the sum of the shortest paths of that node to all other nodes.

The **betweenness centrality** reveals another positional role for controlling flows in the network: it measures not how close a node is but rather how important it is as a go-between in the network. It is determined by calculating the shortest paths from all nodes to all nodes, counting the number of these paths that pass through each node, and then for each node dividing this sum by the total number of shortest paths. Nodes with a high betweenness centrality are those that are frequently passed by paths through the network, whereas nodes with a low score are rarely passed. This has implications for the ability of this node to control the flow of resources in the entire network. In a past social network, for example, a person with a high betweenness centrality will be able to mediate information between different parts of the network; they can take advantage of their position by deciding whether or not to pass on information. Alternatively, in a past road network a settlement with a high betweenness centrality will be well positioned to levy a tax on goods passing through it, since the settlement is a necessary intermediary for much of the flow of resources over this road network.

> A node's *betweenness centrality* is the fraction of the number of shortest paths passing through this node over the number of shortest paths between all pairs of nodes in the network.

The road network examples of closeness and betweenness centrality might come across as rather counterintuitive. Surely the number of road segments to pass is not particularly informative to understand a settlement's ability to obtain information or levy taxes? Is closeness in this context not better expressed in terms of short physical distances? In such spatial archaeological networks where physical distance matters a lot for enabling the hypothesized flow of resources, it is often more interesting to use *weighted versions of closeness and betweenness centrality* (see Peeples and Roberts 2013). In these weighted versions we take the spatial distance of each road segment into account, by considering the shortest path to be the path that has the lowest summed spatial distance (or its inverse as in Fig. 4.11) of its road segments (rather than merely counting the number of road segments). These weighted measures still capture the structuring effect of the spatial network on the flow of resources, while calibrating it with the differential lengths of each edge (for more on spatial networks, see Chapter 6).

4.4.8 *Case Study: Comparing Road Networks with Exploratory Metrics*

In this final case study we illustrate how exploratory network analysis techniques can be used to compare different network definitions and the different results produced by various centrality measures. In the introduction of the Roman roads of the Iberian peninsula case study (see Chapter 2), we argued that different methods could be used to incorporate isolated settlements into the basic network of major Roman roads (Fig. 4.13a). In one network we connected such isolated settlements to all other settlements within 50 kilometers (Fig. 4.13b), and in another network we connected each isolated settlement to one other settlement closest to it (Fig. 4.13c). What are the implications of making such network design decisions on the resulting networks' structures? Does this decision affect the identification of the most central settlements in the network?

Before comparing node-based centrality measures, we will first compare the general network-wide structure of the three networks. We use a combination of degree-, path- and subgroup-based exploratory network analysis measures to get an idea of the effects of our network creation decisions on a diversity of network structural elements. The nearest neighbor approach (Fig. 4.13c) incorporates all isolated settlements in the network, but it also decreases local clustering and increases the diameter and average distance between settlements. This means that, as compared to the basic network, information/goods/people have to traverse more roads on average to disseminate to all settlements, and the resources can

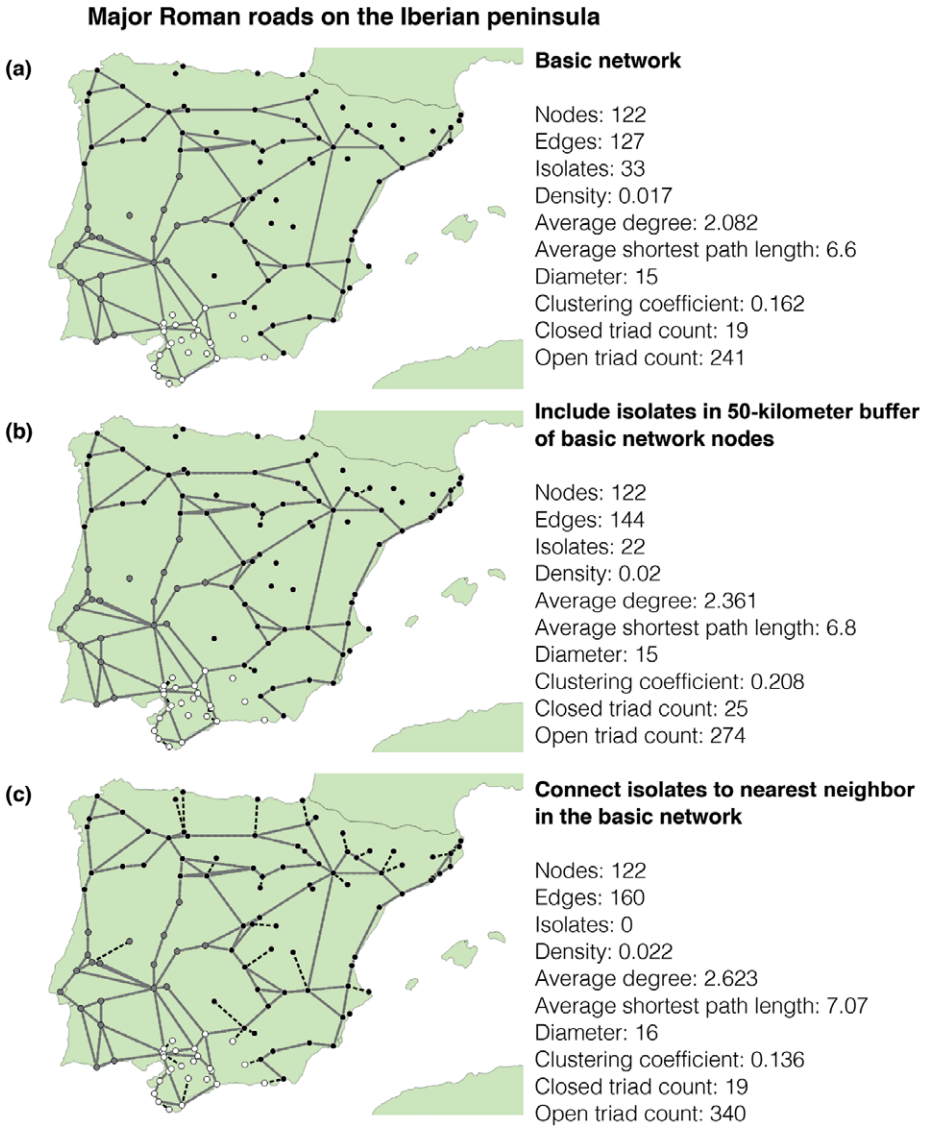


FIG. 4.13 Network representation of the major Roman roads in the Iberian peninsula. In this case study we explore the impact of including the isolated nodes in (a) the basic Roman major road network (b) by connecting isolates within a 50-kilometer buffer or (c) by connecting them to the nearest neighbor in the basic network. The Roman Imperial provinces in which the settlements were located are represented as the node color: black = Tarraconensis, gray = Lusitania, white = Baetica.

disseminate less effectively through direct road connections in clusters. In contrast, incorporating isolated settlements within a 50-kilometer buffer does not affect the average distance between settlements much but increases the local clustering through the creation of more closed triads (Fig. 4.13b). This means that, on average, resources can disseminate more effectively through direct road connections in clusters. However, in this network 22 settlements remain isolated and it is arguably not an appropriate representation of the concept that all settlements are connected by some road or other, no matter how minor.

For evaluating the impact of our network design decisions on the centrality of settlements, we first need to identify the most central settlements in the basic network (Fig. 4.13a) before we can identify how these change when we modify the network structure by incorporating some of the isolates. We will here only show the three top ranked settlements for each centrality measure, mainly to illustrate the differences in the measures' results through a simple and clear example (but see Chapter 5 for statistical methods to compare the robustness of exploratory network analysis results in light of sampling or design decisions). To highlight how the "most central" node can differ depending on the archaeologist's definition of "central," we will use four different centrality measures: degree, eigenvector, closeness, and betweenness. For the road network being studied here we can interpret these measures as follows. Degree centrality results are interpreted for this road network as a settlement's importance in terms of the number of other settlements it is directly connected to. Settlements with high eigenvector centrality are interpreted as being connected to other settlements that are themselves very central. A settlement with a high betweenness centrality is one that has to be frequently passed in all traffic through this road network. Finally, closeness centrality represents a settlement's importance with reference to how well its inhabitants can reach all other settlements over paths in the network.

Crucially, the results in Table 4.1 show that no settlement is in the top three of all centrality measures, reinforcing the need to define "importance" in light of archaeological theory and to select that centrality measure that best expresses this.

The results further suggest that in the basic network representation of the major roads on the Iberian peninsula, Emerita Augusta and Caesaraugusta (present-day Mérida and Zaragoza, respectively) are central according to three out of four measures. They both occupy very strong positions in the network for sharing information directly with many neighboring settlements and as go-betweens in all road-based traffic through the peninsula. In addition, Emerita Augusta is connected to many other particularly central settlements (eigenvector centrality), while Caesaraugusta is close to all other settlements in network terms (closeness).

Table 4.1 *The top three scoring settlements according to degree, closeness, betweenness, and eigenvector centrality (a) for the basic Roman road network (Fig. 4.13a) and the inclusion of isolates in this network, (b) within a 50-kilometer buffer, and (c) by connecting them to the nearest neighbor in the basic network*[a]

(a) Basic network				
Rank	**Degree**	**Closeness**	**Betweenness**	**Eigenvector**
1	Emerita Augusta (L)	Tiltucia (T)	Caesaraugusta (T)	Emerita Augusta (L)
2	Hispalis (B)	Caesarobriga (L)	Emerita Augusta (L)	Tubucci (L)
3	Caesaraugusta (T)	Caesaraugusta (T)	Saltigi (T)	Scallabis (L)
(b) Basic network + 50-kilometer buffer				
Rank	**Degree**	**Closeness**	**Betweenness**	**Eigenvector**
1	Emerita Augusta (L)	Tiltucia (T)	Caesaraugusta (T)	Emerita Augusta (L)
2	Hispalis (B)	Caesarobriga (L)	Emerita Augusta (L)	Hispalis (B)
3	Caesaraugusta (T)	Caesaraugusta (T)	Saltigi (T)	Tubucci (L)
(c) Basic network + nearest neighbor				
Rank	**Degree**	**Closeness**	**Betweenness**	**Eigenvector**
1	Emerita Augusta (L)	Caesaraugusta (T)	Caesaraugusta (T)	Emerita Augusta (L)
2	Hispalis (B)	Saltigi (T)	Saltigi (T)	Tubucci (L)
3	Caesaraugusta (T)	Tiltucia (T)	Emerita Augusta (L)	Scallabis (L)

[a] The Roman Imperial provinces in which the settlements were located are included in parentheses: B = Baetica, L = Lusitania, T = Tarraconensis.

It is interesting to note that each Roman imperial province has one node in the top three degree centrality ranking: settlements with a particularly high number of roads leading directly to other settlements. All three were very important Roman cities within their provinces (they all held the prestigious urban title of Roman colonies), but only Emerita Augusta was a provincial capital. Perhaps this reflects a provincial logic in the structuring of the Roman road network of the Iberian peninsula, where one or few economically or politically important centers acted as focal points in the province with roads radiating out to all corners of the province.

This provincial division is not visible in the top three of other measures, in which the southern province of Baetica is not represented. Arguably this road network

needs to be complemented by riverine and maritime transport routes to understand the integration of this province rich in natural resources (e.g., Carreras and De Soto 2013; Isaksen 2013). The top three central settlements according to eigenvector centrality are located in the province of Lusitania. This is caused by the more clustered nature of the roads there and the measure's definition of highly central places being directly connected to other highly central places.

Comparing the most central settlements in the basic network with those in our two modified networks reveals that these network design decisions have little impact on identifying the top three most central settlements in the Roman road network of the peninsula (but see Chapter 5 for more advanced sensitivity analyses). Including isolated settlements within a 50-kilometer buffer results in the same top ranking for all measures except for eigenvector centrality, which now includes the central Baetican city of Hispalis. Connecting isolated settlements to their nearest neighbor results in top three rankings almost exactly the same as those of the basic network. According to the closeness measures, the Lusitanian settlement of Caesarobriga is replaced by the town of Saltigi in Tarraconensis, but we do not attach much importance to this change because the closeness scores of many settlements in the geographical center of the Iberian peninsula are highly similar and Caesarobriga is still ranked fifth.

4.5 DON'T STOP HERE!

The methods described above are diverse and capture many of the most common tools in the network science toolbox, but in reality, we have barely scratched the surface. There are many more exploratory techniques, concepts, and theories in the broader network science literature beyond archaeology that are quite appropriate for addressing specific archaeological questions. Some of the most useful methods fall outside the typical "off the shelf" techniques commonly discussed in textbooks or implemented in network analytical software. We hope readers of this book will feel confident searching the vast network science literature beyond archaeology to find new appropriate techniques, and potentially to develop techniques that specifically cater to archaeological data. To get you thinking, we outline just a couple of recent examples of such efforts, but we note that we are heartened that such approaches are becoming increasingly common.

Sometimes addressing a specific archaeological question hinges on *finding a very specific appropriate method*. In one study focused on archaeological networks defined based on named references on Maya stelae (stone monuments), Munson

and Macri (2009) were interested in comparing the degree centralization of their networks (a measure of the overall centrality of a network) to track how this network property changed through time. In this case, centralization in network terms was seen as proxy evidence for political centralization. The only problem is that typical centralization measures are quite sensitive to the size (number of nodes) and density of the networks being compared. In order to avoid such issues, Munson and Macri applied a relatively uncommon (but quite useful) renormalized centralization index (rCI) that normalizes centralization scores based on the density and maximum degree of each network. In this way, researchers are afforded the ability to make direct comparisons between networks (between time periods in this case) that are meaningful in absolute terms. Munson and Macri make great use of this technique and are able to demonstrate changes in centralization through time that they relate to the changing political landscape of interactions among Maya elites. This study provides an excellent example of the application of a very specific but somewhat rare network metric appropriate for addressing a very specific archaeological question.

Beyond this, it is also sometimes profitable to *take an existing network metric and modify it for archaeological data*. For example, Peeples and Haas (2013) were interested in exploring the role of a network position called "brokerage" in the long-term trajectories of sites and regions in the late Pre-Hispanic US Southwest. Brokerage is an intermediate network position defined as a node connected to two other nodes that are not themselves connected (e.g., B is a broker if A is connected to B and B is connected to C but A is not connected to C). This is a quite popular measure in social network analysis, based on binary directed ties among nodes (Gould and Fernandez 1989). In this case, however, the data that Peeples and Haas had to evaluate for patterns of interaction and brokerage were similarities in the frequencies of ceramic wares used as proxy for shared patterns of influence and exchange (see Chapter 2). These archaeological network data were both undirected and weighted (rather than directed and binary), so common measures of network brokerage would not work. Thus, in this study, Peeples and Haas developed and implemented a new weighted measure of brokerage that had similar properties to the common social network analysis technique, but which was appropriate for the specific properties of the archaeological data at hand. Using this measure, Peeples and Haas were able to show that sites characterized by a high degree of brokerage tended to be small, short-lived, and in isolated portions of the overall study area, expectations in line with one of two network theories they were evaluating.

4.6 SUMMARY

- Exploratory network analysis is the use of statistical or visual analytical tools to explore the structure of networks by identifying and summarizing network patterning.
- A huge number of exploratory network analysis techniques exist; this chapter introduced some of the most commonly applied ones. For more exhaustive reference works, see the Further Reading.
- Method selection should happen in light of research aims, questions, and archaeological theories. Ask yourself: Am I studying properties of the network, individual nodes, paths, and/or communities? In addition, always perform an initial exploration of a network dataset using a range of different exploratory techniques.
- The interpretation of exploratory network analysis results in archaeological network research will be entirely dependent on the archaeological research context and theoretical framework.
- Methods based on the concept of node degree measure how nodes are directly related to other nodes.
- Methods based on subnetworks identify and measure properties of sets of nodes and edges that are a subset of the network.
- Path-based methods are used to represent or explore the flow of resources over the edges of a network.
- A network is disconnected if not all node pairs are reachable.
- A range of methods exists to identify cohesive subgroups, based on the direct connections between all members of the subgroup, the reachability of members of the subgroup, or the denseness of the subgroup relative to other parts of the network.
- Centrality is a family of measures used to identify the most important or prominent nodes or edges in the network, depending on the different definitions of importance or prominence implemented in the network measure used.
- What "centrality," "importance," or "prominence" means has to be defined explicitly in light of your archaeological theoretical framework before selecting and applying a particular centrality measure.
- Beyond these commonly used methods, browse the network science literature to find specific, less-commonly used methods that are most appropriate to address your archaeological research questions. And if the method you need does not exist, then don't be shy to modify existing methods or develop new ones.

Further Reading

The following references have more exploratory network analysis methods than covered here.

Brandes, Ulrik, and Thomas Erlebach 2005 *Network Analysis: Methodological Foundations.* Springer, Berlin.

Carrington, Peter J., John Scott, Stanley Wasserman, and Katherine Faust 2005. *Models and Methods in Social Network Analysis.* Cambridge University Press, Cambridge.

de Nooy, Wouter, Andrej Mrvar, and Vladimir Batagelj 2018 *Exploratory Social Network Analysis with Pajek: Revised and Expanded Edition for Updated Software.* 3rd ed. Cambridge University Press, New York.

Estrada, Ernesto 2012 *The Structure of Complex Networks: Theory and Applications.* Oxford University Press, Oxford.

Newman, Mark 2010 *Networks: An Introduction.* Oxford University Press, Oxford.

Wasserman, Stanley, and Katherine Faust 1994 *Social Network Analysis: Methods and Applications.* Cambridge University Press, Cambridge.

The above reference works have extensive sections on all topics covered in this chapter. In addition, the following subject-specific review publications can be consulted for centrality, community detection, small-world, and scale-free network structures.

Albert, Réka, and Albert-László Barabási 2002 Statistical Mechanics of Complex Networks. *Reviews of Modern Physics* 74:47–97.

Borgatti, Stephen. P. 2005 Centrality and Network Flows. *Social Networks* 27(1):55–71.

Fortunato, Santo 2010 Community Detection in Graphs. *Physics Reports* 486:75–174.

Frank, Kenneth A. 1995 Identifying Cohesive Subgroups. *Social Networks* 17:27–56.

Freeman, Linton C. 1979 Centrality in Social Networks. I. Conceptual Clarification. *Social Networks* 1:215–239.

EXERCISES

Answers for these exercises can be found in Appendix A.

4.1) You want to identify the most appropriate exploratory network analytical method for your study of the past Inka road network. You theorize goods, people, and ideas can flow over the edges of the network, and you aim to identify the most important intermediaries in this flow of resources. Which one of the following methods is most appropriate for doing so?

a) Eigenvector centrality
b) Clustering coefficient
c) Betweenness centrality
d) Cliques

4.2) You want to identify the most appropriate exploratory network analytical method for your study of a network representing artifact similarities between Iron Age sites in Spain. You theorize that the degree of artifact similarity between a pair of sites is proxy evidence for the frequency of interaction of their past peoples. You theorize that high frequency of interaction led to culturally similar past communities, represented as high similarity values between sites. What two methods are most appropriate for identifying communities in the network with particularly high artifact similarity?
   a) Weighted Louvain modularity
   b) *k*-cores
   c) Average shortest path length
   d) *m*-slices

The following exercises refer to an example network of Caribbean islands, a subnetwork taken from a network created by Angus Mol (2014, fig. 6.1), where a pair of islands is connected if they are less than 45 kilometers distant from each other.

4.3) What is the degree of St. Kitts in Figure 4.14?
4.4) What is the average degree of the network in Figure 4.14?
4.5) Nodes represent islands in Figure 4.14 and edges represent islands located closer than 45 kilometers from each other. The archaeological context of this figure theorizes that islands that are located closer to each other might have had past communities that interacted more frequently than those on islands further away. Interpret the node degree of St. Kitts and how it compares to the network's average degree in terms of the opportunities it provided to its past communities.
4.6) What is the density of the network in Figure 4.14?
4.7) How many closed triads are there in the network in Figure 4.14?
4.8) List the nodes in the shortest path from Saba to Nevis in Figure 4.14.
4.9) What is the diameter of the network in Figure 4.14?
4.10) Is the network in Figure 4.14 connected or disconnected?
4.11) How many components are there in the network in Figure 4.14?
4.12) Are Marie-Galante and Barbuda reachable?
4.13) Are St. Eustatius and Nevis reachable?
4.14) The archaeological context of the network in Figure 4.14 theorizes that pairs of islands that have a low network distance have more frequent exchange of goods and ideas between their past communities than pairs of islands that are separated by a larger network distance. Interpret the diameter and connected/disconnected nature of the network in Figure 4.14 in light of this archaeological theory.

4.15) Name all cutpoints in the network in Figure 4.14.
4.16) Identify the number of cliques in the network in Figure 4.14 and list the nodes that are part of each clique.
4.17) What islands are *not* part of a 2-clique in the network in Figure 4.14?
4.18) Give the names of the islands included in the 2-core of the network in Figure 4.14.
4.19) Which node(s) in Figure 4.14 has the highest degree centrality?
4.20) What is the standardized degree centrality of Antigua in Figure 4.14?
4.21) Which node in the following component of the network in Figure 4.14 has the highest betweenness centrality score? Saba, St. Eustatius, St. Kitts, Nevis.
4.22) The archaeological context of the network in Figure 4.14 theorizes that an important island is one whose past communities can exchange goods and ideas directly with past communities on as many other islands as possible. What is the most appropriate centrality method to represent this theory?

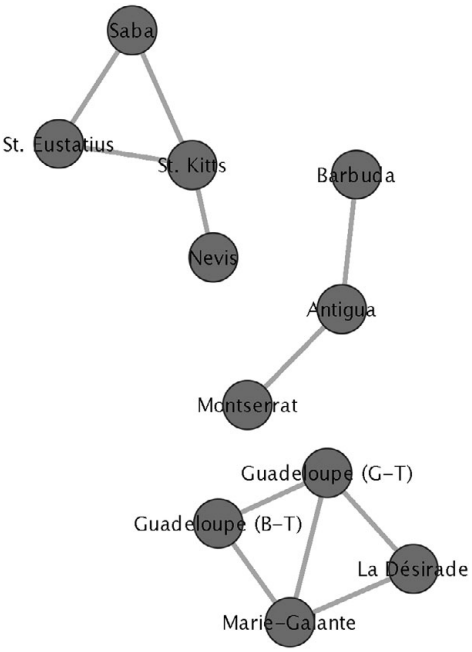


FIG. 4.14 Maximum distance network (see Chapter 6) of Caribbean islands, illustrating a subnetwork taken from a network created by Angus Mol (2014, Fig. 6.1).

5

Quantifying Uncertainty in Archaeological Networks

Are my data good enough to create an archaeological network? What if I am missing some sites or contexts, or I have poor or variable quality information for some observations? Can I still apply network methods and models with incomplete and/or imperfect data, or should I not even attempt to use network methods? At this point, some of you may be asking yourselves questions along these lines. It is good to carefully consider potential data issues when conducting any archaeological analysis, but there are also some specific concerns revolving around sampling and data quality that deserve special attention when dealing with network data. In this chapter, we outline some of the most common issues you will encounter and further offer a generalized approach to identifying and assessing the potential impacts of sources of variation and uncertainty in network data through simulation and resampling.

5.1 Error and Uncertainty in Networks and Beyond

Error is an inevitable component of data collection and analysis, not just in archaeology or network research, but in any field of study. **Error** specifically refers to the difference between the observed/recorded value and the "true value" of interest. However, under most circumstances we have no way of knowing what the "true value" is. Thus, researchers often rely on the concept of **uncertainty**, referring to informed characterizations of variability in a measured or estimated value based on potential sources of error. In other words, uncertainty is a way of formally assessing the degree to which a measurement or estimate reliably represents the likely true value given the known or likely sources of error. For quantitative measurements, this may take the form of a number with a "±" indicating that

the true value likely lies in a range centered on the measured value. An example that archaeologists will be intimately familiar with is a radiocarbon date and associated date range (e.g., 786 ± 27 BP). In other cases, uncertainty cannot be directly quantified in simple terms but instead may arise from vagaries of the data gathering or the analytical process. For example, in faunal analyses individual elements that are small or fragmentary are sometimes placed in categories such as "probable deer" or "possible ungulate." How such tentative identifications are treated in subsequent analyses is a decision with potentially huge substantive impacts. In either case, our goal in any data analysis should be the recognition and management of various sources of uncertainty and error so that inferences drawn from analyses provide robust and veracious characterizations of the event, object, or phenomena in question.

Uncertainty in scientific investigations refers to assessments of the variability of measured or estimated analytical outputs based on the sources of error and variability in analytical inputs.

Error refers to the difference between a measured or estimated value and the likely true value.

Data used to generate network representations are subject to many potential sources of error, driven by problematic observations, incomplete samples, variable quality information, or even simple typos and other human errors. This is true even when networks can be nearly directly observed, such as is the case for email-to-email connections generated and obtained directly through digital databases. Such uncertainty and variability have been shown to sometimes have substantial impacts on the kinds of network structures and positions we wish to track (Chapter 4; e.g., Costenbader and Valente 2003; Frantz et al. 2009; Kossinets 2006; Smith and Moody 2013; Smith et al. 2017, 2022). Thus, one potentially big problem with applying many standard "off-the-shelf" network analytical techniques uncritically is that most common metrics (like assessments of clustering or centrality; see Chapter 4) provide only a single point estimate of the position or characteristic of interest with no way to directly assess potential variability around that estimate. Beyond this, the construction of network representations as abstractions generated from archaeological data often involves assumptions that can have major impacts on network structure and form that too often go unexamined.

In this chapter, we provide a general framework for formal uncertainty assessment and the evaluation of potential sources of error or variability for

archaeological network data through a set of related simulation and resampling methods but note that many of the methods and models we advocate here are generalizable to other archaeological problems and even quantification issues beyond archaeology (see also Peeples 2017; Peeples et al. 2016). The approach we outline here should be seen not as a strict set of rules and tests but instead as a set of general heuristics that can guide you in thinking about the robustness of your network analytical results and how to approach uncertainty and sensitivity analysis in your own work. In the remainder of this section we describe some of the potential sources of error in archaeological network data (and scientific data generally), as well as kinds of uncertainty or variability that can arise due to analytical decisions.

Throughout most chapters in this book, we describe network analytical methods and techniques that can be implemented, perhaps with some slight tweaks or modifications, in a range of common software applications. The methods we describe here are, for the most part, custom tools that are not already included in any of the common platforms. Indeed, many of these approaches require some degree of coding and at least basic programming knowledge. For this chapter, perhaps more than any other in the book, the online companion to the book will provide an essential tool for taking such methods further in your own work. In the online companion we have provided code and examples in the R programming language (R Core Team 2021) to replicate all of the analyses presented here as well as general code and tips to help you implement such approaches using your own data and questions. If you are interested in implementing these methods, we suggest you follow along with those examples as you read this chapter.

The next section of this chapter describes common forms of missing or poor quality data that you may encounter in archaeological network research. With this information as background we then describe our general approach to dealing with missing or poor quality information through simulation and resampling, describing specific approaches for missing nodes/edges, and other potential sources of poor or variable quality data. Finally, we conclude with some recommendations and strategies for coping with uncertainty in archaeological network studies in general.

5.1.1 *Sources of Error*

Sources of error in archaeological network analyses (and data analyses generally) can be divided into two major categories: (1) systematic error and (2) statistical

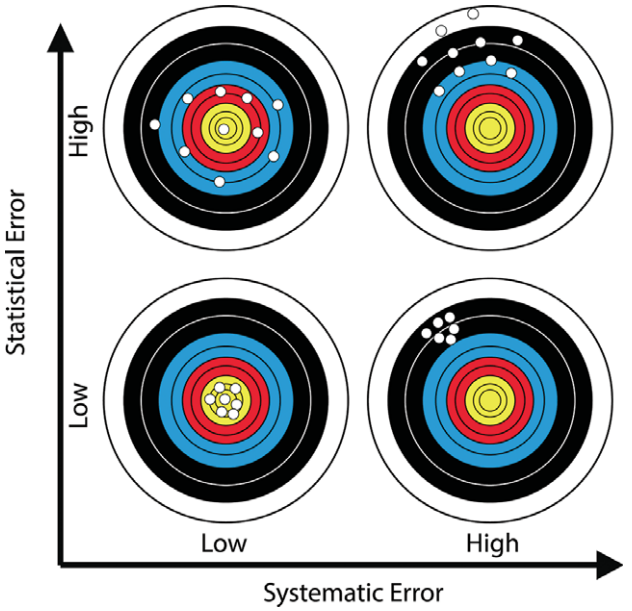


FIG. 5.1 Schematic figure showing the relationship among statistical error, systematic error, and the target "true" value (bullseye). Lower statistical error results in greater precision and lower systematic error results in greater accuracy. Ideally we want measurements to fall in the lower left quadrant.

error (see Fig. 5.1). **Systematic error** refers to deviations from the target value based on flawed or inconsistent methods, instrumentation, or assumptions. Systematic errors lead to lower **accuracy** of estimates of target values. **Statistical error**, sometimes simply glossed as random error, refers to random and unpredictable variation around the target based on the imprecision of measurements or the relative representativeness of samples. Statistical errors generally lead to a lack of **precision** in estimates of target values.

Systematic error: Deviations from the true value based on inconsistent or flawed methods, instrumentation, or assumptions. Systematic errors often reduce the accuracy of estimates of target values.

Statistical error: Random variability around true value based on variation in measurement precision and the representativeness of samples. Statistical error can reduce the precision of estimates of target values.

Accuracy: Accuracy can generally be defined as the distance of a measurement or set of measurements from the true value.

Precision: Precision is the variability of repeated measurements or, in other words, how close repeated measurements of the same thing are to one another.

Accuracy and precision can vary independently, but ideally we want measurements that are both accurate (indicative of the true value) and precise (with little variation).

There are many potential sources of systematic error in archaeological research that we should attempt to identify and guard against where possible. For example, if you were to weigh a collection of artifacts with a scale that measures 0.5 grams when nothing is on the platform, there would be a systematic error across all measurements taken on this instrument, and these measurements would not be comparable to measurements from another instrument without compensation. Systematic errors can also creep into data due to differences in methods or assumptions among researchers. For example, if you were to combine counts of ceramics classified to type from two researchers, and those researchers have different assumptions about the characteristics that define certain types, the data generated may not be directly comparable and variability in any analytical outputs would be impacted by this lack of comparability. As Lyman and Van Pool (2009) illustrate, this can even be true for something as seemingly straightforward as linear measurements taken by different researchers on artifacts. Systematic errors are typically biased in a particular direction. For example, a given instrument may always overestimate a true value or one researcher may consistently identify ceramics to finer typological categories than another. Thus, it may be possible in some cases to directly compensate for such errors, or at least make informed estimates of their potential effects on any substantive interpretations of those data.

Statistical error is an extremely important issue for nearly all aspects of archaeological analysis, though it has not always received the attention it deserves. Of particular concern for archaeological research and networks generally is statistical error in the form of **sampling error**, or the difference between the statistical properties of a sample and the parameters of the population from which that sample is drawn. When a sample, due to size or quality, is not an adequate representation of the target population, sampling error will cause problems for inferences made about that population based on the sample. For many reasons, network data are prone to missing information and small samples, so this can be a particularly important problem. The best solution for dealing with sampling issues is typically increasing the sample size, but where that is not possible or feasible, there are statistical tools for using repeated resampling and related approaches to

provide better assessments of the properties of a network generated from an incomplete sample of nodes and edges or for assessing variability in the underlying archaeological data used to generate network representations (see Section 5.3).

> *Sampling error* is the variability of an estimate of a parameter that is driven by the size and representativeness of the sample from the target population.

5.1.2 *Evaluating Analytical Decisions to Draw Reasonable Inferences*

Network properties and our ability to characterize them are fundamentally influenced by the decisions we make regarding how to generate network representations from archaeological data. As described in greater detail in Chapter 3, creating a network representation involves a number of assumptions, linking arguments and abstractions connecting archaeological data to network processes and network representations. These assumptions should not go unexamined, and we suggest that it is often useful to directly assess the sensitivity of inferences drawn from archaeological network data to such initial assumptions and analytical decisions.

For example, Peeples and colleagues (2016) describe an analysis where archaeological networks are generated based on similarities in the frequencies of certain ceramic wares in the US Southwest using the Southwest Social Networks database (see Section 2.8.2). Importantly, however, there are currently disagreements in the literature on the appropriate date ranges for some types of ceramics that are used to define edges for these network graphs. Specifically, they note that the production of Tanque Verde Red-on-Brown pottery is variously assumed to end by either AD 1350 or AD 1450 by different researchers. Thus, a network built covering the intervening period between AD 1350 and 1450 will differ based on which end date is selected. As Peeples and colleagues suggest, unless a strong argument can be made for one end date or another, any substantive behavioral interpretations of network properties that rely on a particular end date should be presented as preliminary and revisable in light of potential new data. In some cases, presenting more than one alternative and discussing their relative merits may be the best way forward.

Knowing how to properly assess the robustness of the network metrics and properties we rely on to make such inferences is essential, but this is often a difficult task. Indeed, the nature of network structure and size and other

characteristics vary in ways that suggest that different networks are likely to be more or less robust to different kinds of error. As the brief example here suggests, it is very important to always consider what kinds of inferences can reasonably be drawn from a given network (and what kinds cannot) and to limit substantive interpretations of network effects, structures, and dynamics to those that are supportable given the nature of the sample and the properties of the analytical methods to be used. In other words, it is important that you ensure you are not making an argument based on a result that is highly sensitive to an arbitrary or questionable analytical decision or input. The goal of the approaches outlined in this chapter is to help you learn to make informed decisions about what kinds of statements about network properties and positions can be reasonably supported by a given network given the vagaries of data gathering, network representation, and network properties.

5.2 MISSING OR POOR QUALITY INFORMATION IN ARCHAEOLOGICAL NETWORKS

One of the most important features of networks as an analytical framework is the interdependence of observations. Indeed, this is often seen as one of the foundational tenets of network thinking generally (Wasserman and Faust 1994:4). The positions and properties of individual nodes and edges within network structures are always determined by the relative positions and properties of other nodes and edges (see Chapter 3). Thus, ideally to make inferences about a particular node or a particular structure or even graph-level properties, we would want to draw on a well-defined network at a scale that captures all relevant relations. In such a **total network** all nodes and edges are represented (or perhaps nearly so) and the information we use to generate the network is of a consistently high quality across all cases. Although this is a virtually impossible standard to meet at most scales, many of the most common network metrics and methods applied to such data are built around the assumption that the network analyzed represents a well-bounded and well-characterized whole (see discussions in Smith and Moody 2013; Smith et al. 2017; Laumann et al. 1992). This has long been recognized as a potential problem in graph theory and network science (Frank 1971; Granovetter 1976). But as Smith and Moody note (2013, 652), most applications of network methods still proceed assuming complete and consistent population data or at least do little to address the potential impacts of sampling and missing information.

Total networks refer to networks where all actors and all relations are included at a given scale and where data are of a consistent quality across all nodes and edges. Although obtaining a total network or even a near total network is often the goal of many analyses, this is rarely attainable in practice.

Sampled networks refer to networks where the nodes and edges are sampled from a larger population of nodes and edges.

In almost any sort of archaeological network study we could conceive, the high standard of a well-bounded total network is practically unattainable at all but very small scales (see Pailes 2014 for an interesting case study that perhaps approaches a total network). Data can be missing for a variety of reasons, from lack of excavation or survey coverage to site destruction or looting and a lack of chronological control. Archaeologists are certainly accustomed to working with samples of variable quality and designing research sampling strategies to deal with this reality, though the topic has ebbed and flowed in popularity in the literature over the decades (Banning 2021; Orton 2000). One of the problems we must face when dealing with network data, however, is that there is not yet a unified approach or body of theory that provides direct and transferable guidance on dealing with networks that are sampled in different ways or have different topological properties (see Smith and Moody 2013; Smith et al. 2017, 2022 for the first steps toward a general model for networks). Adding to this we also note that, unlike most other fields, many large-scale network studies in archaeology rely on compilations of data from diverse sources that were originally generated to address different kinds of research questions, thus adding to the complexity (e.g., Blake 2014; Coward 2010; Giomi et al. 2021; Golitko et al. 2012; Mills et al. 2013a, 2018). For this reason, sampling and missing data should be topics of great concern for archaeological network research. However, such issues have not received substantial attention in archaeology until recently (see Amati et al. 2018; Gjesfjeld 2015; Lulewicz 2019; Mills et al. 2013a; Peeples 2017; Peeples and Roberts 2017; Peeples et al. 2016).

The processes influencing the exclusion of missing information from a given dataset can typically be described as either a **random sampling process** or a **biased sampling process** (Fig. 5.2). A random sampling process is one in which our observed nodes or edges all have an equal chance of being included in the network. A biased sampling process is one in which some nodes or edges from the target population are more likely to be included in or excluded from the sample than others based on some external consideration. Biased sampling processes are of particular concern when the process driving the inclusion or exclusion of samples is somehow

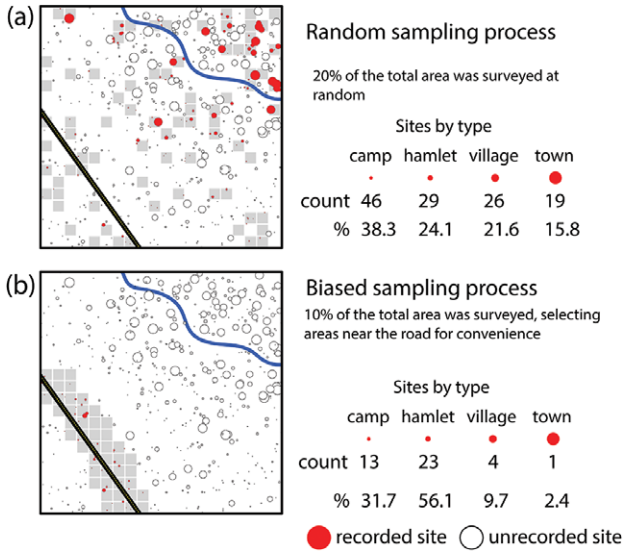


FIG. 5.2 A brief illustration of one example of the distinction between (a) a random sampling process and (b) a biased sampling process and the potential impacts on estimates of population features and distributions. Surveyed areas in each example are shown in light gray.

correlated with or influences the property or metric of interest. Of course, almost any archaeological sample we can imagine exhibits some degree of bias in its construction. For example, we are more likely to recognize and identify large settlements on surveys, whereas smaller hamlets or individual houses may not be as easily identified. When attempting to control for missing information it is essential that we recognize the difference between random and biased processes driving inclusion/exclusion as these processes require somewhat different approaches.

Figure 5.2 illustrates the distinction between a random and biased sampling process. We have in this example a hypothetical archaeological recording project where we survey a portion of a given study area. In this example, all possible sites we could encounter are shown as open circles and the ones we actually encounter are shown as filled circles. In Figure 5.2a, the sampled area is randomly selected and thus includes both the floodplain of the river and areas in the uplands near the road. In Figure 5.2b we are a bit lazy and decide we will choose our sample from areas near the road so that we do not have to walk as far. In this case, we exclude the floodplain and end up missing the largest settlements and the area with the greatest settlement density. Thus the biased sampling process would lead to a biased estimate of site size and the distribution of site types since settlements of various types and sizes are not evenly distributed across the landscape.

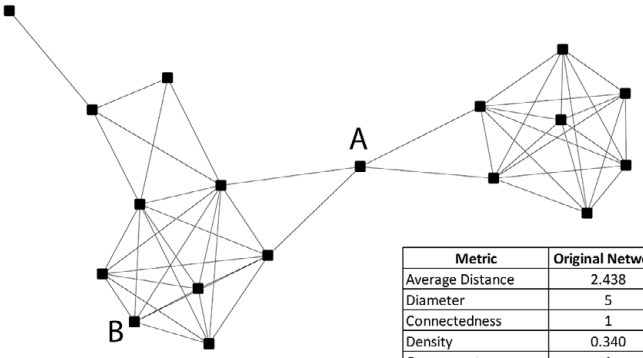
Networks sampled through *random sampling processes* are sampled such that every node and/or edge from the target population has an equal chance of ending up in the sample.

Networks sampled through *biased sampling processes* are sampled such that different nodes and/or edges from the target population are more or less likely to end up in the sample based on some additional criteria, characteristic, or external consideration.

Existing research on the topic of missing data in networks in other fields (Borgatti et al. 2006; Costenbader and Valente 2003; Kossinets 2006; Liu et al. 2017; Martin and Niemeyer 2019; Nakajima and Shudo 2021; Rosenblatt et al. 2020; Smith and Moody 2013; Smith et al. 2017, 2022; Wang et al. 2012) suggests that error in the measurement of network properties does generally increase with more missing information. The relationship is not a simple one, however, and different network properties, structural characteristics, and network types are more or less robust to different sources of missing information in ways that are not yet well understood. Perhaps not surprisingly it seems that when the network property of interest is correlated with missingness in some way our results are likely to be more problematic (Smith et al. 2017), but it is difficult to make many more specific general statements like this. Given this situation, the approach we advocate in this chapter is to directly assess the impacts of missing or poor quality information within the sample at hand in order to better understand the potential properties of the population from which it was drawn. Such an approach has already been put to good use in network studies in a variety of contexts that have similar problems with missing information, such as the study of so-called dark networks of illicit activity (e.g., De Moor et al. 2020; Gerdes 2015; Morris 2012), the networked transmission of infectious disease (e.g., Liu et al. 2017; Maeno 2010), or data generated through complex survey designs or structured interviews where filling in missing or incomplete information after the fact is not feasible (e.g., Burt 1987b; Kossinets 2006; Wang et al. 2016).

5.2.1 *Incomplete Sample of Nodes*

One of the most common problems faced in the construction of archaeological networks is missing nodes. Given the diverse ways that archaeological networks can be constructed (Chapter 3), node-level data may be missing from such datasets for a



Metric	Original Network	Network Missing A	Network Missing B
Average Distance	2.438	1.348	2.441
Diameter	5	3	5
Connectedness	1	0.485	1
Density	0.340	0.353	0.338
Components	1	2	1

FIG. 5.3 Example demonstrating the relationship between node position and the impact of missingness. As the table insert illustrates, if node A were missing, the resulting network would have two separate components and somewhat different topological properties. If node B were missing, some network statistics would change but the overall topology would be largely preserved.

wide variety of reasons. For example, in regional-scale research individual settlements may be missing due to lack of survey coverage, a lack of excavation, being covered by contemporary or more recent development, or having been damaged or destroyed. Modern land ownership and access can also present challenges. In work focused on connections generated based on material culture, we are limited to available and identifiable collections, which may vary in quality and documentation. Beyond this, where chronological information is limited, it may be difficult to determine which nodes date to the target interval of the investigation and which do not.

Missing nodes can potentially cause major problems for characterizing the structure of an archaeological network or the properties of other nodes and edges. For example, in the network shown in Figure 5.3 we have two densely connected clusters of nodes with only one node with connections that span both of those well-connected groups. In this extreme case, if we were missing the node labeled A, we would instead have two wholly disconnected components and dramatically different statistics and properties for the remaining nodes and graph as a whole. On the other hand, if we were instead missing the node labeled B we would still see changes to network properties but they would be less severe, as this node is in a position that is similar to the positions of several other nodes. Thus, missing nodes matter as does the position and properties of those missing nodes. The challenge we typically face, however, is that we have no systematic way of knowing the likely or even potential properties of missing nodes. Indeed, in many cases we may not even be able to directly estimate how many nodes are missing and where within the network structure they might fall.

5.2.2 *Incomplete Sample of Edges*

Like missing nodes, there are equally many processes that can drive missing information on network edges. For example, many spatial networks rely on the identification of features such as roads and trails that may be variably visible depending on terrain and other conditions. In studies of access and line of sight, missing doorways or missing or mischaracterized structure layouts will result in missing edges. In geochemical networks, connections are only possible beyond the geological/geographic boundaries of source materials and thus edges may be absent at certain scales due to the relative granularity of the underlying geologic variability. Where network edges are defined based on material culture, the availability and quality of the sample is also a potential issue.

When edges are missing from a network sample, either at random or due to a biased sampling process, it can be difficult to characterize and evaluate the properties of the underlying total network that is the target of our investigation. For example, Figure 5.4 represents a hypothetical network of trails between sites based on features recorded on the ground. In this simple network, edges are defined between pairs of nodes that are directly connected, and multiple measures of network centrality are shown. In order to evaluate the potential

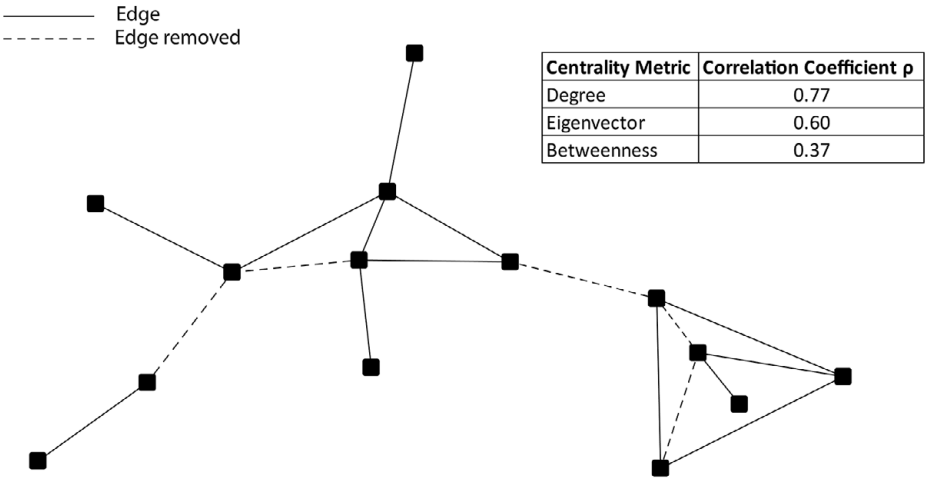


FIG. 5.4 In this example, we created a hypothetical trail network and calculated several centrality statistics and then deleted the edges shown as dashed lines and calculated those same centrality measures again. The table shows the rank-order Spearman's ρ correlation coefficient between the original network and the one with edges removed. Note that some centrality measures are impacted more than others.

Box 5.1 EXPLANATION OF SPEARMAN'S ρ

In many of the analyses in this chapter we use a measure referred to as Spearman's ρ (pronounced rho). Spearman's ρ is a *rank-order* nonparametric measure of correlation between two sets of values (we will call them X and Y here) that measures the degree to which those sets of values are characterized by a monotonic relationship. A monotonic relationship means that when variable X increases, variable Y also increases (or both decrease). Spearman's ρ values range from −1 to 1 where positive values indicate a strong monotonic relationship and negative values indicate an inverse relationship (e.g., when X increases, Y decreases; for a further introduction to the archaeological use of Spearman's ρ, see Fletcher and Lock 2005:123–127). Spearman's ρ is useful for evaluating network metrics because it only evaluates the direction of change of values and does not assume linear relationships or normally distributed values (many network metrics violate this assumption) like comparable common metrics such as Pearson's r.

impact of missing edges, we can randomly delete several of the edges in this graph (shown as dashed lines), simulating the removal of trails that are not visible or have been destroyed. As Figure 5.4 illustrates, removing edges can have a big impact on network structure and properties like centrality, but different measures of centrality are more or less impacted by this perturbation. One way to assess such relationships is to look for correlations in network metrics in networks with and without edges removed. Here we used Spearman's ρ to assess such relationships (Box 5.1). The table inset in Figure 5.4 shows the Spearman's ρ correlation coefficient between the network with all edges and the network with the dashed edge removed for three centrality metrics. Although the signs remain positive for all values of ρ, the rank order of nodes for betweenness centrality in particular seems to be heavily impacted by the removal of edges, which is perhaps not surprising as this is a measure that relies on paths across a network.

5.2.3 *Poor Quality Information*

The final major source of potential problems when creating archaeological network representations is the basic problem of data quality. Where the information used to

define network nodes and edges is not robust, our interpretations of network properties will also suffer. This is not, of course, a network-specific problem but one that we must face for any archaeological data analysis. Given the interrelatedness of observations within a network, however, such problems can be compounded. As we will discuss below, the approaches to assessing the impact of missing nodes and edges overlap in many ways with the approaches to assessing the impact of poor quality information. The primary difference lies in where the assessments and any interventions may lie within the process of creating and analyzing network graphs.

For example, let us say we have a dataset of sourced obsidian objects from a series of archaeological sites and we want to create weighted connections (edges) between those sites (nodes) based on similarities in the proportions of specific sources. In this hypothetical example, however, some of our sites unfortunately have very small sample sizes (say, only three or four obsidian objects) due to both sample availability and analytical costs. How could we be sure that our results are not simply a fluke driven by variability in very small samples? Are the small sizes indicative of an actual low frequency of obsidian or simply low sampling effort? We could eliminate sites that have fewer than a certain number of samples to base our interpretations on the cases where we have the most robust data. Of course, in that case we may be throwing out "true" zeros or low counts from our dataset where we are reasonably confident that we have high quality samples that simply do not include the obsidian objects of interest. At the same time, even if we remove the small samples that represent low sampling effort we are still left with issues caused by missing nodes as described above. Alternatively, we could attempt to assess how much variability we may expect due to variation in sample size (using a resampling procedure or other approaches described in detail below) and then ensure that our interpretations do not hinge on results that are potentially the result of sampling error. In all of these cases, our assessments of poor quality information are made prior to creating network representations of archaeological data (either by removing sites or permuting the underlying archaeological data) but impact our interpretations of those networks.

5.3 A GENERAL APPROACH TO UNCERTAINTY FOR ARCHAEOLOGICAL NETWORK DATA

Given the potential problems outlined above, the task of making meaningful interpretations of archaeological network data and metrics may seem daunting.

Despite these challenges we have found that, with careful consideration, there is usually still much that can be said about archaeological networks even with all of the uncertainty and variability that archaeological data entail. Importantly, however, there are also many other kinds of analyses that are inappropriate, and which specific analyses these are will differ from network to network. The problem we are faced with is how we can use the *sample we have* to understand the properties of the *population from which it was drawn* and limit our interpretations and analyses to those aspects that are reasonably robust to the known or probable shortcomings of our network data and methods.

In this section, we describe and advocate for a general approach to assessing network metrics and properties to the robustness of missing or poor quality information building on similar work in the broader field of social network analysis (e.g., Borgatti et al. 2006; Costenbader and Valente 2003; Kossinets 2006; Smith and Moody 2013; Smith et al. 2017; see also Peeples 2017; Peeples et al. 2016; Roberts et al. 2021). Our approach relies on **Monte Carlo** and related **bootstrapping** resampling methods to simulate and assess variability in our data by generating perturbations that approximate the problems inherent in those data or the alternative treatments of those data and to evaluate their impacts (see Rubinstein and Kroese 2011 and Kowalewski and Novack-Gottshall 2010 for detailed overviews of Monte Carlo and related methods generally). The basic idea behind such approaches is that for problems that are difficult to solve through traditional analytic and mathematical means (which includes assessments of network metric robustness), it is often useful to sample the available data or a model based on those data lots of times to get a better idea of the range of values we might expect for a given metric in the sampling distribution.

Monte Carlo methods refer to a general approach to statistical inference based on the repeated creation of random data and assessment of those data to estimate a quantity or understand characteristics of the population from which they are drawn.

Bootstrapping is a special case of Monte Carlo simulation designed to resample with replacement many times from the observed data to empirically estimate the properties of a sampling distribution.

To show how resampling methods can be used to assess uncertainty, we present a brief thought experiment here to demonstrate the basic tenets of the approach (Fig. 5.5). Say you have a large cloth bag full of an unknown number of red and

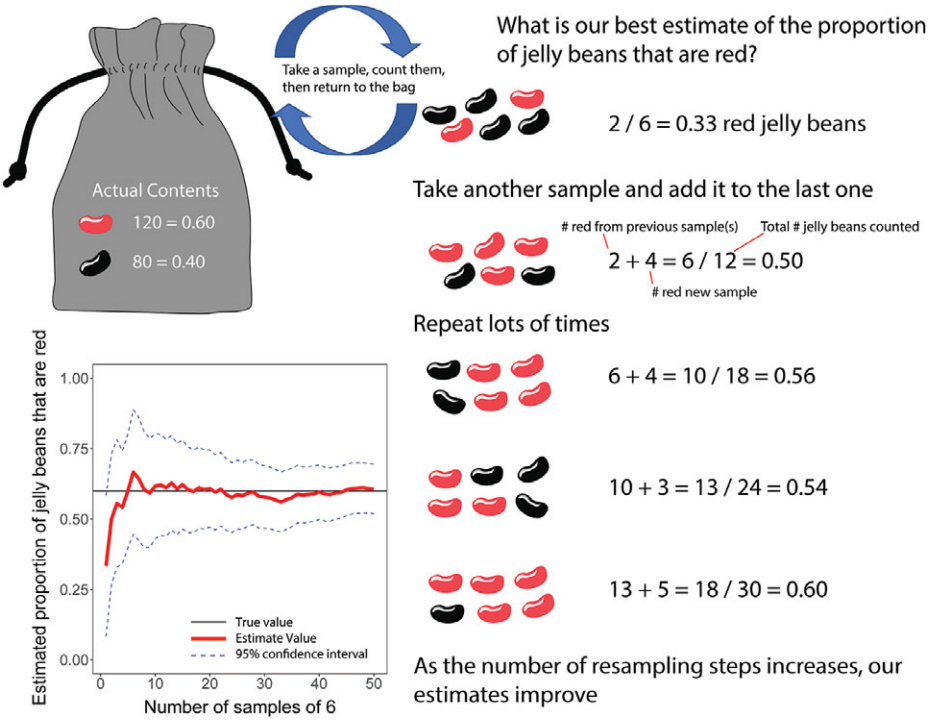


FIG. 5.5 A representation of our jelly bean thought experiment. The plot on the lower left shows how our estimate of the proportion of red jelly beans changes with increased resampling. Note that the confidence interval also shrinks with more sampling.

black jelly beans and you want to know the proportion of jelly beans that are red. If you reach your hand in and pull out a small handful (four black and two red), you might guess based on your sample that two thirds of the jelly beans are black and one third are red. Then if you were to put those six jelly beans back in the bag, shake it up, and pull out another handful (two black and four red this time), you might revise your estimate. Variation in the relative representation of red and black jelly beans between these samples is driven by sampling error. In this case, you could repeatedly take small samples of jelly beans, tally their colors, replace them, and repeat the whole process many times. If you do this enough times, you can combine the results and obtain a better estimate of the true population proportion of each color as well as confidence intervals around that estimate.

You may be asking yourself why would I go through this process of repeated sampling when I could just take a bigger sample and count that. If you were to pull out 100 jelly beans and count them and you got 65 red and 35 black, there is, of course, a good statistical argument to be made that the larger sample provides a

better estimate of the true proportion of red and black jelly beans than any small sample of six (although 100 jelly beans is a lot to hold in your hands at once). In some circumstances, however, repeated sampling provides important additional information that a single larger sample does not. Specifically, repeated sampling at a constant draw size (six in this case) not only allows you to estimate the population parameter of interest (proportion of jelly bean colors in the bag) but also helps you estimate the variability you could expect for a sample of that set size. In the example used here across 50 draws the mean estimate of the proportion of red jelly beans for a draw size of 6 is 0.59 with a 95 percent confidence interval ranging from 0.50 to 0.68. If we were instead to take 20 jelly beans in each draw for 50 draws, we get a mean estimate of 0.60 with a much smaller 95 percent confidence interval ranging from 0.59 to 0.62. As we are often interested in judging variability in a particular observed sample size (see Section 5.3.5), assessments like this can be quite useful.

The jelly bean example described here is a modified version of a common statistical thought experiment called the urn problem and specifically in this case a binomial sampling urn problem. The most important general point to note here is that resampling procedures like this allow us not only to estimate parameters in the population from which samples are drawn but also to assess variability in our samples in relation to our resampling design. Our resampling design can be simple random draws like the example here but can also be designed to capture many different kinds of collection strategies, potential data problems, or other drivers of uncertainty that are of particular interest.

Given the general issues we outline above, related resampling approaches can be adapted to address many questions regarding the robustness of our data to different kinds of perturbations or missingness. The procedures we advocate primarily take the following basic form (see also Smith et al. 2013:653, 2017):

1) Define a network based on the available data, calculate the metrics, and characterize the properties of interest in that network.
2) Derive a large number of modified samples from the network created in step 1 (or the underlying data) that simulate the potential data problem we are trying to address. For example, if we are interested in the impact of nodes missing at random, we could randomly delete some proportion of the nodes in each sample derived from the network created in step 1. If instead we are interested in missing edges and we think that edges that occur over longer distances are more likely to be missing, we can assign probabilities to our edges determining the likelihood that they will be retained or removed in the resampling process to capture such variation.

3) Calculate the metrics and characterize the properties of the features of interest in every one of the random samples created in step 2 and assess central tendency (mean, median) and distributional properties (range, standard deviation, distribution shape, etc.) or other features of the output as appropriate.
4) Compare the distributions of metrics and properties (at the graph, node, or edge level) from the random samples with the "original" network created in step 1 to assess the potential impacts of the perturbation or data treatment. This comparison between the properties of the network created in step 1 and the distribution of properties created in step 3 will provide information directly relevant to assessing the impact of the kind of perturbation created in step 2 on the original network sample and, by extension, the total network from which it was drawn.

The underlying assumption of the approach outlined here is that the robustness or vulnerability of the observed network data to a particular perturbation provides information about the robustness or vulnerability of the unattainable total network to the same kinds of perturbations. For example, if we are interested in exploring the degree distribution of a network and our sampling experiments as described above show massive fluctuations in that distribution in subsamples with only small numbers of nodes removed at random, this would suggest that the particular properties of this network are not robust to nodes missing at random and we should not place much confidence in any results obtained from the original sample as indicative of the total network from which it was drawn in terms of degree. On the other hand, say we instead find that in the resampling experiments the degree distributions in our subsamples are substantially similar to that of the original network sample even when moderate or large numbers of nodes are removed. In that case, we might conclude that our network structure is such that assessments of degree distribution are robust to node missingness within the range of what we might expect for our original sample in relation to the total network from which it was drawn. It is important to note, however, that this finding should not be transferred to any other metrics as any given network is likely to be robust to certain kinds of perturbations *for certain network metrics*, but not to others.

In order to set up resampling experiments like this, you first need to ask several questions of your data and your proposed investigation. What kind of missing data or other data problem am I assessing? Are such data problems generated through a random process or are they biased in some particular way? If the processes involved are biased, what information do I have about the nature of that bias or its probable scale? What network metrics or properties am I interested in exploring? In practice,

you will likely have lots of potential sources of error that you are interested in exploring and may end up doing many experiments. In the following sections, we provide some examples to illustrate how you might approach such considerations.

5.3.1 *Nodes or Edges Missing Due to Random Sampling Processes*

When nodes or edges are missing at random within a dataset, the question we are tasked with addressing is: How much information can we lose before substantial interpretive issues arise? By way of example, we use the Southwest Social Network Project ceramic similarity networks (see Section 2.8.2). Specifically, we focus here on a portion of the database that includes sites with public architectural features related to the Chaco World (see Mills et al. 2018). The Chaco World represents a large-scale social and political system that spanned much of the Colorado Plateau ca. AD 800–1150. This settlement system was marked by the construction of massive public architectural features known as great houses and great kivas. The total inventory of sites with these public architectural features within the Chaco World is relatively well understood so investigators know that the sample of sites for which ceramics are available constitutes approximately 70 percent of large Chacoan structures, and most Chacoan structures are likely known (Mills et al. 2018). The question, then, is: What features of the total network of Chacoan sites can be generalized from a sample that includes only ~70 percent of the possible nodes sampled at random? Since we have a relatively complete inventory of Chacoan sites and know which ones have ceramic data, we have conducted other analyses and determined that the sites without ceramic data are not concentrated regionally, temporally, or in terms of attributes like size, so a random sampling process is appropriate.

In this example, we will explore the impact of nodes missing at random on two measures of node centrality: eigenvector centrality and betweenness centrality. Networks for individual time slices of 50 years were generated using methods described in detail elsewhere (Mills et al. 2018; Ortman 2016), and weighted network ties were generated by calculating similarity in terms of the proportions of wares (using the Brainerd-Robinson metric; see Peeples and Roberts 2013) with unweighted, undirected edges created for this example for all pairs of sites with similarity scores of 0.75 or greater. For this example, we use a single time period marked as the peak of the regional occupation (AD 1050–1100).

Following the procedures for assessing missingness described above, we can calculate the network statistics of interest (eigenvector and betweenness centrality)

for every node. With this in place we can then create 1,000 random versions of the original network by subsampling at sampling fractions from 90 percent down to 10 percent (a total of 9,000 random network subsamples). In other words, we create 1,000 networks that are a random sample of 90 percent of the original network nodes, 1,000 networks that are a random sample of 80 percent of the original network nodes, and so on. We then calculate eigenvector and betweenness centrality for every node in every one of those 9,000 random network subsamples. Finally, we then compare the distributions of values for our original network to the distribution of values obtained for our random samples at each sampling fraction. In order to do that in this case, we calculate simple rank-order correlation (Spearman's ρ or rho) between our original network and each subsample and examine the results. As described above for Spearman's ρ, a value of $\rho = 1$ would mean that every node in the subsample fell in the same rank-order position in the subsample as in the original network, and $\rho = -1$ would indicate a complete reversal of the order (see Box 5.2 for an explanation of the visualizations).

As the results in Figure 5.6 illustrate, the rank-order correlations between the original networks and subsamples are quite high at all but the smallest sampling fractions for eigenvector centrality. Indeed, all correlations are positive with the

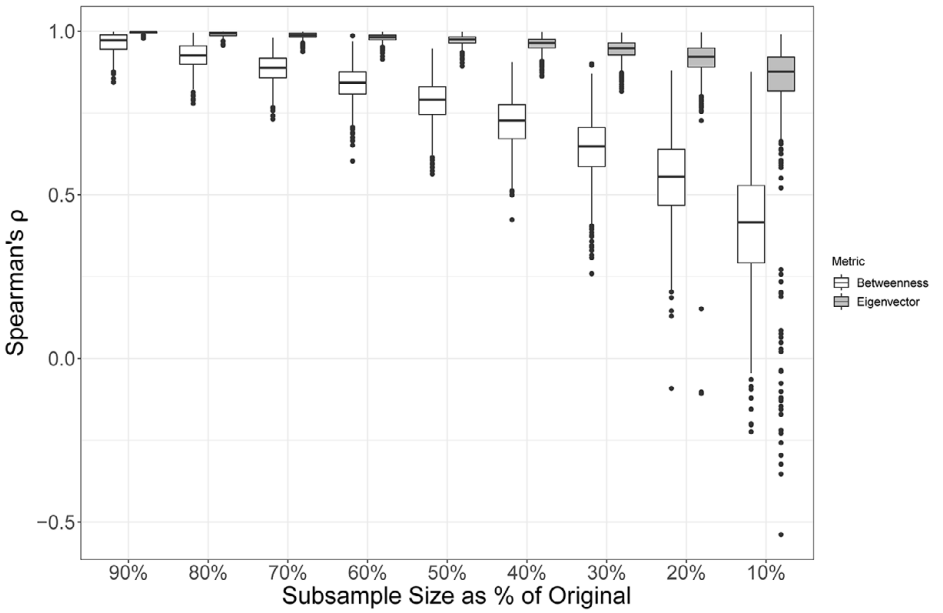


FIG. 5.6 Boxplots showing the Spearman's ρ correlations between the original network and the 1,000 random networks at each sampling fraction from 90 percent to 10 percent.

Box 5.2 BOXPLOTS

Boxplots are visualizations frequently used to display distributions of continuous frequency data and to show central tendencies, skewness, and outliers in the data. There are several different criteria that can be used to define the features of a boxplot, so we describe the definitions we use throughout this book here. The interquartile range (IQR; the area between the lower and upper hinges) represents the middle 50 percent of values in the distribution and the "whiskers" and outliers of the boxplot are defined in relation to this value. Specifically, the upper whisker is defined as the highest value within (1.5 × IQR) + the upper hinge and the lower whisker is (1.5 × IQR) − the lower hinge. Outliers represent all values beyond each whisker in either direction (Fig. 5.7).

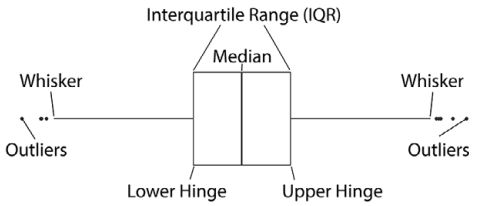


FIG. 5.7 Features of boxplot data visualizations.

exception of the outliers of the 20 and 10 percent fractions and values of $\rho < 0.75$ are not even seen among outliers until the sampling fraction reaches 20 percent. Overall, this suggests that assessments of eigenvector centrality in this network are quite robust to nodes missing at random even at very small sampling fractions. Since we have approximately 70 percent of the total number of known Chacoan sites in our database, this suggests that we can be reasonably confident that our original network sample provides a good approximation of the rank order of eigenvector centrality by node. Looking at betweenness centrality, however, we can see that as the sample fraction decreases, the Spearman's ρ correlation between the original network and the subsamples declines quite rapidly though the positive relationship is retained until the 10 percent subsample comparison ($\rho > 0$). Given this observation, we should be cautious in interpreting small differences in betweenness centrality among nodes as behaviorally meaningful. As this brief example illustrates, the same network can be robust to missing nodes for some network metrics and not others.

5.3.2 *Assessing the Positions or Properties of Individual Nodes or Edges*

In some cases we may be interested not simply in the robustness of a particular network metric to a specific kind of perturbation across all nodes or edges, but also in the potential variability of the position or characteristics of a single node (or group of nodes) due to such perturbations. In order to explore individual nodes, we can employ procedures similar to those outlined above. In this example, we use the technological similarity network of cooking pottery assemblages from sites in the Cibola region of the US Southwest (Section 2.8.3). In this case, say we are particularly interested in the Garcia Ranch site. This is a site that sits on the edge of the Colorado Plateau and contains archaeological material and pottery geochemically sourced to diverse areas to the north and south (Peeples 2018). In the original network, this site was characterized by the second-highest betweenness centrality in terms of ceramic technological similarity, suggesting that it is along the shortest paths connecting several sets of nodes (Fig. 5.8). We want to know the whether or not the position of Garcia Ranch as a node with a high betweenness centrality is robust to nodes missing at random in this network.

In order to conduct this test, we follow the same basic procedures outlined above. We create 1,000 random subsamples of the original network by removing a portion of the nodes at random and recalculating the positions of each node in each random network at different levels of node missingness. The only difference here is that we ensure that our "target" node (Garcia Ranch) is included in each random sample. For every random network we calculate the rank order of betweenness centrality and track it across all 1,000 random subsamples. If we see that Garcia Ranch retains its position as a high centrality node across the random samples at different levels of missingness, we can be somewhat more confident in interpreting the relative position as behaviorally meaningful.

Figure 5.9 shows the results of this test. By way of example we test one subsampling level by removing six of the 30 nodes within this graph at random (20%) in each of 1,000 random subsamples (we could, of course, conduct similar analyses at many subsampling levels). The histogram shows the number of times that Garcia Ranch appeared in each rank-order position. As this plot shows, Garcia Ranch is most often the second-highest ranked node (in over 60% of random subsamples), and in the vast majority of the random subsamples it falls in position one, two, or three, and the rank-order position of 2 (which was the observed value in the original network) is the most common value. Overall, this provides evidence that the rank-order position of this node is fairly robust to moderate amounts of missing information, and thus we may place greater confidence in any interpretations that

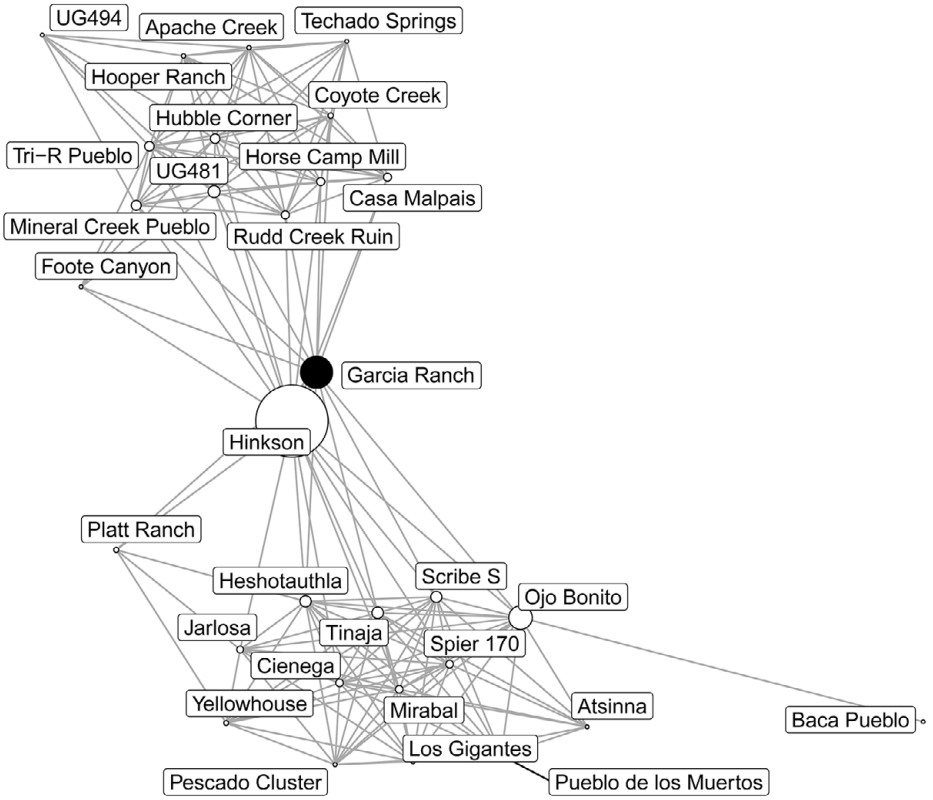


FIG. 5.8 Ceramic technological similarity network based on cooking pottery from the Cibola region. Nodes are scaled based on betweenness centrality, and the target node (Garcia Ranch) is shown in black.

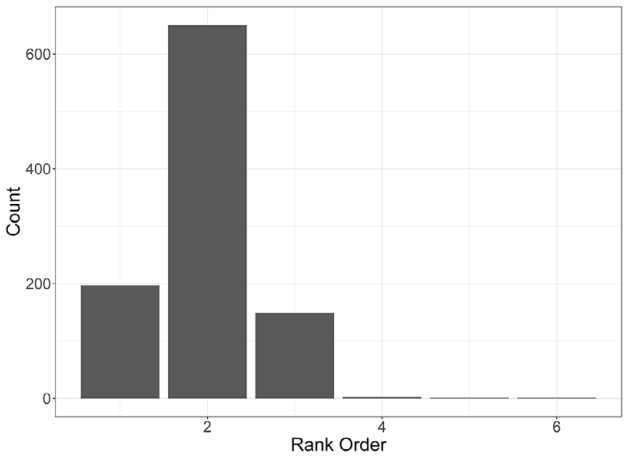


FIG. 5.9 Histogram of the rank-order position of betweenness centrality for Garcia Ranch in each of 1,000 random subsamples.

we draw based on this position. We could similarly conduct this exercise for other nodes in the network to better understand if certain positions are more vulnerable or robust than others.

5.3.3 *Nodes or Edges Missing Due to Biased Sampling Processes*

There is a wide variety of processes that can create bias in our samples of nodes or edges that we might wish to model in our resampling experiments. For example, if for some reason we think that small sites are more likely to be missed than large sites in a given network sample, we could define differential probabilities that large and small site nodes would be retained in replicates of a dataset. To provide an example of how this can work we turn to the coauthorship data on archaeological network publications originally published by Brughmans and Peeples (2017; see Section 2.8.5). In this case, we are creating an author-to-author network where an edge is placed between any two authors who appear in the same publication together.

There are many reasons we might miss publications when compiling a database of coauthorship such as that used here. If a publication is older or in a more obscure venue we may be more likely to miss it when searching academic indexes and library resources. We also might be more likely to miss applications of archaeological network research in compilations or edited books where the overall focus is not specifically on network research. If we wanted to understand the robustness of this coauthorship network to missing nodes, it seems that modeling this as a random sampling process would likely be inappropriate. How, then, should we proceed to encapsulate the hypothesized processes that may govern missing nodes in this case?

As a first take on this issue, we can model a missing node process where the date of the publication is inversely proportional to the probability that it will be included in each random resample. In other words, we can use the same kind of bootstrapping procedure we described above but set up our sampling such that we are more likely to retain recent publications and less likely to retain older publications. For the sake of illustration here we will compare this to the results we would obtain if we simply modeled nodes missing at random without reference to age. In this example we explore betweenness centrality and create 1,000 random simulated datasets for each of the 90, 80, and 70 percent subsample fractions for our "probability by date" model where nodes are removed with a probability inversely related to their recency, as well as another 1,000 for the same subsample fractions with nodes removed at random. For each simulated run we compare betweenness centrality scores to the original observed network once again using Spearman's ρ rank-order correlation coefficient.

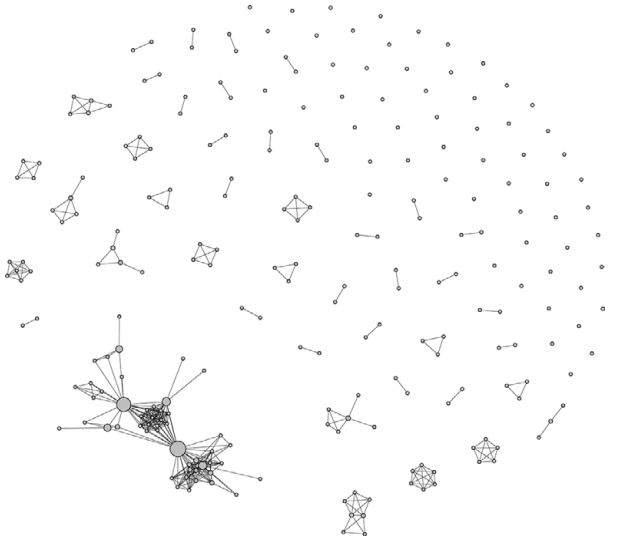


FIG. 5.10 Network graph showing connections among coauthors with nodes scaled based on betweenness centrality.

Figure 5.10 shows the observed coauthorship network and Figure 5.11 shows the results of the resampling analysis as a set of boxplots. As the network graph shows, there are several clusters of authors who coauthor together as well as many isolated nodes (authors that coauthor with no one). Further, there is a single large component that includes the nodes with the highest betweenness centrality. As Figure 5.11 illustrates, the correlation in betweenness centrality declines as the sampling fraction is reduced (as we would expect), but notably, the model where the probability of node removal is inversely proportional to publication date provides a closer fit to the observed across all sampling fractions than the model where nodes were missing at random (and these differences are statistically significant for all comparisons shown here).

In practice, it is unlikely that we would be missing a very large fraction of archaeological network publications, and it is likely that more of our missing publications are older publications (as our probability by date model simulates). Together, this suggests that we could be relatively confident that our rank-order betweenness centrality scores in this network are robust to missing publications. In particular, these results suggest that this metric for our particular network is even more robust to missing data when we model missingness using some realistic assumptions about what publications we would be more or less likely to miss. We could use a similar approach to that briefly described here to model any other sort of process for missing nodes or missing edges we could imagine. As this example

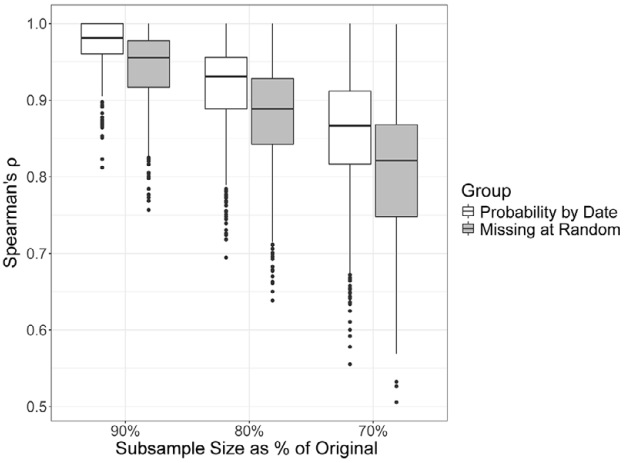


FIG. 5.11 Boxplots showing the rank-order correlation in betweenness centrality for resampled networks and the observed for various subsampling fractions with probability inversely proportional to publication date shown in white and with random removal shown in gray.

shows, modeling missingness to match the processes that are likely driving missingness can produce different results than simply considering missing data as a random sampling process.

5.3.4 *Edge Probability Modeling*

In this section we explore another potential direction for analysis of variability in edges in archaeological networks. To our knowledge, this method has not been explored beyond some simple experiments (Peeples 2017), but we suggest such an approach may offer a useful future direction. The approach we are calling *edge probability modeling* draws particular inspiration from the world of "dark networks" focused on tracking networks of illicit activity (Bright et al. 2012; Everton 2012; Litant 2013; Morris and Deckro 2013). In such networks where source information may come from diverse and variable quality information (sounds familiar, right?), edges are sometimes associated with some qualitative measure of reliability. For example, information on a connection between two individuals provided by an informant may be coded on an ordinal scale (reliable, usually reliable, sometimes reliable, usually unreliable, unreliable). These are essentially qualitative anchors that help an analyst assess the quality of information, and such values can be assigned probability values and further assessed in this way.

Although we are unaware of any archaeological examples where edge "confidence" has been coded on a qualitative scale like this, weighted edges based on similarity are often interpreted in a similar way as representing probabilities of connection rather than weights in a strict sense (see, e.g., Golitko et al. 2012; Mills et al. 2013a; Peeples and Roberts 2013). Thus, under certain circumstances, the weights of edges can be seen as statements of the probability that a tie existed between two nodes given the available evidence.

To illustrate the potential of such an approach, we simulate a small network dataset where every tie is associated with a qualitative anchor assigned a probability along a five-part scale. This could represent connections between sites where edges are defined with a similar qualitative approach to assessing the data that provide information for each edge. In other words, every tie has a weight associated with it based on a qualitative assessment of edge likelihood from the following values: 0.2, 0.4, 0.6, 0.8, 1.0. If we treat these "weights" as probabilities, we can generate random networks that are all "potential" networks supported by the available data. For example, a connection with a probability of 0.2 should be present in about 20 percent of random networks generated with these data. A connection with a probability of 0.8 should be present in about 80 percent of random networks generated. Given the available evidence, we have no strong sense of which network is "better," so simulating lots of potential networks and comparing them is a useful approach. A key assumption here is that the probabilities assigned to each edge are independent of each other. If this were not the case, we would need to model more complex generative models using similar methods.

Figure 5.12a shows a randomly generated network with edge weights defined as described above with nodes scaled based on degree centrality. Treating the edge weights in this network as probabilities, we can produce 1,000 random networks where the probability of the presence or absence of a given edge is proportional to the weight of that edge in the data we have generated. Figure 5.12b shows three example networks. With the data we have available we have no strong reason to prefer one random "candidate" network over another. Thus, we can take all of the random candidate networks based on the available evidence and use them to make inferences about the likely "true" network of interest. For example, if we were interested in robustness of degree centrality scores to variability in edge data quality, we could then calculate degree centrality for every one of the randomly generated networks based on edge weight probabilities and compare the results for individual nodes or the overall distribution across each random network much as we have done in earlier examples in this chapter.

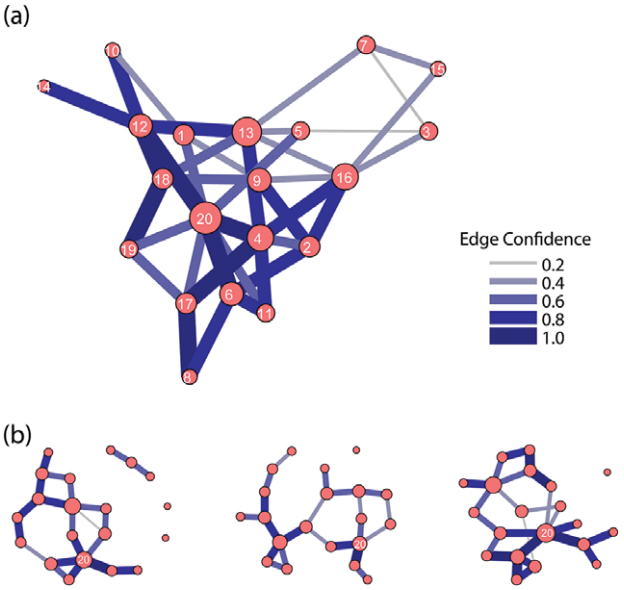


FIG. 5.12 (a) Randomly generated network graph with edge weights assigned probability scores and nodes scaled by degree centrality. (b) Three examples of possible network graphs generated based on assigning ties with probabilities proportional to the weights of edges.

If instead we are interested in the role of a particular node we could compare the metric of interest (degree centrality in this case) across all 1,000 random candidate networks and get a median estimate of degree centrality as well as a standard deviation or confidence interval. For example, the high centrality node labeled "node 20" in this network has a degree centrality of 8 in the original network where all edges are included. Given our assessments of the probability that each edge is present, this value for degree can be thought of as a theoretical maximum and is actually quite unlikely to represent the true value given our assessment of each edge (unless all edges are associated with very high probabilities). If we look at the range of values obtained for the random candidate networks we get a mean value of 4.8 and a 95 percent confidence interval ranging from 2.7 to 7.0, which provides a more realistic estimate of degree for the node in question given our confidence in each edge and the edge probability model we use here. Note that degree centrality for candidate networks for any particular node may not always be lower than that in the original network but it can never be higher. Thus, we are able to leverage the information we have regarding the reliability of particular edges to provide a better estimate of centrality as well as expected ranges around those estimates. We could similarly use this approach to explore other metrics or properties of this network.

5.3.5 *Assessing the Effect of Small or Variable Sample Sizes*

One issue frequently faced by archaeologists in almost any kind of investigation is variability in sample size. Where sample sizes are small or vary considerably from observation to observation, the accuracy of any conclusions regarding the population from which that sample was drawn will likely be negatively impacted. The question we must ask ourselves as analysts is the nature and severity of sampling error in our specific dataset and the kinds of problems that such issues may generate for our analytical results and interpretations. What inferences can we confidently draw given the sample available? What is the degree of variability in our analytical results that may be attributable to sampling error? The approach to addressing these questions and assessing uncertainty in light of sampling error for network investigations can take a form quite similar to the examples focused on nodes and edges outlined above. The major difference here, however, is that the focus of the resampling is now the underlying raw archaeological data used to generate a network representation rather than the network representation itself. This is a classic example of the bootstrap approach.

Say we have an archaeological network where edges are defined between pairs of sites (nodes) across a region based on the similarities of artifact assemblages (e.g., Mills et al. 2013a). Say the samples from each site were taken based on the analyses of available collections that vary considerably in size (some sites may have samples of only a few objects, whereas others may have thousands). The question we must then address is: How does the variability in sample size impact the structure and properties of the network generated using these data?

A well-known property of samples that has been documented and explored thoroughly in archaeology, ecology, and other fields is that the diversity of a sample is dependent in complex ways on the sample size (e.g., Kintigh 1984; Meltzer et al. 1992; Plog and Hegmon 1993; Willis 2019). In general, larger samples will tend to have more categories (more types of artefacts in our example above) and therefore will likely generate different patterns of connections when used to define network relations. In order to explore how we might attempt to at least partially ameliorate the issue of sampling error, we once again turn to an example using the Southwest Social Networks database (Mills et al. 2013a; Peeples 2017; Peeples et al. 2016) and the Chaco World data described in Section 5.3.1 (see Mills et al. 2018). Again, these data consist of ceramic counts by type/ware for hundreds of sites, which include Chacoan architectural complexes and with assemblages apportioned into short chronological intervals. Network edges have been defined here as weighted and undirected based on similarity in terms of the proportions of different ceramic

wares using the Brainerd-Robinson similarity metric. Sample sizes vary considerably from site to site, with some sites having as few as 10 identified ceramic fragments in a given time period (we excluded smaller samples from inclusion so this is an absolute minimum), while others have tens of thousands. For this example, we use the period marked by the maximum extent of the Chaco World (AD 1050–1100).

In order to assess the impact of sampling variability on the network as a whole as well as for individual nodes, we take our original matrix of ceramic counts by site and create a large number of replicates (1,000) of each row with the sample size for each node held constant as the observed sample size for that site. Further, the probabilities that a given sherd will be a given type are determined by the underlying frequency distribution of types at that site. In other words, we pull a bunch of random samples from the site with the probability that a given sample is a given type determined by the relative frequency of that type in the actual data for that site (so we are sampling with replacement from the observed sample). Once this procedure has been completed, we can then assess similarity scores, centrality metrics, or any other graph-, node-, or edge-level property and determine the degree to which absolute values and relative ranks are potentially influenced by sampling error (see Fig. 5.13).

Say we are interested in degree centrality. With the 1,000 replicates produced as described above, we can explore the robustness of degree assessments to sampling error. Figure 5.14 is a histogram of Spearman's ρ rank-order correlation coefficients between the original network and each of the 1,000 random replicates. As this figure illustrates, the rank-order correlations in every resampled replicate are high at a mean of 0.94 and a 95 percent confidence interval ranging from 0.93 to 0.96. Overall, this suggests that the rank order of degree in this particular network is quite robust to sampling error in this dataset.

If we were interested in how this pattern plays out for particular nodes or groups of nodes, we could further assess the variability across the sample using procedures similar to the methods described above. Specifically, Figure 5.15 displays line plots for every node with the order from left to right set by the rank order of degree centrality in the original observed network. The vertical lines in these plots represent the 95 percent confidence interval around the estimates of degree centrality based on assessments of the 1,000 random replicates. As this figure illustrates, there are at least two "plateaus" in the rank-order values of degree, and for sites with very high and very low degree centrality, there is a great deal of agreement between the original network and the 1,000 simulated replicates. For sites with middling degree centrality scores, however, there is considerable variability. Overall, this suggests

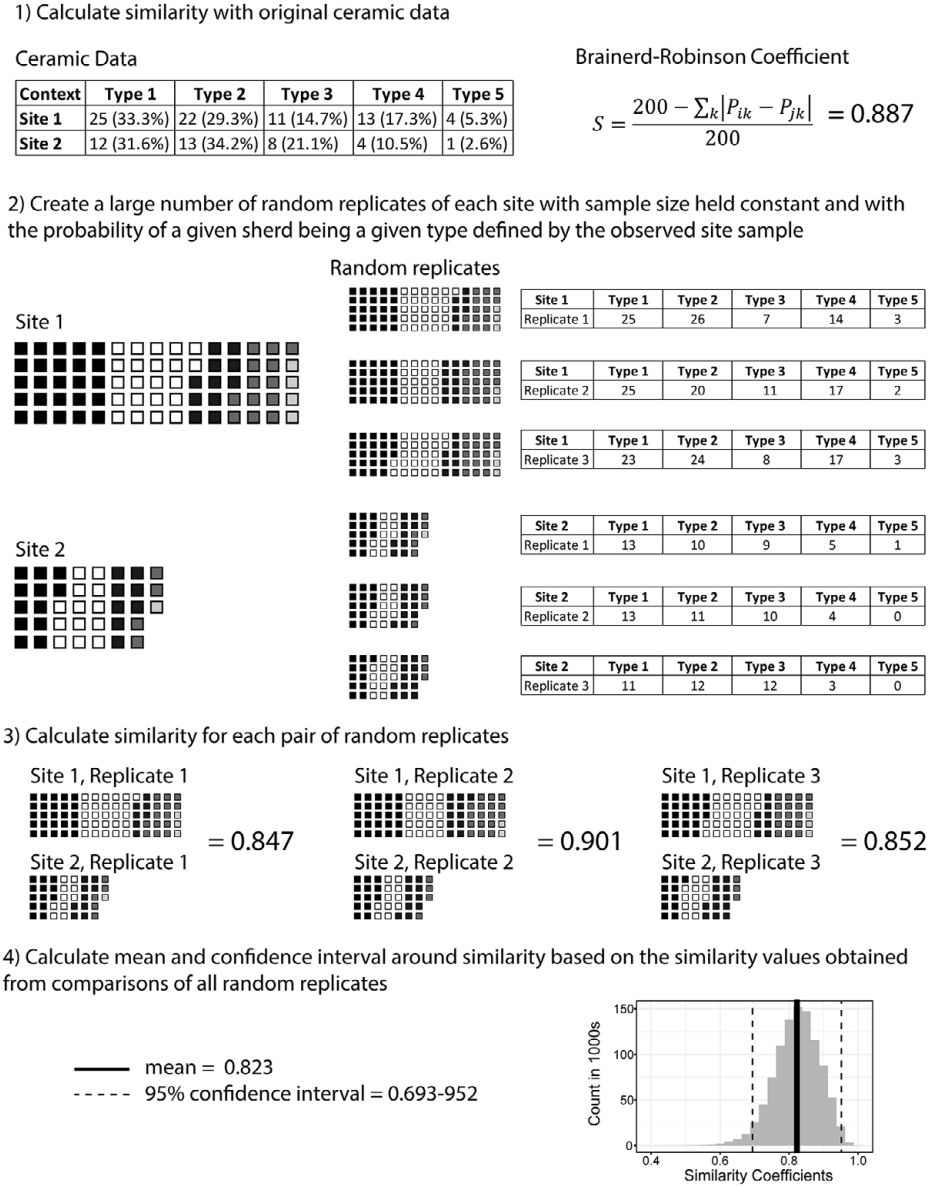


FIG. 5.13 Explanation of the bootstrap resampling procedure used to evaluate sampling error.

that we should be careful in making substantive interpretations about relative degree centrality for sites with middling values, but we can be somewhat confident that the placement of sites with low or high degree centrality is not severely impacted by sampling error. Note that we could also use these replicates to assess

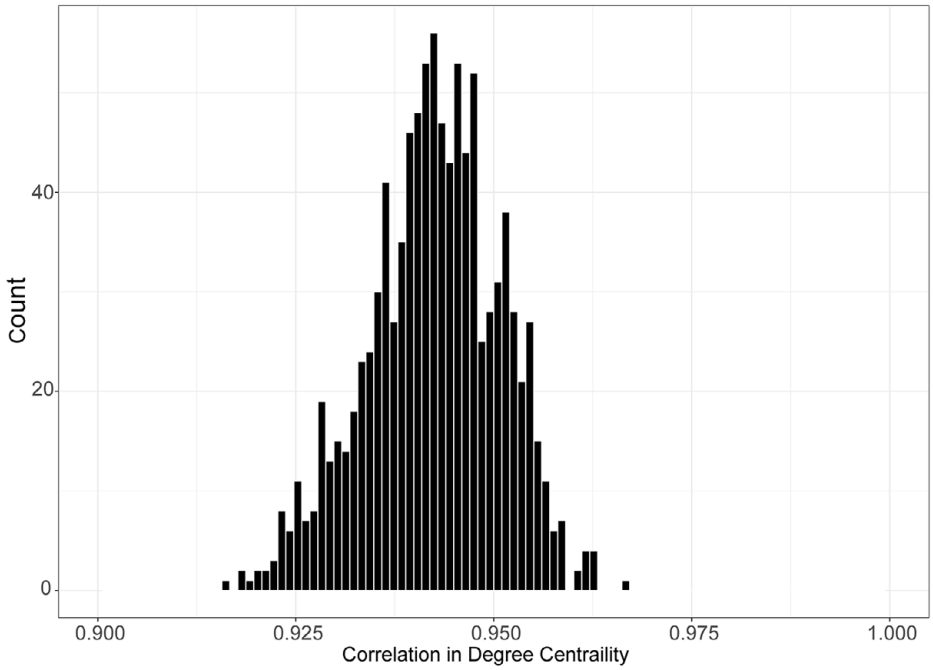


FIGURE 5.14 Histogram of Spearman's ρ rank-order correlation coefficient between 1,000 random runs and the original network.

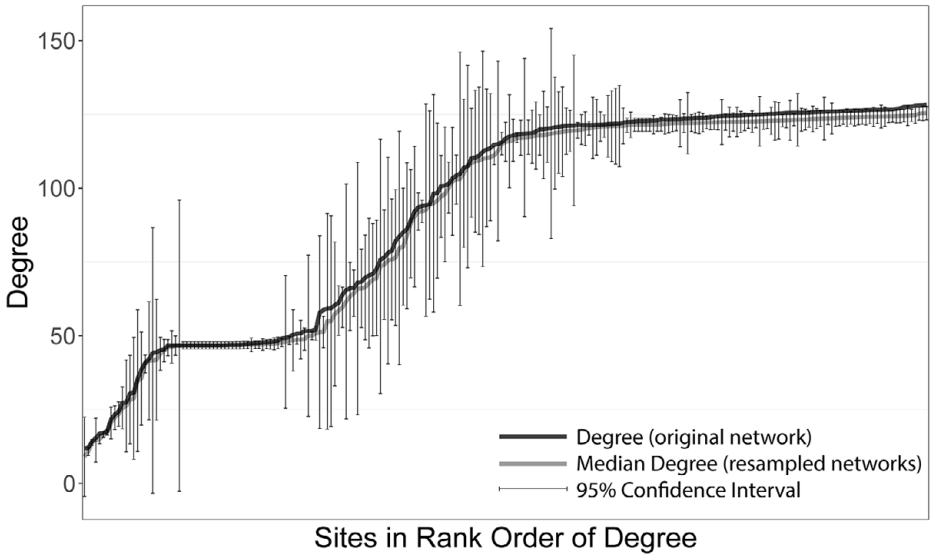


FIG. 5.15 Plot of degree centrality by site for each of the 1,000 replicates of the original dataset. Vertical lines represent the 95% confidence interval for each node. Sites are shown in order from left to right in rank order of degree centrality from the original network.

the variability in the centrality score for any particular node (to calculate a median, standard deviation, etc.).

The example outlined here is but one of many potential approaches to identifying and assessing the impacts of sampling error on network properties (see also Feugnet et al. 2017; Gjesfield 2015; Mills et al. 2013a; Peeples et al. 2016). In your own work, the specific nature of the data, the sample, the network abstraction, and the properties of interest are likely to differ. Thus, the example shown here can be thought of as a guide to how one might assess variability in sampling using similar procedures. Importantly, this bootstrap approach has been shown to produce robust and reliable assessments of sampling variability for similarly structured simulated archaeological data (Roberts et al. 2021).

5.3.6 *Assessing the Sensitivity of Results to Classification Schema and Other General Issues*

There are many other potential sources of uncertainty in networks that are derived generally from the nature and treatment of the archaeological data used to generate networks. It is impossible to be exhaustive here as the potential sources of error are so numerous and often so specific to a single dataset or problem. In order to outline the kinds of issues we are referring to here and potential approaches to dealing with such issues, we present a few basic examples but do not present detailed worked case studies as these are less likely to be of direct relevance to future work than the many examples above.

One of the major sources of uncertainty in many kinds of archaeological analysis comes from classifications. This could, of course, be driven by classification errors (where particular items are clearly misclassified within a given schema) but perhaps just as often this comes from the nature of the categories themselves. How we define types and classes of objects/materials and the extent to which these represent categories that may have been recognizable to the people who made them is a classic problem in archaeology and one that has generated lots of debate and a myriad of approaches (see, e.g., Adams and Adams 1991; Borck et al. 2020; Read 2009; Rouse 1960). It is certainly true that if the same set of objects or sites were classified using a different schema, the results generated through any analysis would likely differ, possibly in substantively important ways. This is true for archaeological networks or any sort of archaeological analysis.

As a brief example, say we are interested in defining networks based on the co-occurrence of distinct styles of projectile point. If we had a classification scheme

that included points classified as type A, type B, and a third category called type A/B (representing points that are identifiable as one of the two types but which cannot be assigned to a single category), what are we to do with the type A/B objects in our analysis? If we treat them as a separate category we may end up getting results that are spurious, or if, say, completeness is driving our ability to classify these objects, we may get patterns that relate to the size and degree of fragmentation of objects across our assemblages. If we ignore that category and remove it from consideration, we may be missing information about the potential boundedness of types or objects with intermediate features. In any case, the decision to be made is likely to have substantive impacts on our analyses, but there may not always be a strong argument to be made for one decision or another for the analysis at hand.

In another example already mentioned briefly above, Peeples and colleagues (2016) conducted an analysis of networks based on ceramic similarity in the Sonoran Desert of southern Arizona. In this context, there is one particular ceramic type, "Tanque Verde Red-on-Brown," which is variously given end dates between AD 1350 and 1450 in the literature with no way of currently resolving this dispute. Thus, if we were to create networks or conduct any other kind of analysis focused on the interval between these two possible end dates for that type, we will have to decide whether to include or exclude Tanque Verde.

In the two examples described here, we have uncertainty that cannot be resolved without some additional information or analysis that is beyond the scope of the current investigation. In such cases, it is often advisable to conduct multiple analyses and compare the results. For example, in the first case, we could generate networks where we treat the type A/B objects as a distinct category and another set where we leave those out and compare them for the properties of interest. If the key metrics we are interested in exploring with our networks change little despite changes in the way we treat those data, we can be reasonably confident that the results are not merely a product of our analytical decision. If, on the other hand, the nature and structure of the network changes dramatically, then any results we obtain should be treated as preliminary until the typological issue is resolved. The same is true of the example where we have temporal uncertainty in the date range of objects used to generate network representations. In the example published by Peeples and colleagues (2016), they found that when alternative date ranges were used for defining connections among settlements in the Sonoran Desert, the resulting networks differed in size, connectedness, and centrality for some nodes. Importantly, however, they found that these changes were limited to just a couple of small areas within the broader study area where Tanque Verde

pottery was common, suggesting that we might treat the results for sites in those smaller areas as preliminary while having greater confidence in the results for other places.

As these brief examples suggest, when there are multiple potentially valid assumptions about how the data used to create network representations could be treated but little reason to favor one treatment over another, it is often advisable to conduct multiple analyses and compare the results. There is, of course, a practical limit to how many different potential issues one could address (see discussion in Section 5.4), but where possible addressing such issues will add clarity to results and interpretations.

5.3.7 *Imputation and Alternative Approaches to Missing Data*

In addition to the approaches outlined above based on resampling and related approaches, there are a number of other promising models for dealing with missing data in networks that merit mention as potential future directions. In particular, methods for network imputation and simulation ranging from relatively simple to extremely complex have shown promise for dealing with certain kinds of network data problems in other fields (see examples in Krause et al. 2018; Smith et al. 2022). Most of these methods have seen relatively little use in archaeological network studies. Although the details are beyond the scope of this chapter, in this section we briefly note some of the most common imputation (and related) methods.

Imputation refers to the process of assigning values to missing observations in a dataset. The array of specific approaches that could be used to impute missing values in networks is enormous. Following Smith and colleagues (2022) most approaches to imputing missing data in networks fall into one of two basic categories: **simple imputation methods** or **model-based imputation methods**. We might also define a third category as **listwise deletion**, which is not an imputation method but simply refers to removing nodes with incomplete information and limiting analyses to only the complete cases.

Simple imputation refers to the use of the observed network data to help fill in missing observations (see examples in Huisman 2009; Krause et al. 2018). This could entail, for example, defining an unobserved edge between a pair of nodes when they share relations to other nodes in the observed dataset. Alternatively, this could also take the form of assigning an edge weight probabilistically using the overall frequency of ties in the observed network sample. Simple imputation can

also be used to add missing nodes when they share other kinds of relations with nodes in the existing sample, such as edges in other layers of multilayer networks.

Imputation refers to the process of assigning values to missing observations in a dataset.

Listwise deletion refers to the practice of removing all cases with partially missing observations and limiting analyses only to complete cases.

Simple imputation methods refer to approaches that use simple characterizations of existing data to define values for missing observations.

Model-based imputation methods refer to more complex methods for filling in missing values in a network based on a particular network generative process or topological property observed or assumed to be present in that network.

Model-based imputation methods are often more complex and involve imputing data relative to some specific observed or expected generative properties of the network at hand. For example, if we expect homophily or closed triads or some other specific structural property in our network, we can impute data with respect to that expectation. Similarly, if we expect that our network follows a gravity model or some other generative model, we could incorporate that into our imputation as well (see Amati et al. 2018). Such an approach often relies on a particular class of model called an exponential random graph model (ERGM) to simulate random networks that retain specific topological properties or positions of interest (see Box 4.1; Koskinen 2010; Lusher et al. 2013). Such models have been used to simulate network generative processes in a small number of archaeological studies (e.g., Amati et al. 2018, 2020; Brughmans et al. 2015; Cegielski 2020; Wang and Marwick 2021), though we are unaware of examples where they are used specifically for the purposes of imputation in archaeological research (but see Amati et al. 2020 for some possibilities).

Another area of recent research that has been used to help deal with the kinds of data issues often faced in archaeological network studies is centered on Bayesian inference and related empirical approaches. Bayesian methods are simply a set of statistical approaches that use Bayes' theorem to assess and update the probability of a hypothesis given the evidence and prior belief. There is a suite of Bayesian approaches to network construction and analyses that are designed to use statistical techniques to reconstruct plausible networks from observational data that are assumed to have missing information and measurement error. Although

approaches to Bayesian inference differ, many such approaches to network models assume there is some "real" network that was generated by the kinds of network generative processes we may be interested in studying, and our observations are related to that "real" network but this relationship is complicated by noise and other sources of uncertainty. Under such a framework, network representations are not simply defined based on the observations alone, but observations are combined with other information about the likely or possible data-generating processes. Based on this information, probability distributions of different models compatible with the empirical data are generated (Linero and Daniels 2018; Young et al. 2021). Although such models are often complex to develop, there are published examples and, importantly, software packages available to implement these methods, which are becoming increasingly common in other fields (e.g., Nemmers et al. 2019; Young et al. 2021).

A related set of approaches that have been used to quantify and ameliorate the effects of error and uncertainty in network models revolves around what are known as empirical Bayesian methods (Farine and Strandburg-Peshkin 2015; Peeples and Roberts 2017). These are similar to the models described above but derive the prior knowledge not from model-based assumptions focused on network-generating processes but instead directly from the data. This might involve, for example, combining the pooled global data for a given property with the data observed for a specific node within a network to estimate the posterior probabilities associated with the network parameter in question. Similar approaches have been used in a limited number of examples in archaeology that deal with sample size variability (Robertson 1999) or in combining multiple sources of data into a single estimate of some important parameter (Ortman et al. 2007), but as far as we are aware, this has been applied to archaeological networks in only one study (Peeples and Roberts 2017). Specifically, Peeples and Roberts (2017), working with the Chaco World data discussed above, used empirical Bayesian methods to improve estimates of ceramic frequency (used to generate networks) incorporating both the data available for each observation and the data available from observations weighted based on the inverse of their geographic distance to the target node. This approach improved the reliability of many network metrics when real and simulated data were subsampled and also provided assessments of the credible intervals of all network properties estimated. We see a great deal of potential in such approaches for archaeological network research and suggest others continue to explore such methods in greater detail in other contexts.

Although work on imputation and related approaches in networks is still relatively new, there are a few initial insights from this work that are relevant for archaeological

applications (as summarized in Smith et al. 2022). First of all, much like the resampling approach we described in the rest of this chapter, the appropriateness and usefulness of a particular imputation method will differ from network to network and from metric to metric in ways that can be difficult to predict. Imputation methods have been shown to greatly reduce bias in assessments of network metrics in many situations, and even simple methods of imputation are often better than listwise deletion. At the same time, many networks are robust to certain kinds of missing information and imputation may not always substantially reduce bias over simply analyzing the observed sample. Finally, it is important to note that missing information has most frequently been modeled in network research beyond archaeology in situations where we have some information about a particular node or edge but lack full information. This could be, for example, a network generated by a survey where some respondents did not fully fill out all questions. In practice, this kind of missing information is probably rare in archaeological networks, which are more often generated based on material cultural samples, observations of physical features like roads, geographic distances, or configurations, or other similar criteria (see Chapter 3). Thus, it is likely that existing imputation methods will require some additional attention and modification for use in archaeological contexts. We argue that this is a direction worth pursuing in the future.

5.4 COPING WITH UNCERTAINTY

This chapter has highlighted many things that can drive uncertainty and error in archaeological networks (and other kinds of archaeological analyses) as well as a general approach to addressing those issues centered on resampling methods and related techniques. In any real archaeological analysis we are likely to be faced with many of the challenges outlined above. Without a clear approach to identifying and dealing with the most pertinent issues, the sheer number of issues can seem overwhelming.

In order to frame a heuristic approach to thinking about these challenges, we borrow a tongue-in-cheek metaphor once used by Barbara Stark to talk about addressing research challenges in general that has long inspired Peeples in thinking about such issues (Stark to Peeples, personal communication 2009). As she suggests, we can think of all of the potential sources of uncertainty and error in our analyses as zombies slowly and relentlessly attempting to surround us, overwhelm us, and add us to their shambling horde. How do we deal with this approaching undead mass? As Stark put it, we are never going to be able to kill all of the zombies, but if we kill the closest zombies, we can sometimes escape (Fig. 5.16).



FIG. 5.16 Our metaphorical data problem zombies are coming to add us to their shambling horde. Illustration by Gavin McCullough.

This metaphor may seem a bit silly, but this insight is actually quite useful in thinking about how we deal with uncertainty and error. We will never address all potential forms of uncertainty, and if we attempted to deal with every source of error, we would likely never be able to move on and complete an analysis. If we, however, identify and attempt to ameliorate the biggest and most relevant problems (i.e., the closest zombies) we can produce useful results and interpretations (i.e., escape) despite some lingering issues.

In practical terms, what this means is that we should not take our results for any archaeological network study (or any analysis for that matter) as simply a given but should attempt to discover and describe whether or not our most important substantive results could be driven by any of the common data/method issues outlined above or others not mentioned here. For example, if we are interested in using eigenvector centrality to explore flows of materials and knowledge across an archaeological network, we should conduct analyses to explore whether or not this particular measure of centrality is robust to the kinds of missing information that characterizes our data. If we are interested in the position of a particular node or edge in a given network as a "bridge" between otherwise distinct components, we may then use some measure like betweenness centrality. In such a case, not only

would we want to know how robust that particular measure is to potential perturbations of the network as a whole; we also would want some way of assessing variability for particular nodes in question and properties like the standard deviation of a metric rather than simply a point estimate. If we are interested in a graph-level structural property of a network, like whether or not the degree distribution of a given network follows a power-law distribution, we would want to be sure that our measure of degree is relatively robust to issues such as missing nodes or edges.

There is no simple means for assessing the results of the kinds of uncertainty analyses outlined here. We can, in some cases, calculate things like confidence intervals or (though we have not highlighted such an approach here) statistical significance of particular relations (see Östborn and Gerding 2014). Such assessments get us only so far, however. Our job as thorough analysts is to convince ourselves and others that we have done a good job of accounting for and assessing the potential sources of uncertainty that are most proximate to our particular argument; if we do, our results and interpretations will be more convincing. *Kill the closest zombies, and you just may survive.*

Given page and word limits for many publications, it is often not feasible to fully publish the details of all of the experiments you conduct to assess the impacts of data problems in the main body of your text, even if we limit ourselves to the most pertinent kinds of uncertainty quantification. As the kinds of assessments outlined above are essential to evaluating any substantive interpretations you make of your archaeological networks, however, we argue that it is *essential* that you provide readers of your work the opportunity to fully evaluate all of the analyses you have conducted, and this includes sharing experiments and their results. In some cases this leads to situations where the supplemental materials are many times longer than the article itself (e.g., Mills et al. 2013a). This should also include directly sharing data and code used to conduct such analyses and sensitivity experiments. There have been increasing calls for a turn to open science in archaeology recently (Marwick et al. 2017) and we see such endeavors as absolutely *necessary* for moving the analysis of archaeological networks (and archaeology in general) forward.

5.5 SUMMARY

- Error refers to differences between observed and recorded values and the true value of interest. Uncertainty refers to assessments of the range of expected variability in observations based on potential sources of error.

- Error can be divided into systematic errors, which come from flawed methods, instrumentation, or assumptions, and statistical error, which refers to variability in estimates of the value of interest due to the limits of precision of measurements and sampling variability.
- The goal of formal uncertainty assessment is to identify and attempt to compensate for potential sources of error and variability in analytical outputs based on both systematic and statistical error.
- Common sources of error in networks are missing data and poor or inconsistent data quality. Because observations are by definition related in networks, the impacts of error can be compounded.
- Monte Carlo methods, bootstrapping, and related resampling methods refer to approaches that simulate different sampling schema or data problems using *the sample we have* to make inferences about the potential effects of similar issues on *the population of interest*.
- Assessments of uncertainty and the sensitivity of a particular network to a particular kind of perturbation cannot be directly transferred to another network or even another metric. For example, if a given network is sensitive to nodes missing at random for betweenness centrality, we cannot then assume that all networks will be sensitive for the same measure nor that other measures of centrality (like degree centrality) will show similar results.
- It is impossible to assess and deal with all sources of error and variability in our networks, so our goal should be to deal with the most proximate data issues and make interpretations that are sensitive to the uncertainty of our estimates.

Further Reading

The following references provide several examples of approaches to assessments of missing data in network analysis from studies focused on social networks, archaeological networks, and so-called dark networks.

Borgatti, Stephen P., Kathleen M. Carley, and David Krackhardt 2006 On the Robustness of Centrality Measures under Conditions of Imperfect Data. *Social Networks* 28(2):124–136.

Costenbader, Elizabeth, Thomas W. Valente 2003 The Stability of Centrality Measure When Networks are Sampled. *Social Networks* 25(4):283–307.

De Moor, Sabine, Christophe Vandeviver, and Tom Vander Beken 2020 Assessing the Missing Data Problem in Criminal Network Analysis Using Forensic DNA Data. *Social Networks* 61:99–106.

Frantz, Terrill L., Marcelo Cataldo, and Kathleen M. Carley 2009 Robustness of Centrality Measures under Uncertainty: Examining the Role of Network Topology. *Computational and Mathematical Organization Theory* 15(4):article 303.

Gjesfjeld, Erik 2015 Network Analysis of Archaeological Data from Hunter-Gatherers: Methodological Problems and Potential Solutions. *Journal of Archaeological Method and Theory* 22(1):182–205.

Huisman, Mark 2009 Imputation of Missing Network Data: Some Simple Procedures. *Journal of Social Structure* 10(1):1–29.

Kossinets, Gueorgi 2006 Effects of Missing Data in Social Networks. *Social Networks* 28 (3):247–268.

Roberts, John M., Yi Yin, Emily Dorshorst, Matthew A. Peeples, and Barbara J. Mills 2021 Assessing the Performance of the Bootstrap in Simulated Assemblage Networks. *Social Networks* 65:98–109.

The following set of articles by Smith and colleagues provide the most detailed available general assessments of the robustness of network properties and positions to different kinds of missing information (both random and nonrandom) as well as overviews of network data imputation methods.

Smith, Jeffrey A., and James Moody 2013 Structural Effects of Network Sampling Coverage I: Nodes Missing at Random. *Social Networks* 35(4):652–668.

Smith, Jeffrey A., James Moody, and Jonathan H. Morgan 2017 Network Sampling Coverage II: The Effect of Non-Random Missing Data on Network Measurement. *Social Networks* 48:78–99.

Smith, Jeffrey A., Jonathan H. Morgan, and James Moody 2022 Network Sampling Coverage III: Imputation of Missing Network Data under Different Network and Missing Data Conditions. *Social Networks* 68:148–178.

EXERCISES

Answers for these exercises can be found in Appendix A.

5.1) Imagine you have defined an archaeological network and you were interested in exploring the relationship between network centrality and site size but you know that you have missing nodes. In general, you think that larger sites have seen more frequent investigation and would more likely be included than smaller sites. Would you model node missingness as a random or biased sampling process in this case? Why? What other information would you want to have to make this decision?

5.2) Say you conducted a resampling experiment like those described in this chapter and found that your particular network showed huge fluctuations in betweenness centrality when only a small fraction of the nodes or edges were removed at random. What would this tell you about the robustness of degree centrality in your particular network to the same kinds of perturbations?

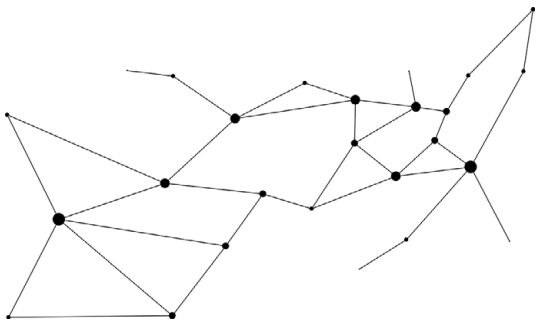


FIG. 5.17 Hypothetical road network with nodes scaled based on degree centrality.

5.3) Imagine you have defined a road network among ancient settlements based on remotely sensed trails and roads across a large region. In your network, nodes are defined by recorded settlements and edges are defined by the roads or trails detected between them (Fig. 5.17). Assume that the region has seen extensive full coverage archaeological survey so you can be reasonably confident that most settlements and features have been recorded, but some may have been destroyed or obscured by contemporary development. In this investigation you are interested in evaluating whether network degree centrality is associated with settlement growth and prominence in this study area. Assume you have attribute data for each settlement, including size and other architectural characteristics.

Questions

a) If you were interested in understanding the potential impact of missing edges in this dataset, would you model this process as a random sampling process or a biased sampling process? Why?

b) Once you have determined how you will model missing edges, describe the approach you would take to assess the impact of missingness on your metric of interest (degree centrality).

c) If you were to find that degree centrality fluctuates considerably in relation to the process of node missingness you model, how would you adjust your assessments of your results and your interpretations?

5.4) Say you conducted a resampling experiment on a large undirected binary network (150 nodes) to evaluate the robustness of the network dataset to *nodes missing at random* (and let us assume for the purposes of this question that "missing at random" is an appropriate choice). For your experiment assume you created 1,000 replicates at subsamples of 90 percent, 80 percent,

and 70 percent of the total nodes. Say you then compare centrality scores for each subsample to the original network using Spearman's ρ rank-order correlations as described in this chapter. If you got the results shown in Figure 5.18, how would this influence your interpretations of variation in centrality among nodes in your network?

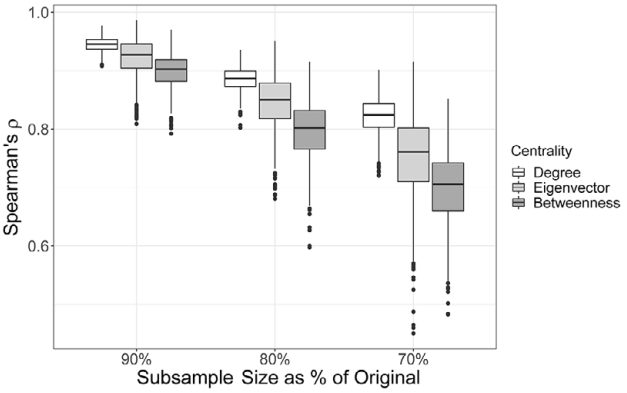


FIG. 5.18 Results of the hypothetical resampling experiment in Exercise 5.4.

6

Network Visualization

6.1 A PICTURE IS WORTH A THOUSAND WORDS

When many people (network researchers included) think about networks, the first thing that pops into their head is the classic network node-link diagram. In its simplest form a network graph is just a collection of points on a page representing entities of some sort with lines drawn to indicate the connections among those entities. Network visuals can be small and include only a few actors and relations where structural patterns and positions can be clearly observed. They can also be dizzyingly complex bundles of thousands, tens of thousands, or more entities and connections where general textures of relations and topological features might be visible but the positions of most nodes and edges are obscured by complexity. In either case, such visuals can paint a fascinating picture of a dataset and help a researcher recognize, interpret, and explain patterns in all manner of relational data that would otherwise be difficult to identify or communicate even with the myriad of network metrics available.

Linton Freeman (2000) published a history of network visualizations in the social sciences and how such images have been used to make substantive arguments about relational data since the emergence of sociometry in the early 20th century. As Freeman notes, some of the earliest network diagrams were simply hand-drawn images with nodes placed as the researcher saw fit (perhaps based on some attribute of each node or their perceived importance within the network) with lines or arrows sketched between them based on the relationships under consideration (Fig. 6.1). These visualizations were increasingly elaborated throughout the 20th century as researchers sought new ways to identify and emphasize particular kinds of network structures or the relative importance of specific actors within networks. Some of the most fascinating efforts even included the construction of physical representations

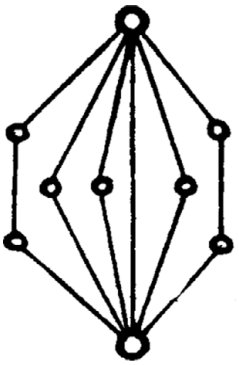


FIG. 6.1 An example of an early hand-drawn network graph (sociogram) published by Moreno (from Moreno 1932:101). Moreno noted that the nodes at the top and bottom of the sociogram have the most connections (highest degree centrality in contemporary network parlance) and therefore represent the nodes of greatest importance. These specific "important" points are emphasized through both their size and their placement.

of networks using creative materials like rubber bands and pushpins (Northway 1953) to complex three-dimensional abacus-like devices that allowed researchers to change the positions of actors for the purposes of exploratory analysis (Chapin 1950; see Freeman 2000:fig. 10). Many early and influential researchers such as Jacob Moreno (1932, 1934) saw the process of creating and exploring visuals as a key part of network analysis that went far beyond painting a pretty picture. As Moreno (1953:95) wrote, a network graph (sociogram as he called it) "is more than merely a method of presentation. It is first of all a method of exploration. It makes possible the exploration of sociometric facts." Thus, from the very beginning, visualization was seen as an *essential* part of the analytical process.

In the years since these early efforts, a wide variety of methods for visualizing networks has been developed focused on both static and dynamic displays. Importantly, there is no single "best" network visualization method and different approaches are better or worse at doing different things. By way of example, Figures 6.2 and 6.3 show several different approaches to visualizing the same network data. The data shown here come from Peeples's (2018) work and nodes represent 13th- and 14th century settlements in the Cibola region of the US Southwest with ties among them based on strong similarities in the technology of cooking pots produced and used at those settlements (see Section 2.8.3). Each of these plots emphasizes different features and attributes of the nodes and edges, different structural positions, different node and edge placement techniques, and different graph-level properties.

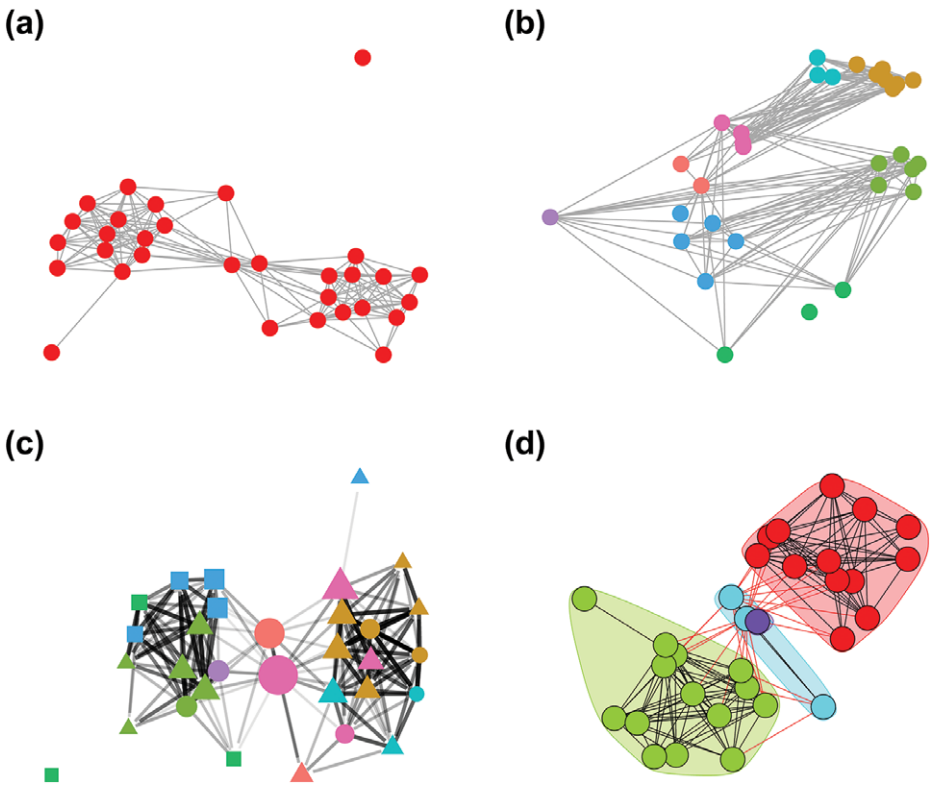


FIG. 6.2 These plots are all different visual representations of the same network data from Peeples's (2018) data where edges are defined based on the technological similarities of cooking pots from each node, which represent archaeological settlements.

Figure 6.2a shows a simple network graph with nodes placed using a force-based layout algorithm (see Section 6.2.4) designed to group nodes with similar edges. Figure 6.2b shows the same data with nodes placed by their geographical locations and color-coded by subregion. Figure 6.2c illustrates how much information can be combined into a single figure. In this network nodes are color-coded based on region, with different symbols for different kinds of public architectural features found at those sites, with node size scaled based on betweenness centrality scores, and with line weight for each edge used to indicate relative tie strength. Figure 6.2d shows one way to illustrate groups or communities within a network with nodes color-coded and outlined based on communities detected using the Louvain community detection algorithm (see Section 4.4.6). Edges within communities are shown in black and edges between communities are shown in red.

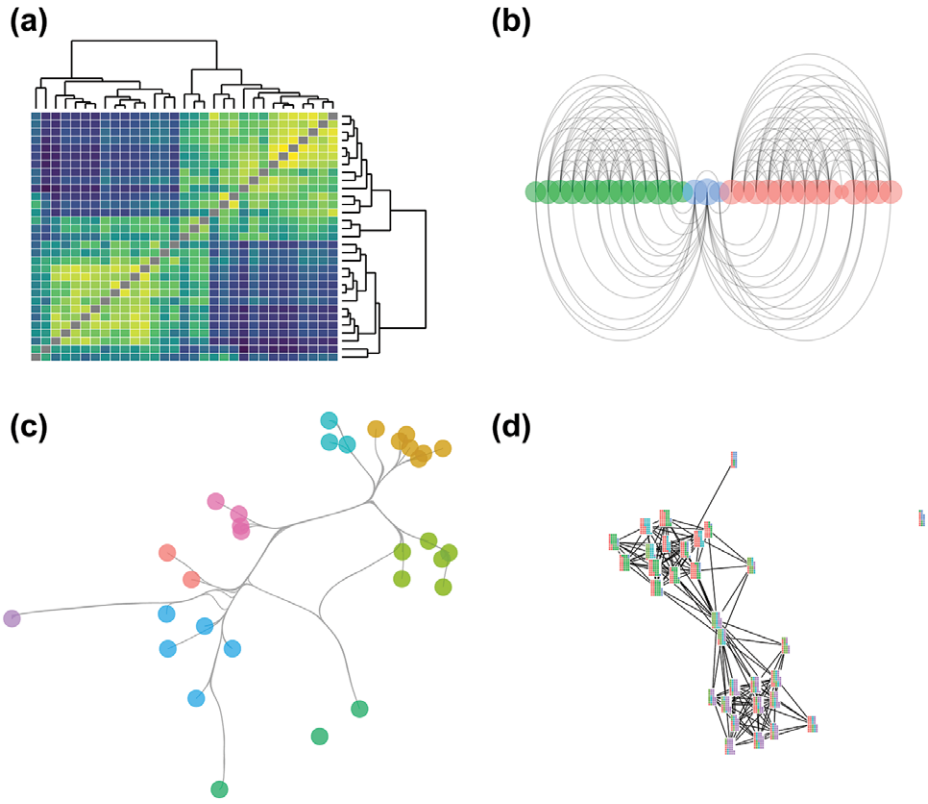


FIG. 6.3 Examples of less common network visuals techniques for Peeples's (2018) ceramic technological similarity data.

To take this example even further, Figure 6.3 illustrates a few of the less frequently employed options for presenting network data. Figure 6.3a is a symmetric heatmap, ordered by a cluster dendrogram shown on each axis, where each square represents the degree of similarity for a given pair of nodes. Lighter colors represent greater similarity. Figure 6.3b is an arc plot where nodes are shown along the horizontal axis with all ties among them displayed as arcs above or below that axis. In this example, nodes are color-coded by Louvain cluster membership. Between-group ties are shown below the line, while within-group ties are shown above. Figure 6.3c has nodes color-coded and positioned by subregion with the edges bundled using an algorithm called hammer bundling, which uses splines to pull together edges that have similar paths. This approach is designed to help reduce some of the visual complexity of networks with many overlapping edges. Finally, Figure 6.3d replaces the simple symbols for nodes with waffle plots that

represent the proportions of the five most common technological categories present at those sites.

The examples briefly shown here are quite diverse despite all being derived from the same underlying data. One might reasonably draw different conclusions about the relations in this setting based on any one of these plots in isolation. The decision regarding which visual(s) we might pick for a given context is largely determined by the questions we ask and the arguments we wish to make with the data.

Although we must often pick a small number of visualizations for presentations and publications, we frequently explore dozens of different approaches to visualizing our network data in the process of analyzing a given dataset. Indeed, network visualization should be, we argue, a dynamic exploratory process. Once you have your relational data gathered and defined, you can start by displaying it with a specific layout algorithm. Does that layout provide any particular insights in relation to the question at hand? Try another layout algorithm. Did you learn anything new? What about changing node symbols based on attributes or based on some network property? What would we learn if we were to explore these data in a different format like a heatmap or adjacency matrix? Having specific hypotheses and network dependencies to test is important, and we would not advocate that network analysis in archaeology be *only* exploratory, but we also suggest that such exploration is still quite valuable. In our experience, the more experimenting with visualization techniques like this you can do, the more familiar you will become with the intricacies of your complex relational data and the better equipped you will be for more detailed statistical analyses and formal testing of network theories.

Beyond this, we have stumbled upon new and interesting research questions that we had not initially considered through visual exploratory analysis. For example, when the Southwest Social Networks Project (see Sections 2.8.2 and 8.1) was first getting off the ground, the team produced a series of simple network visuals to explore and better understand this new database and the complex relations among settlements in terms of ceramic similarity. In the process of examining these early visual experiments, Peeples noticed that the names of many of the sites that appeared to be in intermediate positions between network clusters happened to have "Pass" or "Gap" in their names (e.g., Chavez Pass, Eagle Pass, Largo Gap). This started a new line of inquiry into whether sites in certain kinds of intermediate physiographic locations (hence the Pass/Gap in the names) more often served as brokers that mediated relations among sets of settlements that were otherwise weakly connected. This very simple exploratory insight eventually spun out into multiple formal analyses and publications that tested specific network theories (Peeples and Haas 2013; Peeples and Mills 2018), but it all started with an

exploratory visualization and a conversation among the team members. There is really no substitute for combining visual exploration with your own domain-specific knowledge.

When the time comes for presenting and publishing your network research and sending it out into the world for others, the goal is no longer simply exploration but instead efficient and accurate communication. It is important that you consider what kinds of visuals will provide the best insights into pivotal aspects of your results. Remember, you may have done lots of exploration but your audience is likely far less familiar with your data. What kinds of visuals can you produce that will quickly and efficiently communicate or supplement your arguments while not misleading them or causing them to make unfounded assumptions? The key here is making sure that your visuals, your text or presentation, and your statistical analyses are all in sync. When done well, network visualizations can be both effective tools that provide information to complement other statistical and topological analyses and descriptions as well as beautiful representations of your data (see discussion in Bach and van Garderen 2023).

There are many kinds of network relationships, structures, and topological features that are important in developing interpretations and explaining patterns of interaction in networks that are difficult to capture in traditional network metrics but relatively easy to communicate visually. For example, many networks have nodes that fall into bridging positions that connect sets of entities that would otherwise represent distinct groups. In a large network we can relatively easily identify which nodes are most frequently in such intermediate positions using different kinds of structural position metrics like betweenness and brokerage (see Chapter 4). At the same time, if we wanted to know which specific subgroups were connected by *specific* bridges, a visual is likely to be much more effective and informative than simply more metrics.

For example, Figure 6.4 shows a network of Clovis era sites (ca. 13,350–12,800 cal BP) in western North America with edges defined between pairs of sites with lithic raw materials sources in common (see Buchanan, Chao et al. 2017; Buchanan, Andrews et al. 2019; Buchanan, Hamilton et al. 2019 for the original studies where these data were used). In the plot here, network communities defined using the Louvain modularity clustering method (see Section 4.4.6) have been color-coded, outlined, and labeled, and node size is scaled based on betweenness centrality (Section 4.4.7). The seven sites with the highest centrality scores have been identified with labels. Note that several of the high centrality sites are involved in shortest paths between identified communities, whereas others (Blackwater Draw, Anzick, and Crook County) are along shortest paths within communities. This visual

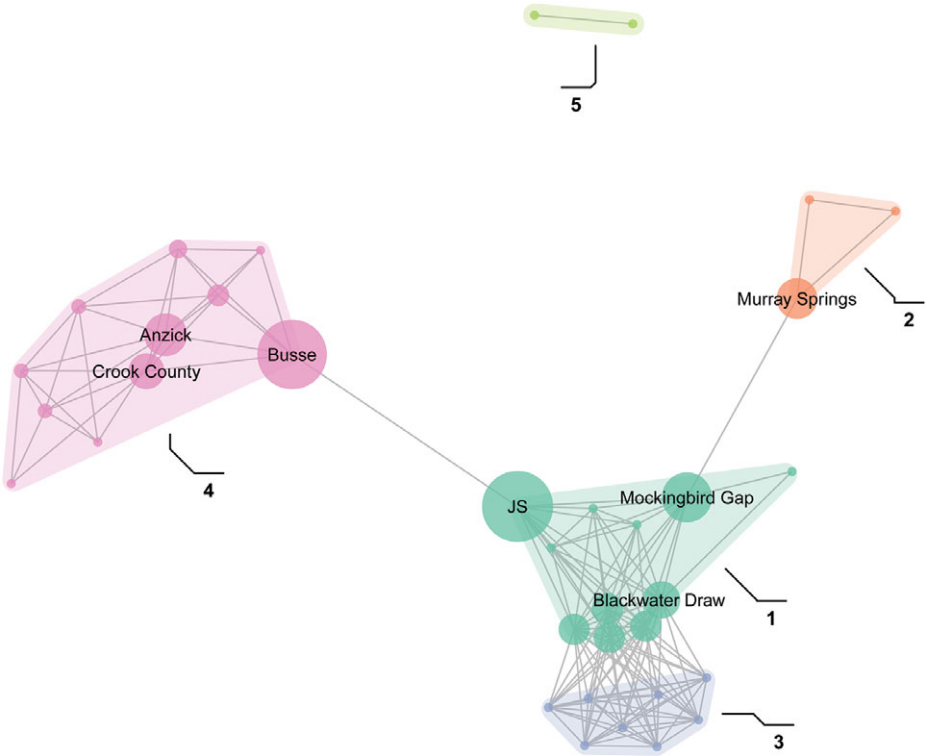


FIG. 6.4 A network among Clovis-era sites in the Western United States with connections based on shared lithic raw material sources. Nodes are scaled based on betweenness centrality with the top seven sites labeled. Color-coded clusters were defined using the Louvain algorithm.

provides an efficient way of communicating this insight, which would be difficult to illustrate with centrality metrics alone. Learning when to rely on visuals and when to rely on metrics is as important a skill as learning how to apply the statistical and descriptive techniques in this book.

The purpose of this chapter is to provide a general introduction to some of the most useful tools and approaches for network visualization you should have at your disposal. The bulk of this discussion is focused on providing guidance for some of the specific decisions you will make in the process of creating a network graph or related visuals: (1) node placement and graph layout, (2) visualizing node and edge properties or attributes, (3) visualizing network communities and groups, (4) visualizing networks through time, and (5) interactive network visualizations. After discussing these issues in detail, we then return to a broader conversation regarding the thought process behind how you might combine these techniques to

effectively communicate your research results using a case study from the Southwest Social Networks Project. Throughout this discussion we hope to emphasize that, although visuals can be stunningly beautiful on their own, they work best when they are clearly connected to and complement a specific research objective and other statistical analyses.

6.2 NODE PLACEMENT AND GRAPH LAYOUT

Graph layout simply refers to the placement and organization in two- or three-dimensional space of nodes and edges in a network. There are many different approaches to both manual and algorithmic graph layouts, and these approaches all have advantages and disadvantages. In this section, we outline some of the most common approaches.

> *Graph layout* refers to the specific assumptions and procedures used to place nodes and edges of a network in two- or three-dimensional space.

6.2.1 *User-Defined Layouts*

Perhaps the simplest approach to defining a graph layout was also the first used by early network researchers in the 20th century. This approach simply entails placing nodes and their corresponding edges manually to produce a layout that emphasizes a particular point of interest. Many contemporary software packages allow the user to do this in real time, facilitating exploration. This approach has some merit in very small networks where a specific point of interest needs to be emphasized or a particular graph aesthetic needs to be achieved (Fig. 6.5), but in larger networks this is usually untenable. In practice, this approach may be used in a minor way even in very large networks to slightly adjust the position of nodes to ensure that labels, arrowheads, and other features of interest are visible even when another approach is used for the general layout.

6.2.2 *Geographic Layouts*

In many archaeological networks the nodes themselves and perhaps also the edges have some physical positions or paths that can be represented in geographic space.

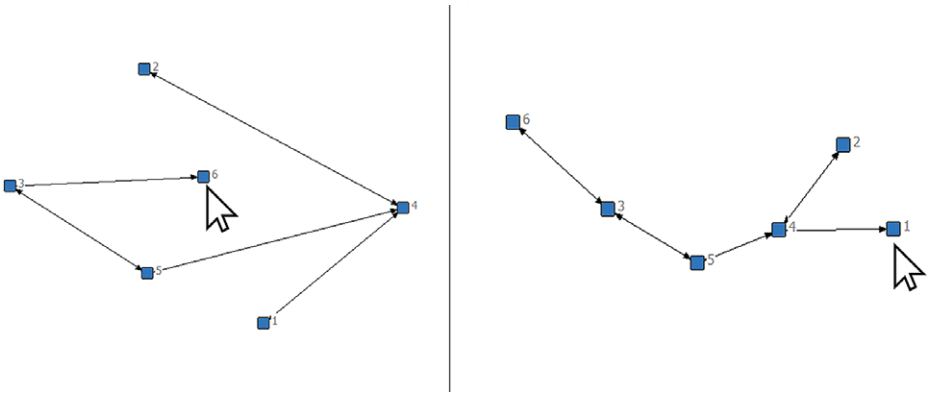


FIG. 6.5 An example of the same network graph with two simple user-defined layouts created interactively.

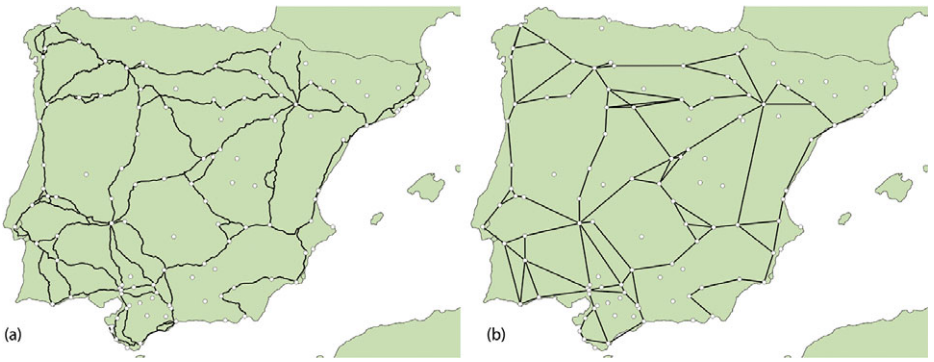


FIG. 6.6 Map of major Roman roads and major settlements on the Iberian Peninsula, with (a) roads mapped along their actual geographic paths and (b) roads shown as simple line segments between nodes.

For example, if nodes in a network represent settlements or structures within a site, placing the nodes in those geographic locations makes some intuitive sense as we are often interested in the geographic character of networks. Edges between such geographically situated nodes can be represented as straight lines between them or paths that follow other geographic features such as roads or trails or even modeled using GIS tools like least cost paths (Fig. 6.6).

Although geographic presentations of networks can provide an excellent means for displaying analyses that relate network properties across space, there are often substantial challenges that need to be dealt with. For example, many of the kinds of geographic features that archaeologists study tend to be situated along linear features like coastlines, rivers, or roads. This presents a particular challenge as

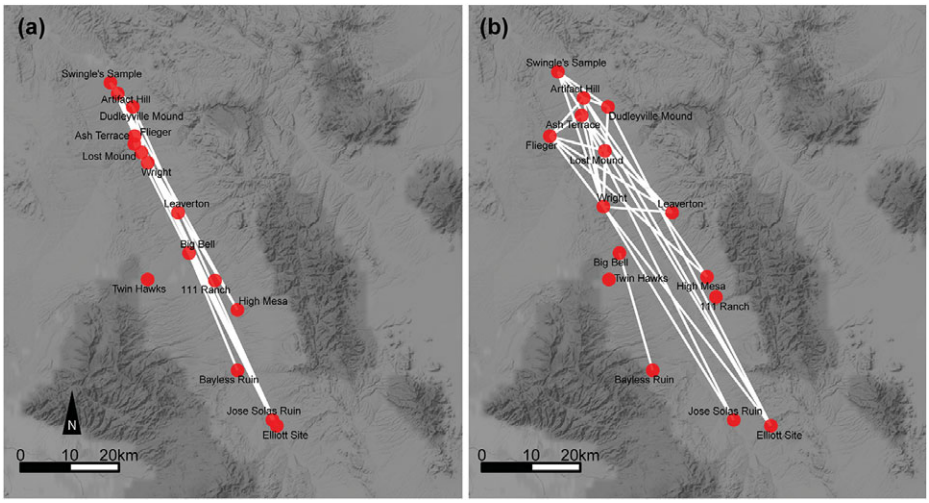


FIG. 6.7 This ceramic similarity network of the San Pedro River Valley in Arizona shows the challenges of creating geographic network layouts. Panel (a) shows sites in their original locations, whereas panel (b) shifts locations to improve the visibility of network structure. Note how the distorted geographic layout retains the basic relationships among the nodes while altering their locations slightly.

interconnections among sets of nodes all falling in a line will often overlap, making it difficult or impossible to determine which nodes are connected by edges and which are not. In such cases, it may be advisable to modify the geographic positions of nodes slightly to ensure visual clarity (Fig. 6.7). This is not unlike the many public transportation maps for buses, trains, and subways you have likely seen that place locations roughly in relative geographic space while modifying and warping that geography or routing edges for the purposes of legibility (Böttger et al. 2008; see also Gastner et al. 2018).

Schöttler and colleagues (2021) provide an essential resource in the consideration of the graphic design for geospatial networks based on a review of 95 recent publications on the subject. In this overview, the authors note that geographic networks range from those that explicitly retain the exact locations and paths, or that are distorted and modified to serve the purpose of visual clarity, to those where spatial relationships are retained only in an abstracted way. Each of these approaches has advantages and disadvantages. For example, **geographically explicit network graphs** preserve locations and can reveal information about real distances among nodes, but they are also prone to visual noise when nodes are clustered in similar locations or along similar axes as discussed above. **Distorted geographic network graphs** retain some of the features of mapped networks such

as the ability to identify patterns among adjacent areas, while they lack some of the geographic specificity of explicit location graphs. Schöttler and colleagues (2021) suggest that such intermediate approaches are often a trade-off between showing location or network topologies with the maximum clarity and accuracy. **Abstract geographic network graphs** refer to network visuals that retain only general geographic information but can sometimes provide insights that would otherwise be difficult to communicate. For example, a network with nodes clustered by region could be useful in exploring the difference within and between regions even though the positions of nodes are not strictly geographical.

Explicit geographic layouts are graph layouts which retain the exact locations of nodes and/or edges in geographic space in two or three dimensions.

Distorted geographic layouts retain certain features of relative spatial relationships among nodes and edges while modifying those locations and relationships for the sake of visual clarity or to emphasize particular topological features of a graph.

Abstract geographic layouts do not explicitly reproduce spatial relationships among nodes and edges but retain features like spatial groupings in determining positions so that geographic insights are still possible.

6.2.3 *Shape-Based Layouts*

Several graph layout approaches are designed to simply place nodes within some specific shape configuration and draw the edges between them either directly or using an approach like edge bundling (see Section 6.3.6). The most common variations of this approach are circular layouts and grid layouts. In general, these simple shape-based layouts have very specific applications where they are useful, but they are rare in other settings. Circular layouts are most frequently used to display dense networks with little substructure, to display simple network topologies like star or ring networks, or to evaluate properties like regularity in small networks, but can also be useful for hierarchical networks as discussed further below. Grid-based layouts place individual nodes on a regular (or modified) grid and draw connections among them. Grid-based layouts are relatively rare but can be useful in displaying data that are planar, meaning that a given graph can be configured such that edges only intersect at nodes (see Section 7.3). Related to these

shape-based algorithms is what is often referred to as an arc diagram or arc plot. This is essentially a linear layout where nodes are placed in a line and connections among them are drawn as smooth arcs that connect those sharing an edge (see Fig. 6.3b). In all of these approaches the specific order of nodes is key to making a visually compelling figure. For example, in the arc plot shown in Figure 6.3b, the sites are ordered based on the Louvain cluster membership of each node such that within-group edges are clearly visible, as are the small number of bridges between them. Although this figure is still visually complex, a different ordering would make this pattern more difficult to recognize.

6.2.4 *Algorithmic Layouts*

Many of the most frequently used graph layouts are defined algorithmically based on relationships within the network itself. There are a variety of different approaches, some based on ordination tools commonly used in other areas of archaeological research for things like seriation and some based on techniques specific to network data. In this section we outline a small number of the most common approaches that are implemented in many network software packages but note that there are many more approaches to choose from. Drawing graphs that are accurate, efficient, and pleasing to the eye is difficult, particularly so with larger graphs. In this section we present a series of examples all using data previously published by Evans, Rivers, and Knappett (2012), which represent a network of distance across land and sea for a series of 39 Bronze Age settlements in the Aegean (see Knappett et al. 2008 and 2011 for published analyses of these settlements). Here we have created a simple network that defines an edge as present for all sites that are less than 124 kilometers from each other over land or sea (this is a maximum distance network as described in Section 7.5). The distance cutoff selected is somewhat arbitrary, but for the purposes of the visualizations shown here we chose the minimum distance required to create a single connected component among all 39 sites. This example was selected because it displays community structure and is a medium-sized network and can help us demonstrate the strengths and weaknesses of different visualization techniques for networks of various sizes. Figure 6.8 shows these data with nodes placed first geographically and then using a range of different algorithmic layouts described below.

One common graph layout method relies on metric **multidimensional scaling** (MDS) to place nodes in space. MDS is a common approach to displaying

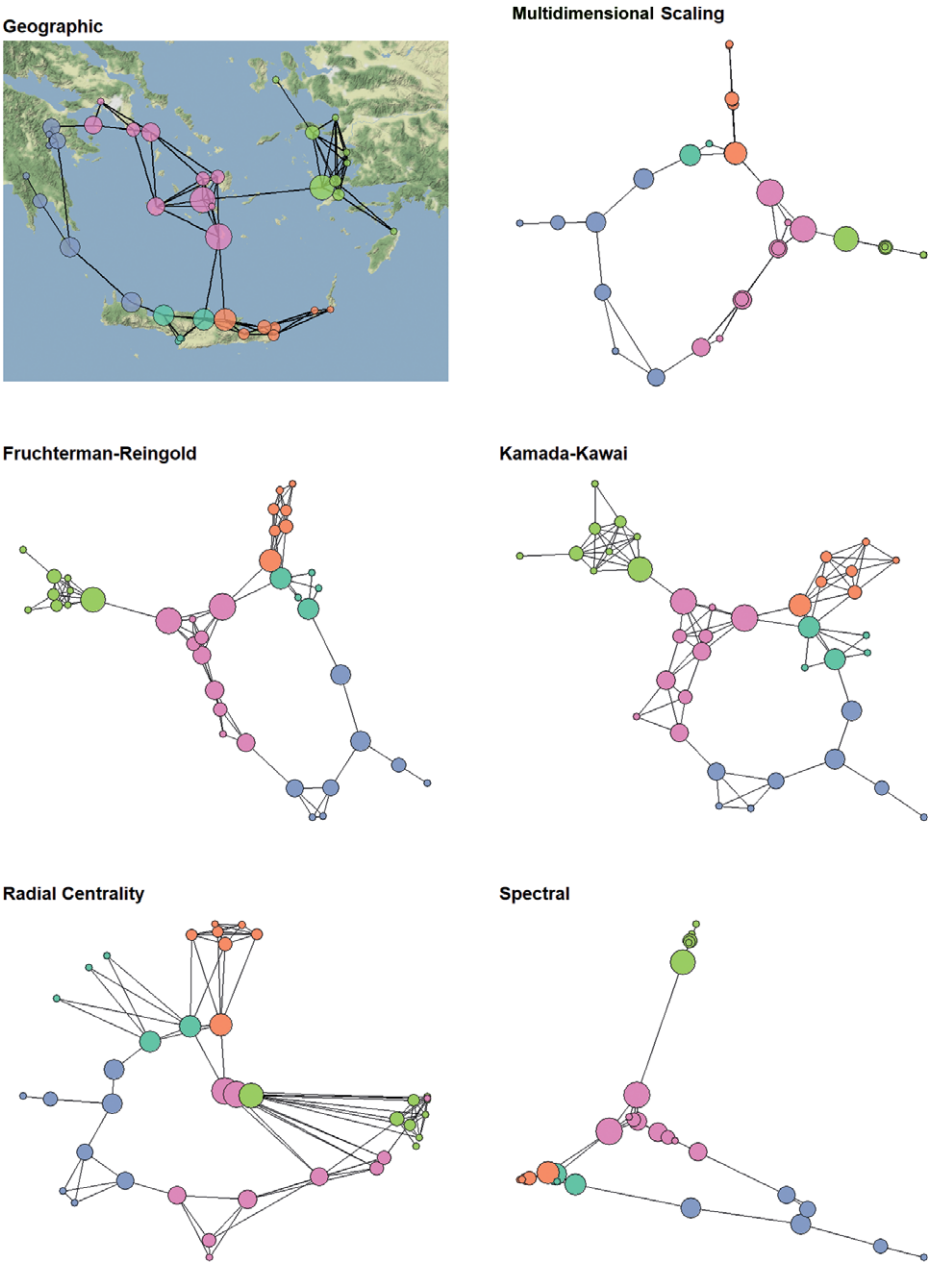


FIG. 6.8 Several different graph layouts all using the Bronze Age Aegean geographic network (Evans et al. 2012). In each graph, nodes are scaled based on betweenness centrality and color-coded based on clusters defined using modularity maximization.

high-dimensional data in a two- or three-dimensional Euclidean space, which attempts to create new axes that retain the relative distances (in terms of network relations) among points across all dimensions to the greatest extent possible (see Cox and Cox 2008). In general, MDS produces clusters of nodes that are similar in terms of their relationships to other nodes. This approach can be useful in large networks in emphasizing group structure, but the "clumping" of nodes that it often produces can sometimes make it difficult to visualize connections within dense clusters of similar nodes.

Another class of graph layouts that are frequently used are referred to as **force-directed algorithms** (see Kobourov 2013). These methods have rapidly grown to prominence in the graph visualization world as they provide results that have proven useful, intuitive, and flexible for networks of a range of sizes from small to large. These approaches often provide reasonable characterizations of topology and density both locally and globally. There are many different kinds of force-directed algorithms – and the specific mathematical details that differentiate them are beyond the scope of this book – but there are a few features that connect all of these approaches worth mentioning here. Specifically, force-directed algorithms typically attempt to create a node layout in two- or three-dimensional space that allows for edges of roughly equivalent length. This is often described using a physical "spring" analogy (Michailidis 2008:109–110). Imagine a set of nodes and the edges connecting them as springs of approximately the same length. The goal of force-directed layout algorithms is to find a configuration of nodes that minimizes the imbalance of stress placed on any particular "spring" in relation to the repelling and attracting forces among nodes in the system both locally and globally under some set of constraints.

These force-directed approaches work by iteratively moving points and calculating the energy or stress in the combined system under a set of rules and then stopping when some threshold value is reached. This means that force-directed algorithms involve a degree of stochasticity and will not necessarily provide identical layouts when run for the same input data twice. In practice, the layouts obtained between multiple runs will typically be similar, though the orientation and direction may vary arbitrarily. Importantly, some software implementations of these algorithms start with the same initialization every time (thus producing more similar results between runs), while others do not, so you should consult the documentation for whatever package you use.

Three commonly used force-directed graph layout algorithms include the Kamada-Kawai algorithm (Kamada and Kawai 1989), the Fruchterman-Reingold

algorithm (Furchterman and Reingold 1991), and the stress majorization algorithm (Gansner et al. 2005). All of these work under the same basic principles but they have slightly different stress or energy functions and constraints. The Kamada-Kawai and stress majorization algorithms are similar in that they both work to find an optimal solution where edges between nodes are proportional to the distance between those nodes in the graph, but the two methods differ in their approaches to optimization (see Gansner et al. 2005). The Fruchterman-Reingold algorithm is an iterative approach that attempts to move nodes and edges in a series of steps such that the forces among nodes are minimized. The computational details of these approaches are complex, but all of these methods produce results that are well suited for visual communication in that nodes with similar sets of edges will usually be close to each other and the number of edge crossings will be minimized to the extent possible. The major drawback of force-directed algorithms, especially for large networks, is that they can be computationally expensive to calculate and it can sometimes be difficult to visualize structure within communities in large graphs while also representing the relationships between those communities.

In some situations, you may be interested in highlighting certain nodes based on a characteristic like centrality. In such cases a **constrained radial layout** focused on the characteristic of interest is often a good fit (see Brandes and Pich 2010). For example, if we are interested in betweenness centrality we can define a layout where the highest centrality nodes are placed in the center and nodes are placed at increasing distances from the center as betweenness declines with nodes arrayed radially along each concentric band such that there are relatively few edges that span across the midline of the network. This approach works well when the primary goal of the visualization is to emphasize the connections for a few key nodes with a given metric in either large or small networks, but this layout potentially makes visualizing specific edges difficult.

Another approach worth considering is the **spectral layout algorithm**. This approach positions nodes using the eigenvectors of the underlying adjacency matrix such that nodes are located in low-dimensional space at the centroid of their neighboring nodes (Koren 2005). This approach tends to produce clusters of nodes with similar ties but, much like with the MDS method described above, often makes visualizing specific relations and local topologies difficult due to the clumping together of similar nodes (Fig. 6.8f). This method is often applied to very large networks where the goal is not necessarily visualizing local structures but instead focusing on overall graph-level properties like clustering or core-periphery structure.

Multidimensional scaling (MDS) is an approach to graph layout that attempts to display high-dimensional network data in low-dimensional space while retaining the relative distances among nodes.

Force-directed graph layout algorithms are a set of approaches to defining graph layouts in low-dimensional space by assigning repelling and attracting forces to nodes and edges, and finding a layout that minimizes their energy under specific constraints.

Constrained radial graph layouts refer to an approach to graph construction where nodes are positioned based on some metric of interest with the highest value in the center with nodes in concentric circles around the center at increasing distances as the value of the metric declines.

Spectral graph layout algorithms use the smallest eigenvector of the adjacency matrix underlying a graph to create a low-dimensional node placement such that nodes are at the centroids of their neighboring nodes.

6.2.5 *Hierarchical Network Layouts*

When visualizing hierarchical networks (see Sections 4.4.4 and 7.3) many of the same approaches described above apply, but there are also several additional options (Fig. 6.9). Hierarchical data can be plotted as a typical network with any of the layouts described above with nodes displaying hierarchical level by color or size (Fig. 6.9a). We can also veer away from the strict graph-based visuals and create tree plots and "circle packing" plots that show the hierarchical nature of the data with nested subgroups scaled based on their relative size (Fig. 6.9b). Hierarchical data are well suited to dendrogram displays that highlight the levels of hierarchy and can also use branch length to provide some indication of the relative similarity or distance between levels of the hierarchy when such data are available (Fig. 6.9c). There are many styles of dendrogram with rectilinear or curvilinear branches, and even those are plotted in radial layouts for visual clarity. In some cases all junctures in the dendrogram can be labeled or perhaps more typically only the lowest level is labeled. One final visualization tool worth mentioning here for hierarchical data is the radial plot. These plots place all of the nodes of a hierarchical graph in a circular pattern and produce arcs showing the connections between levels in the hierarchy (Fig. 6.9d). Using edge bundling (see below) can also help to emphasize the hierarchical nature of the connections and the direction of connections between groups of nodes.

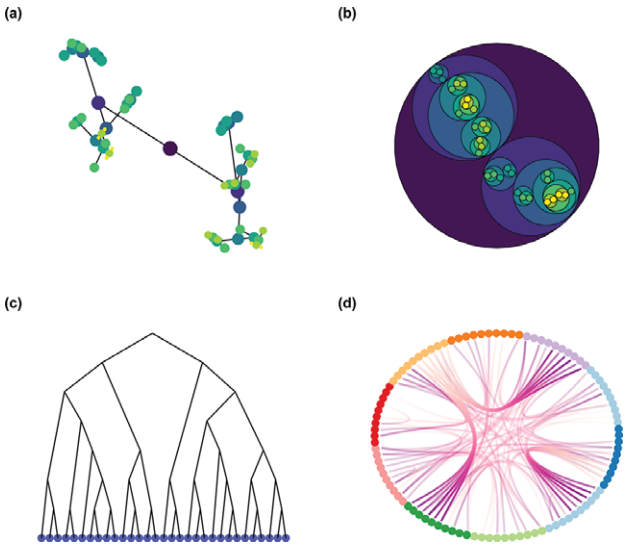


FIG. 6.9 Examples of visualizations based on hierarchical graph data. (a) Graph with nodes color-coded by hierarchical level. (b) Bubble plot where nodes are scaled proportional to the subgroup size. (c) Dendrogram of hierarchical cluster data. (d) Radial graph with edges bundled based on similarity in relations. Edges are color-coded such that they are red at the origin and purple at the destination to help visualize direction.

6.3 VISUALIZING NODE AND EDGE PROPERTIES AND ATTRIBUTES

One of the most common reasons that researchers employ network graph visualizations is to illustrate the identity and/or structural positions of specific key nodes or edges for a given metric or attribute of interest. This typically involves varying features of the network graph such as the size, symbol, or color of the nodes or edges according to some discrete or continuous classification scheme. Although many modern network analysis and visualization tools make it quite easy to make such changes, the specific decisions made can be the difference between a figure that provides novel insights and one that just confuses or misleads viewers. In this section, we discuss some of the most common approaches to visualizing node and edge attributes and properties and try to provide a set of best practices for producing impactful and enlightening visuals. As with all of the methods discussed in this chapter, the "best" approach depends on the question at hand and the kinds of information you hope to communicate (see Box 6.1 for important considerations about the use of color).

Box 6.1 BE KIND TO THE COLOR BLIND!

When selecting color schemes for your network graph it is important to consider the visual color sensitivity of your audience. As many as 5–6 percent of your potential viewership may have some kind of color vision deficiency, and certain combinations of colors are likely to be particularly difficult for these people. Luckily, there are many resources available online these days to help, including color vision–friendly color selectors and color palettes specifically designed to be viewable by people with several different kinds of color vision deficiencies (Adobe.com 2021; Palleton.com 2021; Tol 2021; Wong 2011). There are even ways to simulate different kinds of color vision to ensure you can reach the broadest possible audience. The plots shown in Figure 6.10 illustrate what a given color palette would potentially look like to people with three of the most common color vision deficiencies.

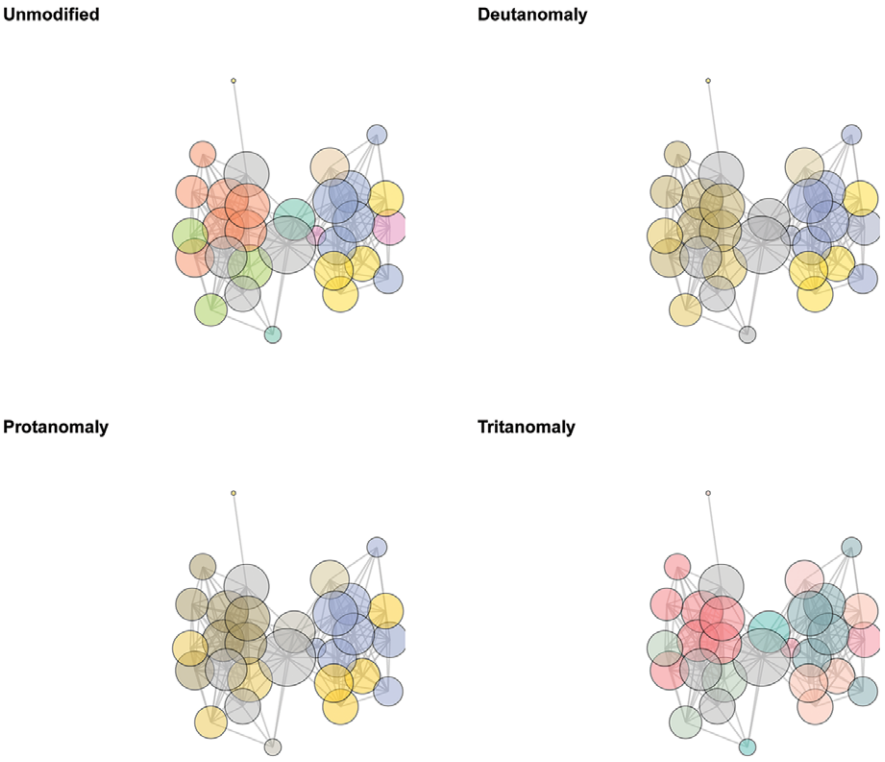


FIG. 6.10 Examples of a simple network graph with color-coded clusters. The top left example shows the unmodified figure and the remaining examples simulate what such a figure might look like to people with various kinds of color vision deficiencies.

6.3.1 *Node Symbol, Color, and Size*

Varying the symbol, color, shading, pattern, size, or other features of nodes in a graph are all useful ways to communicate differences among those nodes. Such differences could be specific network properties of the nodes such as centrality or they could be non-network attributes of those nodes derived from some other source (such as settlement size or date). It is possible, but perhaps not advisable, to vary symbol, color, and size all in the same graph with each indicating a different attribute or metric. For example, in Figure 6.2c, nodes are scaled based on betweenness centrality scores; different colors are used to indicate subregions across the study area; and different symbols are used to indicate the different kinds of public architectural features found at each site. We will be the first to admit that this is too much and not likely to be a useful figure for most purposes. So, with so many options available to us, how do we choose? There is, of course, the huge existing literature on visual communication and design that would be relevant here (e.g., Healy 2018; Lima 2013; Tufte 2001) and we can only scratch the surface in this chapter. There are a few common challenges that occur frequently enough in network graph design specifically, however, that merit some discussion here (Fig. 6.11).

First, when attempting to visualize a continuous metric like centrality or perhaps some attribute or property divided into a number of ordinal classes, it is sensible to use either size or color scale to help viewers grasp variation among nodes. This works best when there are large differences in the metric of interest we hope to communicate. In practice when there are lots of nodes and gradational variation among them, scaling nodes by size or color alone may not provide enough information. In such cases it can be beneficial to use both the size and color of each node together to visualize the same metric or variable (e.g., Fig. 6.11). This approach reinforces similarities and differences among nodes and helps to ameliorate known challenges such as the human difficulty in gauging relative differences in the area of shapes (Cleveland and McGill 1985). Of course, in some publication contexts color is not a possibility. In such cases variations in grayscale may have to do, and we have found combining variation in shade and size (or using textures or patterns) to indicate a common feature of the data to be particularly useful.

Symbol selection is another option for indicating differences among nodes by group or classification. We have found that varying symbols works best when the groups to be visualized are discrete and where no ordinality or relative scale is implied. Symbols then are perhaps most effectively used to differentiate non-network properties of the nodes such as site type or some other external

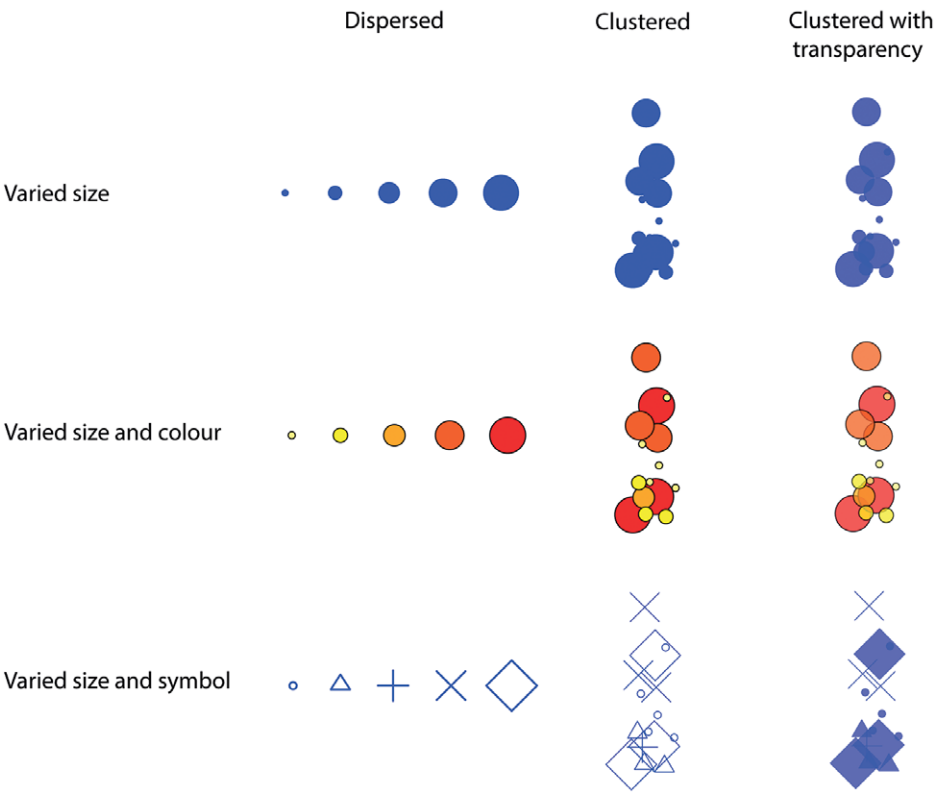


FIG. 6.11 Examples of different node color and symbol schemes. Note how adding color and size eases the identification of particular values, in particular with closely spaced points. Using transparency can similarly aid in showing multiple overlapping nodes.

classification. Importantly, combining different symbols *and* sizes is rarely a good idea in all but very small networks (and we broke this rule in Fig. 6.2c to demonstrate this point). The human eye has difficulty judging relative differences in the size and area of shapes even when the shape is held constant, and with multiple shapes we do not stand a chance. Beyond this, in situations where many nodes partially overlap, it can be difficult to differentiate symbols. The use of multiple symbols is therefore perhaps best for small networks where there are clear differences between the groups symbolized and relatively few overlapping nodes with the selected layout.

Finally, another issue that is frequently encountered in networks with lots of overlapping ties is the clumping together of nodes. This is a particular issue with many algorithmic graph layouts as discussed above as well as with explicit geographic graphs. When nodes are closely spaced it can be difficult to scale them to

indicate a metric of interest without overlap. One approach to addressing this challenge is to vary the *z*-layer on which nodes are drawn based on the attribute in question. This could involve, for example, drawing the highest centrality nodes above lower centrality nodes to emphasize them or to scale nodes based on the metric in question and plot the smaller values on top so that both the large and small nodes are simultaneously visible (as we have done in the examples in Fig. 6.11). This idea can be extended to three-dimensional plots, which can sometimes highlight variation difficult to capture in a single plane. Alternatively, showing overlapping nodes can also sometimes be achieved by varying the transparency of individual points so that we can see the overlaps directly.

6.3.2 *Node Figures or Graphics*

Moving on to some of the more exotic options available for depicting nodes in network graphs, it is also possible to replace nodes with custom graphics or even other data visualizations. For example, nodes in a network need not be simple shapes but could be images or icons that actually represent some feature of the node in question. This has been done particularly effectively in networks of artifact similarity such as the example presented in Figure 6.12, which shows edges representing similarities among a set of carved faces where the nodes in the network represent the specific objects in question (Mol 2014:237). This type of visual communicates a great deal of information efficiently and also allows viewers to directly assess some of the criteria that went into creating the network in the first place. Although we have not seen this approach widely used, graphical nodes like this would likely be particularly applicable for two-mode networks where different sets of nodes often represent different classifications of objects (Fig. 6.13). Why not include a visual cue in the graphic itself of the categories represented in the nodes?

Beyond nodes as graphics, it is also possible to replace nodes with figures or plots of data underlying that node. For example, in Figure 6.3d, the network of ceramic similarity among cooking pots is shown as a simple network graph with individual nodes depicted as waffle plots showing the relative frequencies of the most common technological categories in each assemblage rather than simple shapes. The waffle plots could be replaced by all manner of other plots like bar charts, pie charts, or anything else you can imagine. Figures like this are potentially useful in communicating lots of information in a small space, but such displays should be used sparingly and only in smaller networks as they can also add considerably to visual clutter and noise.

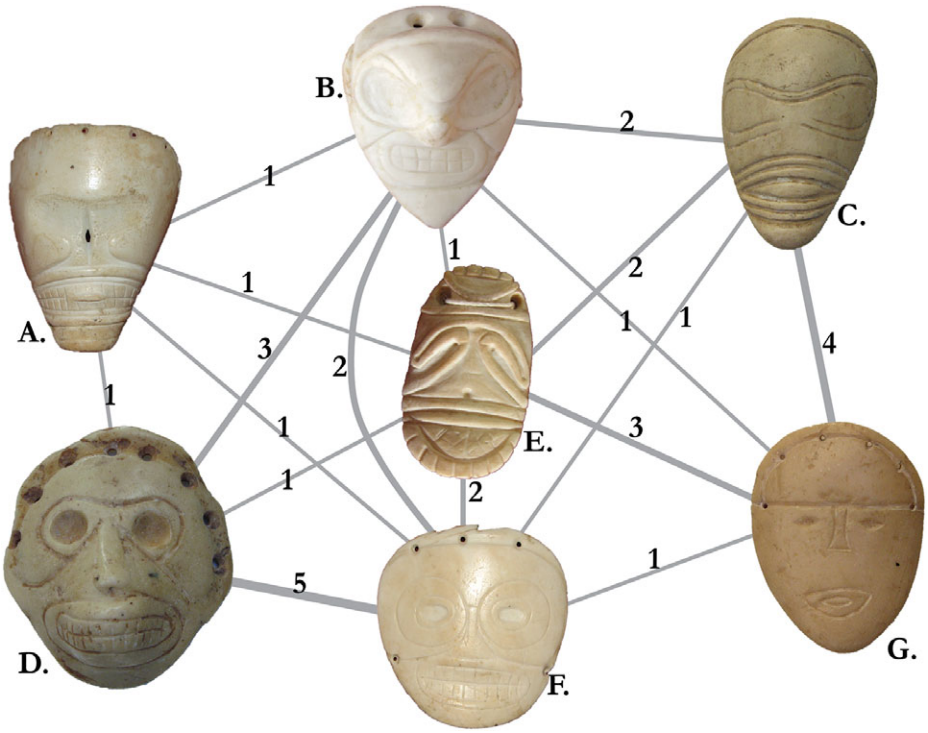


FIG. 6.12 Network graph showing similarity among carved faces from Banés, Holguín province, Cuba (from Mol 2014: fig. 8.6). Nodes are depicted as the objects in question themselves and edges represent shared attributes with numbers indicating the number of shared attributes for each pair of faces.

6.3.3 *Edge Line, Color, and Size*

As we noted for nodes above, we can also vary the line type, color, and size of edges in a graph to help emphasize a particular aspect of specific edges or to emphasize structural properties of the network as a whole. There are many options and we cannot hope to be comprehensive here, but we discuss a few of the most frequently encountered approaches and challenges.

Perhaps the most frequent way that edge variation is used in network visualization is as an indication of tie strength or edge weight. It is relatively common and intuitive to use line width or a color scale to indicate weight, whether this is a continuous or ordinal scale in a given network. Much as we suggested for nodes above, where color visuals are permitted, it is often useful to use *both* line color and line width to indicate weight or whatever other continuous/ordinal attribute is being indicated (Fig. 6.14). Ideally, such a color/weight schema should include an

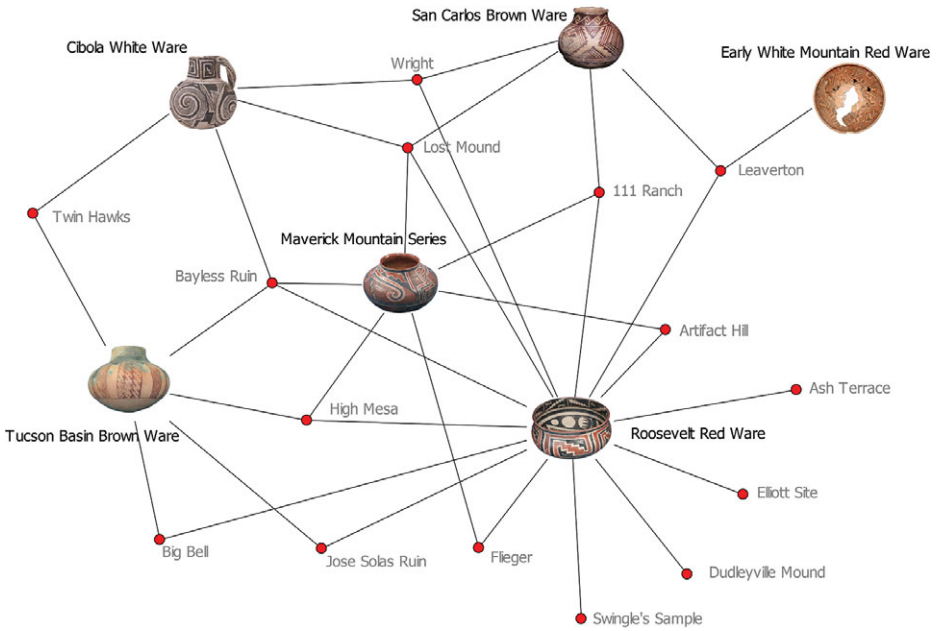


FIG. 6.13 Two-mode network of ceramics and sites in the San Pedro Valley with ceramic ware categories represented by a graphic example of each ware. Photo credits: Cibola White Ware, 2006.60.19.1,Tularosa black-on-white jar. Courtesy of Brigham Young University Museum of Peoples and Cultures; Tucson Basin Brown Ware, ASM 21232, Tanque Verde Red-on-Brown jar from Martinez Hill (AZ BB:13:3[ASM]). Photo by Patrick D. Lyons, courtesy of the Arizona State Museum, University of Arizona; Maverick Mountain Series, 90.5.822, Maverick Mountain Polychrome jar. Photo by Mathew A. Devitt, courtesy of Eastern Arizona College; San Carlos Brown Ware, 2015-517-93, San Carlos Red-on-Brown jar from Polles Mesa from the Robinson Collections. Photo by Kai Little; Roosevelt Red Ware, 88.1.8, Cliff Polychrome bowl. Photo by Mathew A. Devitt, courtesy of Eastern Arizona College; Early White Mountain Red Ware, 2004.2.11052, St Johns Polychrome bowl, Scribe S Pueblo CS12. Photo by Garrett Trask, courtesy of the Center for Archaeology and Society, Arizona State University.

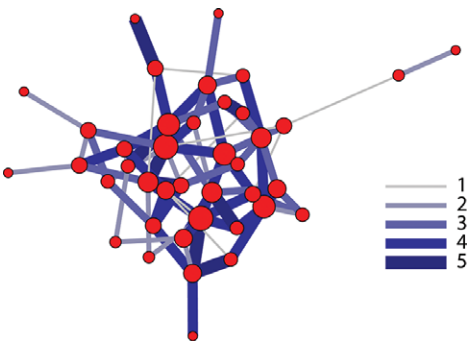


FIG. 6.14 A random weighted graph where edge line thickness and color are both used to indicate weight in five categories.

ordering of the edge *z*-layer according to edge weight and be associated with a clear legend as well so that viewers can quickly understand what these variations mean.

Another context where edge color or line type coding is often used is to differentiate edges within and between communities. In Figure 6.2d, edges within the same community are shown in black, while ties that cross community boundaries are shown in red. Even in relatively complex networks such a color scheme can often be helpful in identifying which nodes are involved in bridging ties between clusters or communities. In settings where color cannot be used, varying the line type (such as using dotted lines to indicate edges between communities and solid lines for edges within them) may be useful. We have found that varying line type works in practice only in small networks with relatively few overlapping edges.

It is also sometimes useful to vary the size or color of edges based on other non-network attributes of those edges. For example, Mills and colleagues (2013a; see also Fig. 6.27) produced a series of explicit geographic network maps of ceramic similarity from the US Southwest, showing edges that are color-coded such that longer distance ties are characterized by darker colors. On these networks they also used *z*-layer to help with the visibility of relations. Specifically, the shortest distance edges were shown above those longer distance ties in an attempt to make them both simultaneously visible when viewing the entire region. Notably, nodes were not shown at all in order to make the texture of relations more apparent and to limit visual clutter. This example is described in more detail below (see Section 6.7).

6.3.4 *Edge Direction*

Throughout this chapter so far we have focused on undirected networks where edges are assumed to extend in both directions. There are, of course, many different kinds of archaeological networks where the directionality of ties comes into play (e.g., the movement of sourced materials, connections through time, diffusion of ideas or materials, lines of sight, or down- and upstream river navigation), and many different possible approaches to visualizing such directed network paths and structures. The simplest solution to visualizing edge directionality is to place an arrowhead pointing to the destination node for a given relation. Just as we discussed above, the weight or some other feature of that relation can also be indicated by the size, color, or style of the line and arrowhead. When ties are bidirectional there will be an arrowhead on both sides of this edge (Fig. 6.15).

Another approach is to use not a single arrow but multiple arrows, each going in a different direction, to indicate edges with a particular direction. These could be parallel lines, or, to make it easier to identify uni- and bidirectional ties, these can also be arcs or

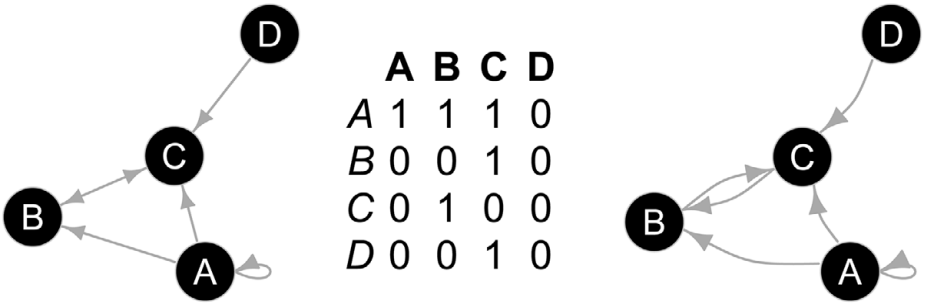


FIG. 6.15 Two methods of displaying directed ties using arrows (left) and arcs (right). Both of these simple networks represent the same relationships shown in the adjacency matrix in the center.

curves of various specifications (Fig. 6.15). In certain situations, self-edges or self-ties (ties from a node to itself) hold meaning and need to be visualized. In such cases it is typical to indicate self-loops with small looped lines that exit and then return to the node in question (with or without an arrowhead). In our experience, visualizing directionality is feasible only in small- to medium-sized networks and becomes increasingly challenging as network layouts create more and more overlapping edges. One potential solution for larger networks is to use color gradients where edges change color along the path from source to destination (see Fig. 6.9d). Such visuals may help to emphasize dramatic differences in direction between different parts of a network but in practice may add to visual noise so should be used only sparingly.

6.3.5 *Edge Binarization*

Many of the kinds of archaeological networks described in this book or found in the current literature are based on weighted data where relations between pairs of nodes in a network are based on some sort of metric of similarity or distance (e.g., Goliko et al. 2012; Mills et al. 2013a; Peeples and Haas 2013; Peeples and Roberts 2013). In such cases network metrics like centrality are often calculated on raw similarity scores among nodes rather than binary ties. In practice, however, it is difficult to visualize raw similarities among large numbers of nodes. In such cases, you may wish to create some threshold or criteria for defining an edge as present or absent *for the purposes of a visual* even if your analyses do not rely on those binary ties. For example, Mills and colleagues (2015) conducted an analysis of centrality by subregion at various scales using raw Brainerd-Robinson similarity scores among sets of sites. These similarity scores range from 0 (indicating no similarity) to 1 (indicating perfect

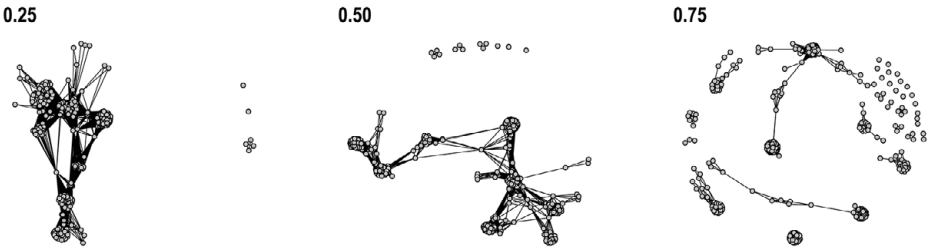


FIG. 6.16 These networks all show the same data based on similarity scores among sites in the US Southwest (ca. AD 1350–1400) but each has a different cutoff for binarization.

similarity). For the purposes of visualization and to aid in the narrative discussion of the patterns in the data for those not familiar with the context in question, Mills and colleagues chose to binarize edges with a similarity score of 0.75 or greater *for the purposes of visualization* while very explicitly noting that analyses were based on raw scores and not the binarized ties.

In many cases binarization is useful for creating compelling visualizations to accompany statistical analysis, but you should be aware that the specific decision you make to decide what kind of cutoff to define ties as either present or absent can have big implications for your network visuals (see also Section 3.3.5). There are no hard and fast rules here, though approaches we have found useful in the past include examining histograms of raw similarity scores to look for breaks, using quantiles of some other distributional cutoff, experimenting with cutoffs until you get a visualization that helps you emphasize your argument the best, or, where possible, simply providing representations of more than one cutoff. As Peeples and Roberts (2013) note, however, choosing a different cutoff will sometimes produce dramatically different visuals that may cause viewers to draw radically different conclusions about the underlying raw similarity data, so this is a decision that should be made with careful consideration (Fig. 6.16).

6.3.6 *Edge Bundling and Visual Complexity*

In recent years researchers working with very large networks have developed new methods to help visualize complex networks with lots of overlapping edges. One set of methods that is currently growing in popularity is referred to collectively as "edge bundling" techniques. This label actually refers to several somewhat different methods that work toward the same goal of reducing the visual noise and complexity of large graphs by joining together similar edges (Holten and Wijk 2009; Nocaj and Brandes 2013; Sun et al. 2019). Briefly, edge bundling refers to approaches that

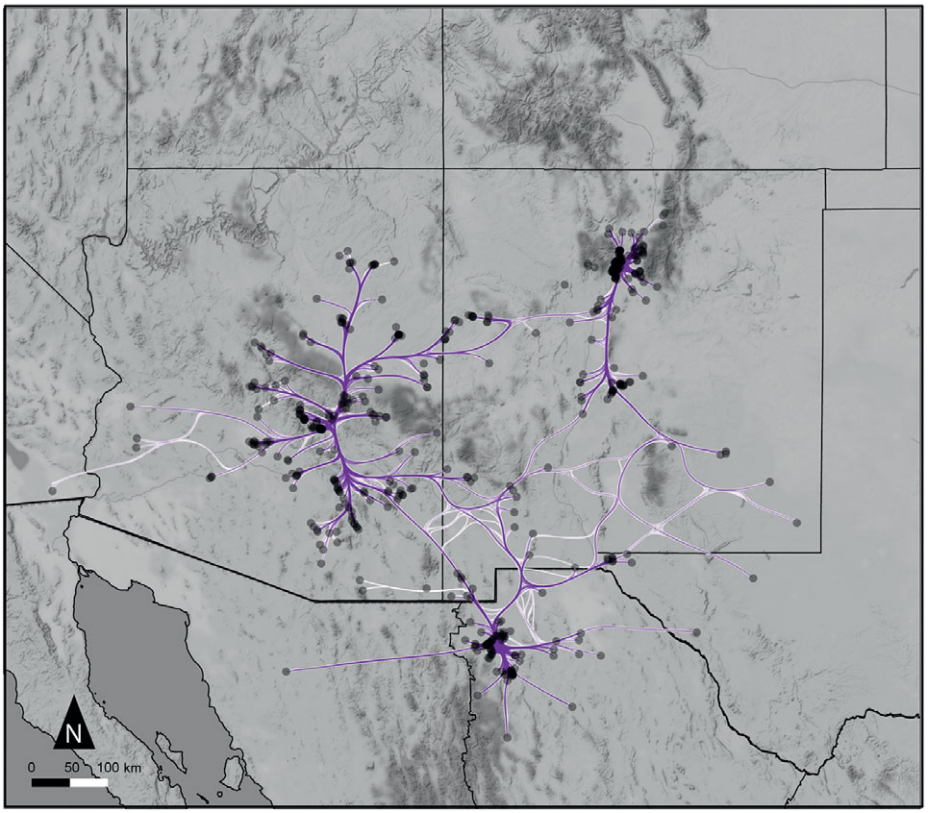


FIG. 6.17 Network map of ceramic similarity from the US Southwest/Mexican Northwest ca. AD 1350–1400 based on the hammer-bundling algorithm.

attempt to algorithmically route edges from nodes with similar sets of relations so that they run together along paths across the network and diverge again near the destination. This is a powerful method of data reduction that can sometimes reveal strong trajectories of connections across large networks that are otherwise buried in visual noise. In particular edge bundling methods have been used to visualize flows across geographically explicit networks to examine potential directional flows of interactions or paths across space.

Figure 6.17 represents a network map using what is called the "hammer" bundling technique to bundle similar edges along trajectories across the map (Schoch 2021). This approach has some promise for helping us deal with such large and complex networks, but we also note that the specific configuration of the edge-bundled map is highly dependent on the parameter values selected for the bandwidth and decay of the bundling algorithm, so this approach should not take the place of other methods of analysis and visualization. Still, we suggest that such approaches show promise and should be explored more in the future.

6.3.7 *Label Placement*

As we have seen in several of the examples in this chapter and throughout the book, it is sometimes useful to label nodes or edges with names or other information so that we can discuss them directly in text or compare visuals to tabular data or statistics. Simply labeling every node works reasonably well for small- to medium-sized networks with up to about a dozen nodes. Once we get bigger than that, however, it is very difficult to add labels without creating too much clutter or overlap. One solution is to label only those nodes that are directly referenced in the text. In Figure 6.4, for example, we indicated betweenness centrality only for the top seven nodes in the network as we were discussing only those high centrality nodes. In Figure 6.7 where we were comparing the geographically explicit network of the San Pedro Valley to the distorted geographic network, we chose to label every node so that readers could easily see the relative positions of each node in both graphs. However, if our goal was to simply show relational structures in the San Pedro Valley, it might have been best to omit the labels to make the edge more visible or to replace full names with letters or numbers and corresponding information in the figure caption.

In addition to labeling nodes, it is also common practice to label edges (e.g., Fig. 6.12) to indicate values of interest associated with them. Again, this can be useful when you want to discuss in detail particular edges or highlight variability in edge weights or other values, but this also very quickly adds to visual complexity and noise. In practice, we have found such labeling schemes work only in very small networks with few overlapping edges. There are, of course, no strict rules for deciding when to label and when to omit them, but in general, the larger the network, the less you probably should be labeling.

6.4 VISUALIZING COMMUNITIES AND GROUPS

Network graphs and related visuals can be quite useful in identifying and highlighting group or community structures (see Section 4.4.6) or evaluating the relationship between some external classification scheme and network topology. We have already seen several examples of this in the chapter so far and noted how different graph layout algorithms help to cluster (or distribute) nodes with similar structural positions. Along with tools for varying the size, color, and shape of nodes already discussed in detail above, graphs and related visuals can provide powerful tools for understanding community structure within a network. In this section, we highlight a few additional approaches to visualizing such groupings.

6.4.1 *Communities and Groups in Networks*

The previous section provides guidance on the selection of color, symbols, and other tools for displaying properties and attributes of nodes and edges in networks no matter what they represent, and those approaches are relevant to the kinds of communities and groupings discussed here. As several examples above show, groups or communities in networks, whether they are based on network topology or external attributes, are frequently indicated using color or symbols in network graphs. It is also common to outline members of such groups to emphasize community structures in a network (see Figs. 6.2d and 6.4). Importantly, displaying groupings effectively requires the selection of an appropriate graph layout algorithm that allows the efficient visualization of relations with and between groups in question and ideally reduces overlap between group boundaries to the extent possible.

In addition to the examples provided in previous sections, we also want to note that there is considerable room for the development of custom tools and layouts to help deal with the complex problem of evaluating groupings based on network topology and groupings based on external attributes at the same time. For example, Rodrigues and colleagues (2011) have developed a layout that they call "group-in-a-box," which places sets of nodes in adjoining boxes based on user-specified groupings or classifications. These boxes are scaled as a tree diagram based on the relative sizes of those groups. Figure 6.18 presents an example of this group-in-a-box technique using a ceramic similarity network with clusters defined based on the Girvan-Newman clustering algorithm (see Section 4.4.6) which are each placed within a box scaled based on the number of nodes in that cluster. In this version, nodes within each box are positioned using a Fruchterman-Reingold algorithm and the boxes themselves are treated as nodes and similarly positioned in relation to one another using this algorithm. Flows between clusters are indicated by lines that combine all group-to-group edges, with thicker lines indicating more connections. The goal of this approach is to filter sets of nodes by the classification of analytical interest while retaining the ability to optimize layouts within each group. Although this method has not yet been widely adopted, we suggest that such custom approaches have considerable potential, in particular for very large and complex networks.

6.4.2 *Adjacency Matrices and Heatmaps*

One of the most daunting challenges in visually representing networks with more than a couple of dozen nodes or those with lots of overlapping ties is what has

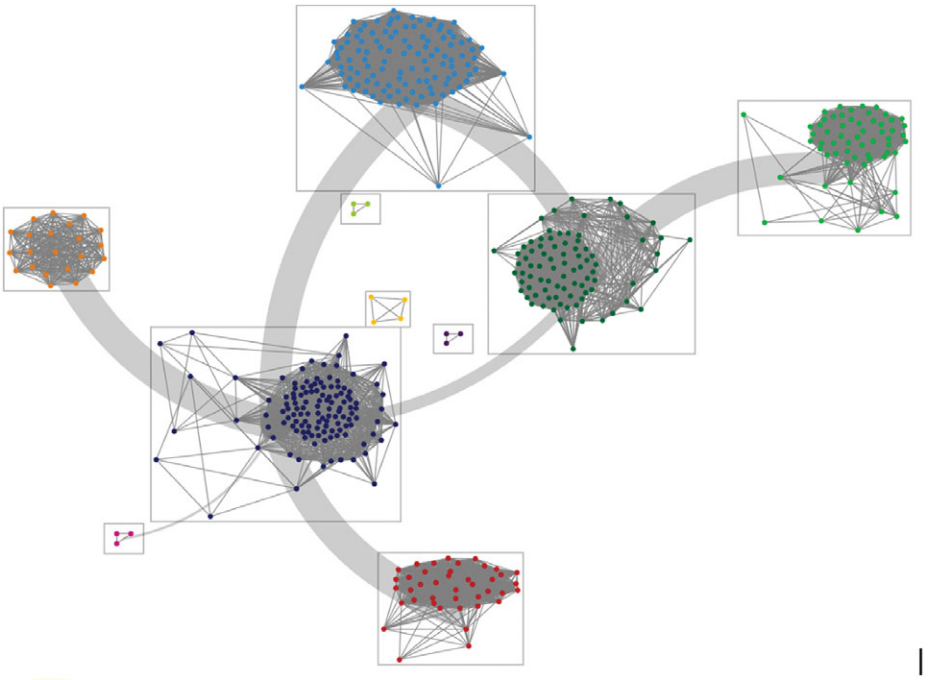


FIG. 6.18 Example of a group-in-a-box custom graph layout created in NodeXL based on ceramic similarity data from the US Southwest/Mexican Northwest ca. AD 1350–1400.

sometimes been called the "hairball" problem. Because networks often contain sets of nodes with similar edges we often end up with complex clumps of nodes and a tangle of overlapping edges that we could never hope to parse (and there are certainly plenty of examples of this in this chapter and elsewhere in this book). Tasks that are simple to do visually in smaller networks, such as estimating the number of edges a given node has or finding missing edges within clusters, become impossible in such hairballs. In such cases, it can sometimes be beneficial to use additional alternative displays such as representations of the adjacency matrix underlying a given graph. There is a considerable and growing literature on how best to design such adjacency matrix visualizations, and this work is encouraging, suggesting that such displays can be designed to be useful for evaluating relational structures within clusters or comparing the properties of different clusters across a network (see examples in Alper et al. 2013; Behrisch et al. 2016; Chang et al. 2017; Ghoniem et al. 2005; Liiv 2010).

The simplest adjacency matrix graphic is a grid in which each row and column represents a node in the network, and the presence or absence of an edge between a pair of nodes is indicated by a solid or empty grid cell, respectively. This can be

extended to include edge weight or other properties using color or transparency (see Fig. 6.19). Such displays are often referred to as heatmaps and are particularly useful in displaying raw similarity or distance matrices in addition to adjacency matrices. In any case, one of the most important analytical decisions that needs to be made is how to order the matrix or heatmap. As we have seen with most of the visualization tools and methods outlined in this chapter, there is no single best option, and the most useful display depends on what you are trying to communicate. There is a sizable literature outlining the options for sorting matrices, and this work extends into other statistical approaches to network analysis such as block-modeling or even seriation methods common in ecology and other areas of archaeology (see Liiv 2010). Although the specifics of the sorting algorithms are beyond the scope of this chapter, we note that most approaches share the common goal of placing nodes with similar sets of edges together such that "blocks" of connected nodes end up along the diagonal of the adjacency matrix with few or no connections further from the diagonal.

By way of example, Figure 6.19 shows a simple network graph of the ceramic technological data from the US Southwest used elsewhere in this chapter with an associated matrix display to the right. This matrix shows filled squares for each edge with the darkness of the colored squares determined by edge weight (with darker squares indicating stronger edges). In the network graph nodes are color-coded based on clusters determined by a Louvain clustering algorithm (see Section 4.4.6), and in the matrix display ties within each cluster are shown using the color of the cluster in question with ties between clusters shown in gray. The ordering of the matrix here is determined by the vertical axis of the associated network graph. In other words, rows in the matrix are ordered from top to bottom according to their vertical locations in the graph layout algorithm so that comparison between the two visuals is easier. This visualization communicates a considerable amount of information in a small amount of space. It is possible in both the graph and matrix to recognize that only three nodes are responsible for all between-cluster ties. The matrix further makes it possible to identify missing ties within each cluster and to compare the relative density between them.

6.4.3 *Hybrid Visualizations*

The adjacency matrix methods described above are useful in that they help to quickly communicate information about community structure in networks, and they also provide a means for assessing missing edges and other features within

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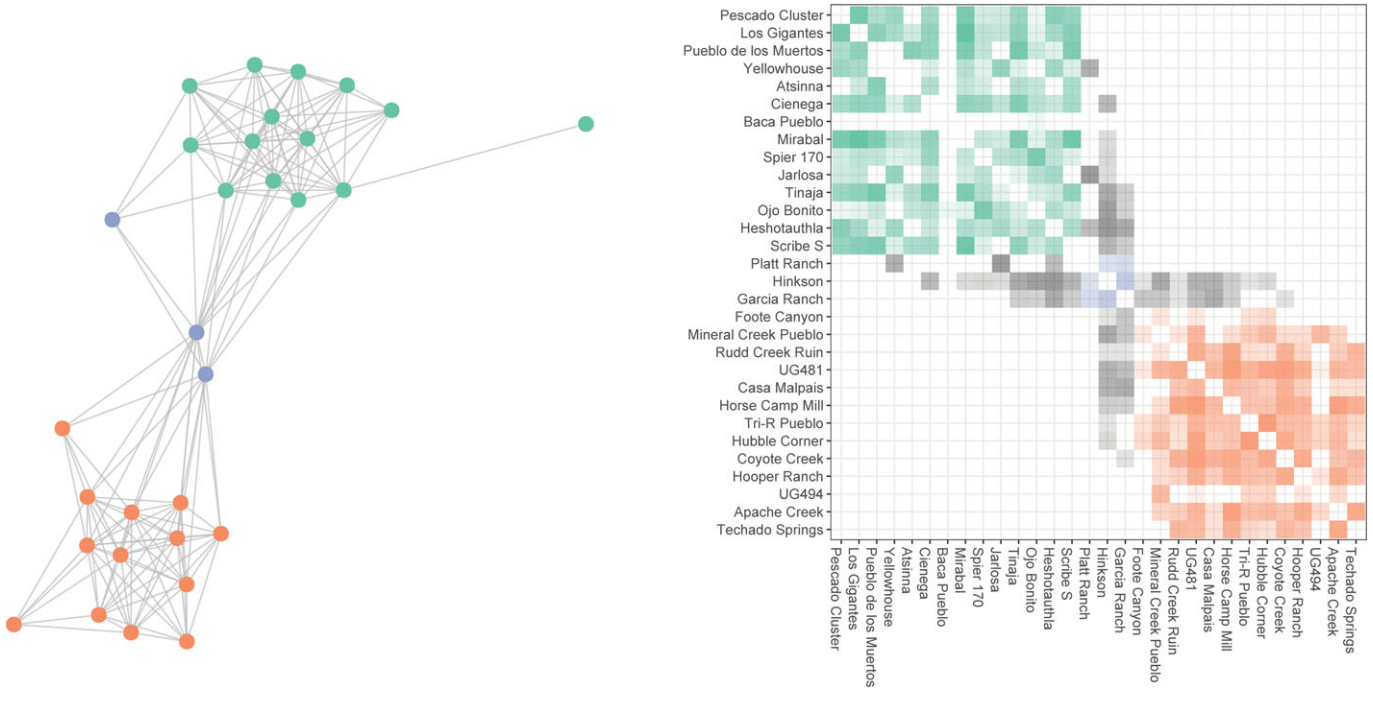


FIG. 6.19 Dual display of a network graph and associated weighted adjacency matrix based on Peeples (2018) ceramic technology data.

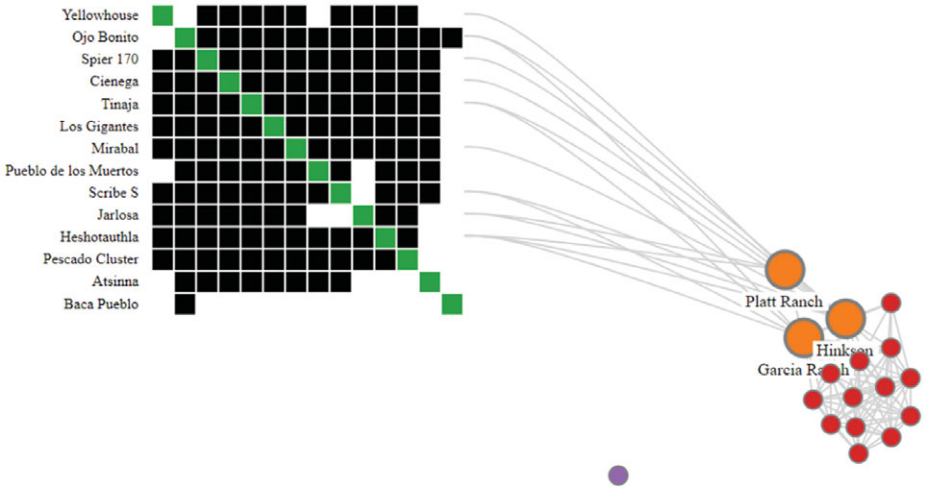


FIG. 6.20 NodeTrix visualization of the Peeples (2018) ceramic technological data showing one dense cluster as an adjacency matrix and the remainder of the graph as a node-link diagram.

clusters that are often difficult to see in a traditional network graph. What these adjacency matrix methods lack, however, is the ability to efficiently communicate information about paths across a network and connections between nodes in distinct groups. Henry and colleagues (Henry and Fekete 2007; Henry et al. 2007) have proposed a method that offers a potential solution to this shortcoming of adjacency matrices that they call NodeTrix (Fig. 6.20). The NodeTrix approach allows researchers to dynamically switch between adjacency matrices and traditional network graphs in real time. The method can help to visualize dense clusters within the network as adjacency matrices with paths between nodes in different clusters as lines. This approach further allows the methods to coexist with some portions of a network shown as node-link diagrams and others as adjacency matrices. We think that hybrid approaches like this have considerable potential and hope that custom visualizations like this will become more common in the near future.

6.5 VISUALIZING NETWORKS THROUGH TIME

Network visuals that display change through time are relatively rare in the broader world of network science but are typical in archaeological network research. We are archaeologists after all. There are a number of different approaches to displaying

network information for different time periods or even between time periods (see Beck et al. 2017 for a comprehensive review and evaluation of many approaches), but as we write today we are aware of relatively few of these having been employed in archaeological network studies. This is an area ripe for new research and development and one where we argue archaeologists might have a role to play in developing tools and techniques that would be of use for the broader network science community.

6.5.1 *The Filmstrip Approach*

By far the most common method employed for visualizing network data through time is what we might call the "filmstrip" approach (see Section 3.4). This simply entails creating a set of ordered snapshots of networks that represent specific periods of time (these can be nonoverlapping or overlapping intervals) and displaying them together in sequence. Such an approach allows a viewer to see changes in network topology and other features across the periods in question. For example, in Figure 6.21 we can very clearly see that the density of the network in question increased through time and also that certain peripheral nodes remained peripheral throughout the period considered. Of course we are not limited to simple network graphs here and could use a sequence of any of the network visualization approaches described above or many others to a similar effect.

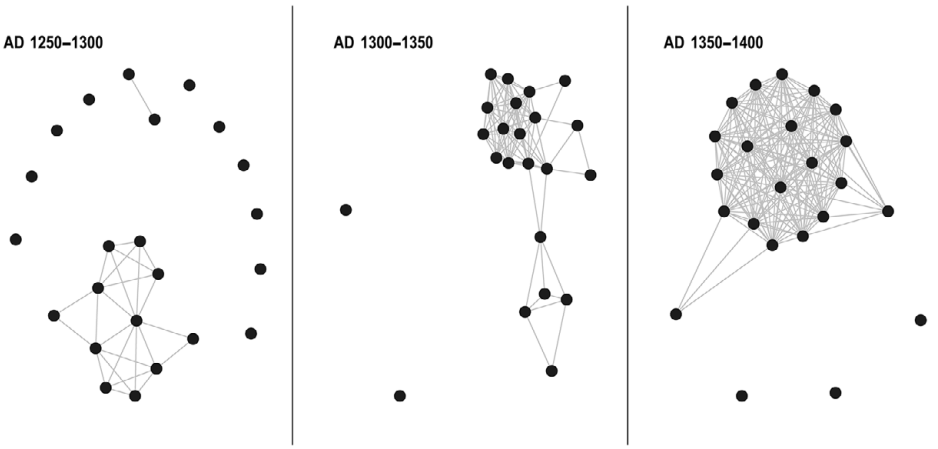


FIG. 6.21 A demonstration of the filmstrip approach to plotting longitudinal network data. These data represent networks of ceramic similarity in the San Pedro Valley of Arizona for three consecutive 50-year intervals.

When the same nodes are present across multiple time steps, it can sometimes be useful to maintain a common graph layout across each interval so that nodes always appear in the same place. This makes tracking patterns through time visually a bit easier. In small networks it may also be possible to use line style or color to indicate nodes or edges that are new in a given time step.

6.5.2 *Simultaneous Display*

Another relatively simple approach to displaying the temporal dimension of networks is to create a single display that shows edges indicating the interval they represent. This method may be viable in small networks but becomes increasingly difficult to visually interpret as the size and complexity of the graph grows. This approach often works best in color where the edges can be given transparency such that it is possible to recognize edges that relate to a single period or more than one period (Fig. 6.22). Although this approach could technically be applied to any number of periods, to avoid adding to visual complexity it is probably best to limit this to two adjacent intervals and relatively small networks. The layout that you choose for such a display is key, and in some circumstances it may be difficult to select a layout that provides a reasonable orientation for both periods.

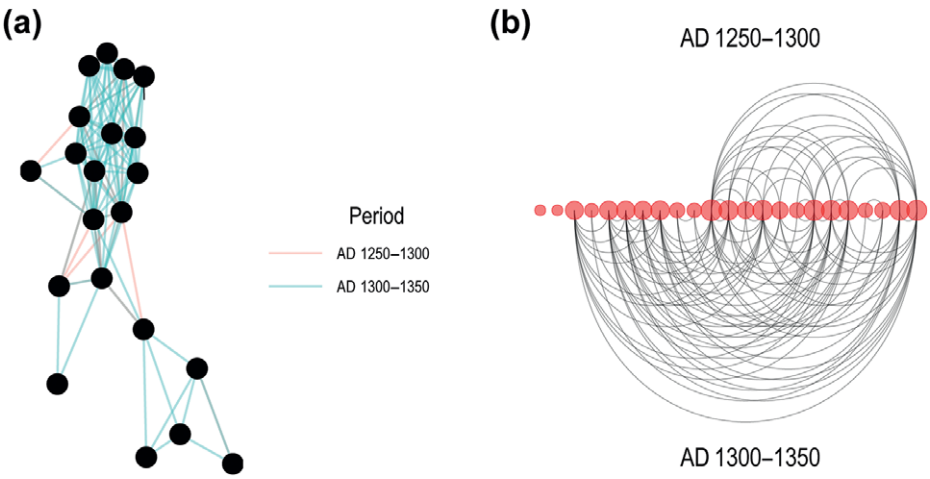


FIG. 6.22 Examples of simultaneous display of two consecutive intervals for the San Pedro Valley ceramic similarity network. (a) A network using the Kamada-Kawai algorithm with edges color-coded based on time period. (b) An arc plot showing ties in consecutive intervals above and below the line.

6.5.3 *Timelines and Time Prisms*

Another useful approach for displaying temporal networks and change through time is a timeline that emphasizes temporal information rather than local or global structural information. When used together with other network visuals designed to reveal network topology, this can be a powerful tool. There are many different options for creating a timeline, but typically these visuals show discrete or continuous time intervals on one axis and information about node or edge relations along the other. For example, Figure 6.23 displays ceramic similarity data from several adjacent subregions of the Sonoran Desert in Arizona. The top plot is a display called a "time prism," which shows slices of a network at a few specific intervals superimposed on top of one another in a three-dimensional projection. The lower plot is a timeline that shows the same information (but for more intervals) with time steps shown along the *x*-axis and with nodes represented as lines distributed along the *y*-axis such that nodes with similar structural positions are grouped together. Note that the *y*-axis has a relative scale and that the positions of lines in relation to one another are meaningful though the absolute height or position on the axis are not. In both of these plots, nodes are color-coded by subregion.

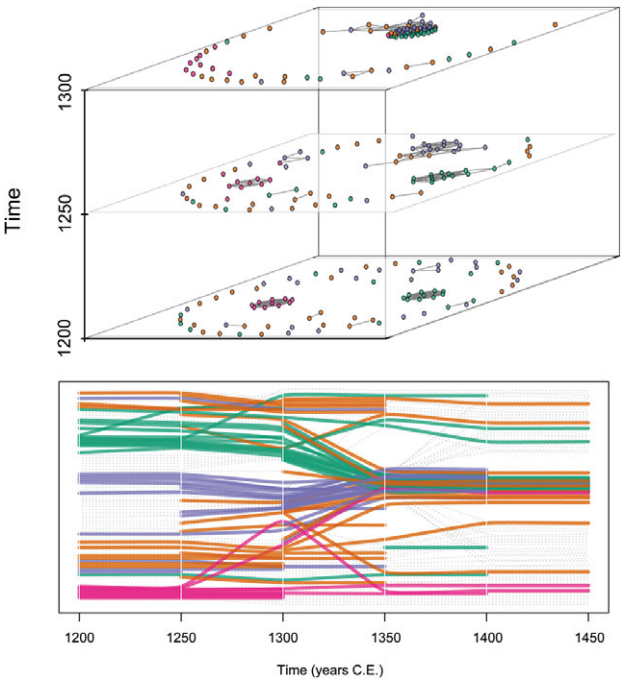


FIG. 6.23 This plot shows two displays of the same ceramic similarity data from the Sonoran Desert in the US Southwest as a time prism (top) and timeline (bottom).

Altogether this plot conveys a substantial amount of information. These plots suggest that earlier in the sequence the network consisted of several dense clusters that were limited to ties within subregions but that through time most nodes became increasingly structurally similar as they clustered together in a large component that included settlements from every region. The timeline is also useful in that it can convey the varying life spans of nodes within a network by limiting lines to a certain span of the *x*-axis. This is but one version of the timeline approach, and we direct readers to Beck and colleagues (2017) for more.

6.5.4 *Animations*

Another useful tool for displaying temporal information is dynamic network animation, which could be created for many of the different network visualizations described above. What makes an animation better than a simple filmstrip is the ability to show the transition between intervals (Bach et al. 2014). It is hard to convey this on a static page, but Figure 6.24 gives some indication of how this works using the same data as in Figure 6.23. The plot on the left in Figure 6.24 represents the network state at T = 1 and the plot on the right shows the state at T = 2. The middle plot is a frame from the transition animation between the two and shows how new edges are drawn first and then nodes are repositioned to their new locations. Being able to observe this transition makes it considerably easier to track changes in network structures between periods. Such animations can be powerful communicative tools in cases where their use is possible. The static images here certainly do not do this animation justice, and we have placed the dynamic version of this animation in the online companion to this book.

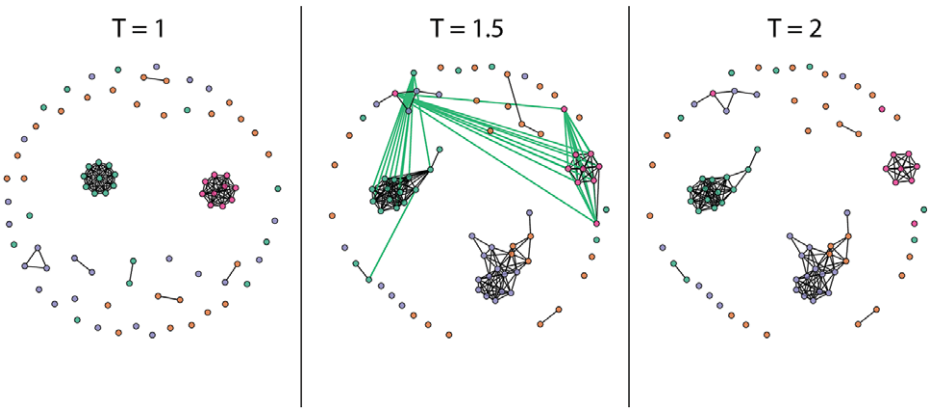


FIG. 6.24 An example of three frames from a network animation.

6.6 INTERACTIVE VISUALIZATIONS

Earlier in this chapter we extolled the virtues of exploration of network data to learn about complex network structures and properties firsthand. There really is no replacement for being able to dynamically investigate different layouts or color schemes or node/edge scales. In our experience, seeing a network change in real time is one of the most intuitive ways to understand the properties of the underlying relational data. Sadly, dynamic and interactive visualizations are not possible in many contexts (like this book), but when viable, we suggest it is worth providing your readers with the opportunity to explore your data on their own.

In recent years, there have been increasing calls for archaeologists to engage in reproducible research by sharing data, code, and other methods used to generate the results we present and publish (see Marwick et al. 2017). The increased availability of tools for doing this provides a whole host of new options for simultaneously facilitating reproducibility and supplementing static published work with beautiful and intuitive dynamic visualizations. Tools like Markdown and Jupyter Notebooks allow you to give your readers the opportunity to actively explore your data and evaluate your arguments and assumptions in real time with real data. This, we suggest, is the wave of the future and we expect that dynamic displays accompanying static articles will be increasingly common in the near future. It is, of course, hard to fully experience the effect of such dynamic visuals on a static page, but Figure 6.25 shows a screenshot of such a visual to help illustrate what is possible. We also direct readers to the supplementary material associated with this book where we provide data, code, and examples for the visualizations in this book (including the dynamic display here).

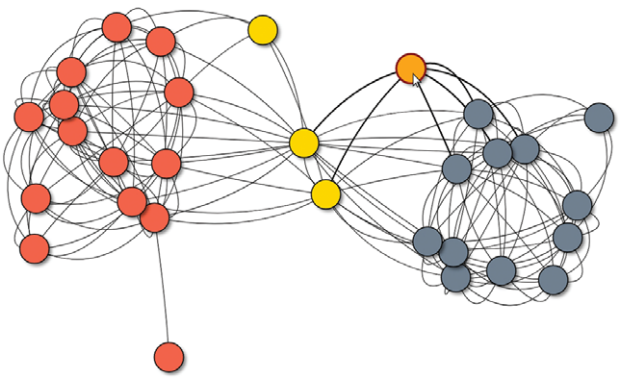


FIG. 6.25 An example of a dynamic network visual created in R. Notice how the nodes and edges are responding to the movement of the edge under the cursor.

6.7 CASE STUDY: VISUALIZING NETWORKS IN THE US SOUTHWEST

In order to pull together the various threads we have woven throughout this chapter so far, we provide a case study focused on the decision-making process involved in the visualization of network data for publication. In this discussion, we evaluate how and why specific decisions about visuals were made and reflect on the effectiveness of these approaches. The data presented here come from the Southwest Social Networks Project and are described in detail in Section 2.8.2. To briefly review, the data presented here represent a series of ceramic similarity networks in consecutive 50-year intervals from settlements in a large portion of the US Southwest.

In the original presentation of these data (Mills et al. 2013a), the authors relied on two different visuals of the ceramic similarity data. First, the authors produced a temporal sequence of traditional node-link diagrams using a Fruchterman-Reingold layout where every node is a site and every edge represented a similarity of greater than 75 percent based on the Brainerd-Robinson metric (Fig. 6.26). Given the size of this network it is not possible to identify specific local relational patterns, but discrete components are quite clear and the force-directed algorithm causes nodes with similar sets of edges to cluster together. Nodes were scaled based on eigenvector centrality. Again, given the number of nodes it is difficult to determine specific values or to directly compare relative values between nodes with similar scores. In this case, however, the distribution of eigenvector centrality scores was such that for each interval there was a set of high-scoring nodes and a set of lower-scoring nodes with relatively few in the middle. As that was all the authors hoped to convey to readers, this size scaling scheme worked reasonably well.

Nodes in this network were also color-coded by subregion so that regional patterns would be more apparent. In the associated article, Mills and colleagues made several arguments with regard to the northern half and the southern half of the study area (roughly divided by a physiographic feature known as the Mogollon Rim). In order to help communicate that distinction, the team chose "cool" colors like blues and greens for the northern half of the study area and "warm" colors for the southern half, noting this convention in the text. Beyond this, individual discrete components were arranged so that they roughly corresponded with geography (this is somewhat similar to the group-in-a-box faceting procedure described above). Finally, isolated nodes that had no ties were shown on the lower left side of each interval, color-coded to their corresponding subregion and scaled based on eigenvector centrality just like all other nodes.

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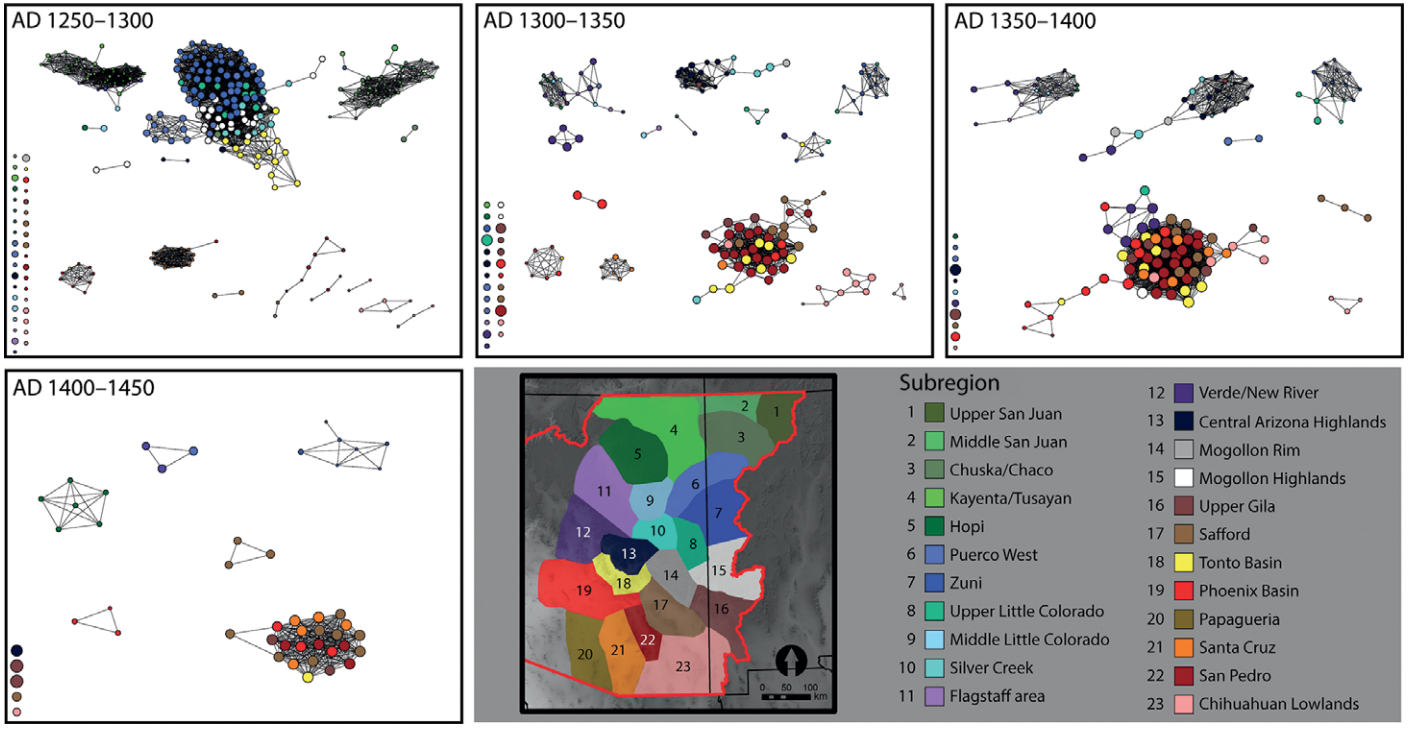


FIG. 6.26 Networks by time for the Southwest Social Networks 1.0 project area (from Mills et al. 2013a).

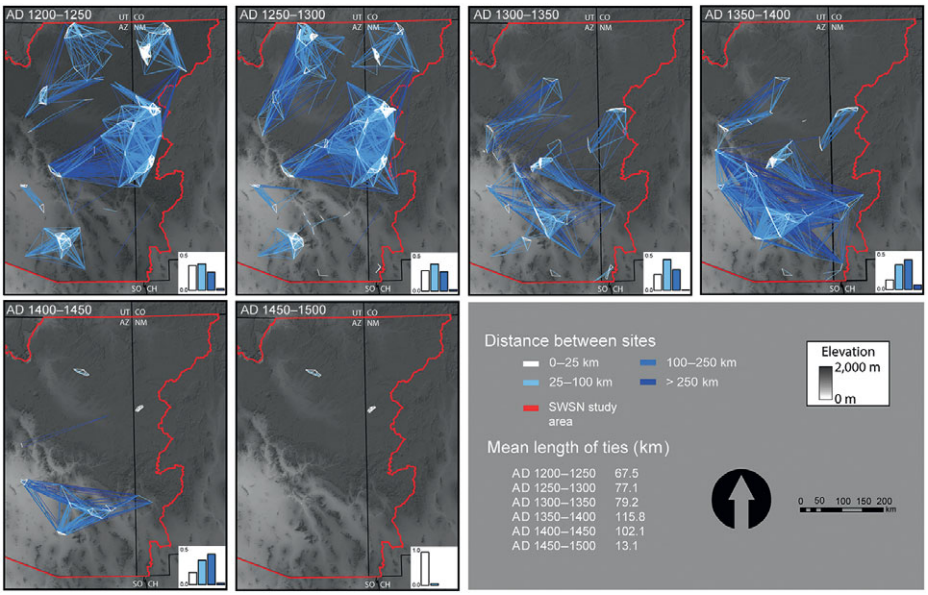


FIG. 6.27 An explicit geographic map network of the Southwest Social Networks 1.0 project area through time (from Mills et al. 2013a).

The second network visual was a temporal sequence of geographically explicit networks (Fig. 6.27). This figure showed only edges representing relations with greater than 75 percent ceramic similarity (nodes were omitted to limit visual clutter) on a grayscale topographic map of the study area. Edges were color-coded in bands of distance such that short-distance ties were shown in white and longer-distance ties were shown in shades of blue. The *z*-layer of edges was organized such that the shorter-distance ties were shown above the longer-distance ties so that both might be visible. These maps helped to illustrate the geographic character of network relations but also showed things that could not be gleaned from Figure 6.26 alone. Specifically, it is quite clear in Figure 6.27 that long-distance ties were at first common only in the northern half of the study area but that beginning in the AD 1300–1350 interval, there is a shift where long-distance ties are increasingly more common to the south (and simultaneously declining in the north) until the final interval considered when only short-distance ties remain. To further emphasize this pattern, the authors also included bar chart inserts in each map showing the relative frequencies of edges of various geographic lengths along with the mean values by interval within the legend of the figure.

Together, these visuals highlight many important topological and geographic properties of the underlying relational data that was the focus of the work by Mills

and colleagues. Specifically, these figures efficiently show changes in the density of relationships in different portions of the study area and how those shifted through time across the early 14th century. In the published article, Mills and colleagues (2013a) relate this transformation to a major period of migration from north to south occurring in the late 13th and early 14th centuries AD. Although the statistical analyses were focused on raw similarity data, the team decided that a set of compelling visuals was needed and, thus, created the binarized data shown here for the purposes of illustration. As the discussion here suggests, these visuals do not show everything, but they do convey important distinctions in centrality scores and the distances associated with ties at the scale of large subregions throughout the study area. Since the subregion was the geographic scale at which Mills and colleagues made their argument, these visuals provided an appropriate backdrop to that discussion that (we hope) helps the reader better understand the complex relational patterns described in the text and statistical analyses.

6.8 SUMMARY

- Network visualization is about more than making beautiful displays of your data and is a fundamental part of the process of exploratory analysis.
- Network visuals should be designed to convey information efficiently and avoid misleading viewers. Ideally, network visuals should be designed with a clear purpose to support narrative discussions in the text and associated statistical analysis.
- There are many options for displaying relational data, and viewers may draw different conclusions from different visuals.
- Available approaches to graph layout are diverse and include everything from simple user-defined layouts where the analyst decides node position to maps where nodes are placed geographically to algorithmic approaches that place nodes and edges based on their properties to special applications like radial plots.
- Different graph layouts have advantages and disadvantages for communicating different kinds of information and should be selected carefully.
- Node and edge color, shape, and size can be used to communicate both network properties (like centrality, direction, or edge weight) or non-network properties (like site size, region, etc.).
- Network visuals can be used to identify or illustrate community structure or the relationships between an external grouping and network topology.

- In large and dense networks with lots of overlapping ties, it may be beneficial to replace the typical node-link diagram with an alternative display like a heatmap or adjacency matrix.
- Displaying change in networks through time can be achieved through static approaches like a set of temporally sequenced networks or timeline or dynamic approaches like animations.
- Interactive visualizations are one of the most intuitive ways to explore relational structures, and when the venue for presentation or publication allows, could be of great use to your readers.
- Although we covered many options in this chapter, this is far from comprehensive. Be sure to explore other tools for network visualization from the resources below.

Further Reading

The following references provide detailed discussions of a variety of visualization techniques or evaluations of different kinds of visualizations for communicating information to a diversity of intended audiences.

Bach, Benjamin, and Mereke van Garderen 2023 Beyond the Node-Link Diagram: A Fast Forward about Network Visualisation for Archaeology. In *The Oxford Handbook of Archaeological Network Research*, edited by Tom Brughmans, Barbara J. Mills, Jessica L. Munson, and Matthew A. Peeples. Oxford University Press, Oxford.

Beck, Fabian, Michael Burch, Stephan Diehl, and Daniel Weiskopf 2017 A Taxonomy and Survey of Dynamic Graph Visualization. *Computer Graphics Forum* 36(1):133–159.

Kandel, Sean, Jeffrey Heer, Catherine Plaisant, Jessie Kennedy, Frank van Ham, Nathalie Henry Riche, Chris Weaver, Bongshin Lee, Dominique Brodbeck, and Paolo Buono 2011 Research Directions in Data Wrangling: Visualizations and Transformations for Usable and Credible Data. *Information Visualization* 10(4):271–288.

Schöttler, Sarah, Yalong Yang, Hanspeter Pfister, and Benjamin Bach 2021 Visualizing and Interacting with Geospatial Networks: A Survey and Design Space. *Computer Graphics Forum* 40(6):5–33.

The following references provide general information on the design of data-based visualizations and design.

Healy, Kieran 2018 *Data Visualization: A Practical Introduction*. Princeton University Press, Princeton, NJ.

Schwabish, Jonathan 2021 *Better Data Visualizations: A Guide for Scholars, Researchers, and Wonks*. Columbia University Press, New York.

Tufte, Edward R. 2001 *The Visual Display of Quantitative Information*. Graphics Press, Cheshire, CT.

Wilke, Claus O. 2019 *Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures*. O'Reilly Media, Sebastopol, CA.

Yau, Nathan 2011 *Visualize This: The FlowingData Guide to Design, Visualization, and Statistics.* John Wiley & Sons, Hoboken, NJ.

EXERCISES

Answers for these exercises can be found in Appendix A.

6.1) Imagine you have a small weighted undirected network of 25 nodes and you are interested in visualizing both degree centrality and community structure. Nodes and edges can be considered to have no natural geographic layout or relative positions. How would you go about designing your visualization? Note that there is not one correct answer to this question, but thinking through each point below will help you think through decisions you will often make when creating network visuals. What graph layout techniques would you try and why? How would you use symbols, color, or size in drawing nodes and edges? What other labels or features would you use to communicate the network properties of interest?

6.2) For this question, consider the two-mode network shown in Figure 6.13. Which of the ceramic wares in the first mode is characterized by the highest degree centrality? If you wanted to emphasize variation in centrality among those ceramic wares, how might you change this figure?

6.3) Figures 6.2 and 6.3 are based on the same underlying dataset (networks of ceramic technological similarity based on cooking pots recovered from settlements in the Cibola region). If you were interested in discussing the network relationships among geographic subregions in this area, which visualization would you use and why?

6.4) Consider Figure 6.8 for this question. Each of the visualizations in this figure shows individual nodes scaled based on betweenness centrality and nodes color-coded based on community membership. If you were primarily interested in making an argument regarding the sites with the highest betweenness centrality, which layout would you select? Why? If you were also interested in exploring community structure, which layout would you select?

7

Spatial Networks and Networks in Space

7.1 WHAT ARE SPATIAL NETWORKS?

Most of the past phenomena we study as archaeologists took place in physical space: individuals lived in homes and towns, and they moved through landscapes; they fought wars on battlefields and they exchanged goods from faraway places. Through our excavations, fieldwork, and literature studies we record spatial information such as the outlines of houses, the locations of sites, the slopes of terrain, or the distance between natural resources and settlements. Many relational phenomena are explicitly geographical, in that the medium of geographical space is an important aspect of the relationship itself. For example, road segments connect pairs of settlements that are close together, and lines of sight connect places from which observers can see features. Such phenomena could be quite straightforwardly represented as **spatial networks** since the nodes and edges are both explicitly embedded in physical space. But for other relational phenomena, space is more like a background feature that can be brought into analyses when relevant but does not feature prominently in the definition of either nodes or edges. For example, past food webs where species are connected through trophic flows or social networks where individuals are connected to their contacts both involve entities (nodes) and relationships (edges) that have spatial properties or attributes, but those spatial properties are not directly invoked in the definition of such networks. We refer to these as **networks in space** in that we could include spatial features into their network representations, but this is not explicitly included in their definition.

Spatial networks are networks where the nodes and/or edges are explicitly defined based on their geographical spatial features.

Networks in space refer to networks where nodes and edges are not explicitly defined based on their locations in space but where those nodes and/or edges still have spatial information embedded in their attributes or representations.

Much of network science beyond archaeology has, until recently, tended to focus on nonspatial features of relational phenomena. In some cases it may indeed be informative to explore the effects and roles of relationships separately from spatial relations. For instance, if we aim to understand the structure of ancient social networks where individuals were connected to each other through documented interactions, it could be useful to temporarily put aside spatial relations to explore the topology of networks of interaction directly rather than the locations where such interactions took place or the origins of the individuals involved. Indeed, most of the examples and analyses presented in this book thus far involve analyses of networks in space with a focus on the nonspatial aspects of network structures and positions.

But very often space matters in archaeological network research. It matters in a way that is perhaps best described by Tobler's first law of geography: "everything is related to everything else, but near things are more related than distant things" (Tobler 1970:236). This law certainly holds for social network studies applied to present-day communities, where it is referred to as the so-called *propinquity* effect: people who are located closer together have a higher probability of creating and maintaining relationships (Festinger et al. 1950). The relational structures we identify with our network techniques are not solely influenced by opportunities the physical environment offers but might be conditioned on what is *possible* within the physical environment. An observer is not able to see a town if it is 500 kilometers removed from them, and humans do not usually move in a direct line between two places separated by a 5,000-meter-tall mountain. Moreover, in our archaeological network research we very often identify network structures or properties that reflect features of the landscape of our study area rather than of the past individuals moving around in it whose behavior, after all, we are trying to understand.

To what extent did past human behavior follow Tobler's law, and in which cases and how did it deviate from this common pattern? How can we differentiate between relational features that are the result of the physical environment and those deriving from purposeful human activities despite the effects of the physical environments and space? Explicitly taking spatial information into account will help us answer such questions, whether it is through studying "spatial networks" or

"networks in space." Both approaches have received considerable attention in archaeology from the very beginning of archaeological network research in the 1960s and 1970s. They have been used to study a range of phenomena, including transportation or flows across roads, rivers, currents, or other cost paths; line-of-sight networks for exploring intervisibility; space-syntax graphs for exploring the accessibility of features, settlements, or broader landscapes; and material culture networks of exchange, interaction, or similarity constrained by the geographic loci of production and consumption of those materials. In this chapter we will explore a range of approaches to incorporating and studying spatial information, and how to apply it in archaeological network research.

Box 7.1 SPACE MATTERS

Archaeological network research has a very long tradition of studying relational phenomena contextualized in a physical landscape or constrained by space (a good overview of spatial relational work in archaeology is offered by Knappett 2011:15–36). The study of the effects of space on networks is also an increasingly prominent topic in the social network analysis community, as revealed through a special issue of the journal *Social Networks* dedicated to the topic. In this issue, the editors argue there is potential for advancing the integration of network and spatial analyses, and they propose five conceptual strategies for this integration. Thanks to its long tradition of spatial network research, we believe archaeology will play an important role in this. To this end, archaeological network researchers could consider the contributions or impact of their work in light of the following possibilities and strategies (from Adams et al. 2012:1):

1. Spatial locations influence social networks,
2. social networks influence spatial location and environments,
3. social and spatial boundaries can be contextually defined,
4. integrated measures of networks and space can be constructed, and
5. combined network and spatial effects can produce social, behavioral or health outcomes and patterns.

7.2 ARCHAEOLOGICAL SPATIAL NETWORKS

Spatial networks are networks where nodes and edges are defined in geographical space and where network topology is explicitly constrained by the spatial

relationships among those nodes and edges (see Brughmans and Peeples 2020). Most commonly, both nodes and edges of spatial networks are explicitly spatially embedded. In a road, river, or rail network, for instance, the nodes refer to specific geographical locations or areas and the edges follow particular paths through a landscape. The same can be said for visibility networks, in which case the line of sight follows a specific path from the observer's eyes to the observed feature. Although it is not a tangible line in the same way as a railroad, a line of sight is explicitly spatially embedded, moving from one specified location to another and having spatial properties. However, not all elements of spatial networks need to be explicitly spatially embedded. In some cases only the nodes have specified locations and the edges merely represent the presence of a relationship rather than the actual spatial path followed by that relationship. A typical example is a spatial representation of a past social network, where individuals are located in their hometowns or places of origins and the edges represent the presence of social relationships among them identified using some nonspatial criteria rather than any explicitly spatial information.

Spatial networks as both analytical tools and visualizations have a long history in archaeology. Most of the earliest forays into archaeological networks were centered on spatial network methods and models largely coming out of geography. Spatial network visualizations were used to illustrate and explicitly model the structure of exchange systems and settlement patterns as early as the 1960s and 1970s (e.g., Doran and Hodson 1975:12–15; Hodder and Orton 1976:68–73; Stjernquist 1967). As described in Chapter 2, the first formal analyses of spatial networks in archaeology drawing on methods coming out of graph theory were also emerging at about the same time (e.g., Irwin 1978; Terrell 1977). Despite these early efforts, it wasn't until the last 15 years or so that such approaches became more common (Brughmans and Peeples 2017), perhaps coinciding with the increasing popularity of GIS tools and applications making such analyses considerably easier to conduct (see Conolly and Lake 2006:234–262). The most common applications can be roughly classified into four groups: transport networks, space syntax, visibility networks, and spatial material culture networks (see Chapter 2 for introductions to these and other archaeological applications).

Spatial information can be represented as network data in a number of ways, but perhaps the most common approach is to include spatial coordinates or other locational information in the node and edge tables associated with such networks so that these data can be used in subsequent representations and analyses (for more information, see Section 3.3.5). Spatial information can include the *x*-, *y*, and *z*-coordinates of the nodes alongside other locational features such as the orientation and aspect of the slope on which it is located or the general landscape type in the immediate environment (e.g., hill, plateau, flatland). The length of an edge can be

added to the edge table, along with other spatial information about it such as the amount of time to complete a journey or the calories expended by those undertaking the journey. By far the most common use of this spatial information is the spatial representation of networks on maps followed by visual interpretation without subsequent formal analyses. In this chapter we show a number of ways in which we can do more than making spatial network maps, by including spatial node and edge information into our formal network analyses and by comparing the archaeologically observed spatial networks with those generated by spatial network models.

7.3 PLANAR NETWORKS

A key feature of many spatial networks is **planarity**: a planar network is one in which no edges cross. Note that planar networks can often be drawn on a plane with their edges crossing, but for a network to be called planar there needs to exist at least one way of drawing it that does not include any crossing edges (Fig. 7.1). For example, the network in Figure 7.1b is planar because even though it has crossing edges, it can be redrawn such that the edges no longer cross (Fig. 7.1a). Figure 7.1c, on the other hand, is not planar, because no matter how we redraw it the crossing edges will remain (Fig. 7.1d).

Planarity is often enforced in spatial networks precisely because nodes and sometimes edges are spatially embedded, and physical space "forces" crossing edges to connect to a node. The typical example of this is a road system (Fig. 7.2a). When a pair of roads in the physical world cross each other, it is rather intuitive to consider that crossing to be an entity in itself where cars driving on the roads can either turn onto the crossing road or drive on. In that sense, crossing roads are represented not as crossing edges (where it would not be possible to move from one edge to the other over the crossing) but rather as a set of edges that come together in a node representing the crossroads (Fig. 7.2b). However, there is a very common exception to planarity in road systems: bridges (Fig. 7.2c). It is typically not possible to move from a bottom road to the road crossing over it via a bridge, in which case this "crossroads" created by the bridge cannot be represented as a node (Fig. 7.2d). Moreover, road systems can equally be represented as network data without having to consider crossroads as nodes. For example, if we are only interested in whether a pair of towns is reachable over the roads without having to drive through a third town, then we can represent only towns as nodes and not the crossroads.

Other common examples of planar networks are rivers (for the same reason as roads: confluences are typically considered nodes) and trees. A **tree** is a special kind of hierarchical network (see Section 4.4.4) commonly used in computer science and

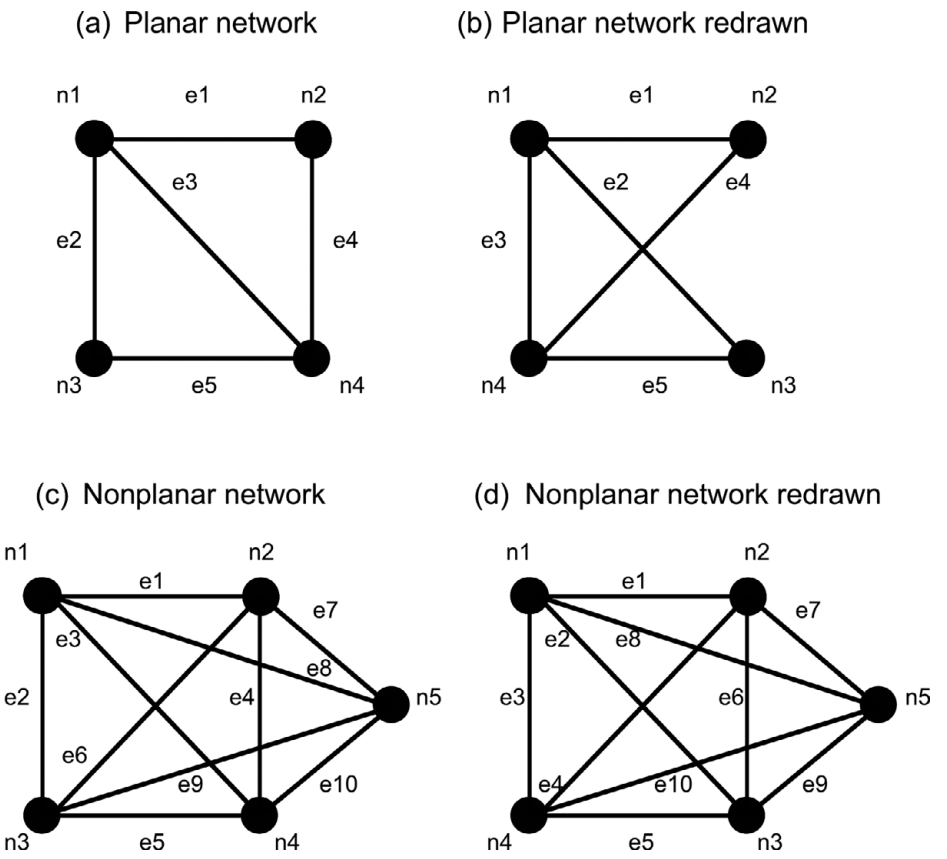


FIG. 7.1 In planar networks (a, b) the edges do not cross in at least one way of drawing it. Nonplanar networks (c, d) cannot be drawn without crossing edges.

mathematics, and we will rely on this concept when introducing the minimum spanning tree below (Section 7.5.2). A tree is a connected network in the sense that it consists of a set of nodes that can all be connected by a path (see Section 3.4.1) and it is acyclic (see Section 4.4.4). A tree is called a **spanning tree** if it includes all nodes of the network under consideration (Fig. 7.3a). Trees contain the minimum number of edges for a set of nodes to be connected, which results in an acyclic network with some interesting properties:

1. Every edge in a tree is a bridge, in that its removal would increase the number of components (see Section 4.4.5).
2. The number of edges in a tree is equal to the number of nodes minus one.
3. There can be only one single path between every pair of nodes in a tree.

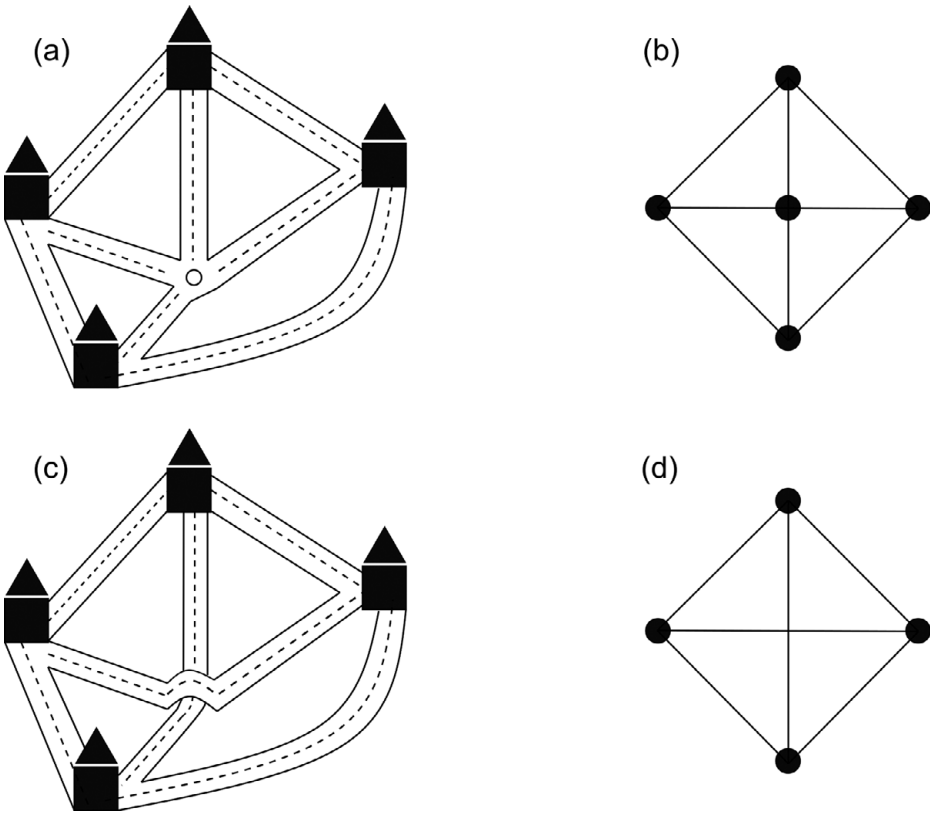


FIG. 7.2 Roads between towns that cross at (a) a crossroads can be represented using (b) a node for the crossroads, whereas roads crossing (c) via bridges can be represented (d) by edges crossing each other. Using nodes to represent crossroads can often lead to planar networks.

The arboreal analogies are rather widespread in tree terminology. A network that consists of components all of which are trees is called a **forest**. And trees themselves are often drawn in a *rooted* manner with *branches* pointing up, where the bottom node is a single *root node* and the top nodes are the *leaves* (Fig. 7.3b).

Such planar spatial networks have traditionally received more attention in network science than nonplanar spatial networks, and many network analysis methods and models have been purposely developed to study planar networks (Barthélemy 2011:4). The main reason for this is that planarity is an extremely common feature in many of the networks studied by network scientists, in particular, infrastructure networks such as roads and rails. In the rest of this chapter we will introduce the methods for exploring the effects of space and the key spatial network models, many of which are developed for planar networks.

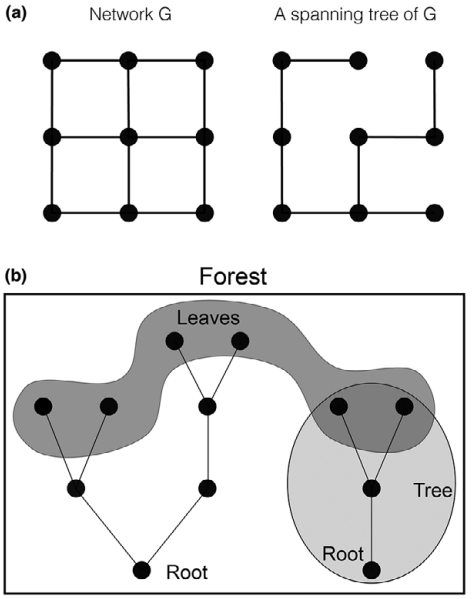


FIG. 7.3 (a) A network G and a spanning tree of network G. (b) A forest consisting of two trees; each tree has a root and leaves.

A *planar network* is a network that can be drawn on a plane where the edges do not cross but instead always end in nodes.

A *tree* is a network that is connected and acyclic.

A *spanning tree* of a network is a tree that includes all nodes of the network.

7.4 EXPLORATORY NETWORK METHODS DESIGNED FOR SPATIAL NETWORKS

In Chapter 4, we outlined a range of common exploratory network analytical methods useful for characterizing the node-, edge-, and graph-level properties and topologies of networks generally. Most of those methods can be readily applied to spatial networks, but there are also a few important modifications or extensions that merit discussion here. In many cases, spatial relationships can be incorporated into network analyses by including geometric distance or related spatial properties of networks into our calculations directly. In other cases we may need to simply recalibrate our expectations or standardize measures in relation to the peculiar properties of spatial networks and the constraints of movement in physical space.

We discuss a few of the most common kinds of network science measures with spatial variants here, some of which will be further explored in the case study below (Section 7.6). We direct readers to the additional cited resources for more detailed overviews of spatial network metrics and properties generally (Barthélemy 2011; Haggett and Chorley 1969; Xie and Levison 2007).

The most common approach for creating a *spatial variant* of a particular network metric is to allow physical distance to directly inform the weight of edges or the length of paths. Specifically, spatially derived attributes such as the absolute distance, time, or effort associated with crossing a given edge can be used to define weights and lengths of those edges such that shorter distances or less time/effort equate to greater weights for a given connection. For example, if we wish to calculate betweenness centrality in a spatial network, we could use the typical nonspatial approach by identifying how many shortest paths (in terms of the number of edges crossed) a given node is involved in. Alternatively, we could create a spatial variant of betweenness by treating distance as a weight associated with each edge (or, more specifically, inverse distance so that shorter edges are more heavily weighted) and defining shortest paths as those that minimize the real physical distance traveled to get between nodes. Degree centrality in a spatial network could similarly be defined as the summed number of edges incident on a node weighted by their summed length. We can apply a range of common exploratory techniques discussed in Chapter 4 by treating inverse distance as an edge weight variable. Such approaches are common in the analysis of archaeological spatial proximity networks and movement networks as well as visibility networks and justified graphs from spatial syntax approaches (Hillier 1996; Hillier and Hanson 1984; Turner et al. 2001).

In some cases, the application of network analytical techniques to spatial networks requires us to alter our metrics due to the special properties of spatial networks. For example, networks that exhibit planarity typically have low density. Indeed, an interesting defining feature of planar networks is that their average degree is always lower than 6, which means that planar networks tend to be sparse and that a graph with an average degree higher than this can never be planar. Thus, if we want to assess density in a planar network as a proportion of the total possible ties, we should consider a maximally connected *planar* network rather than a maximally connected network based purely on the number of nodes and edges (the maximally connected planar graph for a given number of nodes and edges will often be considerably less dense than the maximally connected nonplanar graph). Another typical feature of spatial networks is the limited range of variation in its degree with a degree distribution peaking close to the mean (Amaral et al. 2000).

This stands in sharp contrast to most nonspatial networks that do not have such physical constraints: for example, it is possible to identify ancient writers with many correspondents but less common to find a road junction with more than four roads emanating from it. Similarly, strong differences between spatial and nonspatial networks have been identified in terms of the expected shape of distributions of metrics and features such as betweenness centrality, average shortest path length, and clustering coefficients (Barthélemy 2011). Space offers particularly strong constraints for networks where both edges and nodes are explicitly spatially embedded, which can be identified using exploratory spatial network methods.

7.5 SPATIAL NETWORK MODELS

In our archaeological research, we often wish to represent or explore theories concerning the interactions or interaction opportunities between individuals, sites, or communities with reference to their spatial location. It is a commonly formulated assumption that communities that lived close to each other interacted more frequently or had more opportunities for frequent interaction. We might argue, for example, that certain types or probabilities of interaction tend to take place within a certain maximum distance. In this case we might want to identify what areas lie within this distance for each site and what other sites are located inside these areas, allowing us to identify pairs of sites separated by a maximum distance reflecting our theory of interaction potential. However, we might equally be interested in determining whether certain past communities had important interaction potential that was spatially structured, but relative to all other communities' locations rather than considering a fixed distance. For example, Terrell (1977) argued that the spatial proximity of communities on the Solomon Islands to their three nearest neighbors (rather than all neighboring communities within a certain maximum distance) structured the flow of cultural features between island communities. Such absolute and relative spatial proximity approaches provide helpful ways of representing common theories and visually identifying hypothetical connections, as well as exploring core spatial structuring of a study region.

We can use the approaches discussed in this section to represent such theories and assumptions. These concern techniques developed mainly in computational geometry that allow us to explore patterning in sets of spatially embedded nodes (Fig. 7.4). All we need to apply them is a theory or a spatial relational assumption, and a set of spatial locations. We can then proceed to identify core spatial connectivity between the locations based on our assumptions and discuss the implications of this theory or discuss the plausibility of some of the patterning through

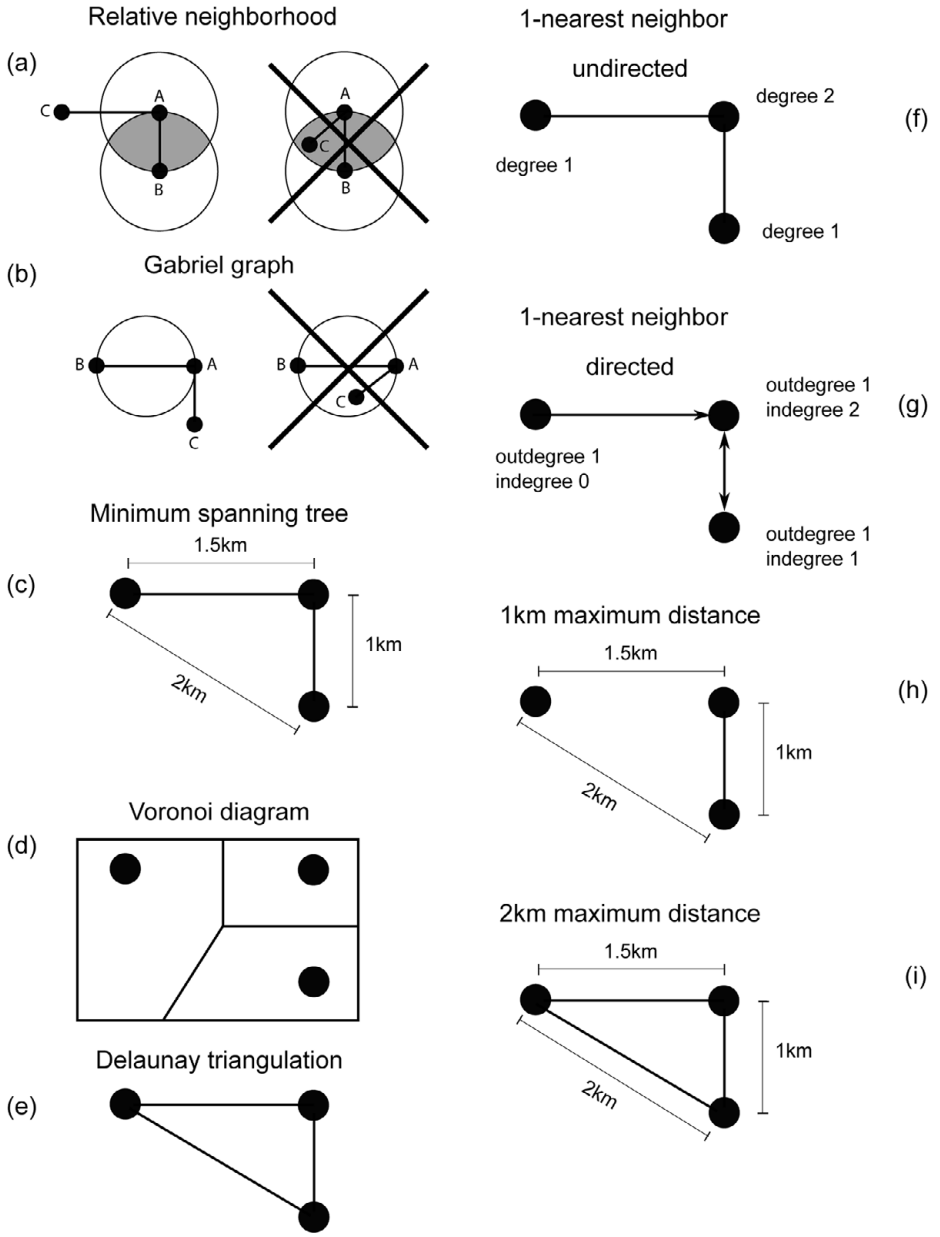


FIG. 7.4 Abstract examples of the spatial network models (and Voronoi diagram) covered in this section.

comparison with empirical observations. We provide a basic introduction to some of the most fundamental techniques, and we refer the reader to much more elaborate discussions of the subject matter in computational geometry and physics handbooks and reviews (Barthélemy 2011; Chorley and Haggett 1967; Jiménez-

Badillo 2012). A good overview of the models in a landscape archaeology context and their comparison with least-cost path networks is provided by Herzog (2013). An example of the application of some of these models in an archaeological study is shown in Section 7.6.1.

7.5.1 *Relative Neighborhood Networks, Beta Skeletons, and Gabriel Graphs*

A pair of nodes are relative neighbors and are connected by an edge in a **relative neighborhood graph** (Fig. 7.4a) if they are at least as close to each other as they are to any other point (Toussaint 1980). It can be derived for a pair of nodes by considering a circle around each with a radius equal to the distance between the nodes. If the almond-shaped intersection of the two circles does not include any other nodes, then the nodes are relative neighbors. This network has an interesting relationship to other models discussed in this section, in that it is a subset of the Delaunay triangulation and contains the minimum spanning tree (both introduced below).

When we apply the same principle to a circular (rather than almond-shaped) region between every pair of nodes, then we obtain a so-called **Gabriel graph** (Fig. 7.4b): if no other nodes lie within the circular region with a diameter equal to the distance between the pair of nodes, then this pair of nodes is connected in the Gabriel graph.

Both the relative neighborhood graph and the Gabriel graph result in a single set of connections each; that is, given a set of nodes and their locations there will be only one relative neighborhood graph and one Gabriel graph. This can be really valuable when aiming to identify core structural properties of a study area. The concept of **beta-skeletons** (or β-skeletons) provides a more flexible approach where the distance at which something is considered a relative neighbor can be varied (Kirkpatrick and Radke 1985). As such, relative proximity can be controlled and varied in interesting ways. It allows for this by varying the diameter of the circle using a parameter, beta, rather than fixing the diameter as in the Gabriel graph. This leads to interesting alternative network structures that are denser with lower values and sparser with higher values, and the beta skeleton equals the Gabriel graph when beta equals 1. This concept was used by Brughmans (2010) in his study of the distribution of red-slipped ceramic tableware in the Roman east. He explored at which value of beta the set of sites where these ceramics were found started to fall apart into different components, which might suggest limitations in opportunities to distribute ceramics between sites belonging to different components, and at

which value of beta denser site clusters appeared that might identify communities that had robust opportunities to distribute ceramics between them.

Relative neighborhood graph: A pair of nodes are connected if there are no other nodes in the area marked by the overlap of a circle around each node with a radius equal to the distance between the nodes.

Gabriel graph: A pair of nodes are connected in a Gabriel graph if no other nodes lie within the circular region with a diameter equal to the distance between the pair of nodes.

Beta skeleton: A Gabriel graph in which the diameter of the circular region between a pair of nodes is controlled by a parameter beta.

7.5.2 *Minimum Spanning Tree*

We consider a set of nodes in the Euclidean plane. To obtain a minimum spanning tree, we create edges between pairs of nodes where each node can be reached by each other node, such that the sum of the Euclidean edge lengths is less than the sum for any other spanning tree (Fig. 7.4c). Per Hage and Frank Harary (1996) applied the model in their studies of kinship networks and descent, the evolution and devolution of social and linguistic networks, and classification systems in Pacific archaeology. Interestingly, Hage and Harary also describe a model that dynamically generates a minimum spanning tree, where one edge is created at each timestep. They use it as an example of a theorized representation of the growth of a past social network.

Minimum spanning tree: In a set of nodes in the Euclidean plane, edges are created between pairs of nodes to form a tree where each node can be reached by each other node, such that the sum of the Euclidean edge lengths is less than the sum for any other spanning tree.

7.5.3 *Delaunay Triangulation*

We could connect a spatial distribution of points with as many triangles as possible, avoiding overlap of edges. This is done, for example, when we want to generate a mesh from a point cloud in photogrammetry as the basis for making

three-dimensional models, or when we interpolate a triangulated digital terrain model from spot measurements of elevation in a landscape. The result is a triangulation network, which has a planar structure due to the absence of crossing edges.

In one way to obtain the **Delaunay triangulation**, we first need to create so-called Voronoi diagrams or Thiessen polygons (Fig. 7.4d). Such a triangulation network is created by considering a pair of nodes to be connected by an edge if and only if their corresponding tiles in a Voronoi diagram (or Thiessen polygon) share a side (Fig. 7.4e). Delaunay triangulation has been widely applied in archaeology, especially for studies of transport systems. For example, Fulminante (2012) studied road and river transport systems between Iron Age towns in Central Italy (Latium Vetus) using this approach, whereas Evans and Rivers (2017) used Delaunay triangulation in their study of the emergence of Greek city-states.

Voronoi diagram or Thiessen polygons: For each node in a set of nodes in a Euclidean plane, a region is created covering the area that is closer or equidistant to that node than it is to any other node in the set.

Delaunay triangulation: A pair of nodes are connected by an edge if and only if their corresponding regions in a Voronoi diagram share a side.

7.5.4 *K-Nearest Neighbors and Maximum Distance Networks*

A particularly popular approach in spatial archaeological network research is ***K*-nearest neighbor networks** (Fig. 7.4f; sometimes referred to as proximal point analysis [PPA] after Terrell 1976, 1977). It is a representation of the idea that each node is connected to the K other nodes closest to it (K is an integer number representing the number of nearest neighbors, e.g., 2-nearest neighbor network, 3-nearest neighbor network). This means that 6-nearest neighbor networks and above (i.e., $K > 5$) give nonplanar networks because planar networks have an average degree strictly less than 6 (see Section 7.3). Crucially, because we identify the nearest neighbors for each node individually, we can make a distinction between the directionality of edges. In a directed nearest neighbor network, we draw edges from the source node to its nearest neighbor (Fig. 7.4g). If that target node in turn has the source node as its nearest neighbor, then the edge is bidirectional. However, there are many cases where a node further removed from spatially clustered nodes only sends edges but does not receive many: all spatially clustered nodes will send edges to closer nodes in the cluster rather than to the node positioned at a greater distance. This leads to a counterintuitive structural feature

of *K*-nearest neighbor networks: the degree of nodes in undirected *K*-nearest neighbor networks will always be at least equal to *K* but can be larger, because some nodes will be selected as targets of other nodes through edges they do not reciprocate (see, e.g., the degree of nodes in Fig. 7.4f). Similarly, in directed networks the indegree and outdegree distributions can differ significantly: all nodes will have an outdegree equal to *K*, but some nodes can have an indegree that is lower than *K* (if they get selected as target less than *K*) or higher than *K* (if they get selected as targets more than *K*; see Fig. 7.4g).

We wish to highlight one further approach that is particularly intuitive and considers absolute distances. As such, it provides an alternative to relative distance approaches. In a **maximum distance network** a node pair is connected if the distance from each other is lower than or equal to a threshold distance value (Fig. 7.4h, i). This concept is comparable to what is known as a random geometric graph in the mathematics literature. In the case of a visibility network we might be interested to identify all nodes located within a theorized maximum viewing distance of 15 kilometers of each other, before we explore which of these node pairs are in fact intervisible. The visibility network is necessarily a subset of this maximum distance network because a pair of nodes have to be at least as close as the maximum viewing distance in order to be intervisible (see the case study below).

K-nearest neighbor and maximum distance models are often used in archaeological research to represent the very commonly formulated assumption that places that are nearer to each other or within a certain distance have more opportunities for interaction and sharing ideas and materials than places further away from each other. In such studies, we see the edges of nearest neighborhood and maximum distance networks used to represent plausible channels for the flow of material or immaterial resources between individuals, settlements, or island communities (Broodbank 2000; Collar 2013; Terrell 1977).

K-nearest neighbor network: Each node is connected to *K* other nodes closest to it.

Maximum distance network: Each node is connected to all other nodes at a distance closer than or equal to a threshold value.

7.5.5 *Spatial Interaction Models*

In ecology, geography, economics, and a broad range of other fields in the social and natural sciences, several related classes of spatial interaction models have

frequently been used to explore or simulate spatial network interaction processes. This includes popular approaches like gravity models (Evans and Rivers 2017), radiation models (Simini et al. 2013), maximum entropy models (Phillips and Dudík 2008), and a range of related tools and extensions of these models. Although the specifics differ from approach to approach and there are several versions of each of these models, in general, such approaches assess or simulate the pairwise movement of resources among entities in relation to potential external predictors of flows along with measures of the distance (in physical space, time, or effort) among them. For example, the classic gravity model as it is often used in economic geography treats the strength of interaction between pairs of nodes in a network as a ratio of their relative population sizes to the square of the distance between them. The basic assumption here is that larger populations can create and maintain more contacts but that impedance to interaction compounds at increasing distances. The specific details of these models are beyond the scope of this manual, but Evans and colleagues (2012; also Evans 2016) provide a good overview of several of the most common options using archaeological data.

Despite their popularity in other fields, spatial interaction models have been less common in archaeology perhaps because it can often be difficult to estimate the parameters required to use many such models (relative flows or exchanges between sites). There are, however, several examples of empirical studies applying this class of interaction model in regions with high-resolution archaeological data as well as many more applications focused on archaeological simulations in contexts where more limited data are available (e.g., Bevan and Wilson 2013; Davies et al. 2014; Evans 2016; Evans and Rivers 2017; Evans et al. 2012; Gauthier 2021; Knappett et al. 2008; Menze and Ur 2012; Paliou and Bevan 2016; Rihll and Wilson 1987). For example, Rihll and Wilson (1987) study the emergence of Greek city-states using a version of the gravity model to infer likely flows among communities. Menze and Ur (2012) use a modified flow model (called an exponentially truncated power function) to explore the expected patterns of interactions among a set of settlements in Northern Mesopotamia in relation to both relative settlement size, settlement locations, and remotely sensed trails among them. Gauthier (2021) uses an approach based on general additive models to identify patterns in spatial material cultural networks that can be explained in part by hydroclimatic variability in the US Southwest. There have also been custom models built explicitly for the purposes of archaeological analysis, including the ARIADNE model, which has been used to study patterns of interaction among island

communities in the Middle Bronze Age Aegean. The ARIADNE model is designed to account for both the costs and benefits of networked connections among communities based on land and sea travel, the relative sizes of communities, and a range of other factors. This model has been used to generate plausible and efficient networks as well as to reconstruct the relative importance of individual settlements within these networks (Knappett et al. 2008). We see models like this which specifically account for the nature of transportation and the nature of archaeological evidence as an important area of development in the future.

Box 7.2 A MODEL OF SETTLEMENT HIERARCHY IN BRONZE AGE CRETE

Bevan and Wilson (2013) used spatial interaction models to explore political and economic hierarchies and road networks between settlements in Bronze Age Crete. These are key topics for second millennium BC Minoan settlements, given ongoing debates about the importance and size of palatial-style buildings. They argue these models can provide such useful insights even in cases where the surviving evidence of settlement patterns is patchy and incomplete, as it is for Bronze Age Crete. This makes the model very attractive for many archaeological and historical cases, because the only input it requires are (1) a set of locations of sites, (2) the distance between sites, and (3) where possible a rough estimate of the attractiveness of a site (parameter α) and of the decay of effective communication with increasing distance (parameter β). The model estimates the spatial distribution of settlement sizes by modeling flows between settlements. These flows should be understood as an abstract composite of the relative intensity of trade, transport, or movement. They do not represent a specific phenomenon like the daily number of people moving between a pair of settlements, nor do they draw on any archaeological observations. Rather, it is an estimate of the interaction between two places based on their location and distance. Figure 7.5 shows the estimated settlement sizes (of one of many outputs of the model) as open circles: the larger the circle, the larger the estimated size. The lines represent the sum of the largest outflows from each settlement to its maximum creditor. The results are argued to reflect the emergence of "peer polities" of comparable size, and they point to a tension between the proximity of central places to the coast and their centrality on the island of Crete itself.

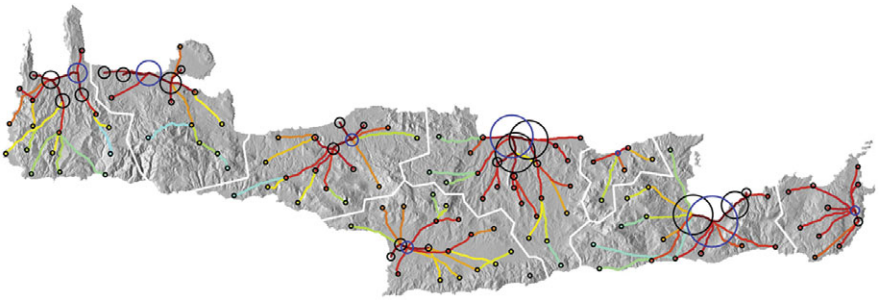


FIG. 7.5 Known and simulated settlements on Bronze Age Crete with estimated settlement sizes (circle size) and modeled major flows (edges) (reproduced with permission from Bevan and Wilson 2013, fig. 6a).

7.6 CASE STUDIES

In this section we provide two brief case studies to demonstrate some of the methods and models described above. The first example, using data from Iron Age Spain, provides a look at how proximity models can be used to generate, compare, and interpret **spatial networks**. The second example uses data from the Southwest Social Networks Project to illustrate how spatial material cultural networks can be analyzed as **networks in space** and the implications of evaluating the role of spatial patterns in nonspatial networks.

7.6.1 *Proximity of Iron Age Sites in Southern Spain*

We will illustrate these spatial network models using a set of 86 sites located in the Guadalquivir river valley in southern Spain. This region has a long history of being densely urbanized, and the 86 urban and rural settlements and agglomerations used there were inhabited during the late Iron Age (known as the Iron Age II period, from ca. the 5th to the 3rd century BC). Most sites seem clustered around the tributaries in the hillier southern part of the Guadalquivir river basin, while a number of sites are dotted along the Guadalquivir itself (Fig. 2.9). Riverine transport of goods and people was a key feature of urban connectivity in this region. Which network model best captures this feature of the sites' locations? Can network models teach us something about how the distribution and proximity of sites in the region are structured?

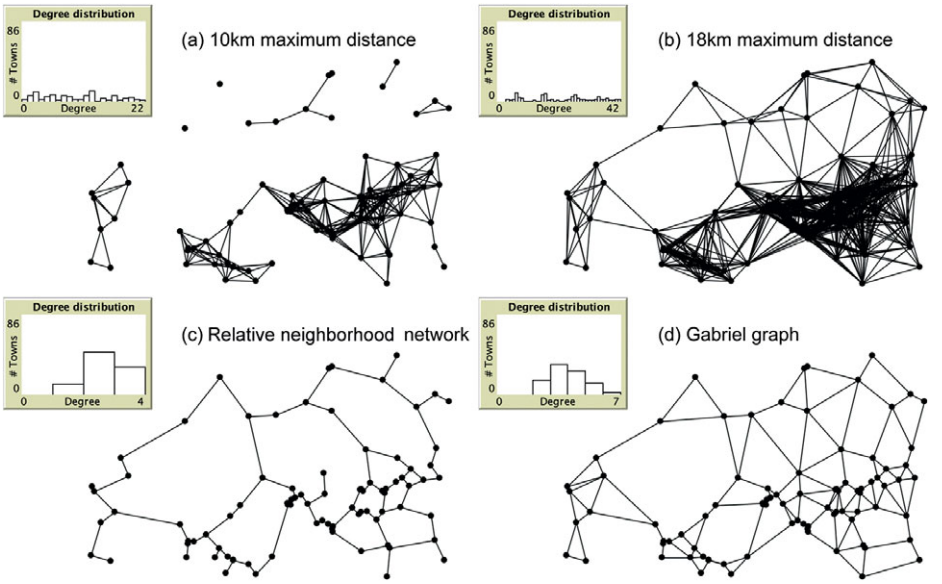


FIG. 7.6 Spatial network models applied to the 86 sites shown in Figure 2.9: (a) 10-kilometer maximum distance network, (b) 18-kilometer maximum distance network, (c) relative neighborhood network, (d) Gabriel graph.

The maximum distance networks reveal the dense clusters of settlements in the southern part of the study area but are less successful at reflecting the riverine structure. At a maximum distance of 10 kilometers, the settlements along the Guadalquivir are very disconnected, which is resolved only when we allow for a maximum distance of up to 18 kilometers (Fig. 7.6a, b).

The nearest neighborhood networks seem more appropriate for reflecting the riverine structure by bridging the often long distances between neighboring riverine settlements (Fig. 7.7). Allowing for only 2 nearest neighbors reveals a very disconnected network, but when each site is connected to its 3 nearest neighbors the Guadalquivir river structure is clearly revealed. However, a higher number of nearest neighbors is needed to connect the Guadalquivir to the southern clusters of sites (4 nearest neighbors to connect the center and east, and 6 nearest neighbors to connect the west).

The relative neighborhood graph picks up a few features that are extremely interesting (Fig. 7.6c). The sites along the Guadalquivir river are connected as a chain, which in turn is connected to the southern clusters through a number of key sites. It creates a link between the Guadalquivir river and the site of Carmona, an Iberian settlement located on a prominent hilltop in the southern area that was also

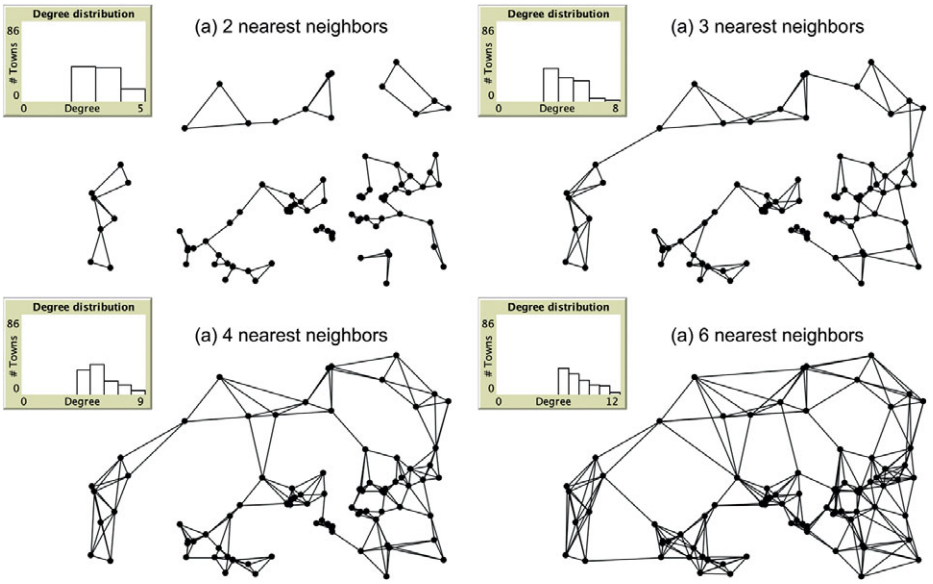


FIG. 7.7 *K*-nearest neighbor models applied to the 86 sites shown in Figure 2.9: (a) 2 nearest neighbors, (b) 3 nearest neighbors, (c) 4 nearest neighbors, (d) 6 nearest neighbors.

an important settlement in the region in the subsequent Roman periods. Moreover, in the southwest it connects to the Guadalquivir river at the site of Seville, an urban settlement that played a major role in the region throughout the ages. The Gabriel graph picks up a similar structure but with more alternative links: it connects sites through a mesh of links rather than the skeletal structure of river settlements connected through key sites as identified by the relative neighborhood graph (Fig. 7.7d).

These models have taught us that models based on the relative distances between settlements are better able to represent the riverine structure behind settlement location, rather than the absolute distances (as reflected by the maximum distance network).

7.6.2 *Networks in Space in the US Southwest*

In this brief case study we use the Southwest Social Networks (SWSN) Project data to illustrate one potential approach toward using spatial network methods to help analyze material culture networks in space as defined above. In the example here we

use a network of ceramic similarity defined for the period from AD 1050 to 1100 where each node is a structure with public architectural features in the US Southwest. Edges in this network are weighted based on the Brainerd-Robinson similarity coefficient in terms of the proportions of ceramic wares present at each pair of sites. We also use the physical locations of sites to create a series of *maximum distance networks* to compare to our weighted ceramic similarity network.

One very common question we might want to ask of network data like these is the degree to which ceramic similarity (a proxy for social interaction) can be explained by the physical distance between pairs of sites. We can assess this using regression models and here we apply the general additive model (GAM), which is an extension of the generalized linear model but is better suited to bounded data (our similarities range from 0 to 1) and other properties of our current dataset. We do not go into great detail on the specifics of this approach here (but see Gauthier 2021 for more) but rather focus on the general process through which we evaluate the relationship between social and spatial patterns. When we build a model to predict ceramic similarity based on distance we find that physical distance is a statistically significant predictor of ceramic similarity ($p < 0.00001$), that ceramic similarity between pairs of sites unsurprisingly declines as the distance between them increases, and that spatial distance accounts for about 37 percent of the variance in ceramic similarity ($r^2 = 0.372$). This model helps us specify the relationship between social and spatial patterns, but what this approach does not tell us is, however, is whether there are certain distances across which the relationships between ceramic similarity and physical distance are more or less pronounced.

To take this further we use maximum distance networks to explore patterns of similarity at a range of distances to identify differences in the relationships between ceramic similarity and distance at different scales. We created a series of maximum distance networks at intervals of 36 kilometers (approximately one day of travel on foot) from 36 kilometers all the way out to nearly 400 kilometers (Fig. 7.8). In other words, we create a series of separate spatial networks where ties are defined as either present or absent if they fall within the selected threshold distances at concentric days of travel from the origin and we compare each of these to the weighted ceramic similarity network defined above. By defining these networks at these concentric differences we are essentially arguing that a trip of one day's travel on foot is somehow qualitatively different from a trip of two days, and so on, rather than viewing distance

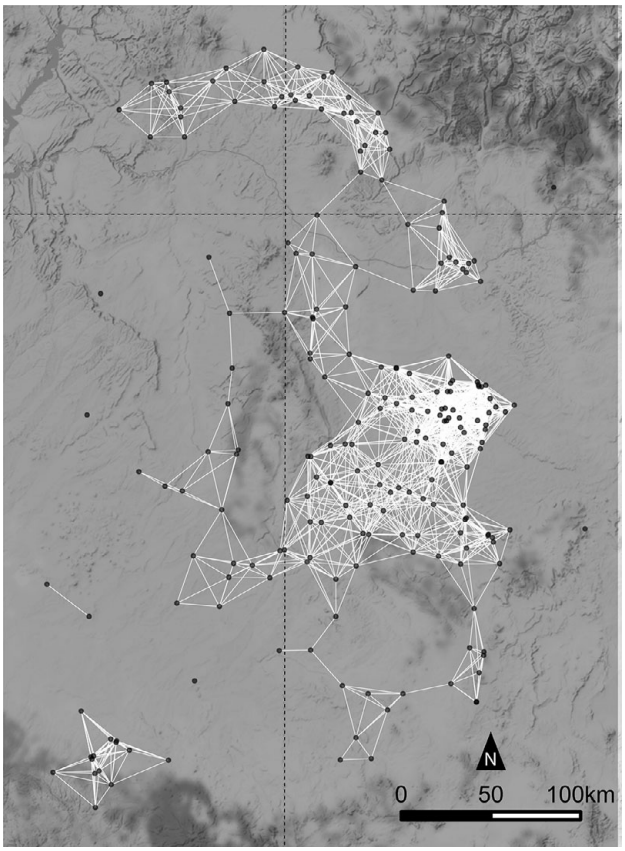


FIG. 7.8 Thirty-six-kilometer maximum distance network map of the Chaco World study area ca. AD 1050–1100.

as simply continuous (e.g., provisioning trips for multiple days involves quite different considerations). Figure 7.9 shows the results for each of these maximum distance networks with the distance threshold on the *x*-axis and the r^2 or proportion of variance in ceramic similarity explained by spatial network ties on the *y*-axis. As this plot shows, just as with raw distance, networks defined across a range of distances help predict weights in the ceramic similarity network across all distances, but the predictions are strongest between about 100 and 180 kilometers (or three to five days' travel on foot). This analysis along with the regression above illustrates that physical distance perhaps plays a stronger role in predicting ceramic similarity (and, by extension, social interaction) at moderate distances rather than very short or very long distances (see also Hill et al. 2015 for a similar argument using cost-equivalent distances).

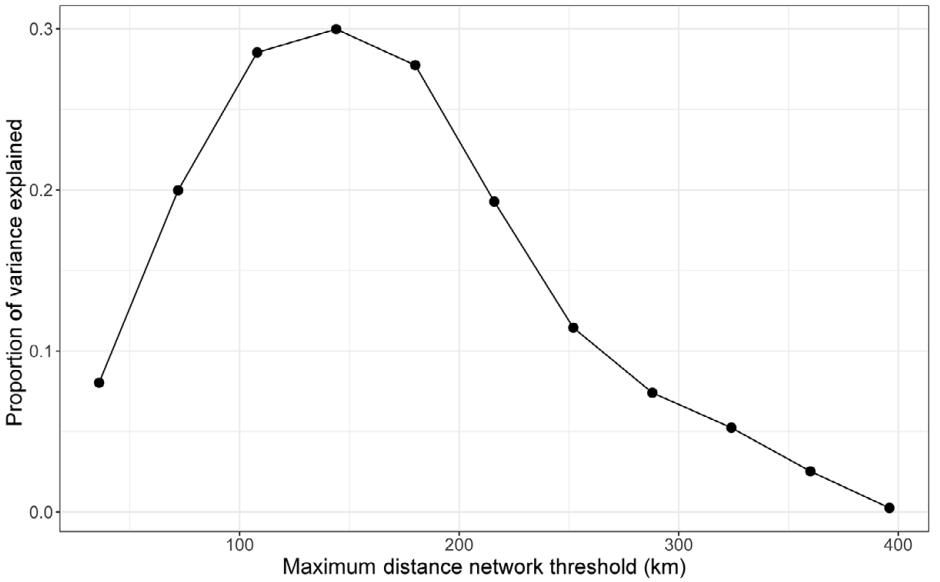


FIG. 7.9 Plot showing the proportion of variance in ceramic similarity explained by network ties for a series of maximum distance networks.

7.7 SUMMARY

- Many relational phenomena studied by archaeologists are inherently spatial, for example, transport systems and visibility. Spatial network methods are invaluable for their study.
- Spatial information can be included in formal studies of many other relational phenomena where at least some of its aspects are spatially embedded, such as social networks or economic transactions.
- In much archaeological network research, spatial information matters in the sense that people/communities/settlements/objects that are located closer together have a higher probability of being related or of having strong relationships.
- The most common use of spatial information in archaeological network research is in studies of transport networks, space syntax, visibility networks, and spatial material culture networks.
- A key feature of many spatial networks is planarity: a planar network is one that can be drawn such that no edges cross.
- Spatial network measures usually take the form of nonspatial network science techniques modified to include a physical distance variable on the edges or nodes.

- In a relative neighborhood graph, a pair of nodes are relative neighbors and are connected by an edge if they are at least as close to each other as they are to any other point.
- If no other nodes lie within the circular region with a diameter equal to the distance between a pair of nodes, then this pair of nodes is connected in the Gabriel graph.
- In a set of nodes in the Euclidean plane, edges are created between pairs of nodes to form a tree where each node can be reached by each other node, such that the sum of the Euclidean edge lengths is less than the sum for any other spanning tree.
- A beta skeleton is a Gabriel graph in which the diameter of the circle is controlled by a parameter, beta.
- In Delaunay triangulation, a pair of nodes are connected by an edge if their corresponding tiles in a Voronoi diagram share a side.
- In a *K*-nearest neighbor network each node is connected to the *K* other nodes closest to it.
- In a maximum distance network, a node pair is connected if the distance from each other is lower than or equal to a threshold distance value.

Further Reading

The following are in-depth overviews of spatial network approaches.

Adams, Jimi, Katherine Faust, and Gina S. Lovasi 2012 Capturing Context: Integrating Spatial and Social Network Analyses. *Social Networks* 34(1):1–5.

Barthelemy, Marc 2011 Spatial Networks. *Physics Reports* 499(1–3):1–101.

Haggett, P., and R. J. Chorley 1969 *Network Analysis in Geography*. Arnold, London.

The following are overviews and introductions to spatial network methods used in archaeology.

Evans, Tim 2016 Which Network Model Should I Use? Towards a Quantitative Comparison of Spatial Network Models in Archaeology. In *The Connected Past: Challenges to Network Studies in Archaeology and History*, edited by Tom Brughmans, Anna Collar, and Fiona Coward, pp. 149–173. Oxford University Press, Oxford.

Conolly, James, and Mark Lake 2006 *Geographical Information Systems in Archaeology*. Cambridge Manuals in Archaeology. Cambridge University Press, Cambridge.

Concerning geographical network visualization, see the following websites.

https://geographic-networks.github.io/poster.pdf
With abstract:
https://geographic-networks.github.io/abstract.pdf
And comprehensive website:
https://geographic-networks.github.io

EXERCISES

Answers for these exercises can be found in Appendix A.

7.1) Which of the networks in Figure 7.10 are planar?

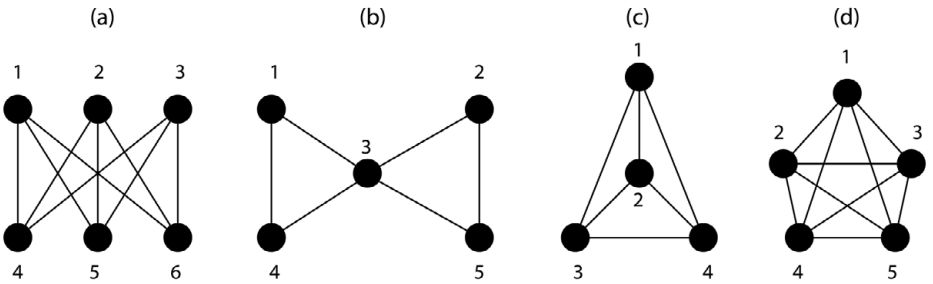


FIG. 7.10 Figure for Exercise 7.1.

7.2) Redraw the network in Figure 7.11 such that no edges cross.

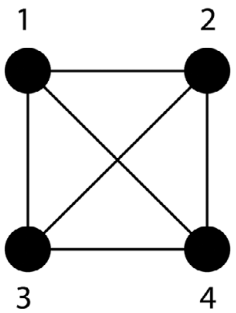


FIG. 7.11 Figure for Exercise 7.2.

7.3) Can the network in Figure 7.12 be redrawn such that no edges cross?

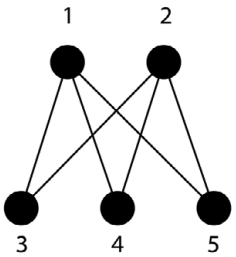


FIG. 7.12 Figure for Exercise 7.3.

7.4) Draw a road network where roads are represented as edges and settlements are represented as nodes, using the Roman roads (thick lines) and Roman settlements (black dots) in the region around Zaragoza (Roman Caesaraugusta) in Spain shown in Figure 7.13.

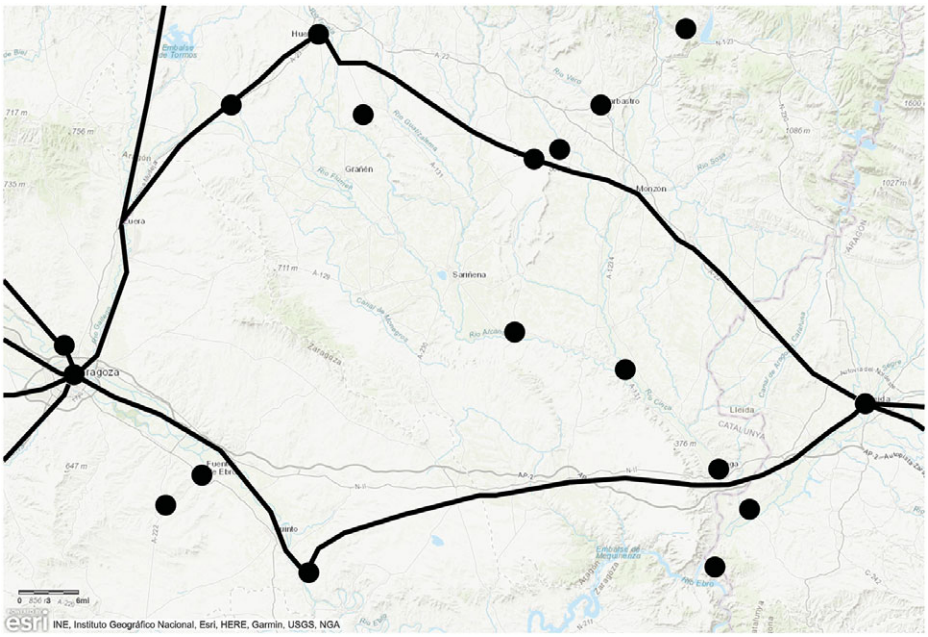


FIG. 7.13 Screenshot from The Digital Atlas of Roman and Medieval Civilizations of the Roman roads and settlements in the area of Zaragoza (https://darmc.harvard.edu). Adapted to enhance visibility of roads and settlements.

7.5) Draw the *undirected* 2-nearest neighbors network of the nodes shown in Figure 7.14.

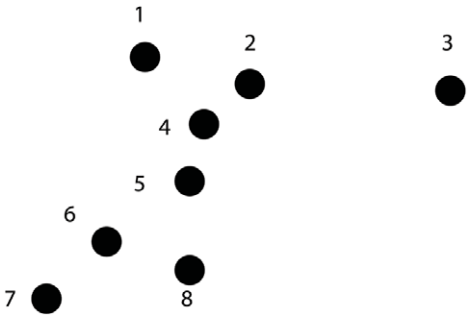


FIG. 7.14 Nodes for Exercises 7.5 and 7.6.

7.6) Draw the *directed* 2-nearest neighbors network of the nodes shown in Figure 7.14.

8

Uniting Theory and Method for Archaeological Network Research

In this chapter we take a step back from our in-depth methodological overview to describe what we see as some of the possible future trajectories of productive and critical network research in archaeology. Networks are already beginning to help us address a broad range of archaeological questions, as we have seen in this book (see discussion in Chapter 2), and networks are certainly useful tools and analytical constructs for many common archaeological tasks. We see the increased importance of network *methods* in archaeology as a trend that is likely to continue. From this vantage point, we ask: What are the profitable next steps that might push network thinking and archaeological network research to the next level? How can network methods and theories help us toward new answers for old archaeological questions or even toward questions we have not yet considered? Can archaeologists contribute to the world of network science beyond archaeology? We believe that network science has transformative potential for archaeological research and that archaeologists can be important players in network science in general, *if and only if* explicitly formulated relational theories drive network research in the field going forward.

In this chapter, we pull together some parting thoughts on how method and theory in networks can (and should) be synchronized in archaeological network research. First, we reflect on the potential of relational thinking and the necessity of a strong and explicit connection between theory- and method-driven research. To explain how this can work in practice, we offer a personal account of sorts to outline how some of our own work has developed in relation to such concerns. In this context we then highlight a few areas of traditional archaeological concern where we think network theories would be able to contribute to archaeological questions. This is by no means meant to be an exhaustive list but provides a hint of what we hope could be on the horizon. Finally, we close on an optimistic note by suggesting

that, despite all of the challenges we still need to face, the future for archaeological network research is looking brighter every day, and we hope this book inspires more interesting and invigorating research in the coming years.

8.1 THE POTENTIAL OF RELATIONAL THINKING IN ARCHAEOLOGY

The prominence of relational thinking in archaeology is in part responsible for the adoption and current popularity of formal network methods in the field. As we described in Chapter 1, relational perspectives are quite diverse and many traditional archaeological concerns and theories involve relational concepts where interactions among units of analyses are thought to be important drivers of the behavior and variability in outcomes for those units. In recent years, influential works in archaeological network research have explored how such relational perspectives and theories can be productively engaged through explicitly relational methods (e.g., Knappett 2011). As a result, a wide array of formal network methods and models has become popular in a variety of archaeological research contexts (see Chapter 2). Although originally inspired by a desire to formalize such relational theories, in practice the recent popularity of networks in archaeology has been perhaps overly method-driven. That is, network science techniques have very often been applied for methodological novelty, largely independently of the relational theoretical debates that inspired their adoption in the first place. At the same time, as relational thinking has increased in popularity in the field there has often not been considerable effort expended in developing such ideas into testable relational theories suitable for formal investigation. We summarize this current state through two observations:

- Relational theories are, and have long been, ubiquitous in archaeological thought but have rarely been formalized using network methods. Further, many relational theories within archaeological research are not communicated in a way that facilitates such formalization.
- Network methods are now commonly applied in archaeological research but are less frequently selected or developed in light of explicitly formulated relational theories or questions.

These brief observations reveal the edges of two related but still somewhat distinct research traditions that are, respectively, theory- and method-driven, but less often both. We believe it is time to rebalance the scales. We suggest that work at the intersection of these two research traditions has the potential to build a new

trajectory of archaeological network research that is more powerful and productive for addressing archaeological and broader social and behavioral science questions. This is not to say that there is not already good work that successfully combines network method and theory in archaeology (and we cite and describe many excellent examples in the book), but if we want to take the field further, we need more critical research in this vein.

Moving forward, we suggest that archaeological applications of network science techniques need to more directly and more explicitly engage with relational theories *from the outset*. It is not enough to simply state that *relations matter*. Critical archaeological network research requires that we evaluate *specific* statements and dependencies among nodes, edges, network properties, generative processes, attributes, and outcomes (or any combination thereof), and these should be derived from clear relational theoretical statements. As a general rule, if you cannot state in a few sentences the specific network process or dependency you are evaluating and the related material proxy, you may need to further develop your relational theory and perspective before jumping into an analysis. On the other hand, we also suggest that it is necessary to expend considerably more effort in developing relational theories that are more explicit, testable, and formal (see Knappett 2020). A useful relational theory for network research needs to define the units of analyses, the expected relationships among them, and the criteria for evaluating the plausibility of the theory (see Smith 2015 for a related general argument about archaeological theory). Further, the use of particular kinds of archaeological proxy evidence to stand in for relational processes or to measure particular outcomes also needs to be well justified. These are areas that have often been given surprisingly little attention in many archaeological network studies.

Our perspective here is firmly rooted in how we introduced archaeological network research as a part of the archaeological process in Chapter 1. Network science in archaeology is the *application* of network science methods to the study of the human past, and not the development of abstract mathematical and computational methods in an interpretative vacuum. The entire process of doing network science in archaeology is necessarily embedded in the archaeological research process (as argued in Fig. 1.3). This means theory and data critique should be the first port of call to motivate the application of network science, the selection of particular concepts to abstract the past phenomena being studied, the appropriate representation of such concepts using network data, the selection or development of network methods, and the analytical decisions being made when applying these methods. Network science merely offers the tools to represent, explore, and test relational theories, and it is a productive laboratory for inspiring new archaeological

theories. For archaeological network research to reach its full potential we must recognize that the relational structures within which individuals and larger groups were embedded offered specific kinds of risks and rewards that need to be studied alongside the more traditional focus on attributes and properties of those individuals and groups. The plausibility of a theory needs to be tested by comparing archaeological evidence against the specific expected outcomes of such theorized risks and rewards. Though relational theories are ubiquitous in archaeological thought, if we cannot formally evaluate them in this way they are of considerably less use. In the next section, we give an example of how theory and methods came together in some of our own work.

8.2 DEVELOPING A TRAJECTORY OF ARCHAEOLOGICAL NETWORK RESEARCH

In light of our discussion of the necessity of combining theory and methods for archaeological network research, we find it useful to provide an anecdote of how this process has played out in some of our own past research. Narratives of how research trajectories develop and change are rarely published, but we suggest they can be quite useful, especially for relatively new fields of study. This example draws on work conducted by Peeples and others as part of the Southwest Social Networks Project (see Section 2.8.2) and, in particular, research focused on a specific kind of intermediate network position called "brokerage" (Peeples and Haas 2013). To briefly review the context, this work is primarily focused on exploring networks of ceramic similarity (and by proxy patterns of potential interaction) among settlements in the US Southwest and Northwest Mexico through time (ca. AD 800–1800). Here we recount how we went from having a vague notion that networks may be useful for studying interaction in the past to a specific trajectory of research focused on the risks and rewards of particular kinds of intermediate structural positions in archaeology and beyond.

A good place to start when thinking about applying network methods and models in archaeology is a specific question involving some sort of *relational processes*. As we described briefly in Section 6.1, this particular research program started with some initial experimental visualizations conducted in the early days of the Southwest Social Networks Project. We knew that we wanted to use similarities in material culture to explore potential patterns of interaction at various scales, but in those early days the specific form that such analyses would take was still somewhat unclear. When our first experimental network graphs were produced,

Peeples noticed that many nodes that apparently fell in intermediate positions in networks of similarity appeared to be settlements that fell in intermediate physiographic positions (along mountain passes or on the edges of other major landforms) across the broader study area. This very simple observation raised a number of new questions: Is there a consistent relationship between geographic intermediacy and network intermediacy? What are the risks and rewards of intermediate network positions for sites in such positions? Do sites in intermediate positions tend to remain in those positions or do network structures cycle through time? These are all *questions about relational phenomena* and the *potential outcomes associated with network positions*.

With these and other questions in mind, the team turned to the vast existing literature on network models in sociology and related fields to see if there were any particular relational theories or models that would inspire us with new approaches or more specific expectations to help address our questions. We were generally interested in how intermediate network positions might have variously facilitated or constrained action or the flow of information or resources in ways that would likely have had material impacts on the settlements that found themselves in such positions. What kinds of studies have already contributed to this or similar questions? Are there particular models or contexts that mirror our own data and/or questions?

We first found a large body of research coming out of diverse fields including sociology, institutional analysis, economics, and international relations focused on "structural holes" as hallmarks of strategically useful positions within networks (Brass 2009; Burt 1995, 2000, 2007; Cowan and Jonard 2007; see also Grannovetter 1973 for a related argument on "the strength of weak ties"). The basic argument in much of this literature is that for many kinds of interactions it is beneficial for an actor (whether an individual or a larger organization/collective) to be connected to lots of other actors who are not connected to each other. The argument is that when an actor is located in an intermediate structural position that actor will, by virtue of their position, have greater access to certain information and resource flows across that network that can be a source of social capital, influence, and advantage. This certainly seems like an interesting model to consider in relation to our questions in that it links intermediate network positions to long-term advantage.

While this structural holes model was appealing, we also encountered another body of research in our search that seemed to offer somewhat different insights regarding similar network positions and processes. Researchers including James Coleman (1988) noted that *closure* (or the concentration of ties among actors with similar relations) was sometimes a source of advantage and that actors in

intermediate positions called *brokers* can sometimes be viewed with suspicion by those in tightly knit groups on either side of the structural holes they span. In certain contexts, it would seem, intermediate actors can be seen as at odds with the cooperative cultural values of the group. A number of studies inspired by Coleman and also in response to the published work on structural holes by Burt and others revealed that in social settings where collectivist values are paramount, actors in intermediate positions do not tend to accrue advantages based on their positions (e.g., Batjargal 2007; Bian 1997; Kwon et al. 2020; Xiao and Tsui 2007). The distinction here perhaps suggests that cultural values and histories are shaping network outcomes (see also Pachucki and Breiger 2010).

So here we have two apparently competing models regarding the risks and rewards associated with intermediate network positions like brokerage. One model (which we called the individualist model) suggests that intermediate positions should be a source of advantage and influence because actors in such positions have access to diverse and exclusive information and resources. The other model (which we call the collectivist model) suggests that intermediate positions are not associated with advantage when collectivist values are particularly salient because actors outside tightly knit groups may be viewed with suspicion or may not be able to maintain their intermediate positions as new coalitions form. This distinction appears in the literature whether the actors in question are individuals or larger collectives. The collectivist model further suggests that we might expect to see closure through time, where nodes that are connected by an intermediate should become increasingly connected themselves. At this point we realized that this was an opportunity to flip the lens and to use network structural patterns to provide insights into the cultural values potentially at work in the distant past. Specifically, we could document intermediate network positions and their associations and evaluate which model better fit the evidence in our case study. This would further provide insights into the individualist or collectivist values that structured risks and rewards in that setting more broadly.

With this *theoretical model* and related *expectations* in hand, we now need to find an appropriate *method*. We had, at this point, already developed a model using similarity in ceramics as a proxy for likely or potential patterns of interaction among settlements at various scales through weighted similarity networks (Mills et al. 2013a, 2013b). From here, there are many existing methods available for identifying nodes in positions characterized by a high degree of brokerage (e.g., Gould and Fernandez 1989; Stovel et al. 2011; Stovel and Shaw 2012). In general, such models explore triads and identify how often a given node connects two other nodes that are not connected to one another (e.g., B is a broker for a particular

relation if A is connected to B and B is connected to C but A is not connected to C). This seems like just what we needed, but we still had more hurdles to jump. The existing methods are all designed to work with binary edges that are either present or absent, but we also had information on edge weight based on the similarities in ceramic assemblages between pairs of sites. Thus, we created a modified version of the network brokerage measure for weighted ties that was suited to the particular structure of our archaeological network data. Finally, we needed a way to evaluate any apparent advantages or disadvantages accrued by nodes based on their structural positions. To accomplish this, we used population growth and longevity as proxies for relative advantage (assuming that sites in positions of relative advantage will outgrow and outlast their peers). If our data fit the individualist model, we would expect to see sites characterized by a high degree of brokerage, exhibiting higher levels of settlement growth and longevity. If instead the collectivist model was a better fit, we would expect sites with the highest degree of brokerage to be among the smallest and shortest-lived sites, and further, we would expect evidence for closure (sites connected by a broker to become more similar through time).

We were now ready to apply our *network theory and methods in tandem*. We found a pair of theoretical models that helped us evaluate the risks and rewards of brokerage positions and we developed archaeological expectations for each model. We calculated site-level brokerage scores for a series of 50-year intervals from 1200 to 1400 and evaluated the evidence (Peeples and Haas 2013). We found that sites characterized by the highest structural potential for brokerage were typically relatively small, in low population density areas, in geographically peripheral or intermediate areas, and were among the shortest-lived sites in our sample. Overall this provides evidence pointing toward the collectivist model. Beyond this, we explored evidence for closure addressing whether sites that were connected by a broker in one period were more likely to be more similar to one another in the subsequent period. Indeed, we also found strong evidence for closure. Altogether, our analysis suggests that the risks and rewards associated with brokerage positions in our US Southwest networks mirror those in highly collectivist contemporary settings, and thus, we might infer similar values at play in our archaeological study area.

The story does not stop here. The research we conducted in this initial investigation of brokerage (Peeples and Haas 2013) led us to many *new questions*. In particular, our group continued to investigate associations between sites and areas characterized by a high potential for brokerage and other social processes, such as migration and ethnic co-residence (Peeples and Mills 2018). In this work we are beginning to see evidence that the risks and rewards of network positions may even have changed through time in our study area, in particular in relation to changes in

political organizational strategies. This is an area where archaeology may have important contributions to make to the broader field of network science. In particular, we have the opportunity to explore the long-term trajectories of networks and the changing risks and rewards of network positions at a temporal scale that only archaeology can provide. Thus, we are hopeful that such work will soon be able to contribute insights to the broader social sciences and network science investigations of how human networks transform across generations.

In this brief example, we highlight all of the steps we took between coming up with a question to new revelations about relational phenomena. We started with a simple question that was prompted by an exploratory visualization. We explored the network literature and found a set of similar relational theories that could provide answers to our questions. We then developed specific archaeological expectations related to the network theories we wished to test and then developed a method and metric catered to the specifics of our archaeological data. Finally, we tested the model and evaluated the network theories, providing new insights that relate to ongoing debates not just in archaeology but in the broader world of the social sciences. Importantly, we also made the data and analytical code/tools we developed to conduct this research publicly available so that others could use our data, methods, and models to further their own analyses (and this has already begun happening; see Hart et al. 2019).

8.2.1 *Advice: What Have We Learned?*

The research trajectory we outline here is certainly not the only way forward, but, as this example illustrates, the road can be a long and winding one. Although we cannot predict where your own research will take you, we can offer some advice in light of this discussion and our other experiences.

First, we suggest that one of the most important steps toward critical archaeological network research is centering relational thinking, relational questions, and relational theories from the beginning. Although there will probably be some natural back-and-forth between questions and analyses as your research develops, we strongly encourage you to start with questions and theories rather than a method or analysis. There is a tendency to treat shiny new methods like toys and to simply jump into an analysis to see what shakes out without first having a specific question in mind (and we could certainly be accused of some of this in our own work). Although exploration is useful, in our experience, the sooner you can get to an explicit relational theory to test, the sooner you will be able to find your way toward novel insights.

Beyond this, consider that the world of network science in the social and behavioral sciences is vast and has much to offer archaeologists. In the example here, we noted a pattern in our archaeological networks and sought examples of similar relational contexts and processes in the broader network literature; this effort paid off when we found two related models connecting intermediate network positions to relative advantage. While there are certainly plenty of relational questions and theories that come directly out of archaeological research, it can often be useful to see what others have done in similar contexts to avoid reinventing the wheel or to identify questions you did not even know you had (which is very much what happened in the example here).

Finally, as this example illustrates, you should be prepared to modify "off the shelf" methods and models for archaeological data. The nature of archaeological network data often requires tweaks to methods or more careful consideration of underlying data and assumptions (see Chapter 5). You will likely be better served by altering a method to address your particular question than by simply applying an existing method that may have been designed for a very different context or type of data.

8.3 RELATIONAL THEORIES HERE, THERE, AND EVERYWHERE

As we have argued throughout this book, relational theories are everywhere in archaeology and the broader social and behavioral sciences. Such theories provide considerable food for thought to stimulate work on the intersection of method- and theory-driven archaeological network research like we advocate here. In this section, we expand on this point by briefly touching on a few areas of archaeological concern where network models and relational theories may have an important role to play. Some of these topics have already seen considerable archaeological network research, while others have as of yet seen very little. This is far from an exhaustive list of topics but we hope that, in light of the discussion and narrative above, this might help to inspire some new formal network research or at least provide examples of interesting relational questions.

8.3.1 *Environmental Variability and Social Networks*

Archaeologists have long focused on topics revolving around patterns of interaction and environmental variability at various scales. There are many common ecological and archaeological models with relational expectations for how interaction should

be structured by the nature and scope of ecological and physiographic variability in a given environment. For example, there is a long anthropological tradition of studying networks of exchange as risk-buffering mechanisms that help to facilitate the flow of resources and information while also providing knowledge of and contacts in other areas as potential options for relocation when local environmental conditions decline (e.g., Allen 2004; Braun and Plog 1982; Cordell et al. 2007; Rautman 1993; Sobel and Bettles 2000; Wiessner 1983, 1998; Wobst 1974). This body of research offers numerous potential directions for exploring the complex relationships between social networks and the environment and many compelling relational questions. At what scales are networks of interaction most influenced by environmental variability? At what scales should we expect the movement of resources versus the movement of people in response to changing or variable environmental conditions? Are certain kinds of environments or risk profiles more amenable to networks as buffering mechanisms than others? Are certain kinds of network structural properties more or less robust to climate challenges or environmental variability? Can we use environmental variables to predict likely patterns of interaction where other information is lacking?

This is an area of research that has already seen some formal archaeological network analyses, but there is certainly much more to do. For example, Borck and colleagues (2015) explored the degree to which network ties, defined in terms of ceramic similarity in the US Southwest, fell within a single region or spanned beyond traditional regional boundaries using measures of network embeddedness. In this work, they found that settlements in regions with diverse ties extending beyond the local environment were more likely to weather severe climate shocks in place than settlements in regions with highly localized networks. In another series of studies, researchers studying hunter-gatherer populations in the Kuril Islands in the northwest Pacific have used formal network analyses based on obsidian and ceramic exchange to explore the role that networks played as sources of information in an unpredictable environment (Fitzhugh et al. 2011; Gjesfjeld 2015; Gjesfjeld and Phillips 2013; Phillips 2011). This work, drawing on network expectations based on ethnographic studies of contemporary hunter-gatherer populations, has explored the different kinds of network topologies that might be expected with varying degrees of environmental variability and interaction costs. This is a topic that we expect will see more formal network research in the future as the role of long-distance interaction for information exchange is a classic topic in the study of hunter-gatherer populations and a research area replete with relational theories ripe for formal investigation (see chapters in Whallon et al. 2011; Wiessner 1983). Beyond this, Fox and colleagues (2017) argue that such network risk-buffering

strategies were fundamental processes in human evolutionary trajectories. They refer to humans as "social network engineers" and argue that the creation of information and risk-buffering networks was a prerequisite for the expansion of the human niche and the creation of resilient communities that ultimately led to the increasing social and organizational complexity of our species. The timing, scale, and scope of interactions and relational processes in the evolution of our species is certainly a topic that we expect would benefit from formal network methods and relational theories.

Another area where network methods may be particularly relevant is work focused on complementarity in the risk profiles of different regions and how such patterns may have influenced interregional interaction over the long term. Rautman (1993), for example, explores patterns of long-distance exchange in central New Mexico and suggests that the geographic "shape" of networks of exchange should include a directional component influenced by differential environmental risk. In other words, if networks were important as risk-buffering mechanisms, we should expect areas that experience environmental risk out of sync with one another to develop strong interactions over the long term, an argument with clear potential for network research (see also Cordell et al. 2007). There have been a small number of formal network studies that have attempted to directly test this relational theory. This includes work by Gauthier (2021) in the US Southwest, who uses cutting-edge retrodicted climate models and ceramic similarity networks to demonstrate that regions that experienced complementary environmental conditions (i.e., when conditions were favorable for settled agriculturalists in one region, they were unfavorable in the other) were characterized by a greater than expected degree of ceramic similarity, suggesting frequent interaction. At a smaller scale Strawhacker and colleagues (2020) focus in depth on two regions of the US Southwest to show that the relationship between social networks and risk-buffering may also vary in relation to quite local conditions, including physiographic features, precipitation regimes, and surface water availability. Overall, this suggests that this is a topic that would benefit from further investigation in a wider variety of environmental and social settings.

Risk-buffering through social networks is a topic that has recently garnered attention among ethnographers and evolutionary anthropologists. Archaeologists interested in network processes would likely be well served by engaging with the implications of this work. For example, work by Pisor and colleagues (e.g., Pisor and Gurven 2016; Pisor and Jones 2021; Pison and Surbeck 2019) suggests that the utility of social networks and long-distance connections as risk management strategies should be expected to vary in relation to the relative costs and benefits

of all available adaptations and even the social values placed on in-group and out-group interaction in a given context. This is a body of work that certainly has implications for future archaeological network research, and we hope that researchers will further engage with it.

8.3.2 *Cultural Evolution and Cultural Transmission Processes*

The study of cultural evolution and cultural transmission processes is an extremely diverse topic with applications across many social and behavioral science fields, including archaeology. Most models of cultural transmission processes involve relational components or considerations of social network structure to varying degrees, and thus, such work potentially provides many avenues for exploration using formal network methods. There is experimental, ethnographic, and other evidence to suggest that cultural transmission both shapes and is shaped by social network topology, size, and the underlying network generative processes (e.g., Cantor et al. 2021; Derex and Boyd 2016; Henrich 2004; Migliano et al. 2017; Romano et al. 2020; Shennan 2001; Smolla and Akçay 2019). Many cultural evolutionary studies offer a broad array of relational theories and questions and quite clear models and expectations that could potentially form the basis of archaeological network research. What is the relationship between network connectivity and the rate of innovation for a population? What is the relationship between biased and unbiased cultural transmission and the degree distribution or other properties of underlying social networks? How does the nature and complexity of a given technology influence the structural properties of the cultural transmission network through which people learn that technology? What network topological properties or structures are associated with punctuated versus gradual innovation? Is network modularity and segmentation related to technology loss over the long term?

Connecting cultural evolution models to archaeological data is particularly appealing as archaeological evidence provides opportunities for exploring cultural transmission processes at temporal scales that are far longer than those available to other lines of evidence. At the same time, there are considerable challenges in applying or testing specific cultural transmission models using most kinds of archaeological evidence. Specifically, being able to differentiate between learning frameworks or different kinds of biased or unbiased cultural transmission often requires a specificity of evidence that can be hard to achieve given the temporal and other uncertainty associated with much archaeological data (see Perrault 2019).

Perhaps due to such challenges one of the most popular approaches to the applying of cultural evolutionary models to archaeological research involves the development and comparison of mathematical models and simulations that are sometimes evaluated in relation to archaeological data. For example, Bentley and Shennan (2003) develop a model of stochastic network growth and define material culture expectations for biased versus unbiased cultural transmission (see also Neiman 1995). They then evaluate the plausibility of this network model using ceramic stylistic data from Linearbandkeramik settlements in western Germany, suggesting that it is possible to differentiate modes of cultural transmission using such stylistic frequency data. More recently Crema and colleagues (2016) expanded on this and related work to further simulate biased and unbiased transmission in situations where the mode of cultural transmission can change through time. They explore their model in relation to the same Linearbandkeramik dataset and demonstrate that if modes of cultural transformation are not at equilibrium, evaluations of transmission processes for any particular time can be misleading. We suggest that approaches such as this (see also Kandler and Caccioli 2016) that use network models and generative principles to create expectations for evaluation through archaeological assemblages have considerable potential for future archaeological network studies.

Beyond this, there is also considerable potential for archaeological network investigations of cultural evolutionary processes through the application of material culture and technology networks like we have discussed throughout this book. Cochrane and Lipo (2010), for example, explore evolutionary relationships among archaeological assemblages for Lapita pottery in the South Pacific using a network of ceramic stylistic similarity and use this model to argue for the importance of horizontal transmission in explaining regional patterns of stylistic variation. Despite the potential for such approaches, studies like this that use networks generated from archaeological assemblages to directly model cultural transmission processes are still relatively rare. We feel that there is considerable unrealized potential here for archaeologists to engage in broader conversations on the nature of social networks and cultural transmission in particular as many cultural evolutionary studies offer clear testable relational statements. For example, Derex and Boyd (2016) provide an experiment that suggests that groups of individuals characterized by partially connected networks produce more diverse complex cultural traits than groups within fully connected networks. This is a relational statement with specific test implications for archaeological network research, yet explicit tests of models like this have been rare. There is some indication that the picture is beginning to change, however. Romano and colleagues (2020) recently outlined

what they call a "multilevel analytical framework for studying cultural evolution in prehistoric hunter-gatherer societies" that draws on cultural evolutionary theory broadly to present concrete questions and, importantly, likely material correlates for evaluating such theories using archaeological evidence. We hope this is the beginning of a new and productive line of research.

8.3.3 *Past Economies*

Another area where archaeological network research has transformative potential concerns the study of the flow of goods, people, and information. What were the interpersonal connections that allowed for commercial information like the availability of products to be shared within households, settlements, communities, and states? How did these flows of information structure the flow of goods? How did the physical proximity and distribution of resources and settlements, and the available infrastructure, shape the physical transport of people, primary products, and goods? And how were social and physical space interdependent: How did the environment structure the flow of information, and how did interpersonal interactions structure the distribution of material culture?

The concept of economic integration and its study in ancient Mediterranean economies reveal this potential. Economic integration refers to both the degree to which members of different communities had free access to each other's markets and to the diversity of political, economic, physical, social, and military ways in which this is expressed (Balassa 1961). The highest degree of integration is achieved when trade and commercial competition is completely free, when there are no obstacles to the movement of people, goods, and/or information. In practice, this extreme is never achieved, and past societies have always been characterized by a mix of different modes of integration: political union, military union, free movement, monetary union, and a common market. The degree of economic integration of the ancient economy and particularly of the Roman imperial economy is the subject of a long-standing debate among archaeologists and historians (Bang 2008; Finley 1973; Temin 2013; Wilson et al. 2012). The Roman Empire was huge, covering at its height an area comparable to the present-day European Union, and there is strong evidence to suggest that the Roman imperial economy was highly integrated. Under the *pax Romana*, all the communities in the Empire were part of a political, economic, and monetary union, where male citizens in theory had the ability to move freely from Britain to Syria. The Roman military presence throughout the empire tended to restrict warfare to the border regions, and the

many campaigns against piracy allowed the Mediterranean to flourish as a nexus for long-distance trade. The huge numbers of ceramic amphora containers provide tangible evidence of foodstuffs being produced in industrial quantities on one end of the Mediterranean to supply the demands of communities on the other side. However, there is no doubt that economic integration was more limited than in our present-day world of ultra-fast global transport and immediate communication. The physical infrastructure of Roman roads and major seaports was a great improvement on what was there before but must have structured and limited the flow of goods, people, and information in a very particular way: some places were very well connected with well-maintained infrastructure; other places were basically unreachable. A further limiting factor was the speed at which information and goods could travel with the available shipping and transport technology. Despite the *pax Romana* introducing a long period of relative peace within the borders of the Empire during the 1st century AD, most of the Roman Empire's history was dominated by frequent, hugely disruptive civil wars.

Many aspects of this debate concern relational phenomena, and a large number of relational theories to explain these phenomena have been developed by archaeologists and historians alike: the role of down-the-line trade of foodstuffs as compared to direct distribution to consumption centers (Arnaud 2005; Nieto 1997), the impact of large population centers as consumption and redistribution hubs (Abadie-Reynal 1989; Hanson 2016), the structure of the physical transport system, and the risks and rewards associated with the structural position of urban settlements within that system (Adams 2007; Carreras and De Soto 2013; Scheidel 2014). In recent years the debate has focused strongly on the role of social networks in facilitating economic integration in the Roman world. For example, Peter Bang (2008) has argued that the role of communities was crucial to structure the flow of commercial information within and between markets. These could be guild-like communities of traders, religious communities, kinship communities, or communities of foreign individuals with the same home region. But all of these diverse types of social relationships are theorized to be media for the flow of commercial information and to structure that flow in a specific way: reliable commercial information is preferentially shared with members of a community, who tend to be protectionist and to disadvantage outsiders. This relational historical theory about the role of social networks in Roman imperial economic integration has been explored using network science methods, by representing the theorized structure as network data (a so-called small-world network structure) and by simulating economic processes over this network structure that produce distributions of simulated bricks and ceramics that were subsequently compared with excavated

distributions of bricks and ceramics (Brughmans and Poblome 2016; Graham and Weingart 2015).

These examples illustrate that network methods are necessary tools for representing and exploring the many relational theories surrounding past economies. But they will also prove invaluable for addressing some of the socioeconomic phenomena that pose grand challenges for archaeology (Kintigh et al. 2014). How and why do market systems emerge, and how do they evolve over long time periods? Why do agricultural economies emerge and how do they spread?

8.3.4 *Interpersonal Social Networks*

Historically, the main area of application of network science methods in the social sciences has been to study human interpersonal interactions. Formal studies of past social networks are also common in history (Ahnert et al. 2020), thanks to interpersonal relations documented in textual sources (e.g., marriage and death registers, correspondences) and the inspiration of how to analyze these sources offered by the thriving social network analysis research tradition. But formal studies of past interpersonal social networks are very rare in archaeology and perhaps most common in historical archaeology (e.g., Graham 2006). This is due to the challenge posed by identifying past interpersonal relationships through material data sources alone.

Curiously, this challenge has not prevented archaeologists from theorizing about past interpersonal relationships. The above examples revealed that such theories are inherent to our studies of cultural transmission and economies, in which social network structures often serve as a medium for the flow of information between individuals. Moreover, archaeologists have developed enormously creative approaches to identifying possible interpersonal relationships in fragmentary material data, such as identifying marriage relationships for pairs of individuals depicted together on funerary sculpture (Brughmans et al. 2021) or assuming higher probabilities of intercommunity interactions based on material culture similarity of sites (Mills et al. 2013a). The history of archaeological research actually offers quite a few leads for formal social network studies.

We believe that the study of past interpersonal social networks is one of the most promising areas for future archaeological network research. Through the study of material remains, our discipline offers the only way into understanding the vast majority of human history, theoretically enabling us to gain insights into the long-term evolution of interpersonal relationships between members of our species.

Network research is the approach needed to gain such insights, because it enables representation of network structures and models for theorized long-term social network change, facilitated through the context-independence of its approaches. What was the structure of social networks of early humans, and what functions could they serve? How has this structure changed over human history? What technologies, institutions, or climatic factors have caused significant changes in human social network structures, or in humans' abilities to sustain integrated social networks of certain sizes? Moreover, insights into the development of human social networks could in turn be used to critically reflect on present-day social networks and their future development over the long term. The rise of the internet and online social networks has significantly changed the structure and nature of human social networks. As a species, we are now more tightly connected on a planetary scale than ever before. These factors shape the structure of interpersonal interactions in the 21st century, and no doubt make present-day social networks significantly different from earlier human networks. But what exactly is that difference? How do the structures of past social networks differ from 21st-century ones? And what are the long-term implications for our species? How will our new interpersonal network structures evolve over long time periods, and what are the implications of these structures for future societies? Such research questions strike at the core of how archaeology can contribute to a better understanding of the past, present, and future of humanity. Addressing this key topic will connect archaeological network research to the broader comparative social sciences and provide a platform for integrating archaeological perspectives into ongoing interdisciplinary debates.

Unlocking the potential of this important research theme will require archaeologists (1) to comprehensively document the material residues of past social networks and (2) to make more use of computational modeling techniques for representing theorized network structures and their long-term change.

A crucial first step is to identify our empirical basis. We need to perform a systematic survey of how archaeologists have been able to deduce interpersonal interactions and relationships of different kinds: kinship, marital, institutional, conversational. What are the material and textual data types that reveal these types of interactions, how can their fragmentary nature be addressed, and how can we represent these using network approaches? Moreover, we should review the diverse ways in which archaeologists commonly formulate assumptions about material data observations aimed at revealing potential for past interpersonal interactions: material culture similarity, spatial proximity, resource distributions, transport infrastructure. How exactly do we tend to formulate assumptions that consider these patterns proxy evidence for past interpersonal interaction potential, and can we derive

inspiration from present-day observations about data patterns correlating with potential for interpersonal interaction? How can we more explicitly attach probabilities to past interaction potential based on material culture observations and our assumptions? Such work would result in a more systematic and explicit overview of the empirical basis for interpersonal interactions and the theories surrounding them: a sort of encyclopedia of the material residues of social networks.

A second area for future research investment is to make more use of the computational modeling approaches that allow us to formally represent our theories of long-term network change. How did human social networks change from one observed structure to another, and what processes drove that change? Scholars in a range of disciplines have developed network models, exponential graph models, and agent-based models of change in present-day social networks, providing excellent formal frameworks to represent our own theories. Do such models reveal different behaviors or data patterns over very long time periods that are beyond the scope of studies in other disciplines? Observations of present-day social networks in a diverse range of societies will prove invaluable for informing archaeological modeling efforts (e.g., Migliano et al. 2017).

8.4 ONWARD AND UPWARD

The time is right for archaeological network research to become more ambitious. Networks should be considered one of our many tools of the trade that can constructively contribute to enhancing our understanding of past human behavior. We have demonstrated throughout this book that network approaches can do so in a unique way that distinguishes them from other tools in our toolbox. As archaeological network researchers, we should lean in to this distinct functionality, and explore it more explicitly and with a greater diversity of approaches than we have done so far. We can do this by formulating archaeological relational theories, by identifying relational theories in the rich history of archaeological thought as this chapter has sought to do, and by letting these theories guide archaeological network research. The human past is shaped by relationships of many kinds, and its study reveals a diversity of relational phenomena. Archaeological network research is the subdiscipline that represents and explores these phenomena using the formal methods designed to do just that. We should do more of it, and we hope this book supports that ambition.

Appendix A

Answers for Exercises

This appendix provides answers and explanations for the exercises at the end of Chapters 3–7.

CHAPTER 3

1) Figure A.1 shows examples of correct answers to questions a, b, and c (note there is not one correct answer: any drawing with the right number of river segments, confluences, and sites will be correct).

   Answer to questions: How do you interpret the differences between these three network representations? What different types of relationships of the same river system do they highlight? What kinds of archaeological research questions would you ask of each of these types of representations?

   a) Network representation (a) stays closest to revealing the course of the river system. It includes all confluences, all relationships between the river segments, and the direction of their streams. Relationships between sites need to be inferred from this. This representation is appropriate for asking research questions about the structure of the river system or how the sites are separated by the confluences and river segments, for example, How many confluences and river segments need to be crossed for the community at site 1 to reach the community at site 3?
   b) Representation (b) has simplified the structure of the river system to focus instead on the potential connections between the site communities and the directions of potential interaction. Which site communities can reach which other site communities, when one can only travel downstream? Do other sites need to be passed?
   c) Representation (c) reveals how river sections are connected by confluences and is arguably a less intuitive representation of the same dataset. It does

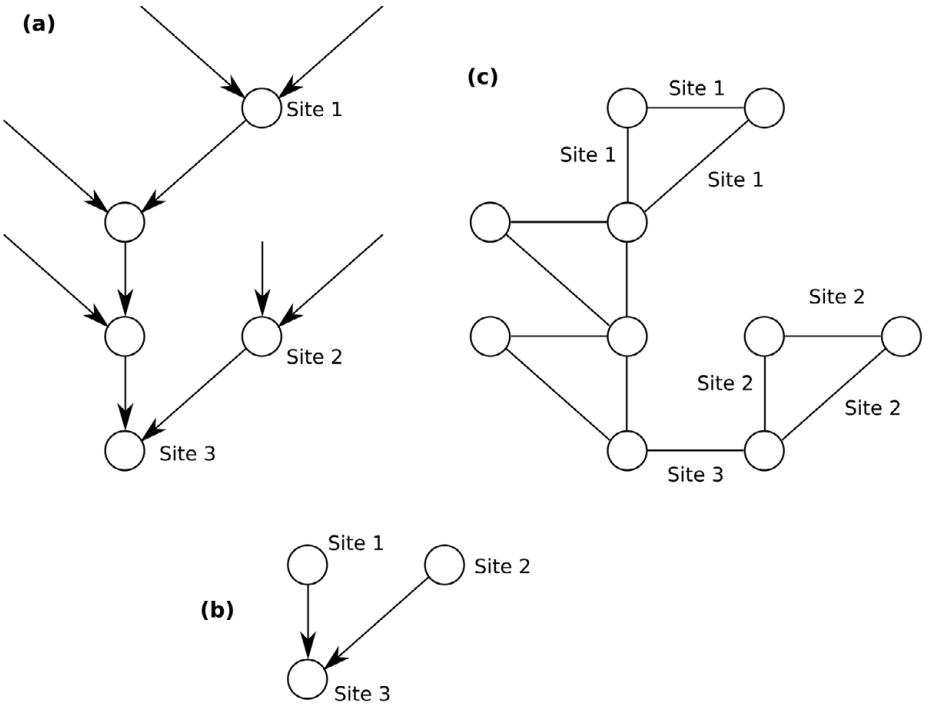


FIG. A.1 Answers to Chapter 3, Exercise 3.1.

not reveal directionality anymore and cannot be used to explore research questions where the distinction between upstream and downstream is of significance.

2) See Figure A.2.

(a)

from node, to node, year
n3,n1,1996
n3,n2,1996
n4,n3,2000
n5,n3,2000
n6,n3,2011
n6,n4,2011
n7,n1,2017
n7,n4,2017
n8,n3,2018
n8,n6,2018
n9,n5,2017

(b)

n3, n1,n2
n4,n3
n5,n3
n6,n3,n4
n7,n1,n4
n8,n3,n6
n9,n5

(c)

	n1	n2	n3	n4	n5	n6	n7	n8	n9
n1			1				1		
n2			1						
n3				1	1	1		1	
n4						1	1		
n5									1
n6								1	
n7									
n8									
n9									

(d)

node, author
n1,Hamilton
n2,Duprez
n3,Franken
n4,Lee
n5,Cortez
n6,Johnson
n7,Gupta
n8,Wu
n9,Assado

FIG. A.2 Answers to Chapter 3, Exercise 3.2.

3) a) Gaius C., Baebius, Galeo, Gaius M.
   b) Gallia Lugdunensis, Africa, Hispania Baetica, Asia, Dalmatia
   c) See Figure A.3c.
   d) See Figure A.3d.
   e) See Figure A.3e.
   f) The one-mode projection of governors reveals how governors are connected in some way through a shared experience or post: a pair of governors is connected if they governed the same province at some point. We can use this to answer questions, such as, which governors held many of the same governor offices and which governors didn't? The one-mode projection of provinces reveals how provinces are connected in some way because they were governed by the same governor at some point. Which provinces tend to be frequently awarded to the same individuals and which provinces don't? Do we have other information available to use that correlates with these patterns, such as the imperial or senatorial status of the provinces, the number of other awards and posts held by these individuals, the cumulative wealth that could be extracted through taxation, or the stage of a governor's career at which they were awarded additional provinces?

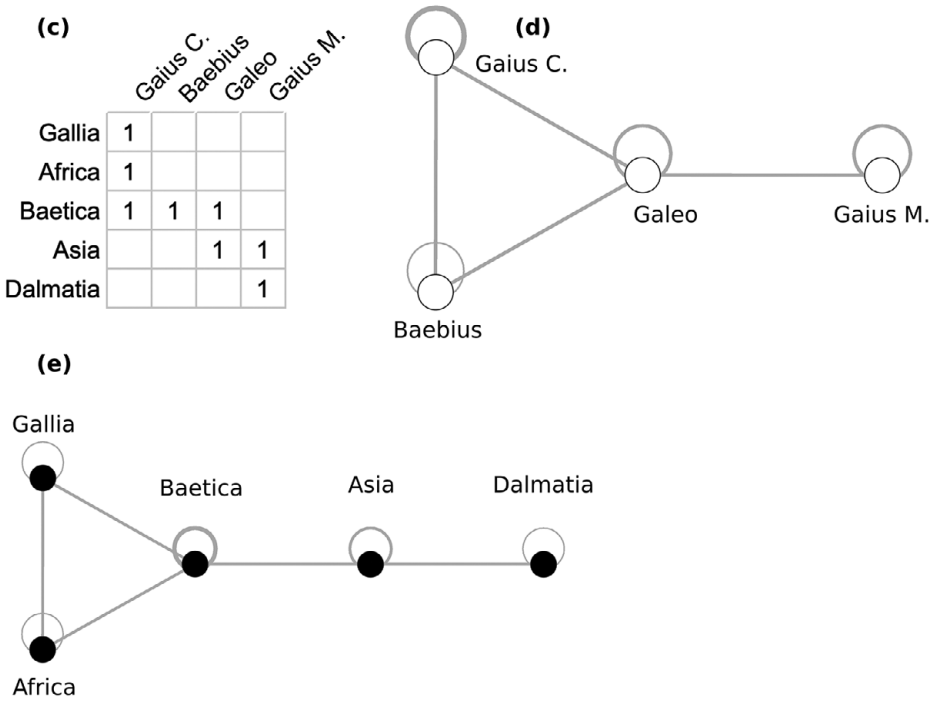


FIG. A.3 Answers to Chapter 3, Exercises 3.3c, d, e.

CHAPTER 4

1) c
2) a and d
3) 3
4) 2
5) St. Kitts is located within a 45-kilometer distance of three other islands. In light of our theory, its past communities had a higher frequency of interaction with past communities on Saba, St. Eustatius, and Nevis than with communities on other islands. The degree of St. Kitts is higher than that of most other nodes in the network: it is higher than the network's average degree. Its past communities were better located to obtain and exchange diverse goods or information than those on islands with a lower degree.
6) 0.2
7) 3
8) Saba, St. Kitts, Nevis
9) 2 edges
10) Disconnected
11) 3
12) No
13) Yes
14) The diameter is the largest network distance between island pairs. It equals 2 for this network and four island pairs are at distance 2 from each other. Past communities at these four island pairs had less frequent exchanges of goods and ideas than past communities separated by a network distance of 1. The network is disconnected, which means that past communities cannot exchange goods and information with other past communities on islands that are part of different network components. Islands that are part of different components are not reachable.
15) St. Kitts and Antigua
16) There are three cliques:
    i) Saba, St. Eustatius, St. Kitts
    ii) Guadeloupe (G–T), Guadeloupe (B–T), Marie-Galante
    iii) Guadeloupe (G–T), Marie-Galante, La Désirade
17) None: all islands are part of at least one 2-clique because the network diameter is 2.

18) Saba, St. Eustatius, St. Kitts, Guadeloupe (G–T), Guadeloupe (B–T), Marie-Galante, La Désirade
19) 3 nodes have a degree centrality of 3: St. Kitts, Guadeloupe (G–T), Marie-Galante
20) 0.2
21) St. Kitts
22) Degree centrality

CHAPTER 5

1) In this case, since you have some information to suggest that site size is related to the likelihood of detection, it makes sense to try to model this as a biased sampling process. There are a number of approaches you might consider here and you probably would want to gather further information to make this decision. For example, you could simply resample such that the larger a site is, the more likely it is to be retained. If you had more information on site distributions and, say, a portion of your study area that has been subject to full coverage survey, you may have other options. For example, if you find that only sites under a certain size threshold are over-represented in the full coverage survey area, you could associate only sites below that threshold with lower probabilities of retention (to simulate smaller sites missing) in the resampling experiments. There is not one "correct" answer to this question, but this discussion helps to highlight the things you will want to consider.
2) This is somewhat of a trick question (sorry). If you conducted an experiment focused on the robustness of degree centrality for a given network that does not provide information which could help you evaluate the robustness of betweenness centrality to the same kinds of perturbations. This is why assessments of network properties need to be focused on the specific measure to be used. As the examples in Chapter 5 show, many networks are robust to missing data for one measure but not others.
3) a) When considering whether to model the situation described in Exercise 5.3 as a random or biased sampling process, it is important to consider all of the information you have about the nature of your sample in relation to the likely underlying population. In this case, you have some indication

that you may be missing settlements that have been destroyed by contemporary development but no evidence to suggest that particular settlements are more likely to be missing than others. Thus, it makes sense to model this as a random sampling process (nodes and/or edges missing at random).

b) In order to assess the robustness of degree centrality to nodes missing at random you could design a resampling experiment to subsample the full dataset at a series of steps (perhaps 90%, 80%, and so on down to 10%) and create 1,000 random subsamples. You could then calculate degree centrality for every node in every one of those random samples and then compare the rank-order distribution of each subsample to the "original" network using Spearman's ρ correlation coefficient. If those correlations were high despite reasonable amounts of missing data, you could have greater confidence in the robustness of that measure to nodes missing at random.

c) If you found that degree centrality fluctuated considerably with only small amounts of subsampling, you should be careful to not make any substantive interpretations of network processes that rely heavily on degree centrality alone. If you were interested in the centrality of a particular settlement, you could further implement the approach to assessing individual nodes as described in Section 5.3.2.

4) The boxplots shown in Figure 5.17 suggest that the rank-order correlations between the original observed network and the subsamples decline for each centrality metric (as would be expected). Beyond this, it is clear that correlations for both eigenvector centrality and betweenness centrality decline more quickly than for degree. Indeed, even at the 90 percent subsample there are a number of correlations for random samples that are relatively low for both eigenvector and betweenness. Based on this evidence, you might be wary of the robustness of eigenvector and betweenness centrality and likely would not want to make any substantive interpretations of a given dataset that rely heavily on those measures alone. Correlations for degree centrality also decline (though much less steeply) with more missing data, but even at the 70 percent sample most correlations were greater than 0.8. Based on this one could argue that degree centrality is robust to moderate numbers of nodes missing at random. Thus, if you expect that you are missing nodes at random in your observed network, these data suggest that this will not necessarily hamper your analysis of degree. Of course, how much variability in relation to missing data you are willing to accept is a question that involves consideration of the archaeological context and details beyond what you have been given here.

CHAPTER 6

1) There are many potential ways you might approach a visualization like that requested in this exercise, and some exploratory visualization would probably be useful. Since this sample includes a relatively small number of nodes, it is probably reasonable to produce a typical node-link diagram. With a small number of nodes like this it might be possible to position nodes manually, but it would be more efficient to experiment with various algorithmic approaches like force directed algorithms, spectral algorithms, or other approaches outlined in Chapter 6. Again, there is no one correct answer here, but since you are also interested in community structure, choosing an approach like Fruchterman-Reingold or Kamada-Kawai force-directed algorithms that place nodes with similar relations near each other makes sense. Since you are interested in variation in degree centrality, you could scale nodes based on their degree centrality and perhaps also use color to further emphasize variation in degree. Finally, to symbolize potential community structure you could use a community detection algorithm like Louvain clustering to define clusters and then use color or circles to highlight them. The specific decisions you make in this case are likely to vary based on the density of the network, the number of potential communities, and the overall similarity in relations among sets of nodes. By way of example, Figure A.4 is a simple 25-node network that highlights degree centrality and network communities.

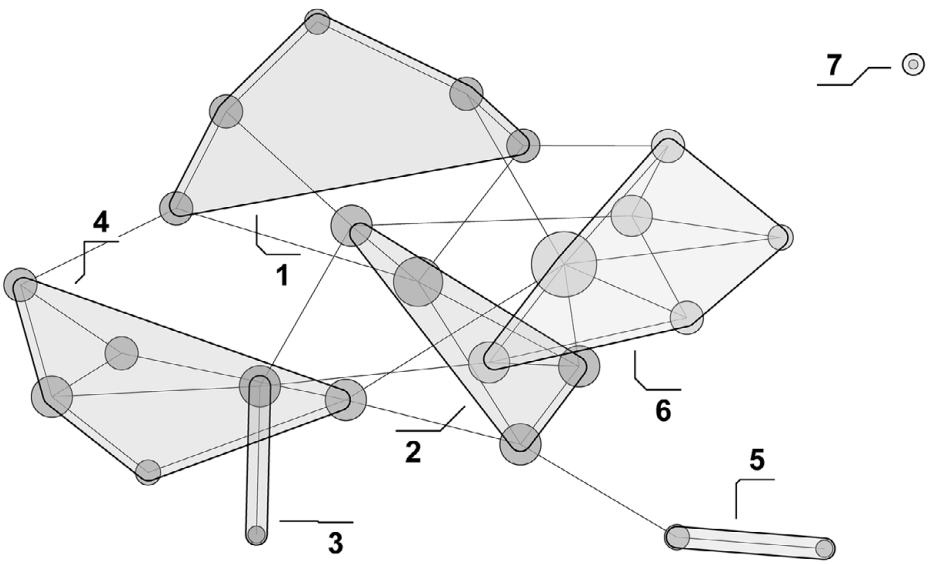


FIG. A.4 Example of a simple undirected network showing degree centrality by node size and with Louvain communities indicated by labels and outlines.

2) In Figure 6.12 Roosevelt Red Ware is the node with the highest degree centrality. If you wanted to emphasize differences in centrality you could scale the symbols based on the desired metric.
3) If you were interested in exploring connections among geographic subregions the figure that likely makes the most sense would be Figure 6.2b. This figure shows sites in their geographic location, color-coded subregions, and thus makes it reasonably easy to count ties among sites in different subregions. Figure 6.3c also shows sites in their geographic locations but in this case the edge-bundling algorithm makes it difficult to track the total volume of flows between areas. Another option that might make sense, especially for larger networks, would be to generate something like the "group-in-a-box" visualization shown in Figure 6.17.
4) For a network of this size it is relatively easy to create a layout that shows variation in centrality without obscuring nodes or other structural features of the network. Since identifying and displaying the highest centrality nodes is of particular interest, the radial centrality layout is a good option. In this case three nodes are clearly much higher in centrality (closer to the center) than others and this is an effective way to demonstrate this. If you were also interested in exploring network communities, the radial layout still works reasonably well as the communities are color-coded and clustered by that algorithm. A few of the other layouts such as the stress majorization, Fructerman-Reingold, and Kamada-Kawai layouts are also reasonable for a network of this size and structure as it is still relatively easy to identify the highest centrality nodes and there is little overlap among nodes due to point clustering. Any of these would be potentially useful options. The spectral and multidimensional scaling layouts, however, are probably not a good choice here as points are quite clustered, and, although you can still identify high centrality nodes, overall network structure and many less central nodes are obscured.

CHAPTER 7

1) b and c
2) See Figure A.5.

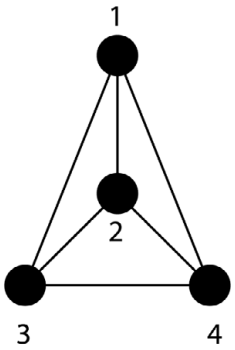


FIG. A.5 Answer to Chapter 7, Exercise 7.2.

3) Yes, see Figure A.6.

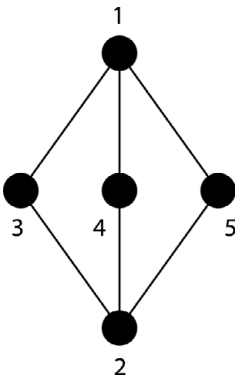


FIG. A.6 Answer to Chapter 7, Exercise 7.3.

4) See Figure A.7.

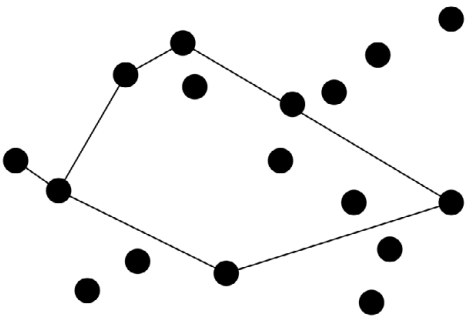


FIG. A.7 Answer to Chapter 7, Exercise 7.4.

5) See Figure A.8.

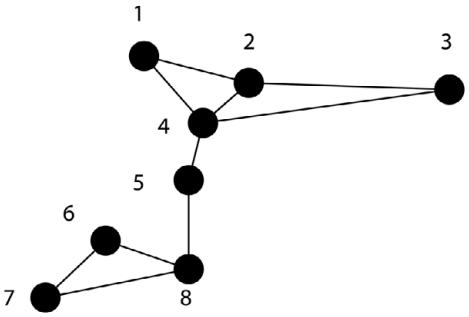


FIG. A.8 Answer to Chapter 7, Exercise 7.5.

6) See igure A.9.

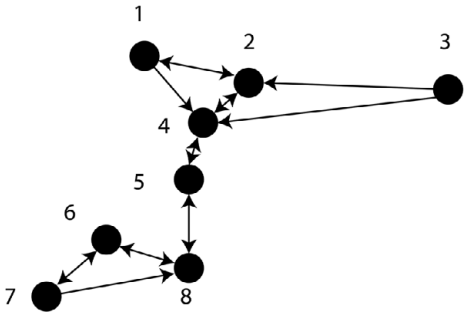


FIG. A.9 Answers to Chapter 7, Exercise 7.6.

Appendix B

Software

There are numerous software packages available for creating and analyzing networks, and the landscape of options is constantly changing. We list a few of the most common platforms here noting those in particular that we used in the creation of this book. We will keep an updated list of software packages on the online companion associated with this book.

Gephi: Gephi is a software package designed to visualize and explore networks large and small (Bastian et al. 2009). Gephi is free and open-source software that has a relatively easy to learn graphical user interface and the ability to conduct most common kinds of network analyses. There is a large user community and many resources for learning and troubleshooting. Gephi has a particularly robust set of options for customizing network visualizations.

Pajek: Pajek is a software package developed by Andrej Mrvar and Vladimir Batagelj for the analysis and visualization of networks, and it is free for noncommercial use (Mrvar and Batagelj 2016). It is particularly potent for the study of very large networks with up to one billion vertices. It has been continually developed for over 20 years, including a huge range of network techniques published in the journal *Social Networks*. It contains key techniques for the analysis of social, bibliometric, and genealogical networks. Its graphical user interface and native file format are bespoke and require some time to master, but this is worth it given the performance and diversity of the tools you'll have at your disposal. The learning process is significantly enhanced thanks to excellent documentation, including a highly accessible manual (de Nooy et al. 2018) and introductions in many languages.[1]

UCInet: UCInet is a software package designed by sociologists Linton Freeman, Martin Everett, and Stephen Borgatti (Borgatti et al. 2002) for the analysis of social

[1] http://mrvar.fdv.uni-lj.si/pajek/ (accessed March 30, 2022).

network data. This is a commercial software package for purchase with discounted rates for academic licenses. UCInet has an easy to use graphical user interface and can be used to conduct many of the most common social network analysis techniques, matrix algebraic calculations, and even advanced permutation-based statistical techniques. The software comes bundled with the free program called NetDraw (see below) for visualization.

NetDraw: NetDraw is a free graph visualization tool that is bundled with UCInet or can be downloaded alone. This tool allows you to visualize small- to medium-sized graphs using a variety of tools for customizing edge and node shape, size, color, and features. NetDraw also has the ability to create graph layouts using many of the most common force-directed and algorithmic approaches.

Visone: Visone is a free software package created by Ulrik Brandes, Dorthea Wagner, and others at the University of Konstanz and Karlsruhe Institute of Technology (Brandes and Wagner 2004). Visone allows for the analysis and visualization of network data of a range of sizes and includes an easy to use graphical user interface and implements many of the most common network metrics and visualization tools.

NodeTrix: NodeTrix is a custom Javascript program created by Nathalie Henry Riche, Michael McGuffin, Anastasia Bezerianos, and Jean-Daniel Fekete (Henry et al. 2007) that allows users to create interactive custom hybrid network visualizations that integrate traditional node-link diagrams and matrix visualizations (see Section 6.4.3). A free version of this program is available on GitHub (https://github.com/IRT-SystemX/nodetrix). This software requires some knowledge of Javascript to use.

The Vistorian: The Vistorian is an online platform, currently under development (Bach et al. 2015), that is designed to create high-impact visualizations of various kinds of networks. This platform allows users to create custom visualizations including traditional node-link diagrams, as well as time arcs, network maps, and various matrix visualizations. The platform is free and open-source.

R Project for Statistical Computing (R Core Team 2021): R is a powerful open-source statistical programming language that includes a robust ecosystem of packages designed to help create, manage, and analyze network data. R is a console-based functional programming language, meaning the user inputs commands as text, which are then executed and output as visualizations, text displays, or exported files. R is capable of reading and exporting virtually any network data format and also has powerful tools for almost any network analytical method or visualization. R is particularly useful for creating custom analyses (like those discussed in Chapter 5 of this book) and reproducible analyses. When data and R code are shared, others can easily replicate analyses.

R was used for many of the analyses in this book, using the following R packages in particular:

- igraph (Csardi and Nepusz 2006): analytical routines for simple graphs and graph analysis
- statnet (Krivitsky et al. 2020): a suite of packages designed for the management and statistical analysis of networks
- tsna (Bender-deMoll and Morris 2021): tools for temporal social network analysis
- tnet (Opsahl 2009): a package containing several related routines for analyzing two-mode networks
- intergraph (Bojanowski 2015): a set of routines for coercing objects between common network formats in R
- edgebundle (Schoch 2021): package that implements a simple edge bundling/flow model for network data
- ndtv (Bender-DeMoll 2019): package designed to create dynamic temporal network visualizations and animations.

Python: Python is a powerful, easy to read, general purpose programming language, and one of the most commonly used programming languages in the world. Like R, it includes a vast ecosystem of existing packages for statistics, data analysis, and visualization. Key packages for network research include:

- NetworkX (Hagberg et al. 2008): creation, manipulation, and study of the structure, dynamics, and functions of complex networks
- igraph (Csardi and Nepusz 2006): graph analysis package (also in R)
- Matplotlib (Hunter 2007): creating static, animated, and interactive visualizations
- Numpy (Harris et al. 2020): scientific computing.

GLOSSARY

Abstract geographic layout Abstract geographic layouts do not explicitly reproduce spatial relationships among nodes and edges but retain features like spatial groupings in determining positions so that geographic insights are still possible (see **geographic layouts**).

Acyclic network A network that includes no cycles (see **cycle**).

Adjacency list A network data format consisting of a set of rows, where the first node in each row is connected to all subsequent nodes in that same row.

Adjacency matrix A network data format consisting of a matrix of size $n \times n$, with a set of rows equal to the number of nodes, and a set of columns equal to the number of nodes. When a pair of nodes is connected by an edge (i.e., when they are *adjacent*), then the corresponding cell will have an entry.

Affiliation Refers in social network analysis to data relating to the presence/absence or weight of relationships between two sets of nodes in a network.

Affiliation networks Networks where nodes and edges are defined based on affiliation data.

Authority In a directed network, an authority is a node with a high indegree that has incoming edges from many hubs (see **hub**).

Average degree The average degree of a network is the sum of the degrees of all nodes in the network divided by the number of nodes. Average degree is calculated as:

$$\bar{d} = \frac{\sum d}{g}$$

where

$\bar{d}$ = average degree
$\sum d$ = the sum of degree for all nodes
g = the total number of nodes.

Average shortest path length The average of all shortest path lengths in a network (see **shortest path length**), calculated as

$$\bar{l}_G = \frac{\sum_n \text{dist}(n_i, n_j)}{\sum_n \text{path}(n_i, n_j)}$$

where

$\bar{l}_G$ is the average shortest path length in network G
$\sum_n \text{dist}(n_i, n_j)$ is the sum of values for all shortest paths from every pair of nodes i and j
$\sum_n \text{dist}(n_i, n_j)$ is the sum of values for all paths within the graph from every pair of nodes i and j.

Beta skeleton A Gabriel graph in which the diameter of the circular region between a pair of nodes is controlled by a parameter, beta (see **Gabriel graph**).

Betweenness centrality A node's betweenness centrality is the fraction of the number of shortest paths passing through this node over the number of shortest paths between all pairs of nodes in the network (see **shortest path**).

Betweenness centrality for binary networks can be calculated as

$$x_i = \sum_{j<k} \frac{p_{jk} n_i}{p_{jk}}$$

where

x_i = Betweenness centrality of node i
$p_{jk} n_i$ = number of shortest paths between nodes j and k passing through node i ($i \neq j$ or k)
p_{jk} = total number of shortest paths between nodes j and k present in the network.

Betweenness centrality for weighted networks can be calculated by defining shortest paths not in terms of the number of edges crossed but in terms of the path that takes the highest weight edge at each juncture. Strongest paths in this way are calculated as

$$d^w(j,k) = \min\left(\frac{1}{w_{jh}} + \cdots + \frac{1}{w_{hk}}\right)$$

where

$d^w(j,k)$ = the strongest path between node j and k
w_{jh} = the weight of connection between the origin j and an intermediate node h on the path between starting node j and target node k.

With the definition of strongest paths defined above, weighted betweenness centrality is calculated as

$$x_i = \sum \frac{d^w_{jk} n_i}{d^w_{jk}}$$

where

x_i = weighted betweenness centrality of node i
$d^w_{jk} n_i$ = number of strongest paths between nodes j and k passing through node i ($i \neq j$ or k)
d^w_{jk} = total number of weighted strongest paths in the network.

Biased sampling process Networks sampled through biased sampling processes are sampled such that different nodes and/or edges from the target population are more or less likely to end up in the sample based on some additional criteria, characteristic, or external consideration (see **random sampling process**; **sampled network**).

Bond Term often used in physics to refer to an edge (see **edge**).

Bootstrapping Bootstrapping is a special case of Monte Carlo simulation designed to resample with replacement many times from the observed data to empirically estimate the properties of a sampling distribution (see **Monte Carlo methods**).

Bridge An edge the removal of which results in a network with a higher number of components (see **component**).

Categorized network A categorized network is a network where edges are classified according to some nominal category that does not necessarily represent an opposition (see also **signed network**).

Centrality Family of measures to implement node rankings based on different ways of conceptualizing node importance (e.g., being connected to a high number of edges, being close to all other edges or nodes, being a crucial go-between).

Centralization Centralization is the graph-level summary of centrality defined as the ratio of the sum of the differences from the maximum observed centrality to all

other node-level centrality scores to the theoretical maximum possible sum of differences. Centralization measures the degree to which the node is focused on one or a small number of highly central nodes.

For a given centrality metric centralization can be calculated as:

$$x_G = \frac{\sum_g (Cn^* - Cn_i)}{\max \sum_g (Cn^* - Cn_i)}$$

where

x_G = the graph-level centralization score for a given metric for network G
$\sum_g (Cn^* - Cn_i)$ = the sum of values for the most central node minus all other node scores
$\max \sum_g (Cn^* - Cn_i)$ = the theoretical maximum difference between the most central node and all other nodes.

Clique A set of nodes in which each node is directly connected to all other nodes in the set.

Closed triad A set of three nodes with three undirected edges between them.

Closeness centrality The closeness centrality of a node is the number of other nodes divided by the sum of the shortest paths of that node to all other nodes (see **shortest path**). Closeness can only be calculated for connected graphs.

Normalized closeness centrality is calculated as

$$x_i = \frac{g}{\sum_i d(i,j)}$$

where

x_i = closeness centrality for node i
g = the total number of nodes
$d(i,j)$ = the graph distance between nodes i and j.

Clustering coefficient 1. A node's clustering coefficient is the fraction of all the edges between the nodes it is directly connected to over the maximum number of edges that can exist between these neighbors. 2. A network's clustering coefficient is the mean of all nodes' clustering coefficient scores in a network.

The clustering coefficient for a given node in a directed network is defined as

$$c_i = \frac{2e_i}{n_i(n_i - 1)}$$

where

c_i = the clustering coefficient for node i
n_i = the number of nodes connected to node i
e_i = the number of edges among the nodes connected to node i.

Cohesive subgroup Set of nodes among which there are relatively strong or direct relationships, as compared to their relationships with nodes outside the set.

Community detection Approaches to identifying cohesive subgroups (see **cohesive subgroup**; **modularity**).

Completely connected network A network where all node pairs are connected.

Component A subset of a network in which any pair of nodes can be connected to each other via at least one path, and where there can be no paths to any nodes outside this subset (see **path**).

Connected network A network where all node pairs are reachable (see **reachable**).

Constrained radial graph layout An approach to graph construction where nodes are positioned based on some metric of interest with the highest value in the center with nodes in concentric circles around the center at increasing distances as the value of the metric declines.

Cutpoint A node the removal of which results in a network with a higher number of components (see **component**).

Cycle A walk of at least three nodes that begins and ends in the same node, in which no edges are repeated and no nodes are repeated except for the starting/ending node (see **walk**).

Degree The degree of a node is the number of edges incident to this node.

Degree centrality A node's degree centrality is the number (or weight) of edges incident to this node (i.e., a node's degree).

For binary networks degree centrality is calculated as

$$x_i = \sum_j A_{ij}$$

where

x_i = degree centrality of node i
A_{ij} = presence/absence of relation between node i and node j (0 or 1) in adjacency matrix A.

For weighted networks degree centrality (sometimes called strength) is calculated as

$$x_i = \sum_j w_{ij}$$

where

x_i = degree centrality of node i
w_{ij} = weight of relation between node i and node j.

Degree distribution Frequency distribution of all nodes' degrees in the network.

Delaunay triangulation A network in which a pair of nodes are connected by an edge if and only if their corresponding regions in a Voronoi diagram share a side (see **Voronoi diagram**).

Dense network A dense network contains a high proportion of the possible edges for a set of nodes (see also **sparse network**).

Density The density of a network is the fraction of the number of edges that are present to the maximum possible number of edges in the network.

For an undirected network, density is calculated as

$$D_G = \frac{e}{g(g-1)/2}$$

where

D_g = the density of graph G
e = the total number of active edges in the network
g = the total number of nodes in the network

For a directed network, density is calculated as

$$D_g = \frac{e}{g(g-1)}$$

where

D_g = the density of graph g
e = the total number of active edges in the network
g = the total number of nodes in the network.

Diameter The diameter of a network is the largest shortest path between any pair of nodes in the network (see **shortest path**).

The diameter of a network can be denoted as

$$\text{dia.} = \max_{ij}\left\{p_{ij}\right\}$$

where

$\max_{ij}\{p_{ij}\}$ = the number of edges in the maximum shortest path p between all nodes i and j.

Directed network A network consisting of a set of nodes and edges connecting them for which the orientation or direction is specified.

Disconnected network A network where not all node pairs are reachable (see **reachable**).

Distorted geographic layout Distorted geographic layouts retain certain features of relative spatial relationships among nodes and edges while modifying those locations and relationships for the sake of visual clarity or to emphasize particular topological features of a graph (see **geographic layouts**).

Dyad A set of two nodes and the edges between them.

Edge 1. A relationship connecting a pair of nodes or connecting a single node to itself. 2. A structural variable measured on a set of nodes.

Edge list A network data format consisting of a list of connected node pairs, $E = \big((n_1, n_2), (n_1, n_3), (n_1, n_4), \ldots, (n_i, n_j)\big)$. It can also be represented as a matrix with two columns for source and target nodes, respectively, and with one edge per row.

Ego network A network including a focal node, the set of nodes the ego is connected to by an edge, and the edges between nodes in this set.

Eigenvector centrality Centrality measure following the concept that high-centrality-scoring neighbors contribute more to a node's eigenvector centrality than low-centrality-scoring neighbors (Bonacich 1972).

For a given adjacency matrix A, eigenvector centrality is defined as

$$\lambda x_i = \sum_{j=1}^{g} a_{ij} x_j$$

where

x_i and x_j are the eigenvector centrality for nodes i and j, respectively
λ = the first eigenvalue of the adjacency matrix A
a_{ij}= the value of the relation between node i and node j in adjacency matrix A.

Eigenvector centrality is calculated the same way for both binary and weighted networks in that the values defining a_{ij} can be either a binary variable representing

the presence or absence of a tie or the weight of the relationship between node *i* and node *j*.

Eigenvector centrality scores are often standardized such that the sum of the squared scores is equal to the number of nodes in that network to facilitate comparisons between different networks.

Empty network A network containing no edges.

Error Refers to the difference between a measured or estimated value and the true value (see **statistical error**; **systematic error**).

Explicit geographic layout Explicit geographic layouts are graph layouts that retain the exact locations of nodes and/or edges in geographic space in two or three dimensions (see **geographic layouts**).

Exploratory network analysis The use of statistical or visual analytical tools to explore the structure of networks by identifying and summarizing network patterning.

Fat-tailed degree distribution A *fat-tailed degree distribution* is a degree distribution with a long tail skewed toward a high degree.

Force-directed graph layout algorithms A set of approaches to defining graph layouts in low-dimensional space by assigning repelling and attracting forces to nodes and edges, and finding a layout that minimizes their energy under specific constraints.

Gabriel graph A network where a pair of nodes are connected if no other nodes lie within the circular region with a diameter equal to the distance between the pair of nodes.

Geodesic Often used in network science to refer to the topological shortest path (see **shortest path**).

Geodesic length Often used in network science to refer to the topological shortest path length (see **shortest path length**).

Geographic layouts 1. Explicit geographic layouts are graph layouts that retain the exact locations of nodes and/or edges in geographic space in two or three dimensions. 2. Distorted geographic layouts retain certain features of relative spatial relationships among nodes and edges while modifying those locations and relationships for the sake of visual clarity or to emphasize particular topological features of a graph.

3. Abstract geographic layouts do not explicitly reproduce spatial relationships among nodes and edges but retain features like spatial groupings in determining positions so that geographic insights are still possible.

Girvan-Newman community detection A measure that calculates the betweenness centrality of each edge, removes the edge with the highest betweenness centrality value, and iteratively repeats this process to identify nested communities, ranging from the top level where the entire network is connected to the bottom level where each node makes up its own community (Girvan and Newman 2002; see **betweenness centrality**).

Graph Term often used in mathematics and computer science to refer to a network (see **network**).

Graph layout The specific assumptions and procedures used to place nodes and edges of a network in two- or three-dimensional space.

Hierarchical network A special class of network where nodes can be placed in an ordered set of levels where edges flow only from higher levels to lower levels with no cycles. Such networks are sometimes referred to as trees due to their stem and branch-like structure.

Hub In a directed network, a hub is a node with a high outdegree that has outgoing edges to many authorities (see **authority**).

Imputation The process of assigning values to missing observations in a dataset (see **simple imputation methods**).

Incidence matrix A network data format consisting of a matrix of size $n \times e$, with a set of rows equal to the number of nodes, and a set of columns equal to the number of edges. An entry is made in a cell if the corresponding node and edge are connected. Each column in the incidence matrix has two entries.

Indegree The indegree of a node in a directed network is the number of incoming incident edges.

***k*-core** A maximal subnetwork in which each vertex has at least degree k within the subnetwork (see **maximal subnetwork**).

***K*-nearest neighbor network** A network in which each node is connected to K other nodes closest to it.

Lambda set A cohesive subgroup that is hard to disconnect by the removal of lines from the subnetwork (Borgatti et al. 1990; see **cohesive subgroup**). Formally,

this is the largest subset of nodes in a network that have more paths between them than to other nodes in the network (see **path**).

Layout See **graph layout**.

Link Term often used in computer science to refer to an edge (see **edge**).

Listwise deletion The practice of removing all cases with partially missing observations and limiting analyses to only complete cases.

Longitudinal network A network that includes information about the change of network structure through time.

Louvain community detection A modularity-optimizing community detection algorithm (Blondel et al. 2008; see **community detection**; **modularity**).

Louvain modularity in a weighted network is defined as

$$Q = \frac{1}{2m}\sum_{ij}\left[A_{ij} - \frac{w_i w_j}{2m}\right]\delta(c_i, c_j)$$

where

A_{ij} = the value in the adjacency matrix A between nodes i and j
w_i, w_j = the sum of edge weights for nodes i and j, respectively
m = the sum of all edge weights in the network
c_i, c_j = the communities for nodes i and j, respectively
δ = the Kronecker delta function defined such that $\delta = 1$ if the values are equal and $\delta = 0$ if the values differ.

The algorithm iterates through community definitions c_i, c_j for nodes to determine the optimal modularity value Q for a given network.

***m*-slice** In a weighted network an m-slice is a maximal subnetwork containing the lines with a weight equal to or greater than m and the nodes incident with these lines (see **maximal subnetwork**).

Maximal subnetwork A subnetwork to which no node can be added without altering its defining characteristic (see **subnetwork**).

Maximum distance network A network in which each node is connected to all other nodes at a distance closer than or equal to a threshold value.

Minimum spanning tree In a set of nodes in the Euclidean plane, edges are created between pairs of nodes to form a tree where each node can be reached by each other node, such that the sum of the Euclidean edge lengths is less than the sum for any other spanning tree (see **tree**).

Mode A distinct set of nodes in a network.

Modularity The observed density of links in a community should be significantly higher than the density we would expect in case the network was formed by a random process (Newman and Girvan 2004).

Monte Carlo methods A general approach to statistical inference based on the repeated creation of random data and assessment of those data to estimate a quantity or understand characteristics of the population from which they are drawn.

Multidimensional scaling (MDS) layout algorithm An approach to graph layout that attempts to display high-dimensional network data in low-dimensional space while retaining the relative distances among nodes.

Multilayer network A network where a single set of nodes is connected by two or more sets of edges that each represent a different kind of relationship among the nodes.

***n*-clique** A maximal subnetwork such that the distance of each pair of its nodes is not larger than *n* (see **maximal subnetwork**).

Network A formal system of interdependent pairwise relationships among a set of entities. Networks involve the formal definition of *nodes* as the entities in question and *edges* as the relationships among them.

Network data Consist of a set of nodes and at least one structural variable (represented by a set of edges) connecting pairs within this set of nodes.

Network projection Procedure used to convert a two-mode network into two one-mode networks each focused on a distinct category (mode) of nodes.

Network representation A formal abstraction of a network created for the purposes of visualization or analysis. In this book, network representations are simply referred to as networks.

Network science The study of the collection, management, analysis, interpretation, and presentation of relational data.

Network science in archaeology The study of network models and network theories developed for an archaeological research context, and formally expressed and tested using network science methods.

Network structure The general properties of a network including the overall patterns of relations, the presence/absence and nature of subgroups, the variation in

the positions of actors within that network, and a broad range of other potentially salient features of organization.

Network theories Formal and testable expressions of dependencies among nodes, edges, attributes, outcomes, or global network structures or any combination thereof. They express why and how relationships matter in a certain research context.

Networks in space Networks where nodes and edges are not explicitly defined based on their locations in space but where those nodes and/or edges still have spatial information embedded in their attributes or representations (see **spatial networks**).

Node A unique bounded entity in a network that can be connected to others via edges.

Node variable Node variables or node attributes are data of any form (categorical, presence/absence of a particular feature, measurements, etc.) pertaining to individual nodes within a network.

Open triad A set of three nodes with two undirected edges between them.

Outdegree The outdegree of a node in a directed network is the number of outgoing incident edges.

PageRank Centrality measure following the concept that a node's centrality score is proportional to the sum of the node's direct neighbors divided by their outdegree (see **outdegree**).

Calculation of PageRank involves a complex set of iterative calculations that are beyond the scope of this book but can be briefly described as the probability of being at a particular node at any point as the algorithm iterates. It can be calculated using the algorithm

$$P_i = \frac{(1 - z)}{g} + z \sum_{j \to i} \frac{P_j}{o_j}$$

where

P_i, P_j = the PageRank values for nodes i and j, respectively
z = a constant value known as the damping factor, which determines the rate at which a random jump to some node will occur
o_j = outgoing edges from node j
g = the number of nodes in the network.

This metric can be calculated iteratively until convergence. For the first iteration, P_i is set as $1/n$. There are several different approaches to defining convergence and calculating this measure that we do not discuss here.

Path A walk between a pair of nodes in which no nodes and edges are repeated (see **walk**).

Planar network A network that can be drawn on a plane where the edges do not cross but instead always end in nodes.

Power law degree distribution A degree distribution that follows a power law, where the probability of a node with a certain degree is proportional to a power of that degree (see **degree distribution**).

Random sampling process Networks sampled through random sampling processes are sampled such that every node and/or edge from the target population has an equal chance of ending up in the sample (see **biased sampling process**; **sampled network**).

Reachable A node pair is reachable if it can be connected by a path (see **path**).

Relational perspective The notion that the structural properties of networks and variation in the positions of nodes and edges in a network are just as important for explaining or predicting the behavior of the actors of that network as the attributes of the social actors themselves.

Relative neighborhood graph A network in which a pair of nodes are connected if there are no other nodes in the area marked by the overlap of a circle around each node with a radius equal to the distance between the nodes.

Sampled network A network where the nodes and edges are sampled from a larger population of nodes and edges.

Sampling error The variability of an estimate of a parameter that is driven by the size and representativeness of the sample from the target population (see **error**; **statistical error**).

Scale-free network A network with a power law degree distribution.

Shortest path A shortest path between a pair of nodes is the path between with the shortest path length (see **path**).

Shortest path length The shortest path length of a pair of nodes is the number of edges in the shortest path between the pair of nodes (see **path**; **shortest path**).

Signed network A network where the edges carry a positive or negative sign, indicating some opposed property of relations in the network.

Similarity networks Networks where edges are defined or weighted based on a quantitative metric of similarity or distance based on node attributes or artifact assemblages.

Simple imputation methods Approaches that use simple characterizations of existing data to define values for missing observations (see **imputation**).

Simple network A set of nodes and a set of edges with no additional information about them.

Small-world network A network in which the average shortest path length is almost as small as that of a uniformly random network with the same number of nodes and density, whereas the clustering coefficient is much higher than in a uniformly random network (see **average shortest path length**; **clustering coefficient**; **density**; **uniformly random network**).

Sociomatrix Term sometimes used in the social sciences to refer to an adjacency matrix (see **adjacency matrix**).

Sparse network A sparse network contains a small proportion of the possible edges for a set of nodes (see also **dense network**).

Spatial networks Networks where the nodes and/or edges are explicitly defined based on their geographical spatial features (see **networks in space**).

Spectral graph layout algorithms Use the smallest eigenvector of the adjacency matrix underlying a graph to create a low-dimensional node placement such that nodes are at the centroids of their neighboring nodes.

Standardized degree centrality A node's standardized degree centrality is the number of edges incident to this node divided by the number of nodes in the network minus 1.

This can be calculated as

$$k_i = \frac{x_i}{g - 1}$$

where

k_i = standardized degree centrality for node i
x_i = degree centrality for node i
g = number of nodes in the network.

Statistical error Random variability around true value based on variation in measurement precision and the representativeness of samples. Statistical errors can reduce the precision of estimates of target values.

Strong community LS-set A subnetwork such that the degree of all nodes within the subnetwork is higher than their degree outside the subnetwork (see **degree**; **subnetwork**).

Structural variables Descriptions of features or elements of pairs of nodes, where each node can be part of multiple node pairs.

Subnetwork A subnetwork (or subgraph) is a subset of the entire network, consisting of a set of nodes and a set of edges that connect these nodes.

Systematic error Deviations from the true value based on inconsistent or flawed methods, instrumentation, or assumptions. Systematic errors often reduce the accuracy of estimates of target values.

Thiessen polygons See **Voronoi diagram**.

Tie Term often used in the social sciences to refer to an edge (see **edge**).

Total network A network where all actors and all relations are included at a given scale and where data are of a consistent quality across all nodes and edges.

Trail A walk between a pair of nodes in which no edges are repeated but nodes can be repeated (see **walk**).

Transitivity A network's transitivity is three times the number of closed triads over the total number of triads in a network (see **clustering coefficient**; **triad**).

Transitivity can be calculated as

$$T = \frac{3t_c}{t_n}$$

where

T = transitivity for a given network
t_c = the number of closed triads
t_n = the total number of triads.

Tree A network that is connected and acyclic (see **acyclic network**; **connected network**).

Triad A set of three nodes and any edges between them.

Triad census A set of counts of each triad state included in a network.

Two-mode network A network where two separate categories of nodes are defined with a structural variable (edges) only between these categories.

Uncertainty In scientific investigations, uncertainty refers to assessments of the variability of measured or estimated analytical outputs based on the sources of error and variability in analytical inputs.

Uniformly random network A network in which each edge exists with a fixed probability p.

Vertex A term often used in mathematics, computer science, and physics to refer to a node (see **node**).

Voronoi diagram For each node in a set of nodes in a Euclidean plane, a region is created covering the area that is equidistant from or closer to that node than it is to any other node in the set.

Walk Any sequence of nodes connected through edges that has that pair of nodes as end points. Nodes and edges can be repeated in a walk.

Walktrap algorithm Approach to defining communities in binary or weighted networks by generating a large number of short random walks and determining which sets of nodes consistently fall along the same short random walks (see **walk**).

Weighted degree The weighted degree of a node in a weighted network is the sum of weights of incident edges.

Weak community LS-set A subnetwork such that the summed degree of nodes in the subnetwork is higher than their summed degree outside the subnetwork (see **degree**; **subnetwork**).

Weighted network A network in which the edges carry a nonbinary value that indicates the strength of a given relationship.

GRAPH THEORETIC NOTATION

Description	Notation
Node/vertex i	n_i
Edge m	e_m
Set of all nodes in a network	$N = \{n_1, n_2, \ldots, n_g\}$
Network/graph G with nodes N and edges E	$G(N, E)$
Adjacency matrix	A
Element of adjacency matrix	A_{ij}
Walk	$w = \{n_0, e_1, n_1, e_2, \ldots\}$

GLOSSARY

Abstract geographic layout Abstract geographic layouts do not explicitly reproduce spatial relationships among nodes and edges but retain features like spatial groupings in determining positions so that geographic insights are still possible (see **geographic layouts**).

Acyclic network A network that includes no cycles (see **cycle**).

Adjacency list A network data format consisting of a set of rows, where the first node in each row is connected to all subsequent nodes in that same row.

Adjacency matrix A network data format consisting of a matrix of size $n \times n$, with a set of rows equal to the number of nodes, and a set of columns equal to the number of nodes. When a pair of nodes is connected by an edge (i.e., when they are *adjacent*), then the corresponding cell will have an entry.

Affiliation Refers in social network analysis to data relating to the presence/absence or weight of relationships between two sets of nodes in a network.

Affiliation networks Networks where nodes and edges are defined based on affiliation data.

Authority In a directed network, an authority is a node with a high indegree that has incoming edges from many hubs (see **hub**).

Average degree The average degree of a network is the sum of the degrees of all nodes in the network divided by the number of nodes. Average degree is calculated as:

$$\bar{d} = \frac{\sum d}{g}$$

where

$\bar{d}$ = average degree
$\sum d$ = the sum of degree for all nodes
g = the total number of nodes.

Average shortest path length The average of all shortest path lengths in a network (see **shortest path length**), calculated as

$$\bar{l}_G = \frac{\sum_n \text{dist}(n_i, n_j)}{\sum_n \text{path}(n_i, n_j)}$$

where

$\bar{l}_G$ is the average shortest path length in network G
$\sum_n \text{dist}(n_i, n_j)$ is the sum of values for all shortest paths from every pair of nodes i and j
$\sum_n \text{dist}(n_i, n_j)$ is the sum of values for all paths within the graph from every pair of nodes i and j.

Beta skeleton A Gabriel graph in which the diameter of the circular region between a pair of nodes is controlled by a parameter, beta (see **Gabriel graph**).

Betweenness centrality A node's betweenness centrality is the fraction of the number of shortest paths passing through this node over the number of shortest paths between all pairs of nodes in the network (see **shortest path**).

Betweenness centrality for binary networks can be calculated as

$$x_i = \sum_{j<k} \frac{p_{jk} n_i}{p_{jk}}$$

where

x_i = Betweenness centrality of node i
$p_{jk} n_i$ = number of shortest paths between nodes j and k passing through node i ($i \neq j$ or k)
p_{jk} = total number of shortest paths between nodes j and k present in the network.

Betweenness centrality for weighted networks can be calculated by defining shortest paths not in terms of the number of edges crossed but in terms of the path that takes the highest weight edge at each juncture. Strongest paths in this way are calculated as

$$d^w(j,k) = \min\left(\frac{1}{w_{jh}} + \cdots + \frac{1}{w_{hk}}\right)$$

where

$d^w(j,k)$ = the strongest path between node j and k
w_{jh} = the weight of connection between the origin j and an intermediate node h on the path between starting node j and target node k.

With the definition of strongest paths defined above, weighted betweenness centrality is calculated as

$$x_i = \sum \frac{d^w_{jk} n_i}{d^w_{jk}}$$

where

x_i = weighted betweenness centrality of node i
$d^w_{jk} n_i$ = number of strongest paths between nodes j and k passing through node i ($i \neq j$ or k)
d^w_{jk} = total number of weighted strongest paths in the network.

Biased sampling process Networks sampled through biased sampling processes are sampled such that different nodes and/or edges from the target population are more or less likely to end up in the sample based on some additional criteria, characteristic, or external consideration (see **random sampling process**; **sampled network**).

Bond Term often used in physics to refer to an edge (see **edge**).

Bootstrapping Bootstrapping is a special case of Monte Carlo simulation designed to resample with replacement many times from the observed data to empirically estimate the properties of a sampling distribution (see **Monte Carlo methods**).

Bridge An edge the removal of which results in a network with a higher number of components (see **component**).

Categorized network A categorized network is a network where edges are classified according to some nominal category that does not necessarily represent an opposition (see also **signed network**).

Centrality Family of measures to implement node rankings based on different ways of conceptualizing node importance (e.g., being connected to a high number of edges, being close to all other edges or nodes, being a crucial go-between).

Centralization Centralization is the graph-level summary of centrality defined as the ratio of the sum of the differences from the maximum observed centrality to all

other node-level centrality scores to the theoretical maximum possible sum of differences. Centralization measures the degree to which the node is focused on one or a small number of highly central nodes.

For a given centrality metric centralization can be calculated as:

$$x_G = \frac{\sum_g (Cn^* - Cn_i)}{\max \sum_g (Cn^* - Cn_i)}$$

where

x_G = the graph-level centralization score for a given metric for network G
$\sum_g (Cn^* - Cn_i)$ = the sum of values for the most central node minus all other node scores
$\max \sum_g (Cn^* - Cn_i)$ = the theoretical maximum difference between the most central node and all other nodes.

Clique A set of nodes in which each node is directly connected to all other nodes in the set.

Closed triad A set of three nodes with three undirected edges between them.

Closeness centrality The closeness centrality of a node is the number of other nodes divided by the sum of the shortest paths of that node to all other nodes (see **shortest path**). Closeness can only be calculated for connected graphs.

Normalized closeness centrality is calculated as

$$x_i = \frac{g}{\sum_i d(i,j)}$$

where

x_i = closeness centrality for node i
g = the total number of nodes
$d(i,j)$ = the graph distance between nodes i and j.

Clustering coefficient 1. A node's clustering coefficient is the fraction of all the edges between the nodes it is directly connected to over the maximum number of edges that can exist between these neighbors. 2. A network's clustering coefficient is the mean of all nodes' clustering coefficient scores in a network.

The clustering coefficient for a given node in a directed network is defined as

$$c_i = \frac{2e_i}{n_i(n_i - 1)}$$

where

c_i = the clustering coefficient for node i
n_i = the number of nodes connected to node i
e_i = the number of edges among the nodes connected to node i.

Cohesive subgroup Set of nodes among which there are relatively strong or direct relationships, as compared to their relationships with nodes outside the set.

Community detection Approaches to identifying cohesive subgroups (see **cohesive subgroup**; **modularity**).

Completely connected network A network where all node pairs are connected.

Component A subset of a network in which any pair of nodes can be connected to each other via at least one path, and where there can be no paths to any nodes outside this subset (see **path**).

Connected network A network where all node pairs are reachable (see **reachable**).

Constrained radial graph layout An approach to graph construction where nodes are positioned based on some metric of interest with the highest value in the center with nodes in concentric circles around the center at increasing distances as the value of the metric declines.

Cutpoint A node the removal of which results in a network with a higher number of components (see **component**).

Cycle A walk of at least three nodes that begins and ends in the same node, in which no edges are repeated and no nodes are repeated except for the starting/ending node (see **walk**).

Degree The degree of a node is the number of edges incident to this node.

Degree centrality A node's degree centrality is the number (or weight) of edges incident to this node (i.e., a node's degree).

For binary networks degree centrality is calculated as

$$x_i = \sum_j A_{ij}$$

where

x_i = degree centrality of node i
A_{ij} = presence/absence of relation between node i and node j (0 or 1) in adjacency matrix A.

For weighted networks degree centrality (sometimes called strength) is calculated as

$$x_i = \sum_j w_{ij}$$

where

x_i = degree centrality of node i
w_{ij} = weight of relation between node i and node j.

Degree distribution Frequency distribution of all nodes' degrees in the network.

Delaunay triangulation A network in which a pair of nodes are connected by an edge if and only if their corresponding regions in a Voronoi diagram share a side (see **Voronoi diagram**).

Dense network A dense network contains a high proportion of the possible edges for a set of nodes (see also **sparse network**).

Density The density of a network is the fraction of the number of edges that are present to the maximum possible number of edges in the network.

For an undirected network, density is calculated as

$$D_G = \frac{e}{g(g-1)/2}$$

where

D_g = the density of graph G
e = the total number of active edges in the network
g = the total number of nodes in the network

For a directed network, density is calculated as

$$D_g = \frac{e}{g(g-1)}$$

where

D_g = the density of graph g
e = the total number of active edges in the network
g = the total number of nodes in the network.

Diameter The diameter of a network is the largest shortest path between any pair of nodes in the network (see **shortest path**).

The diameter of a network can be denoted as

$$\text{dia.} = \max_{ij}\left\{p_{ij}\right\}$$

where

$\max_{ij}\left\{p_{ij}\right\}$ = the number of edges in the maximum shortest path p between all nodes i and j.

Directed network A network consisting of a set of nodes and edges connecting them for which the orientation or direction is specified.

Disconnected network A network where not all node pairs are reachable (see **reachable**).

Distorted geographic layout Distorted geographic layouts retain certain features of relative spatial relationships among nodes and edges while modifying those locations and relationships for the sake of visual clarity or to emphasize particular topological features of a graph (see **geographic layouts**).

Dyad A set of two nodes and the edges between them.

Edge 1. A relationship connecting a pair of nodes or connecting a single node to itself. 2. A structural variable measured on a set of nodes.

Edge list A network data format consisting of a list of connected node pairs, $E = \left((n_1, n_2), (n_1, n_3), (n_1, n_4), \ldots, \left(n_i, n_j\right)\right)$. It can also be represented as a matrix with two columns for source and target nodes, respectively, and with one edge per row.

Ego network A network including a focal node, the set of nodes the ego is connected to by an edge, and the edges between nodes in this set.

Eigenvector centrality Centrality measure following the concept that high-centrality-scoring neighbors contribute more to a node's eigenvector centrality than low-centrality-scoring neighbors (Bonacich 1972).

For a given adjacency matrix A, eigenvector centrality is defined as

$$\lambda x_i = \sum_{j=1}^{g} a_{ij} x_j$$

where

x_i and x_j are the eigenvector centrality for nodes i and j, respectively
λ = the first eigenvalue of the adjacency matrix A
a_{ij}= the value of the relation between node i and node j in adjacency matrix A.

Eigenvector centrality is calculated the same way for both binary and weighted networks in that the values defining a_{ij} can be either a binary variable representing

the presence or absence of a tie or the weight of the relationship between node *i* and node *j*.

Eigenvector centrality scores are often standardized such that the sum of the squared scores is equal to the number of nodes in that network to facilitate comparisons between different networks.

Empty network A network containing no edges.

Error Refers to the difference between a measured or estimated value and the true value (see **statistical error**; **systematic error**).

Explicit geographic layout Explicit geographic layouts are graph layouts that retain the exact locations of nodes and/or edges in geographic space in two or three dimensions (see **geographic layouts**).

Exploratory network analysis The use of statistical or visual analytical tools to explore the structure of networks by identifying and summarizing network patterning.

Fat-tailed degree distribution A *fat-tailed degree distribution* is a degree distribution with a long tail skewed toward a high degree.

Force-directed graph layout algorithms A set of approaches to defining graph layouts in low-dimensional space by assigning repelling and attracting forces to nodes and edges, and finding a layout that minimizes their energy under specific constraints.

Gabriel graph A network where a pair of nodes are connected if no other nodes lie within the circular region with a diameter equal to the distance between the pair of nodes.

Geodesic Often used in network science to refer to the topological shortest path (see **shortest path**).

Geodesic length Often used in network science to refer to the topological shortest path length (see **shortest path length**).

Geographic layouts 1. Explicit geographic layouts are graph layouts that retain the exact locations of nodes and/or edges in geographic space in two or three dimensions. 2. Distorted geographic layouts retain certain features of relative spatial relationships among nodes and edges while modifying those locations and relationships for the sake of visual clarity or to emphasize particular topological features of a graph.

3. Abstract geographic layouts do not explicitly reproduce spatial relationships among nodes and edges but retain features like spatial groupings in determining positions so that geographic insights are still possible.

Girvan-Newman community detection A measure that calculates the betweenness centrality of each edge, removes the edge with the highest betweenness centrality value, and iteratively repeats this process to identify nested communities, ranging from the top level where the entire network is connected to the bottom level where each node makes up its own community (Girvan and Newman 2002; see **betweenness centrality**).

Graph Term often used in mathematics and computer science to refer to a network (see **network**).

Graph layout The specific assumptions and procedures used to place nodes and edges of a network in two- or three-dimensional space.

Hierarchical network A special class of network where nodes can be placed in an ordered set of levels where edges flow only from higher levels to lower levels with no cycles. Such networks are sometimes referred to as trees due to their stem and branch-like structure.

Hub In a directed network, a hub is a node with a high outdegree that has outgoing edges to many authorities (see **authority**).

Imputation The process of assigning values to missing observations in a dataset (see **simple imputation methods**).

Incidence matrix A network data format consisting of a matrix of size $n \times e$, with a set of rows equal to the number of nodes, and a set of columns equal to the number of edges. An entry is made in a cell if the corresponding node and edge are connected. Each column in the incidence matrix has two entries.

Indegree The indegree of a node in a directed network is the number of incoming incident edges.

***k*-core** A maximal subnetwork in which each vertex has at least degree k within the subnetwork (see **maximal subnetwork**).

***K*-nearest neighbor network** A network in which each node is connected to K other nodes closest to it.

Lambda set A cohesive subgroup that is hard to disconnect by the removal of lines from the subnetwork (Borgatti et al. 1990; see **cohesive subgroup**). Formally,

this is the largest subset of nodes in a network that have more paths between them than to other nodes in the network (see **path**).

Layout See **graph layout**.

Link Term often used in computer science to refer to an edge (see **edge**).

Listwise deletion The practice of removing all cases with partially missing observations and limiting analyses to only complete cases.

Longitudinal network A network that includes information about the change of network structure through time.

Louvain community detection A modularity-optimizing community detection algorithm (Blondel et al. 2008; see **community detection**; **modularity**).

Louvain modularity in a weighted network is defined as

$$Q = \frac{1}{2m}\sum_{ij}\left[A_{ij} - \frac{w_i w_j}{2m}\right]\delta(c_i, c_j)$$

where

A_{ij} = the value in the adjacency matrix A between nodes i and j
w_i, w_j = the sum of edge weights for nodes i and j, respectively
m = the sum of all edge weights in the network
c_i, c_j = the communities for nodes i and j, respectively
δ = the Kronecker delta function defined such that $\delta = 1$ if the values are equal and $\delta = 0$ if the values differ.

The algorithm iterates through community definitions c_i, c_j for nodes to determine the optimal modularity value Q for a given network.

***m*-slice** In a weighted network an m-slice is a maximal subnetwork containing the lines with a weight equal to or greater than m and the nodes incident with these lines (see **maximal subnetwork**).

Maximal subnetwork A subnetwork to which no node can be added without altering its defining characteristic (see **subnetwork**).

Maximum distance network A network in which each node is connected to all other nodes at a distance closer than or equal to a threshold value.

Minimum spanning tree In a set of nodes in the Euclidean plane, edges are created between pairs of nodes to form a tree where each node can be reached by each other node, such that the sum of the Euclidean edge lengths is less than the sum for any other spanning tree (see **tree**).

Mode A distinct set of nodes in a network.

Modularity The observed density of links in a community should be significantly higher than the density we would expect in case the network was formed by a random process (Newman and Girvan 2004).

Monte Carlo methods A general approach to statistical inference based on the repeated creation of random data and assessment of those data to estimate a quantity or understand characteristics of the population from which they are drawn.

Multidimensional scaling (MDS) layout algorithm An approach to graph layout that attempts to display high-dimensional network data in low-dimensional space while retaining the relative distances among nodes.

Multilayer network A network where a single set of nodes is connected by two or more sets of edges that each represent a different kind of relationship among the nodes.

***n*-clique** A maximal subnetwork such that the distance of each pair of its nodes is not larger than *n* (see **maximal subnetwork**).

Network A formal system of interdependent pairwise relationships among a set of entities. Networks involve the formal definition of *nodes* as the entities in question and *edges* as the relationships among them.

Network data Consist of a set of nodes and at least one structural variable (represented by a set of edges) connecting pairs within this set of nodes.

Network projection Procedure used to convert a two-mode network into two one-mode networks each focused on a distinct category (mode) of nodes.

Network representation A formal abstraction of a network created for the purposes of visualization or analysis. In this book, network representations are simply referred to as networks.

Network science The study of the collection, management, analysis, interpretation, and presentation of relational data.

Network science in archaeology The study of network models and network theories developed for an archaeological research context, and formally expressed and tested using network science methods.

Network structure The general properties of a network including the overall patterns of relations, the presence/absence and nature of subgroups, the variation in

the positions of actors within that network, and a broad range of other potentially salient features of organization.

Network theories Formal and testable expressions of dependencies among nodes, edges, attributes, outcomes, or global network structures or any combination thereof. They express why and how relationships matter in a certain research context.

Networks in space Networks where nodes and edges are not explicitly defined based on their locations in space but where those nodes and/or edges still have spatial information embedded in their attributes or representations (see **spatial networks**).

Node A unique bounded entity in a network that can be connected to others via edges.

Node variable Node variables or node attributes are data of any form (categorical, presence/absence of a particular feature, measurements, etc.) pertaining to individual nodes within a network.

Open triad A set of three nodes with two undirected edges between them.

Outdegree The outdegree of a node in a directed network is the number of outgoing incident edges.

PageRank Centrality measure following the concept that a node's centrality score is proportional to the sum of the node's direct neighbors divided by their outdegree (see **outdegree**).

Calculation of PageRank involves a complex set of iterative calculations that are beyond the scope of this book but can be briefly described as the probability of being at a particular node at any point as the algorithm iterates. It can be calculated using the algorithm

$$P_i = \frac{(1 - z)}{g} + z \sum_{j \to i} \frac{P_j}{o_j}$$

where

P_i, P_j = the PageRank values for nodes i and j, respectively
z = a constant value known as the damping factor, which determines the rate at which a random jump to some node will occur
o_j = outgoing edges from node j
g = the number of nodes in the network.

This metric can be calculated iteratively until convergence. For the first iteration, P_i is set as $1/n$. There are several different approaches to defining convergence and calculating this measure that we do not discuss here.

Path A walk between a pair of nodes in which no nodes and edges are repeated (see **walk**).

Planar network A network that can be drawn on a plane where the edges do not cross but instead always end in nodes.

Power law degree distribution A degree distribution that follows a power law, where the probability of a node with a certain degree is proportional to a power of that degree (see **degree distribution**).

Random sampling process Networks sampled through random sampling processes are sampled such that every node and/or edge from the target population has an equal chance of ending up in the sample (see **biased sampling process**; **sampled network**).

Reachable A node pair is reachable if it can be connected by a path (see **path**).

Relational perspective The notion that the structural properties of networks and variation in the positions of nodes and edges in a network are just as important for explaining or predicting the behavior of the actors of that network as the attributes of the social actors themselves.

Relative neighborhood graph A network in which a pair of nodes are connected if there are no other nodes in the area marked by the overlap of a circle around each node with a radius equal to the distance between the nodes.

Sampled network A network where the nodes and edges are sampled from a larger population of nodes and edges.

Sampling error The variability of an estimate of a parameter that is driven by the size and representativeness of the sample from the target population (see **error**; **statistical error**).

Scale-free network A network with a power law degree distribution.

Shortest path A shortest path between a pair of nodes is the path between with the shortest path length (see **path**).

Shortest path length The shortest path length of a pair of nodes is the number of edges in the shortest path between the pair of nodes (see **path**; **shortest path**).

Signed network A network where the edges carry a positive or negative sign, indicating some opposed property of relations in the network.

Similarity networks Networks where edges are defined or weighted based on a quantitative metric of similarity or distance based on node attributes or artifact assemblages.

Simple imputation methods Approaches that use simple characterizations of existing data to define values for missing observations (see **imputation**).

Simple network A set of nodes and a set of edges with no additional information about them.

Small-world network A network in which the average shortest path length is almost as small as that of a uniformly random network with the same number of nodes and density, whereas the clustering coefficient is much higher than in a uniformly random network (see **average shortest path length**; **clustering coefficient**; **density**; **uniformly random network**).

Sociomatrix Term sometimes used in the social sciences to refer to an adjacency matrix (see **adjacency matrix**).

Sparse network A sparse network contains a small proportion of the possible edges for a set of nodes (see also **dense network**).

Spatial networks Networks where the nodes and/or edges are explicitly defined based on their geographical spatial features (see **networks in space**).

Spectral graph layout algorithms Use the smallest eigenvector of the adjacency matrix underlying a graph to create a low-dimensional node placement such that nodes are at the centroids of their neighboring nodes.

Standardized degree centrality A node's standardized degree centrality is the number of edges incident to this node divided by the number of nodes in the network minus 1.

This can be calculated as

$$k_i = \frac{x_i}{g - 1}$$

where

k_i = standardized degree centrality for node i
x_i = degree centrality for node i
g = number of nodes in the network.

Statistical error Random variability around true value based on variation in measurement precision and the representativeness of samples. Statistical errors can reduce the precision of estimates of target values.

Strong community LS-set A subnetwork such that the degree of all nodes within the subnetwork is higher than their degree outside the subnetwork (see **degree**; **subnetwork**).

Structural variables Descriptions of features or elements of pairs of nodes, where each node can be part of multiple node pairs.

Subnetwork A subnetwork (or subgraph) is a subset of the entire network, consisting of a set of nodes and a set of edges that connect these nodes.

Systematic error Deviations from the true value based on inconsistent or flawed methods, instrumentation, or assumptions. Systematic errors often reduce the accuracy of estimates of target values.

Thiessen polygons See **Voronoi diagram**.

Tie Term often used in the social sciences to refer to an edge (see **edge**).

Total network A network where all actors and all relations are included at a given scale and where data are of a consistent quality across all nodes and edges.

Trail A walk between a pair of nodes in which no edges are repeated but nodes can be repeated (see **walk**).

Transitivity A network's transitivity is three times the number of closed triads over the total number of triads in a network (see **clustering coefficient**; **triad**).

Transitivity can be calculated as

$$T = \frac{3t_c}{t_n}$$

where

T = transitivity for a given network
t_c = the number of closed triads
t_n = the total number of triads.

Tree A network that is connected and acyclic (see **acyclic network**; **connected network**).

Triad A set of three nodes and any edges between them.

Triad census A set of counts of each triad state included in a network.

Two-mode network A network where two separate categories of nodes are defined with a structural variable (edges) only between these categories.

Uncertainty In scientific investigations, uncertainty refers to assessments of the variability of measured or estimated analytical outputs based on the sources of error and variability in analytical inputs.

Uniformly random network A network in which each edge exists with a fixed probability p.

Vertex A term often used in mathematics, computer science, and physics to refer to a node (see **node**).

Voronoi diagram For each node in a set of nodes in a Euclidean plane, a region is created covering the area that is equidistant from or closer to that node than it is to any other node in the set.

Walk Any sequence of nodes connected through edges that has that pair of nodes as end points. Nodes and edges can be repeated in a walk.

Walktrap algorithm Approach to defining communities in binary or weighted networks by generating a large number of short random walks and determining which sets of nodes consistently fall along the same short random walks (see **walk**).

Weighted degree The weighted degree of a node in a weighted network is the sum of weights of incident edges.

Weak community LS-set A subnetwork such that the summed degree of nodes in the subnetwork is higher than their summed degree outside the subnetwork (see **degree**; **subnetwork**).

Weighted network A network in which the edges carry a nonbinary value that indicates the strength of a given relationship.

GRAPH THEORETIC NOTATION

Description	Notation
Node/vertex i	n_i
Edge m	e_m
Set of all nodes in a network	$N = \{n_1, n_2, \ldots, n_g\}$
Network/graph G with nodes N and edges E	$G(N, E)$
Adjacency matrix	A
Element of adjacency matrix	A_{ij}
Walk	$w = \{n_0, e_1, n_1, e_2, \ldots\}$

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INDEX

A:shiwi, 56
accuracy, 152, 177, 203, 308
actor, 4–7, 24, 36, 79–80, 156, 193, 267–268, 306, 308
Actor-Network Theory, 7
actor-oriented model, 96
adjacency list, 69, 98, 101, 294
affiliation, 29, 79–80, 83, 85, 294
affiliation network. *See* network, types of
agent-based modelling, 50, 280
animation, 229, 235, 293
Aotearoa, 32, 83
ARIADNE, 44, 78, 252
attribute, 4, 6–7, 9–10, 14, 19, 24, 28–30, 33–35, 56–57, 65, 68, 70, 84, 117, 129, 167, 191, 193–194, 197, 199, 209, 211, 213–214, 216, 221, 237–238, 245, 265–266, 305–306
  edge, 45, 53, 58, 101, 209
  node, 13, 64, 66, 85, 101, 307

Bayesian analysis, 95, 184–185
beta skeleton, 248, 260, 295
bi-directional network. *See* network, types of
binarization, 217–218
bootstrapping. *See* resampling
boundary, 66, 98, 160, 216, 221, 239, 272
boxplot, 169, 173–174, 286
Brainerd-Robinson similarity metric, 85, 123, 167, 178, 217, 231, 257
bridge, 123–125, 198, 204, 241–242
brokerage, 144, 197–198, 266, 268–269
brokers. *See* brokerage
Bronze Age, 20, 33, 38, 42, 44, 46, 78, 204–205, 253–254

categorized network. *See* network, types of
centralization, 133
centrality, 8–10, 14–15, 32–33, 37, 104, 132–133, 135, 137, 139, 141, 144–146, 148, 150, 160, 167, 170, 176, 178, 181–182, 187, 189–190, 192, 198–199, 207, 211, 213, 217, 220, 234, 236, 253, 286, 288, 296, 305
  betweenness, 13–14, 46, 132–133, 135, 137–139, 141, 146, 148, 161, 167, 169–173, 187, 189–190, 195, 198–199, 205, 207, 220, 236, 245–246, 285–286, 295–296, 302
  closeness, 130–131, 133, 135, 137–139, 141, 143, 297
  degree, 77, 81, 133, 135–137, 141–142, 144, 148, 175–176, 178–179, 189–191, 194, 236, 245, 285–287, 298, 307
  eigenvector, 133, 137, 141–143, 146, 167–169, 187, 231, 286, 300
ceramics, 15, 32–35, 38, 43, 53, 55–56, 66, 85, 87–89, 95, 131, 144, 153–154, 167, 170–171, 177–178, 182–183, 185, 196, 202, 213, 216, 219, 221–228, 231, 233, 236, 248, 257–258, 266, 268, 272–273, 275, 277–278
Chaco Canyon, 55–56, 167, 169, 177–178, 185, 258

Cibola, 56–57, 89, 170–171, 194, 215, 236
citation, 31, 49, 59, 73, 75, 100, 120, 136–137
clique, 103, 126–127, 146, 148, 284, 297
closeness. *See* centrality
cluster, 6, 32, 35, 46, 61, 70, 78, 83, 112, 114–116, 122, 139, 141, 143, 150, 159, 173, 196–199, 202–207, 209–210, 216, 220–223, 225, 229, 231, 246, 249–250, 254–255, 287–288
clustering coefficient, 114–115, 122, 146, 297, 307
co-authorship, 31, 49, 58–59, 79, 82, 123, 172–173
cohesive subgroup, 105, 126–129, 131, 145, 298, 302
color vision deficiency, 210
complete network. *See* network, types of
complex network. *See* network, types of
complex systems, 2, 19
complexity science, 16, 19, 24
component, 7, 9, 72, 87, 112, 115, 122–125, 147–149, 159, 173, 187, 204, 229, 231, 242, 248, 273–274, 284, 296, 298
correlation coefficient, 160–161, 172, 178, 286
critical network research, 263
crossroad, 30, 36, 241
cutpoint, 124–125, 148, 298
cyberSW project, 53, 55–56
cycle, 120–121, 267, 294, 298, 302

dark network. *See* network, types of
degree, 107–109, 111, 115–116, 128–129, 131, 135–136, 142, 144–145, 147, 176, 188, 245, 250, 284, 286–287, 294, 298, 308–309
degree distribution, 109, 116, 166, 188, 245, 251, 274, 299, 301, 306
Delaunay triangulation, 43, 248–250, 260, 299
dendrogram, 196, 208–209
dense network. *See* network, types of
density, 72, 89, 104, 107, 111, 115, 122, 126, 129, 132, 144, 147, 157, 206, 223, 226, 234, 245, 269, 287, 294, 304, 307
dependency, 9–10, 24, 65, 72, 78, 93, 98, 197, 265, 305
diameter, 121–123, 139, 147, 248, 260, 284, 295, 299, 301
digital elevation model, 45, 61
directed network. *See* network, types of
dissimilarity, 85
distance, 14, 30, 32, 36–38, 42–45, 56, 59, 61, 67, 71, 77–78, 84–85, 87–88, 116, 118, 121–123, 127–130, 138–139, 141, 147–148, 152, 165, 185–186, 202, 204, 206–208, 216–217, 223, 233, 237, 244–246, 248–251, 253, 255–259, 272–273, 277, 284, 297–298, 301, 303–304, 306–307
dyad, 65, 104, 112, 114, 126, 300

edge list, 67–68, 70–71, 74, 97–98, 101, 300
ego network. *See* network, types of
empty network. *See* network, types of
error, 45, 61, 97, 149–151, 155, 158, 167, 181, 184–189, 301, 309
  sampling, 153–154, 162, 164, 177–179, 306
  statistical, 152–153, 189, 307
  systematic, 151–153, 189, 308
Etowah mounds, 35
Euler, Leonhard, 16
event, 6, 70, 79–80, 150
exponential random graph models, 46, 96, 114, 117–118, 184

Florence, Italy, 2–3
food web, 47, 121, 237
force-directed algorithm, 206–207, 231, 287
Fruchterman-Reingold algorithm, 207, 221

Gabriel graph, 248, 256, 260, 295, 301
general additive model, 252, 257
general similarity network. *See* network, types of
generalized linear model, 257
geodesic, 121, 301
Girvan and Newman clustering, 131, 138, 221, 302

graph theory, 16–17, 24, 155, 240
gravity models, 30, 43, 184, 252

Hamming distance, 87
hierarchical network. *See* network, types of
Hohokam, 37, 50, 87
homophily, 117, 184
Hopi, 33
hub, 53, 107, 133–134, 136, 277, 294, 302

illicit network. *See* network, types of
imputation, 183
imputation method, 183–186, 190, 302
incidence matrix. *See* matrix
indegree, 107–108, 135–136, 251, 294, 302
Inka, 146
isotope, 38

k-core, 128, 131, 147, 302
K-nearest neighbor network. *See* network, types of

label, 69–70, 159, 176, 198–200, 208, 218, 220, 236, 287
Lambda set, 131, 302
line-of-sight network. *See* network, types of
longitudinal network. *See* network, types of
Louvain modularity, 59, 132, 147, 195–196, 198–199, 204, 223, 287, 303

Mahalanobis distance, 87
Manhattan distance, 85
mathematical notation, 5, 67
matrix, 40–41, 69–70, 81, 89, 97, 111, 178, 223, 292, 294, 300, 302
  adjacency, 69–70, 74, 77, 81, 85, 98, 101, 111, 197, 207–208, 217, 222–225, 235, 294, 298, 300, 303, 307, 310
  incidence, 70, 81, 83, 85, 87–88, 98, 102, 302
matrix algebra, 52, 81, 292
maximal subnetwork, 112, 127–128, 131, 302–304
maximum distance network. *See* network, types of
Maya, 32, 48, 52, 76–77, 95, 143–144
  inscriptions, 48, 76
  stelae, 143
Medicis, the, 2–3
Mesoamerica, 8
Mesopotamia, 37, 44, 252
missing data, 156, 158, 166, 173–174, 183, 189, 285–286
  edges, 126, 151, 160, 165, 173, 191, 222–223
  nodes, 151, 158–159, 162, 169, 172–173, 183, 188, 190, 286
modularity, 132, 205, 274, 303–304
Monte Carlo method. *See* resampling
Morisita Index, 87
Mousterian, 52
m-slices, 129, 131, 147, 303
multi-dimensional scaling, 204, 208, 288, 304
multilayer network. *See* network, types of
museological networks, 31, 50–51
museum, 50–51, 215
  Metropolitan Museum of Art, 51
  Pitt Rivers Museum, 51

n-clique, 112, 127–128, 304
neighborhood, 30, 40, 43, 248–249, 251, 255, 260
network data, xii, 5, 9, 13, 15, 17–18, 20–23, 27–28, 45–46, 52–53, 64, 66–67, 69–73, 75, 78, 88, 92, 97–100, 103–105, 117–118, 144–145, 149, 151, 153–154, 156, 162–163, 166, 175, 183, 190–191, 194, 196–197, 204, 208, 226, 230–231, 240–241, 257, 265, 269, 271, 277, 292, 294, 300, 302, 304
network science, 1–2, 4, 8–10, 12, 14–16, 20–26, 47, 53, 58–59, 67, 69–70, 72, 75, 91–92, 94, 96, 104, 106, 112, 116, 118–121, 124, 126–127, 131–132, 143, 145, 155, 225, 238, 243, 245, 259, 263–265, 270–271, 277–278, 301, 304
network structure, 6, 8–10, 17, 23, 37–38, 84, 93, 96, 122, 141, 146, 150, 154–155, 159, 161, 166, 193, 202, 229, 238, 248, 267, 274, 277–279, 288, 303–305

network theory, 7, 9, 14, 269
network, types of
  acyclic, 60, 75, 120, 242, 244, 294, 308
  affiliation, 80
  bi-directional, 75
  categorized, 77–79, 98, 296
  complete, 72–73, 131
  complex, 19, 216, 218–219, 221, 230, 293
  dark, 50, 80, 158, 174, 189
  dense, 70, 72–73, 126, 203, 235, 299
  directed, 47, 69, 73–75, 98, 100, 104, 107–109, 111, 113, 120, 135–137, 216, 251, 294, 297, 299–300, 302, 305
  ego, 89–90, 99, 300
  empty, 73, 98, 301
  general similarity, 29
  hierarchical, 120, 203, 208, 241, 302
  illicit, 31, 50, 158, 174
  K-nearest neighbor, 42, 250–251, 256, 260, 302
  line-of-sight, 45–46, 69, 93, 239
  longitudinal, 92, 94–96, 99, 226, 303
  maximum distance, 251, 255, 257
  multilayer, 91–92, 304
  one-mode, 46, 72, 81, 83–84, 304
  ontological, 50
  road, 12, 37–39, 52–53, 72–75, 91, 107, 119–123, 126–129, 138–139, 141–143, 146, 191, 253, 262
  scale-free, 109, 146, 306
  signed, 75–76, 79, 92, 98, 296
  similarity, 57, 84–85, 87–88, 123, 257, 307
  simple, 72–73, 75, 98, 160, 195, 197, 203–204, 210, 213, 217, 223, 226, 307
  simulated, 34
  small-world, 19, 122, 277, 307
  sparse, 72–73, 307
  spatial, 22, 39, 45–46, 58, 62, 124, 129, 139, 160, 202, 237–241, 243–247, 252, 254–257, 259–260, 307
  total, 155–156, 160, 166–167, 308
  two-mode, 44, 51, 71, 79, 81, 83–84, 98, 101, 213, 236, 293, 304, 309
  undirected, 69, 74, 108, 111–112, 135, 216, 236, 287, 299
  uni-directional, 74, 98
  valued, 77–78, 131
  weighted, 77–79, 85, 88, 92, 98, 104, 107–108, 129, 131–132, 167, 295, 299–300, 303, 309
New Zealand. *See* Aotearoa
node variable, 64–66, 98, 100, 305

obsidian, 8, 10, 32, 44, 83–84, 95, 162, 272
Oceania, 17, 42
one-mode network. *See* network, types of
one-mode projection, 81, 83–85, 98, 102, 283
ontological network. *See* network, types of
ORBIS model, 37–38
Ostia, 37, 40
outdegree, 107–108, 135–137, 251, 302, 305

path, 9, 13, 21, 30, 36–40, 42, 46, 62, 66, 72–73, 103–105, 118–120, 122–124, 133, 135, 137–139, 141, 145, 161, 196, 200, 202, 216–217, 219, 225, 239–240, 242, 245, 248, 296, 298, 303, 306
  shortest path, 46, 121–123, 138, 147, 170, 198, 245–246, 295, 297, 299, 301, 306–307
Pompeii, 12, 14
pottery. *See* ceramics
power law, 109–110, 306
precision, 152–153, 189, 307
probability distribution, 109, 185
Proximal Point Analysis, 30, 42, 250

reachability, 19, 39, 78, 122, 124, 126–128, 145, 147, 241, 277, 284, 298, 300, 306
relational perspective, 6–7, 24, 51, 264, 306
relative neighborhood graph, 306
resampling, 89, 149, 151, 153, 162–163, 165–166, 172, 174, 177–179, 183, 186, 189–191, 285–286, 296
  bootstrapping, 163, 172, 177, 179, 181, 189, 296
  Monte Carlo, 163, 189, 304

road network. *See* network, types of
Roman Empire, 7, 37, 42, 54, 276
  Roman roads, 23, 37, 52–53, 65, 107, 119, 121–123, 126–127, 129–130, 139, 142–143, 201, 262, 277
  Romans, 12, 14, 53, 128, 131
sampling, 22, 141, 149, 153, 155–156, 162–166, 168, 170, 172–173, 178, 181, 189, 286, 296
  biased, 156–158, 160, 172, 190–191, 285, 296, 306
  random, 156–158, 166–168, 170, 172, 174, 178, 191, 286, 296, 306

scale-free network. *See* network, types of
seriation, 51–52, 87, 204, 223
signed network. *See* network, types of
similarity, 29–30, 32–35, 43, 47, 51, 53–54, 56, 66, 72–73, 78–79, 84–85, 87–89, 96, 106, 118, 123, 126, 131–132, 135, 147, 167, 170–171, 175, 177, 182, 196–197, 202, 208–209, 213, 216–219, 221–223, 226–228, 231, 233–234, 236, 239, 257–258, 266–268, 272–273, 275, 278–279, 287, 307
similarity network. *See* network, types of
simple network. *See* network, types of
simulated network. *See* networks, types of
small-world network. *See* network, types of
social network analysis, 1, 16, 18, 20, 24, 80, 105, 144, 163, 239, 278, 292–294
social relationship, 1, 7, 80, 106, 240, 277
social system. *See* social relationship
sociogram, 194
sociometry, 18, 193
Southwest Social Networks Project, 18, 53, 87, 154, 167, 177, 197, 200, 231–233, 254, 256, 266
space syntax, 39–40, 47, 240, 259
sparse network. *See* network, types of
spatial analysis, 15
spatial network. *See* network, types of
Spearman's rank correlation, 161, 169, 172, 178, 192, 286
Spearman's rho, 160–161, 168
strength of weak ties, 77, 267
structural variable, 64, 66, 71, 79, 81, 98, 100, 300, 304, 308–309
subnetwork, 71, 90, 112–115, 119, 122–123, 126–128, 130–131, 145, 302–303, 308–309

Thiessen polygon. *See* Voronoi diagram
threshold, 43, 88–89, 206, 217, 251, 257, 260, 285, 303
timeline, 228, 235
total network. *See* network, types of
total viewshed, 46
trail network, 160
transitivity, 112, 114–115, 308
tree, 121, 208, 221, 241, 244, 248–249, 260, 302–303, 308
triad, 76, 112, 114–115, 126, 141, 147, 184, 268, 297, 305, 308
triad census, 114, 309
two-mode network. *See* network, types of

uncertainty, 12, 22, 66, 149–151, 162–163, 165, 177, 181–182, 185–189, 274, 309
undirected network. *See* network, types of
uni-directional network. *See* network, types of
unreachable. *See* reachability

valued network. *See* network, types of
variable, 64, 66, 161, 211, 245, 259, 272, 300
vertex, 4–5, 128, 291, 302, 309–310
visualization, 5–6, 8, 18, 22–23, 27, 42, 50–51, 64, 70, 72, 84, 88–89, 168–169, 235, 240, 260, 266, 270, 287–288, 291–293, 304
  color, 210–212, 214–217, 220, 223, 227–228, 230–231, 234, 236, 287–288, 292
  scaling, 131, 211, 231
Voronoi diagram, 43, 247, 250, 260, 299, 308–309

walk, 118–121, 132, 157, 298, 306, 308–309
walktrap clustering, 132, 309
weighted network. *See* network, types of

Zuni. *See* A:shiwi